

SingleRAN

Green Site Feature Parameter Description

Issue	Draft B
Date	2026-01-30



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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

Technical Changes

None

Editorial Changes

Revised MML command examples for iSDU06D/iSDU03D support for connection with a third-party power system. For details, see [17.1.4.1.2 Using MML Commands](#).

1.2 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 07 (2025-12-14).

Technical Changes

Change Description	Parameter Change	Base Station Model
Added voltage-based port-level hierarchical power-off. For details, see: • 14.1 Principles • 14.3.3 Hardware • 14.3.4 Others • 14.4 Operation and Maintenance (eGBTS/NodeB/eNodeB/gNodeB)	Added parameters: • PDPATTR.CUSLOADSH UTDOWNMODE (5G gNodeB, LTE eNodeB) • PDPATTR.CUSLOADSH UTDOWNVOLTAGE (5G gNodeB, LTE eNodeB)	3900 and 5900 series base stations (macro base stations)
Added applicable scenarios of SitePower Backup Policy Management. For details, see: • 16.3.3 Hardware • 16.3.4 Others • 16.4.1.1 Data Preparation • 16.4.1.2-Using MML Commands • 16.4.2 Activation Verification	None	3900 and 5900 series base stations (macro base stations)
Modified the impact relationships of low power consumption mode (LTE) and SOC level-based low power consumption mode with CMAS. For details, see 11.3.2.3 Function Impacts and 13.3.2.3 Function Impacts.	None	3900 and 5900 series base stations (macro base stations)
Modified the impact relationships of low power consumption mode (LTE) and SOC level-based low power consumption mode with ETWS. For details, see 11.3.2.3 Function Impacts and 13.3.2.3 Function Impacts.	None	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	Base Station Model
Added coordinated power backup of the entire site in combination of the base station and microwave. For details, see 23 Coordinated Power Backup of the Entire Site in Combination of the Base Station and Microwave .	Modified parameter: Added the BKP_PSAV_EVALUATE_DATA_SW option to the LLDPGLOBAL.BTSMWCOS W (5G gNodeB, LTE eNodeB) parameter.	3900 and 5900 series base stations (macro base stations)
Added compatibility of SOC level-based low power consumption mode with 16T16R. For details, see 13.3.4 Cells .	None	3900 and 5900 series base stations (macro base stations)
Modified the function switch for LTE-NR multi-dimensional scheduling optimization. For details, see 11.3.2.3 Function Impacts and 13.3.2.3 Function Impacts .	None	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between L/NR Inter-band Deep Coordination and low power consumption mode (NR). For details, see 12.3.2.3 Function Impacts .	None	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	Base Station Model
Added mutually exclusive relationships of iLiBattScan with the following functions: PSU intelligent shutdown, PSU intelligent coordinated energy saving, automatic PSU voltage optimization, automatic voltage optimization for intelligent lithium batteries, staggering electricity usage, and SOC level-based low power consumption mode. For details, see 4.3.2.2 Mutually Exclusive Functions , 5.3.2.2 Mutually Exclusive Functions , 6.3.2.2 Mutually Exclusive Functions , 7.3.2.2 Mutually Exclusive Functions , 8.3.2.2 Mutually Exclusive Functions , and 13.3.2.2 Mutually Exclusive Functions .	None	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between iLiBattScan and Power Backup Perception (trial). For details, see 19.3.2.3 Function Impacts .	None	3900 and 5900 series base stations (macro base stations)
Added an impact relationship of NCR cells with SOC level-based low power consumption mode. For details, see 13.3.2.3 Function Impacts .	None	3900 and 5900 series base stations (macro base stations)
Added impact relationships of SOC level-based low power consumption mode with RF module deep dormancy, RF module fast deep dormancy, and RF module super dormancy. For details, see 13.3.2.3 Function Impacts .	None	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	Base Station Model
Added an impact relationship between RF module fast deep dormancy and low power consumption mode (LTE). For details, see 11.3.2.3 Function Impacts .	None	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between UMTS Static Split Beam and UMTS SOC level-based low power consumption mode. For details, see 13.3.2.3 Function Impacts .	None	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between RF channel dynamic muting in the presence of TM9 UEs and low power consumption mode (LTE). For details, see 11.3.2.3 Function Impacts .	None	3900 and 5900 series base stations (macro base stations)
Put iSDU06D/iSDU03D used with a third-party power system into commercial use. For details, see 15.3.3 Hardware and 17 iSDU06D/iSDU03D Support for Connection with a Third-Party Power System .	None	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between "cell RX channel interference and noise query" and low power consumption mode (LTE). For details, see 11.3.2.3 Function Impacts .	None	3900 and 5900 series base stations (macro base stations)
Added impact relationships of low power consumption mode and SOC level-based low power consumption mode with LNR CRS muting. For details, see 11.3.2.3 Function Impacts and 13.3.2.3 Function Impacts .	None	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	Base Station Model
Optimized the DSP PSU command output after PSU dormancy takes effect. For details, see 4.4.2 Activation Verification and 5.4.2 Activation Verification .	None	3900 and 5900 series base stations (macro base stations)
Added support for base station hierarchical power-off and intelligent coordinated voltage boosting power distribution by the iSDU03T. For details, see 3 Overview , 14 Base Station Hierarchical Power-Off , and 21 iSDU03S/iSDU03T Intelligent Coordinated Voltage Boosting Power Distribution .	None	3900 and 5900 series base stations (macro base stations)
Added automatic voltage optimization for intelligent lithium batteries. For details, see 3 Overview and 7 Automatic Voltage Optimization for Intelligent Lithium Batteries .	None	3900 and 5900 series base stations (macro base stations)
Added support for PSU intelligent coordinated energy saving by MTS series cabinets. For details, see: 5.1 Principles 5.3.3 Hardware 5.3.4 Others 5.4.1.1 Data Preparation 5.4.1.2 Using MML Commands 5.4.2 Activation Verification	Added parameters: IPWRM.LRPSW (5G gNodeB, LTE eNodeB) IPWRM.MINLR (5G gNodeB, LTE eNodeB) IPWRM.MAXLR (5G gNodeB, LTE eNodeB)	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	Base Station Model
Added requirements on ICC series power systems. For details, see: • 4.3.4 Others • 5.1 Principles • 5.3.3 Hardware • 9.1 Principles • 9.3.3 Hardware • 9.3.4 Others • 14.1.1 Working Principles	None	3900 and 5900 series base stations (macro base stations)
Added support for the precise backup power saving policy by third-party power systems. For details, see: • 11.1.3.1 Power Saving Policy • 11.1.3.2 Power Saving Measure • 11.2.2 Impacts • 11.3.3 Hardware • 12.1 Principles • 12.2.2 Impacts • 12.3.3 Hardware	None	3900 and 5900 series base stations (macro base stations)
Optimized recovery of the automatically recognized topology in the case of iSDU power-on reset. For details, see 22.1 Principles.	None	3900 and 5900 series base stations (macro base stations)

Editorial Changes

Added descriptions of the impact relationship between inter-cell cable sequence detection and low power consumption mode (LTE). For details, see [11.3.2.3 Function Impacts](#).

Added descriptions of the impact relationship between RF channel dynamic muting in TM4 and low power consumption mode. For details, see [11.3.2.3 Function Impacts](#).

Revised descriptions in [3 Overview](#).

Revised descriptions of the iSDU06D/03D topology recognition function. For details, see [22 iSDU06D/iSDU03D Topology Recognition](#).

Added descriptions of the impact relationship between UMTS SOC level-based low power consumption mode and RuralLink antenna solution (UMTS). For details, see [13.3.2.3 Function Impacts](#).

Added suggestions for parameter settings in [10.4.1.1 Data Preparation](#) and [15.4.1.1 Data Preparation](#).

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

Trial Features

Trial features are features that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these features can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial features shall contact Huawei and enter into a memorandum of understanding

(MoU) with Huawei prior to an official application of such trial features. Trial features are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial features may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial features but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial features are free, Huawei is not liable for any trial feature malfunctions or any losses incurred by using the trial features. Huawei does not promise that problems with trial features will be resolved in the current version. Huawei reserves the rights to convert trial features into commercial features in later R/C versions. If trial features are converted into commercial features in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial features. If a customer fails to purchase such a license, the trial feature(s) will be invalidated automatically when the product is upgraded.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and NR.

2.3 Features in This Document

This document describes the following features.

RAT	Feature ID	Feature Name	Chapter/Section
GSM	GBFD-111608	PSU Smart Control	4 PSU Intelligent Shutdown
UMTS	WRFD-020121	Intelligent Power Management	
LTE FDD	LOFD-001056	PSU Intelligent Sleep Mode	
LTE TDD	TDLOFD-001056	PSU Intelligent Sleep Mode	
NR	FBFD-031003	PSU Intelligent Shutdown	4 PSU Intelligent Shutdown
GSM	MRFD-191201	Intelligent Coordinated PSU Energy Saving (GSM)	5 PSU Intelligent Coordinated Energy Saving
UMTS	MRFD-191211	Intelligent Coordinated PSU Energy Saving (UMTS)	6 Automatic PSU Voltage Optimization

RAT	Feature ID	Feature Name	Chapter/Section
LTE FDD	MRFD-191221	Intelligent Coordinated PSU Energy Saving (LTE FDD)	
LTE TDD	MRFD-191231	Intelligent Coordinated PSU Energy Saving (LTE TDD)	
NR	MRFD-191261	Intelligent Coordinated PSU Energy Saving (NR)	
GSM	MRFD-191202	Intelligent Peak Staggering (GSM)	8 Staggering Electricity Usage
UMTS	MRFD-191212	Intelligent Peak Staggering (UMTS)	
LTE FDD	MRFD-191222	Intelligent Peak Staggering (LTE FDD)	
LTE TDD	MRFD-191232	Intelligent Peak Staggering (LTE TDD)	
NR	MRFD-191262	Intelligent Peak Staggering (NR)	
NR	FOFD-021203	gNodeB Power Saving	12 Low Power Consumption Mode (NR)
LTE FDD	LOFD-210202	LTE SOC Level-Based Low Power Consumption Mode	13 SOC Level-based Low Power Consumption Mode
LTE TDD	TDLOFD-210612	LTE SOC Level-Based Low Power Consumption Mode	
UMTS	WRFD-270201	UMTS SOC Level-Based Low Power Consumption Mode	
GSM	GBFD-270201	GSM SOC Level-Based Low Power Consumption Mode	
GSM	GBFD-111601	BTS Power Management	14 Base Station Hierarchical Power-Off
UMTS	WRFD-031400	Power off the equipment level by level	

2.4 Differences

Differences Between LTE FDD and LTE TDD

Function Name	Difference	Chapter/Section
PSU intelligent shutdown	None	4 PSU Intelligent Shutdown
PSU intelligent coordinated energy saving	None	5 PSU Intelligent Coordinated Energy Saving
Automatic PSU voltage optimization	None	6 Automatic PSU Voltage Optimization
Staggering electricity usage	None	8 Staggering Electricity Usage
Low power consumption mode (LTE)	None	11 Low Power Consumption Mode (LTE)
SOC level-based low power consumption mode	None	13 SOC Level-based Low Power Consumption Mode

Differences Between NR FDD and NR TDD

Function Name	Difference	Chapter/Section
PSU intelligent shutdown	None	4 PSU Intelligent Shutdown
PSU intelligent coordinated energy saving	None	5 PSU Intelligent Coordinated Energy Saving
Automatic PSU voltage optimization	None	6 Automatic PSU Voltage Optimization
Staggering electricity usage	None	8 Staggering Electricity Usage
Low power consumption mode (NR)	None	12 Low Power Consumption Mode (NR)

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
PSU intelligent shutdown	None	4 PSU Intelligent Shutdown
PSU intelligent coordinated energy saving	None	5 PSU Intelligent Coordinated Energy Saving
Automatic PSU voltage optimization	None	6 Automatic PSU Voltage Optimization
Staggering electricity usage	None	8 Staggering Electricity Usage
Low power consumption mode (NR)	None	12 Low Power Consumption Mode (NR)

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
PSU intelligent shutdown	Supported in both high and low frequency bands, with the following differences: In low frequency bands, this function works in both SA and NSA networking. In high frequency bands, this function works only in NSA networking and in FWA scenarios in SA networking.	4 PSU Intelligent Shutdown

Function Name	Difference	Chapter/Section
PSU intelligent coordinated energy saving	Supported in both high and low frequency bands, with the following differences: In low frequency bands, this function works in both SA and NSA networking. In high frequency bands, this function works only in NSA networking and in FWA scenarios in SA networking.	5 PSU Intelligent Coordinated Energy Saving
Automatic PSU voltage optimization	Supported in both high and low frequency bands, with the following differences: In low frequency bands, this function works in both SA and NSA networking. In high frequency bands, this function works only in NSA networking and in FWA scenarios in SA networking.	6 Automatic PSU Voltage Optimization
Staggering electricity usage	None	8 Staggering Electricity Usage
Low power consumption mode (NR)	Supported only in low frequency bands	12 Low Power Consumption Mode (NR)

3 Overview

With continuous development of mobile communication technologies, multi-band and multi-RAT networks will coexist for a long time. While embracing green development, operators are faced with the following challenges on their wireless networks: an increase in energy consumption caused by rapid traffic growth and lack of a complete system for evaluating network energy efficiency.

Huawei Wireless Network provides multiple site-level intelligent features and advances green evolution of 5G to help operators reduce network energy consumption while meeting network development requirements, lead a systematic evaluation, and guide green development of networks.

These features are described in the following table.

NOTE

This document only describes site-level energy saving features. For more energy saving features, see related documents in the feature documentation of the corresponding RAT, for example, *Energy Conservation and Emission Reduction in 5G RAN Feature Documentation*.

Table 3-1 Energy-saving power supply

Category	Feature/ Function Name	Chapter/ Section	Feature Introduction
PSU shutdown for energy saving	PSU intelligent shutdown (GSM/NR) Intelligent Power Management (UMTS) PSU Intelligent Sleep Mode (LTE)	4 PSU Intelligent Shutdown	When the network is lightly loaded, the base station dynamically shuts down PSUs and accommodates the network load with a minimum number of PSUs. This enables PSUs to maintain high power conversion efficiency, reducing the energy consumption of the base station and prolonging the PSU lifetime.
PSU shutdown for energy saving	Intelligent Coordinated PSU Energy Saving (GSM/UMTS/LTE/NR)	5 PSU Intelligent Coordinated Energy Saving	With this function, the base station measures the total power consumption in real time, calculates the number of working PSUs required to retain the power system at the maximum conversion efficiency, and dynamically adjusts the number of working PSUs, to ensure that the conversion efficiency of each working PSU is within the optimal efficiency range.
Automatic voltage optimization	Intelligent Coordinated PSU Energy Saving (GSM/UMTS/LTE/NR)	6 Automatic PSU Voltage Optimization	The power system automatically identifies the backup battery type and PSU type, and automatically adjusts the PSU busbar voltage to -57 V DC if certain requirements are met. In this case, the current on the power cables is reduced through a voltage increase to reduce power loss on power cables.
Automatic voltage optimization	Automatic voltage optimization for intelligent lithium batteries (GSM/UMTS/LTE/NR)	7 Automatic Voltage Optimization for Intelligent Lithium Batteries	In scenarios where a third-party power system is used with Huawei intelligent lithium batteries, when the mains supply is disconnected, the base station used with the PMU 15A(Ver.B) in the power cabinet can keep the voltage of those batteries at a relatively high level through the automatic voltage optimization for intelligent lithium batteries function, thereby finding the optimal voltage and reducing power loss during cable transmission.

Category	Feature/ Function Name	Chapter/ Section	Feature Introduction
Power supply mode-based energy saving	Intelligent dynamic peak staggering (GSM/UMTS/LTE/NR)	8 Staggering Electricity Usage	In scenarios where the electricity price varies depending on peak and off-peak hours, lithium batteries are charged during off-peak hours; during peak hours, the output voltage of the power system is decreased and lithium batteries are used to supply power, reducing electricity costs.
Power supply mode-based energy saving	PV deployment for base stations (GSM/UMTS/LTE/NR)	9 PV Deployment for Base Stations	Solar supply units (SSUs) can now be used in addition to the existing PSUs and batteries to provide power supply. With such a photovoltaics (PV) deployment function, the SSUs are preferentially used when there is sunlight. If the generated power is insufficient, the SSUs and PSUs work together to supply power when the mains supply is available, and the SSUs and batteries work together to supply power when the mains supply is unavailable.
Power supply mode-based energy saving	Third-party green power access measurement and O&M management (GSM/UMTS/LTE/NR)	10 Third-Party Green Power Access Measurement and O&M Management	When the third-party green power source is photovoltaic power, the green power input unit (GIU) can be used to perform intelligent photovoltaics (IPV) management, including PV-related energy yield measurement, basic alarm management, and O&M management.
iSDU functions	iSDU06D/iSDU03D support for connection with a third-party power system (GSM/UMTS/LTE/NR)	17 iSDU06D/iSDU03D Support for Connection with a Third-Party Power System	A third-party power system is connected to the iSDU06D/iSDU03D for protocol processing and conversion, and then connected to a wireless BBU through the iSDU06D/iSDU03D for component management.

Category	Feature/ Function Name	Chapter/ Section	Feature Introduction
iSDU functions	iSDU03D circuit breaker capacity adjustme nt	18 iSDU03D Circuit Breaker Capacity Adjustme nt	The iSDU03D can adjust the circuit breaker capacity of the intelligent power distribution ports LOAD0 to LOAD2. In this case, it can supply power to RF modules regardless of whether their power consumption is greater than 800 W. The circuit breaker capacity can be specified based on the power consumption of RF modules connected to the intelligent power distribution ports.
iSDU functions	iSDU03S/ iSDU03T intelligent coordinat ed voltage boosting power distributio n	21 iSDU03S/ iSDU03T Intelligen t Coordina ted Voltage Boosting Power Distributi on	iSDU03S/iSDU03T intelligent coordinated voltage boosting power distribution enables intelligent power distribution ports of an iSDU03S/iSDU03T to perform voltage adjustment separately, and provides different power supply policies based on the capabilities of the RF modules connected to these ports.
iSDU functions	iSDU06D/ iSDU03D topology recognitio n	22 iSDU06D/ iSDU03D Topology Recogniti on	This function enables the iSDU03D or iSDU06D to perform topology pairing with the help of topology recognition. Topology pairing hereby refers to the pairing between power distribution ports on the iSDU and connected RF modules in terms of power supply.

Table 3-2 Backup power saving

Category	Feature/ Function Name	Chapter/ Section	Feature Introduction
Low power consumption mode	Low power consumption mode (LTE)	11 Low Power Consumption Mode (LTE)	In the scenario where both a power supply system and batteries are configured, the available duration of backup power for basic coverage services will be impacted if the power supply system becomes faulty while the deployed batteries are insufficient. Low power consumption mode can be used to extend the duration of basic coverage service provisioning in this scenario. When an alarm indicating power supply insufficiency or failures is reported, cells with high power consumption enter low power consumption mode to reduce the power consumed by RF modules.
Low power consumption mode	Low power consumption mode (NR)	12 Low Power Consumption Mode (NR)	In the scenario where both a power supply system and batteries are configured, the available duration of backup power for basic coverage services will be impacted if the power supply system becomes faulty while the deployed batteries are insufficient. Low power consumption mode can be used to extend the duration of basic coverage service provisioning in this scenario. When an alarm indicating power supply insufficiency or failures is reported, cells with high power consumption enter low power consumption mode to reduce the power consumed by RF modules.
Low power consumption mode	SOC level-based low power consumption mode	13 SOC Level-based Low Power Consumption Mode	When the remaining backup power is insufficient, SOC level-based low power consumption mode enables cells to shut down carriers and enter the low power consumption mode. This reduces the power consumption of RF modules and extends the service time of the base station.

Category	Feature/ Function Name	Chapter/ Section	Feature Introduction
Backup power management for energy saving	BTS Power Management (GSM) Power off the equipment level by level (UMTS)	14 Base Station Hierarchical Power-Off	Hierarchical power-off allows a base station to shut down the RF modules and BBU level by level in the event of mains power failures, increasing the service time of storage batteries and the battery life.
Backup power management for energy saving	Adaptive Power Backup (GSM/UMTS/LTE/NR)	15 Adaptive Power Backup	The Adaptive Power Backup feature is a power backup solution tailored for the entire site for the sake of the lossless running duration and target backup power use duration. When the backup power configuration cannot meet the target backup power use duration, the Adaptive Power Backup feature shuts down non-primary cell-level services at a proper time based on the primary cell-level services set by the operator, historical power consumption, and service load trend. In this way, the target backup power use duration can be ensured for primary services.
Backup power management for energy saving	SitePower Backup Policy Management	16 SitePower Backup Policy Management	SitePower Backup Policy Management provides an intelligent power backup solution. When the backup power configuration cannot meet the target backup power use duration, this solution shuts down low-priority services at a proper time through module power-off or carrier shutdown based on the service priorities defined by the operator, historical power consumption, service load trend, and remaining battery capacity. In this way, the service time of sites is extended and high-priority services can operate for the target backup power use duration with a small amount of or no additional backup power investment.

Category	Feature/ Function Name	Chapter/ Section	Feature Introduction
Backup power management for energy saving	Power Backup Perception (GSM/UMTS/LTE/NR)	19 Power Backup Perception (Trial)	Power Backup Perception predicts the backup power use duration of a site based on its power backup capability, real-time power consumption, and estimated power consumption. In addition, thresholds are configured based on the remaining backup power use duration to present sites with potential risks to customers. Customers can arrange site visits for O&M in advance to prevent site disconnection.
Coordination between base stations and other products	Coordinated power backup of the entire site in combination of the base station and microwave	23 Coordinated Power Backup of the Entire Site in Combination of the Base Station and Microwave	If the mains supply is disconnected and the base station enters the backup power saving state, the microwave device works with the base station so that the power saving status of the base station can be transmitted to the microwave device. In this way, the microwave device also enters the power saving state, increasing the backup power use duration of the entire site.

Table 3-3 Other energy saving functions

Category	Feature/ Function Name	Chapter/ Section	Feature Introduction
Energy-efficient temperature control	PMU support for associated temperature control of battery cabinets (GSM/UMTS/LTE/NR)	20 PMU Support for Associated Temperature Control of Battery Cabinets	This function controls air conditioner cooling of the air-conditioned battery cabinet based on the temperature and service status to save energy.

4 PSU Intelligent Shutdown

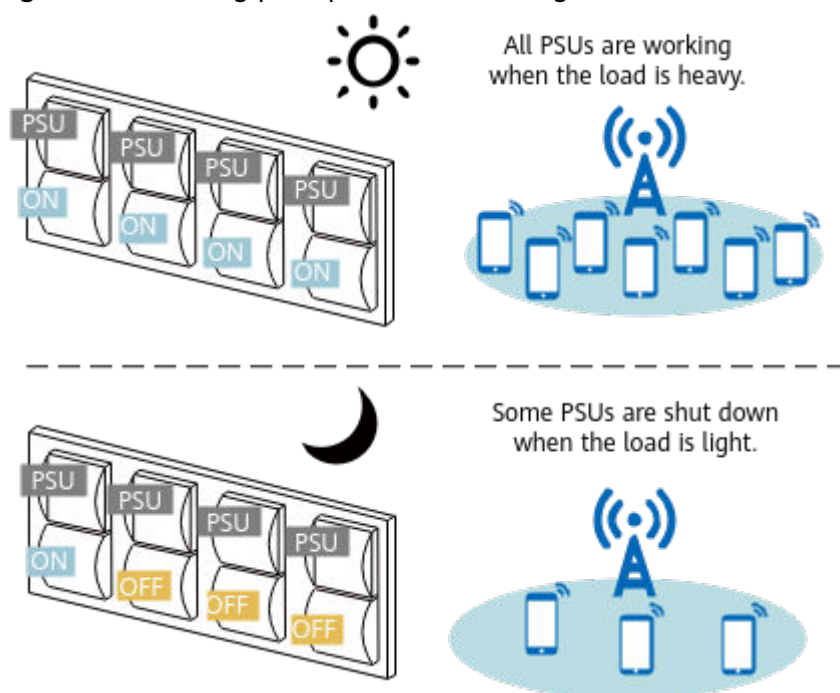
4.1 Principles

A PSU converts 110 V or 220 V alternating current (AC) power into –48 V direct current (DC) power. Generally, an AC-powered base station is configured with multiple PSUs. The number of PSUs required is determined based on the maximum energy consumption of the base station. This ensures that the base station operates properly, even when it is fully loaded.

However, a base station seldom operates at its full load. As a result, the PSUs usually work at partial output power. The power conversion efficiency of PSUs is directly proportional to their output power, meaning that efficiency decreases as output power reduces, increasing the energy consumption of the base station.

The PSU intelligent shutdown function is introduced to maintain proper operation of the base station at reduced energy consumption by enabling PSUs to enter or exit the dormancy state. [Figure 4-1](#) shows its working principles.

Figure 4-1 Working principles of PSU intelligent shutdown



PSU intelligent shutdown can be enabled by turning on the switch specified by the **PSUIS.PSUISS** parameter. If a base station is configured with multiple PSUs for power supply:

- When the network is lightly loaded, the base station dynamically shuts down one or more PSUs and accommodates the network load with a minimum number of PSUs. This enables PSUs to maintain high power conversion efficiency, reducing the energy consumption of the base station and prolonging the PSU lifetime.
- When the network load becomes heavy, the base station dynamically activates one or more PSUs to ensure its proper operation.

4.2 Network Analysis

4.2.1 Benefits

When a base station is configured with multiple PSUs for power supply and the network load is light, PSU intelligent shutdown reduces the base station energy consumption and prolongs the PSU lifetime.

The energy reduced by this function is closely related to the difference in the PSU power conversion efficiency before and after this function is activated. Assuming that the output power remains unchanged, and the PSU power conversion efficiencies before and after the function activation are C_0 and C_1 , respectively, energy is reduced by $(1 - C_0/C_1)$.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

This function has no impact on the network.

4.3 Requirements

4.3.1 Licenses

RAT	Feature ID	Feature Name	Model	Sales Unit
GS M	GBFD-111608	PSU Smart Control	LGMIDSPSURE S (control item ID)	per TRX
UM TS	WRFD-020121	Intelligent Power Management	LQW9PSUISS01 (control item ID)	per NodeB
LTE FD D	LOFD-001056	PSU Intelligent Sleep Mode	LT1S00PISM00 (SBOM)	per Cell
LTE TD D	TDLOFD-0010 56	PSU Intelligent Sleep Mode	LT1STPSUSM00 (SBOM)	per Cell
NR	FBFD-031003	PSU Intelligent Shutdown	None	N/A

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
GSM UMTS LTE FDD LTE TDD NR	PSU intelligent coordinated energy saving	IPWRM. <i>PSUIC</i> ESS	5 PSU Intelligent Coordinated Energy Saving	- In co-MPT scenarios, PSU intelligent shutdown cannot be enabled together with PSU intelligent coordinated energy saving. That is, if PSU intelligent coordinated energy saving is enabled for any RAT of GSM, UMTS, LTE, and NR, PSU intelligent shutdown cannot be deployed for NR at the same time. On the contrary, if PSU intelligent shutdown is enabled for any RAT of GSM, UMTS, LTE, and NR, PSU intelligent coordinated energy saving cannot be deployed for NR at the same time. - In separate-MPT scenarios, if PSU intelligent shutdown and PSU intelligent coordinated energy saving are enabled on

RAT	Function Name	Function Switch	Reference	Description
				two main control boards separately, only PSU intelligent coordinated energy saving takes effect.
NR LTE FDD LTE TDD UMTS GSM	iLiBattScan	BATCTPA.ILIBATT SCANSW (5G gNodeB, LTE eNodeB)	<i>Power Supply Management</i>	None

4.3.2.3 Function Impacts

None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

No requirements

RF Modules

No requirements

4.3.4 Others

- The cabinet of a base station is configured with a power monitoring unit (PMU), at least two PSUs, and storage batteries. The storage batteries are properly configured and are not in equalized charging mode. No battery alarms are generated.
- The base station is AC powered. The mains supply is normal. A Huawei PMU that supports this function is used to manage the power system. [Table 4-1](#) describes requirements of PSU intelligent shutdown on the PMU.

Table 4-1 Requirements of PSU intelligent shutdown on the PMU

Power System Type	PMU Model	PMU Software Version
ETP	PMU 11A	No requirements
	PMU 11B	No requirements
	APM5930 (CCUC)	No requirements
	APM5950H/APM5950H-L (PMU 11C)	No requirements
	OPM200 (PMU 16A)	No requirements
	OPM200-02/OPM200-02 (Ver.B) (PMU 16B)	No requirements
	CCUB	No requirements
	PMU 15A	Not supported
	OPM50M	Not supported
	OPM30M	Not supported
	OPM30A	Not supported
APM30	PMU 01B	No requirements
	PMU 11A	No requirements
	PMU 11B	No requirements
	CCUB	No requirements
	CCUC	No requirements
	EPMU01	138 or later
EPS4890	PMU 01B	No requirements
	PMU 11A	No requirements
	PMU 11B	No requirements
	EPMU01	138 or later
EPS4815	PMU 11A	No requirements
	PMU 11B	No requirements
	EPMU03	Not supported
EPU	None	Not supported
SC48200	None	Not supported

Power System Type	PMU Model	PMU Software Version
SMU	MTS series or ICC series power systems, with the SMU02C (02312MML-009) serving as the PMU	Not supported

NOTE

You can query the software version of the PMU by selecting the corresponding gNodeB in the MML command window of the MAE-Access and running the **DSP BRDVER** command.

For the EPMU01:

- If the software version is 138 or later, you do not need to re-activate the software.
- If the software version is earlier than 138, run the **ACT SOFTWARE** command to activate the software.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

[Table 4-2](#) describes the parameters used for function activation. Function optimization configurations are not involved.

Table 4-2 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Energy Consumption Measurement Point	ENERGYCON.MP	Set this parameter to AC .
PSU Intelligent Shutdown Switch	PSUIS.PSUISS	It is recommended that this switch be turned on when a base station configured with multiple PSUs is lightly loaded.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual

network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Setting the energy consumption measurement point to AC  
SET ECMP: MP=AC;  
//Enabling PSU intelligent shutdown  
SET PSUISS: PSUISS=ENABLE;
```

Optimization Command Examples

N/A

Deactivation Command Examples

```
//Disabling PSU intelligent shutdown  
SET PSUISS: PSUISS=DISABLE;
```

NOTE

In separate-MPT scenarios, PSU intelligent shutdown can be deactivated for the entire base station only when this switch is turned off for all main control boards on which this function has been enabled.

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

- Using MML commands
 - a. Run the **DSP BATTERY** command to query the status of storage batteries.
 - b. Run the **LST PSU** command to verify that at least two PSUs are configured.
 - c. Run the **LST PSUISS** command to verify that the value of the **PSU Intelligent Shutdown Switch** parameter is **Enable**.
 - d. Run the **DSP PMU** command to verify that the value of the **PSU Intelligent Shutdown Switch** parameter is **Enable**.
 - e. Run the **DSP PSU** command. If the value of the **Running Status** parameter is **Dormancy** for a PSU, PSU intelligent shutdown has taken effect.

Alternatively, check the PSU indicators locally to determine its working status. If a PSU is in the shutdown state, PSU intelligent shutdown has taken effect. For details about the indicator status, see *PSU* in *3900 & 5900 Series Base Station Product Documentation*.

 NOTE

In separate-MPT scenarios, **PSU Intel Coordinated Energy Saving Status = Enable** is displayed in the output of the **DSP PMU** command executed on the main control board with the function disabled if the following conditions are met: (1) Two main control boards are configured with one PMU. (2) PSU intelligent shutdown or PSU intelligent coordinated energy saving is enabled on one main control board but disabled on the other main control board. (3) The function has taken effect on the entire base station.

- Using performance counters

Check the PSU shutdown duration through the **VS.PSU.ShutdownTime** counter on the MAE-Access to determine whether PSU intelligent shutdown has taken effect.

- If the value of this counter is not 0, PSU intelligent shutdown has taken effect. If the value of this counter is greater than the measurement period multiplied by a factor of N minus 1 and less than or equal to the measurement period multiplied by N , the number of PSUs that are shut down is N , which is greater than or equal to 1.
- If the value of this counter is 0, PSU intelligent shutdown has not taken effect.

4.4.3 Network Monitoring

NR

On the live network, the one-week PSU input and output power obtained before and after PSU intelligent shutdown is enabled can be compared to assess the energy saving gains of this function.

- Given the same PSU output power, if the PSU input power before function activation is much higher than that after function activation, greater gains are provided.
- Given the same PSU input power, if the PSU output power before function activation is much higher than that after function activation, greater gains are provided.

The following provides methods for observing PSU input and output power. The observation result may be inaccurate. The performance counter data on the MAE-Access is for reference only.

- PSU input power

The PSU input power can be indirectly obtained using the **VS.EnergyCons.BTS.Measuring.NR** counter on the MAE-Access.

- In NR-only scenarios, the value of the counter equals the PSU input power.
- In multimode scenarios, this counter indicates the energy consumption of the NR part of the base station. The PSU input power equals the counter value plus the values of counters for other RATs.

- PSU output power

The PSU output power can be indirectly obtained using the **VS.EnergyCons.BTS.Adding.Service.NR** or **VS.EnergyCons.BTS.Adding.NR** counter on the MAE-Access.

- In NR-only scenarios, the value of the counter equals the PSU output power.
- In multimode scenarios, this counter indicates the energy consumption of the NR part of the base station. The PSU output power equals the counter value plus the values of counters for other RATs.

LTE

Performance counters on the MAE-Access are used to monitor and evaluate the performance of this function.

On the live network, you can compare one-week eNodeB power consumption before and after the function is enabled to evaluate its energy saving gains.

- How long the PSU intelligent sleep mode has been in effect
It is indicated by the **VS.PSU.ShutdownTime** counter on the MAE-Access. For the meaning of this counter, see *eNodeB Performance Counter Reference*.
 - If the value of this counter is not 0, PSU intelligent sleep mode has taken effect. If the value of this counter is greater than the measurement period multiplied by a factor of N minus 1 and less than or equal to the measurement period multiplied by N , the number of PSUs that enter the sleep mode is N , which is greater than or equal to 1.
 - If the value of this counter is 0, PSU intelligent sleep mode has not taken effect.
- PSU input power
It is indicated by the **VS.EnergyCons.BTS.Measuring.LTE** counter on the MAE-Access after the **SET ECMP** command is executed with the **Energy Consumption Measurement Point** parameter set to **AC**. For the meaning of this counter, see *eNodeB Performance Counter Reference*.
 - In LTE-only scenarios, the value of the counter shows the PSU input power.
 - In multimode scenarios, this counter value must be added to the values of the counters used for other RATs to derive the PSU input power.
- PSU output power
It is indirectly indicated by the **VS.EnergyCons.BTS.Adding.Service.LTE** or **VS.EnergyCons.BTS.Adding.LTE** counter on the MAE-Access.
 - In LTE-only scenarios, the value of the counter equals the PSU output power.
 - In multimode scenarios, this counter value must be added to the values of the counters used for other RATs to derive the PSU output power.

If the PSU output power remains unchanged, a larger input power decrease indicates larger function performance gains. If the PSU input power remains unchanged, a larger output power decrease indicates larger function performance gains.

NOTE

The observation results for PSU input and output power may be inaccurate. The performance counter data on the MAE-Access is for reference only.

UMTS

Performance monitoring for Intelligent Power Management on live networks is difficult due to ever-changing traffic loads and fluctuating power consumption. You are advised to select site models for different areas (such as densely populated urban areas, suburban areas, and rural areas) and monitor the performance of this function for these site models in the lab.

Use an external instrument or check the power-saving statistics (obtained by using the Intelligent Power Measurement feature) to monitor the performance. You are advised to monitor the performance for two days with similar traffic volume.

NOTICE

Confirm that batteries are fully charged before monitoring the performance of this function because battery charging affects power consumption. If the **DSP PMU** or **DSP BATTERY** command output indicates that the storage battery current is close to 0, the batteries are fully charged.

- **Using an external instrument**

The performance monitoring procedure is as follows:

Step 1 Perform the following without enabling this function:

1. Use a current measurement instrument and a voltage measurement instrument to continuously measure the current and voltage of the base station, respectively.
 - a. The voltage is stable, and therefore measure the voltage only once.
 - b. The current changes constantly, and therefore measure the base station current for a 24-hour period at a fixed interval such as 5 minutes. Note: The measured current is the base station current, not the current for the equipment room.
2. Use the average value of the recorded current values during an hour as the average current for the hour. Multiply the recorded voltage and the average current of an hour to obtain the power consumption during that hour. Calculate the power consumption during each hour in a day.

Step 2 Enable this function, and repeat the operations in [Step 1](#) to obtain the power consumption in a day.

Step 3 Compare the power consumption values obtained on the two days. The total power consumption of the base station is expected to decrease after the function is enabled.

----End

- **Using the Intelligent Power Measurement feature**

For details about WRFD-020406 Intelligent Power Measurement, see *Green BTS in RAN Feature Documentation*.

The performance monitoring procedure is as follows:

- Step 1** Activate WRFD-020406 Intelligent Power Measurement by referring to section "WRFD-020406 Intelligent Power Measurement" in *Green BTS*.
- Step 2** Select the NodeB to be configured in the **MML Command** window of the MAE. Run the **SET ECMP** command with **Energy Consumption Measurement Point** set to **AC**.
- Step 3** Disable the Intelligent Power Management feature and collect the values of the **VS.EnergyCons.BTS.Measuring.UMTS** counter for one day to obtain the power consumption.
- Step 4** Enable the Intelligent Power Management feature and collect the values of the **VS.EnergyCons.BTS.Measuring.UMTS** counter for one day to obtain the power consumption.
- Step 5** Compare the power consumption values obtained on the two days. The total power consumption of the base station is expected to decrease after the function is enabled.

----End

GSM

To observe the PSU shutdown duration, subscribe to the following counters on the MAE-Access:

- GBTS: SITE.PSU.TURNING.OFF.DUR
- eGBTS: VS.PSU.ShutdownTime

5 PSU Intelligent Coordinated Energy Saving

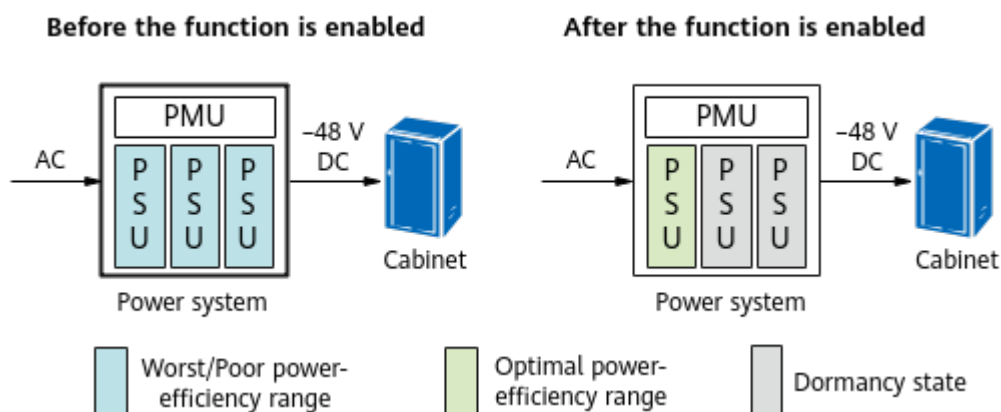
5.1 Principles

PSU intelligent coordinated energy saving indicates the procedure where the power system controls whether a PSU enters or exits the dormancy state using the maximum efficiency algorithm. In this way, the power system retains its maximum conversion efficiency within the full power change range of the base station.

Principles

The base station measures the total power consumption in real time, calculates the number of working PSUs required to retain the power system at the maximum conversion efficiency, and dynamically adjusts the number of working PSUs, to ensure that the conversion efficiency of each working PSU is within the optimal efficiency range, as shown in [Figure 5-1](#). PSU intelligent coordinated energy saving is enabled by setting the **IPWRM.PSUICESS (5G gNodeB, LTE eNodeB)** parameter to **ON**.

Figure 5-1 Working principle of PSU intelligent coordinated energy saving



If MTS series or ICC series power systems are used, the **IPWRM.LRPSW (5G gNodeB, LTE eNodeB)** parameter can be used to specify whether to enable PSU intelligent coordinated energy saving based on the load rate.

- If **IPWRM.LRPSW (5G gNodeB, LTE eNodeB)** is set to **ON**, this function is enabled based on the load rate. The load rate range is [**IPWRM.MINLR (5G gNodeB, LTE eNodeB)**, **IPWRM.MAXLR (5G gNodeB, LTE eNodeB)**].
 - If the PSU load rate is within the load rate range, no adjustment is performed.
 - If the PSU load rate is beyond the load rate range, the base station dynamically adjusts the number of working PSUs so that the load rate of each working PSU is within the load rate range.
- If **IPWRM.LRPSW (5G gNodeB, LTE eNodeB)** is set to **OFF**, the base station dynamically adjusts the number of working PSUs by enabling PSU intelligent coordinated energy saving not based on the load rate, so that the conversion efficiency of each working PSU is optimal.

Related Concepts

PSU load rate = Current PSU load/Rated capacity

The PSU and PMU are the power supply and power monitoring units, respectively. The following describes details about the PSU and PMU.

- The PSU is a power supply unit that can convert 110 V AC or 220 V AC into – 48 V DC.

Generally, each AC-powered base station is configured with multiple PSUs for AC/DC conversion. The number of PSUs is generally determined based on the maximum power consumption of the base station. This ensures that the base station operates properly even with the maximum load. The base station, however, seldom operates at its full load. As a result, each PSU in the base station usually works at partial output power. Since the efficiency at which a PSU converts power is directly proportional to its output power, low PSU power conversion efficiency will affect the overall power consumption of the base station.

When a base station is configured with multiple PSUs, one or more of them can be shut down based on the actual load, improving the conversion efficiency of PSUs.

PSU intelligent coordinated energy saving reduces power consumption by maintaining high PSU power conversion efficiency. Power consumption reduction is directly related to the PSU power conversion efficiency changes before and after PSUs are shut down. Assuming that the overall output power of PSUs remains unchanged, and the PSU power conversion efficiencies before and after PSU shutdown are $c0$ and $c1$, respectively. Then the power consumption reduction proportion is $(1 - c0/c1)$. In addition, PSU shutdown extends the life span of the PSUs.

- The PMU is a power supply monitoring unit that provides a complete power system and functions such as battery management, power distribution detection, and alarm reporting.

5.2 Network Analysis

5.2.1 Benefits

PSU intelligent coordinated energy saving controls PSUs' entry into and exit from dormancy to maximize the PSUs' conversion efficiency within the full power range, reducing the power consumption of the base station.

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

This function has no impact on the network.

5.3 Requirements

5.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
MRFD-191261	Intelligent Coordinated PSU Energy Saving (NR)	NR0S00CPES00	per gNodeB
MRFD-191221	Intelligent Coordinated PSU Energy Saving (LTE FDD)	LT1SNCPEF10	per eNodeB
MRFD-191231	Intelligent Coordinated PSU Energy Saving (LTE TDD)	LT4S0ICPES00	per eNodeB
MRFD-191211	Intelligent Coordinated PSU Energy Saving (UMTS)	LQW9ICPES01	per NodeB
MRFD-191201	Intelligent Coordinated PSU Energy Saving (GSM)	LGB3ICPES01	per Site

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

None

5.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD LTE FDD LTE TDD UMTS GSM	PSU intelligent shutdown	PSUIS.PSUISS (5G gNodeB, LTE eNodeB)	<i>Green Site</i>	- In Co-MPT scenarios, PSU intelligent shutdown cannot be enabled together with PSU intelligent coordinated energy saving. That is, if PSU intelligent coordinated energy saving is enabled for any RAT of GSM, UMTS, LTE, and NR, PSU intelligent shutdown cannot be deployed for NR at the same time. On the contrary, if PSU intelligent shutdown is enabled for any RAT of GSM, UMTS, LTE, and NR, PSU intelligent coordinated energy saving cannot be deployed for NR at the same time. - In separate-MPT scenarios, if PSU intelligent shutdown and PSU intelligent coordinated energy saving are enabled on

RAT	Function Name	Function Switch	Reference	Description
				two main control boards separately, only PSU intelligent coordinated energy saving takes effect.
NR LTE FDD LTE TDD UMTS GSM	iLiBattScan	BATCTPA.ILIBATT SCANSW (5G gNodeB, LTE eNodeB)	<i>Power Supply Management</i>	None

5.3.2.3 Function Impacts

None

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

Boards

No requirements

RF Modules

No requirements

Power Supply

- If the base station is AC powered, the Huawei-provided cabinets or power systems listed in [Table 5-1](#) are required, and software versions must meet the corresponding requirements. For details about the mapping between PMUs and physical cabinets, see "Mapping Between Physical Cabinets and PTYPE Values" in "Hardware" in section "Basic Power Supply Management Functions" in *Power Supply Management*.

Table 5-1 Version requirements on cabinets and power systems

Cabinet or Power System	Version
APM30H (Ver.D) cabinet APM30H (Ver.E) cabinet APM5930(AC) cabinet APM5950 series cabinet OPM200 series blade power module	Base station software version: BTS3900 V100R019C10 or later
MTS series power system ^a	SMU version: iSitePower-A V100R024C10 or later Base station software version: BTS3900 V100R022C00 or later.
ICC series power system ^a	SMU version: iSitePower-A V100R024C10 or later Base station software version: BTS3900 V100R022C00 or later.
a: Only the MTS series and ICC series power systems with the SMU02C (02312MML-009) serving as the power monitoring module support this function. For details about cabinets, see "Mapping Between Physical Cabinets and PTYPE Values" in "Hardware" in section "Basic Power Supply Management Functions" in <i>Power Supply Management</i> .	

 NOTE

If the cabinet type and power system type do not meet the requirements, this function may be adversely affected. You are advised to run the **SET IPWRM** command with **IPWRM.PMUTYPECHKSW (5G gNodeB, LTE eNodeB)** set to **ON** to exclude the cabinet and power system types that do not meet the requirements. This ensures that this function works properly.

- The PMU software version must be 2350 or later, which matches the base station software version BTS3900 V100R019C10. If the PMU software version is not upgraded to 2350 or later, this function cannot be enabled when the **DSP PMU** command output shows that the value of **PSU Intel Coordinated Energy Saving Capb** is **Not Supported**.
 - The PMU software version can be queried by running the **DSP BRDVER** command.
 - The PMU software version can be activated by running the **ACT SOFTWARE** command.
- At least three PSUs are required in an MTS series or ICC series power cabinet, and at least two PSUs are required in other power cabinets.
- The base station must be configured with batteries (which can be added by running the **ADD BATTERY** command) when this function is being activated or used.
- The circuit breakers between the batteries and the cabinet are properly switched on, and no power-supply-related alarm (such as battery-related

alarm) is reported. No temperature alarm or PMU-/PSU-related alarm is reported in the cabinet.

5.3.4 Others

None

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

[Table 5-2](#) describes the parameters used for function activation. Function optimization configurations are not involved.

Table 5-2 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
PSU Intelligent Coordinated Energy Saving SW	IPWRM.PSUICES (<i>5G gNodeB, LTE eNodeB</i>)	It is recommended that this parameter be set to ON when PSU intelligent coordinated energy saving is required.
Load Rate Priority Enable Switch	IPWRM.LRPSW (<i>5G gNodeB, LTE eNodeB</i>)	Select this option if PSU intelligent coordinated energy saving needs to be enabled based on the load rate.
Minimum Load Rate	IPWRM.MINLR (<i>5G gNodeB, LTE eNodeB</i>)	Set this parameter based on the network plan. This parameter takes effect only when PSU intelligent coordinated energy saving is enabled based on the load rate and the value of this parameter must be less than the maximum load rate specified by Maximum Load Rate .
Maximum Load Rate	IPWRM.MAXLR (<i>5G gNodeB, LTE eNodeB</i>)	Set this parameter based on the network plan. This parameter takes effect only when PSU intelligent coordinated energy saving is enabled based on the load rate and the value of this parameter must be greater than the minimum load rate specified by Minimum Load Rate .

5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling PSU intelligent coordinated energy saving
SET IPWRM: PSUICESS=ON;
//(Optional) Setting PSUICESS to ON, MINLR to 40, and MAXLR to 90 for MTS series or ICC series power systems
SET IPWRM: PSUICESS=ON, LRPSW=ON, MINLR=40, MAXLR=90;
```

NOTE

For details about the PMU software upgrade, see *Power Supply Management*.

Deactivation Command Examples

```
//Disabling PSU intelligent coordinated energy saving
SET IPWRM: PSUICESS=OFF;
```

NOTE

In separate-MPT scenarios, PSU intelligent coordinated energy saving can be deactivated for the entire base station only when this switch is turned off for all main control boards on which this function has been enabled.

5.4.1.3 Using the MAE-Deployment

For detailed operations, see *Feature Configuration Using the MAE-Deployment*.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

- Using MML commands
 - a. Run the **DSP BATTERY** command to query the status of storage batteries.
 - b. Run the **LST PSU** command to verify that at least three PSUs are configured for the MTS series or ICC series power system, at least two PSUs are configured for other power systems, and the running state of each PSU is normal.
 - c. Run the **LST IPWRM** command to verify that the value of **PSU Intelligent Coordinated Energy Saving SW** is **ON**. For MTS series or ICC series power systems, check the value of **Load Rate Priority Enable Switch** as well.

- d. Run the **DSP PMU** command to verify that the value of **PSU Intel Coordinated Energy Saving Status** is **Enable**. For MTS series or ICC series power systems, check the value of **PSU Intel Coord Eng Sav Load Rate Sw Status**. If the value is **Enable**, PSU intelligent coordinated energy saving is enabled based on the load rate.
- e. Run the **DSP PSU** command. If the value of the **Running Status** parameter is **Dormancy** for a PSU, PSU intelligent coordinated energy saving has taken effect.

Alternatively, check the PSU indicators locally to determine its working status. If a PSU is in the shutdown state, PSU intelligent coordinated energy saving has taken effect. For details about the indicator status, see *PSU in 3900 & 5900 Series Base Station Product Documentation*.

NOTE

In separate-MPT scenarios, **PSU Intel Coordinated Energy Saving Status = Enable** is displayed in the output of the **DSP PMU** command executed on the main control board with the function disabled if the following conditions are met: (1) Two main control boards are configured with one PMU. (2) PSU intelligent shutdown or PSU intelligent coordinated energy saving is enabled on one main control board but disabled on the other main control board. (3) The function has taken effect on the entire base station.

- Using performance counters
Check the PSU shutdown duration through the **VS.PSU.ShutdownTime** counter on the MAE-Access to determine whether PSU intelligent coordinated energy saving has taken effect.
 - If the value of this counter is not 0, PSU intelligent coordinated energy saving has taken effect. If the value of this counter is greater than the measurement period multiplied by a factor of N minus 1 and less than or equal to the measurement period multiplied by N , the number of PSUs that are shut down is N , which is greater than or equal to 1.
 - If the value of this counter is 0, PSU intelligent coordinated energy saving has not taken effect.

5.4.3 Network Monitoring

On the live network, you can compare the indicator values related to the input to the PSU within one week before and after PSU intelligent coordinated energy saving is enabled to evaluate its benefits.

- Method 1: Use the AC power input to the PSU (indicated by **VS.PMUBoard.ACInputEnergy**).
 - In NR-only scenarios, the value of this counter shows the power input to PSUs.
 - In multi-RAT scenarios, the value of this counter indicates the total power consumption of all RATs in the base station, that is, the total AC power input to PSUs.
- Method 2: If only one power system is used, use the result of the following formula to indirectly observe the PSU conversion efficiency:

$$\frac{(\text{VS.EnergyCons.BTS.Measuring.GSM} + \text{VS.EnergyCons.BTS.Measuring.UMTS} + \text{VS.EnergyCons.BTS.Measuring.LTE} + \text{VS.EnergyCons.BTS.Measuring.NR} + \text{VS.EnergyCons.BTS.Measuring.RFA})}{\text{VS.PMUBoard.ACInputEnergy}}$$

 NOTE

If only some RATs exist in method 2, the values of the counters corresponding to the absent RATs are 0.

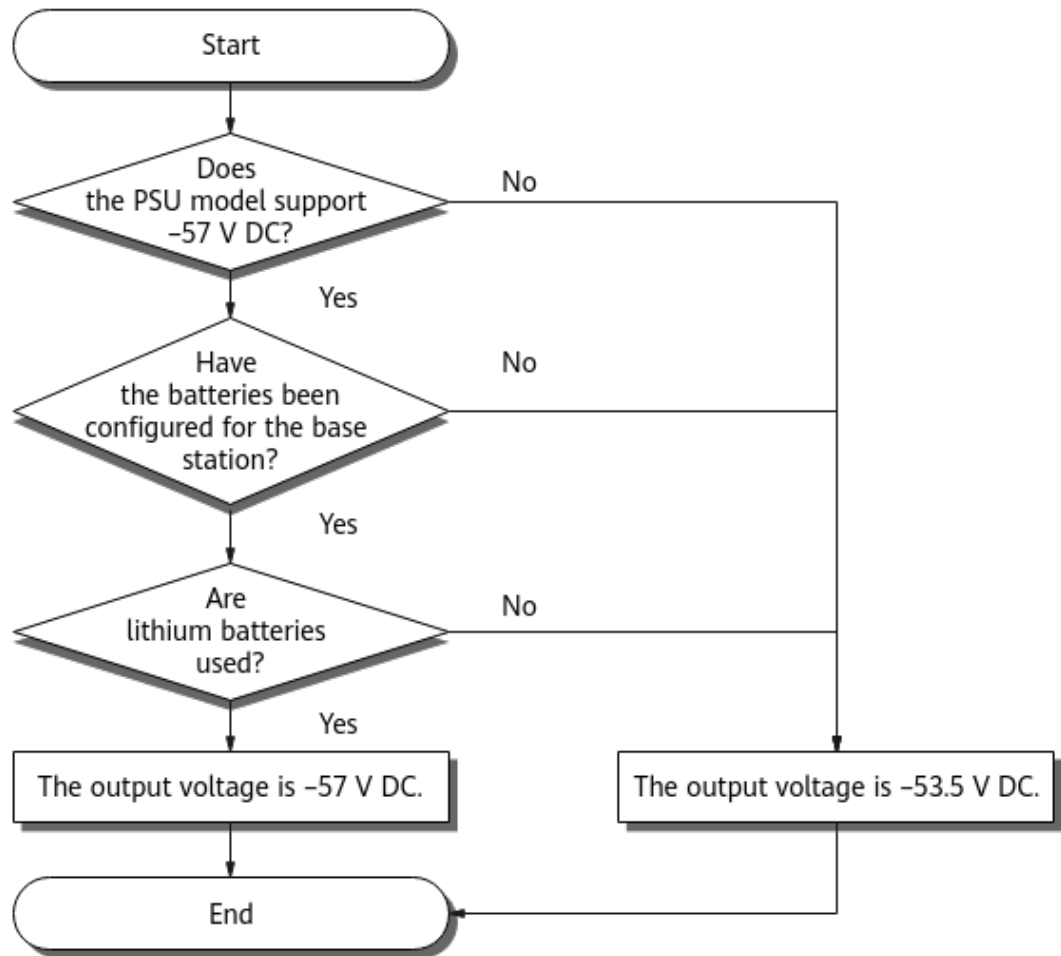
It is recommended that the benefits be observed when the power consumption is stable within one week before and after this function is enabled. If the power consumption fluctuates greatly on the live network, the benefits may be unstable.

6 Automatic PSU Voltage Optimization

6.1 Principles

When RF modules are remotely deployed, the base station experiences severe power loss on power cables. To reduce such power loss, the current on the power cables can be decreased by increasing the voltage. Automatic PSU voltage optimization automatically identifies the battery type and PSU type, measures the output voltage, and changes the voltage to -53.5 V DC or -57 V DC, as shown in [Figure 6-1](#). This function is enabled when the **PMU.BVR (5G gNodeB, LTE eNodeB)** parameter is set to **AUTO**.

Figure 6-1 Working principle of automatic PSU voltage optimization



1. The base station checks whether the PSU model supports -57 V DC. If this voltage is supported, the base station performs 2. If this voltage is not supported, the output voltage is -53.5 V DC.
2. The base station checks whether the battery supports -57 V DC as follows:
 - If the base station is configured with batteries:
 - If **BATTERY.BTYPE** is set to **LI_BAT**, the output voltage is -57 V DC.
 - Otherwise, the voltage is -53.5 V DC.
 - If the base station is not configured with batteries, the output voltage is -53.5 V DC.

6.2 Network Analysis

6.2.1 Benefits

Automatic PSU voltage optimization automatically identifies the battery and PSU types and adjusts the voltage of the batteries and PSUs to -57 V DC if this voltage is supported. The power loss on power cables is reduced. In typical scenarios (50

m remote deployment), the energy saving gains of the entire base station is about 1%. A longer cable length and smaller cable diameter bring greater gains.

6.2.2 Impacts

The impacts between this function and other functions are described in [6.3.2.3 Function Impacts](#).

This function has no impact on the network.

6.3 Requirements

6.3.1 Licenses

None

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

None

6.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
NR LTE FDD LTE TDD UMTS GSM	iLiBattScan	BATCTPA.LIBATT SCANSW (5G gNodeB, LTE eNodeB)	<i>Power Supply Management</i>	None

6.3.2.3 Function Impacts

None

6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

Boards

No requirements

RF Modules

No requirements

6.3.4 Others

- The base station uses AC power supply and the APM5930 AC/APM5950H/APM5950H-L cabinet provided by Huawei. For details about the mapping between PMUs and physical cabinets, see "Mapping Between Physical Cabinets and PTYPE Values" in "Hardware" in section "Basic Power Supply Management Functions" in *Power Supply Management*.
- The base station must be configured with batteries.

NOTE

- You can query the software version of the PMU by selecting the corresponding gNodeB in the MML command window of the MAE-Access and running the **DSP BRDVER** command. If the PMU software version is earlier than 2530 (the version that matches the base station software version SRAN19.1), run the **ACT SOFTWARE** command to activate the PMU software.
- If both lead-acid batteries and lithium batteries are configured for a base station and only lithium batteries are configured on the BBU side, PSUs may automatically adjust the voltage to -57 V DC after automatic PSU voltage optimization is enabled. If it occurs, the service life of lead-acid batteries is affected. Therefore, you are advised to disable automatic PSU voltage optimization in this scenario.

6.4 Operation and Maintenance

6.4.1 Data Configuration

6.4.1.1 Data Preparation

Table 6-1 describes the parameters used for function activation. Function optimization configurations are not involved.

Table 6-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Busbar Voltage Reference	PMU.BVR (5G gNodeB, LTE eNodeB)	The value AUTO is recommended when automatic PSU voltage optimization is required.

6.4.1.2 Using MML Commands

Before using MML commands, refer to [6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Querying the PMU software version
DSP BRDVER;;
//Activating the PMU software
ACT SOFTWARE: OT=BOARDTYPE, BT=PMU;
//Enabling automatic PSU voltage optimization
MOD PMU: CN=0, SRN=7, SN=0, PTYPE=ETP, BVR=AUTO;
```

NOTE

For details about the PMU software upgrade, see *Power Supply Management*.

Deactivation Command Examples

```
//Disabling automatic PSU voltage optimization
MOD PMU: CN=0, SRN=7, SN=0, PTYPE=ETP, BVR=DEFAULT;
```

6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.2 Activation Verification

Step 1 Run the **LST PMU** command to verify that the value of **Busbar Voltage Reference** is **AUTO**.

Step 2 Run the **DSP PMU** command to verify that the value of **Busbar Voltage(V)** is **57**.

----End

6.4.3 Network Monitoring

On the live network, you can compare the AC power input to PSUs within one week before and after the automatic PSU voltage optimization function is enabled to evaluate its benefits.

The AC power input to PSUs can be obtained using the **VS.PMUBoard.ACInputEnergy** counter on the MAE-Access.

- In NR-only scenarios, the value of this counter shows the power input to PSUs.
- In multi-RAT scenarios, the value of this counter indicates the total power consumption of all RATs in the base station, that is, the total AC power input to PSUs.

7 Automatic Voltage Optimization for Intelligent Lithium Batteries

7.1 Principles

When RF modules are remotely deployed, the base station experiences severe power loss on power cables. To reduce such power loss, the current on the power cables can be decreased by increasing the voltage. The following describes the processing methods in scenarios with different power systems:

- When Huawei power systems are used, some of them can increase the voltage through the automatic PSU voltage optimization function to reduce the current on the power cables, thereby reducing the power loss during cable transmission. For details about automatic PSU voltage optimization, see [6 Automatic PSU Voltage Optimization](#).
- If a third-party power system is used with Huawei intelligent lithium batteries, automatic PSU voltage optimization cannot be used. However, when the mains supply is disconnected, the base station used with the PMU 15A(Ver.B) in the power cabinet can keep the voltage of those batteries at a relatively high level through the automatic voltage optimization for intelligent lithium batteries function, thereby finding the optimal voltage and reducing power loss during cable transmission. Huawei intelligent lithium batteries are a must, and the combination of Huawei intelligent lithium batteries and other batteries such as lead-acid batteries is acceptable.

Automatic voltage optimization for intelligent lithium batteries is controlled by the **PMU.BVR (5G gNodeB, LTE eNodeB)** parameter. If this parameter is set to **AUTO**, this function is enabled.

 NOTE

If Huawei intelligent lithium batteries and third-party lead-acid batteries are used together, it is recommended that battery temperature compensation be disabled for lead-acid batteries when automatic voltage optimization for intelligent lithium batteries is enabled.

Automatic voltage optimization for intelligent lithium batteries is not recommended if only Huawei intelligent lithium batteries are used. In this case, if this function is required, it must be enabled when the mains supply is normal. If this function is enabled when the mains supply is abnormal (for example, the mains supply disconnection or undervoltage occurs), the output voltage cannot reach the optimal value during the period when the first abnormality persists.

7.2 Network Analysis

7.2.1 Benefits

Automatic voltage optimization for intelligent lithium batteries reduces power loss on power cables. A longer cable length and smaller cable diameter bring greater gains. In addition, with this function enabled, common lithium batteries and lead-acid batteries can be used together with intelligent lithium batteries, enabling reuse of existing batteries.

7.2.2 Impacts

The impacts between this function and other functions are described in [6.3.2.3 Function Impacts](#).

This function has no impact on the network.

7.3 Requirements

7.3.1 Licenses

None

7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.2.1 Prerequisite Functions

None

7.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
NR LTE FDD LTE TDD UMTS GSM	iLiBattScan	BATCTPA.LIBATT SCANSW (5G gNodeB, LTE eNodeB)	<i>Power Supply Management</i>	None

7.3.2.3 Function Impacts

None

7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

Boards

No requirements

RF Modules

No requirements

Power Supply

- The base station must be configured with the PMU 15A(Ver.B). The PMU software version can be queried by running the **DSP BRDVER** command.
- The base station must be configured with Huawei intelligent lithium batteries. For details, contact Huawei engineers. (ESM-100B1, ESM-150B1, ESM-100A7, ESM-150A3, and ESM-200A1).
- If Huawei intelligent lithium batteries are used, they can be used together with lead-acid batteries and common lithium batteries without affecting the automatic voltage optimization for intelligent lithium batteries function.
- If Huawei intelligent lithium batteries and third-party lead-acid batteries are used together, it is recommended that battery temperature compensation be disabled for lead-acid batteries when automatic voltage optimization for intelligent lithium batteries is enabled.

7.3.4 Others

None

7.4 Operation and Maintenance

7.4.1 Data Configuration

7.4.1.1 Data Preparation

Table 7-1 describes the parameters used for function activation. Function optimization configurations are not involved.

Table 7-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Busbar Voltage Reference	PMU.BVR (5G gNodeB, LTE eNodeB)	Set this parameter to AUTO if automatic voltage optimization for intelligent lithium batteries is required in the case of the mains supply disconnection in scenarios where Huawei intelligent lithium batteries are used.

7.4.1.2 Using MML Commands

Before using MML commands, refer to **6.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling automatic voltage optimization for intelligent lithium batteries  
MOD PMU: CN=0, SRN=7, SN=0, PTYPE=ETP, BVR=AUTO;
```

 NOTE

For details about the PMU software upgrade, see *Power Supply Management*.

Deactivation Command Examples

```
//Disabling automatic PSU voltage optimization  
MOD PMU: CN=0, SRN=7, SN=0, PTYPE=ETP, BVR=DEFAULT;
```

7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.4.2 Activation Verification

Step 1 Run the **LST PMU** command to verify that the value of **Busbar Voltage Reference** is **AUTO**.

Step 2 Run the **DSP PMU** command to check the value of **Busbar Voltage(V)**. After automatic voltage optimization for intelligent lithium batteries is enabled, the voltage of Huawei intelligent lithium batteries does not continuously decrease and is slightly lower than the busbar voltage.

----End

7.4.3 Network Monitoring

This function ensures basic network performance and does not need to be separately monitored.

8 Staggering Electricity Usage

8.1 Principles

In areas with quality power grids, lithium batteries are often in the float-charging state. In most cases, operators do not benefit from the capital expenditure (CAPEX) of lithium battery power backup. Staggering electricity usage applies to scenarios where the electricity price varies depending on peak and off-peak hours. With this function, lithium batteries are charged during off-peak hours; during peak hours, lithium batteries are used to supply power. This reduces the overall electricity costs of the base station and protects operators' investment in lithium batteries.

Staggering electricity usage consists of a basic function and an enhanced one, that is, fixed staggering electricity usage and intelligent dynamic peak staggering, respectively. Both functions must be configured and activated on the MAE-Deployment for NEs.

- Operators can set parameters as follows to adjust fixed staggering electricity usage for individual NEs. For details, see [8.4.1.2 Using MML Commands](#).
 - Set the **EQUIPMENT.STAGGEREDPOWERSWITCH (LTE eNodeB, 5G gNodeB)** parameter to **ON**.
 - Set the **EQUIPMENT.SPCEM (LTE eNodeB, 5G gNodeB)** parameter to **FIXED**.
 - Set the **EQUIPMENT.STAGGEREDDISCHARGESTOPTHLD (5G gNodeB, LTE eNodeB)** parameter.
- Operators can set parameters as follows to adjust intelligent dynamic peak staggering for individual NEs. For details, see [8.4.1.2 Using MML Commands](#).
 - Set the **EQUIPMENT.STAGGEREDPOWERSWITCH (LTE eNodeB, 5G gNodeB)** parameter to **ON**.
 - Set the **EQUIPMENT.SPCEM (LTE eNodeB, 5G gNodeB)** parameter to **DYNAMIC**.
 - Adjust the target backup power use duration by setting the **EQUIPMENT.TGBKD (LTE eNodeB, 5G gNodeB)** parameter.
- Operators can run the **LST STAGGEREDPOWERRECORD** command to query configuration information for staggering electricity usage.

Intelligent Dynamic Peak Staggering

Intelligent dynamic peak staggering is introduced based on fixed staggering electricity usage. It predicts the site load power consumption, and intelligently and dynamically calculates the battery capacity that can be discharged during peak hours while ensuring backup power use duration.

- How intelligent dynamic peak staggering works on NEs after it is activated is basically the same as how fixed staggering electricity usage works on NEs, as described in [Working Mechanism of Staggering Electricity Usage](#). The difference lies in that, before the output voltage of the power system is decreased and lithium batteries are used to supply power, the site power consumption is predicted based on the historical power consumption and the battery discharge capacity is then calculated based on the target backup power use duration.
- After intelligent dynamic peak staggering is enabled, it takes at least one day for the PMU to collect power consumption data to predict the required power consumption before intelligent dynamic peak staggering can take effect. After the PMU resets or restarts, the power consumption data is cleared and needs to be collected again.

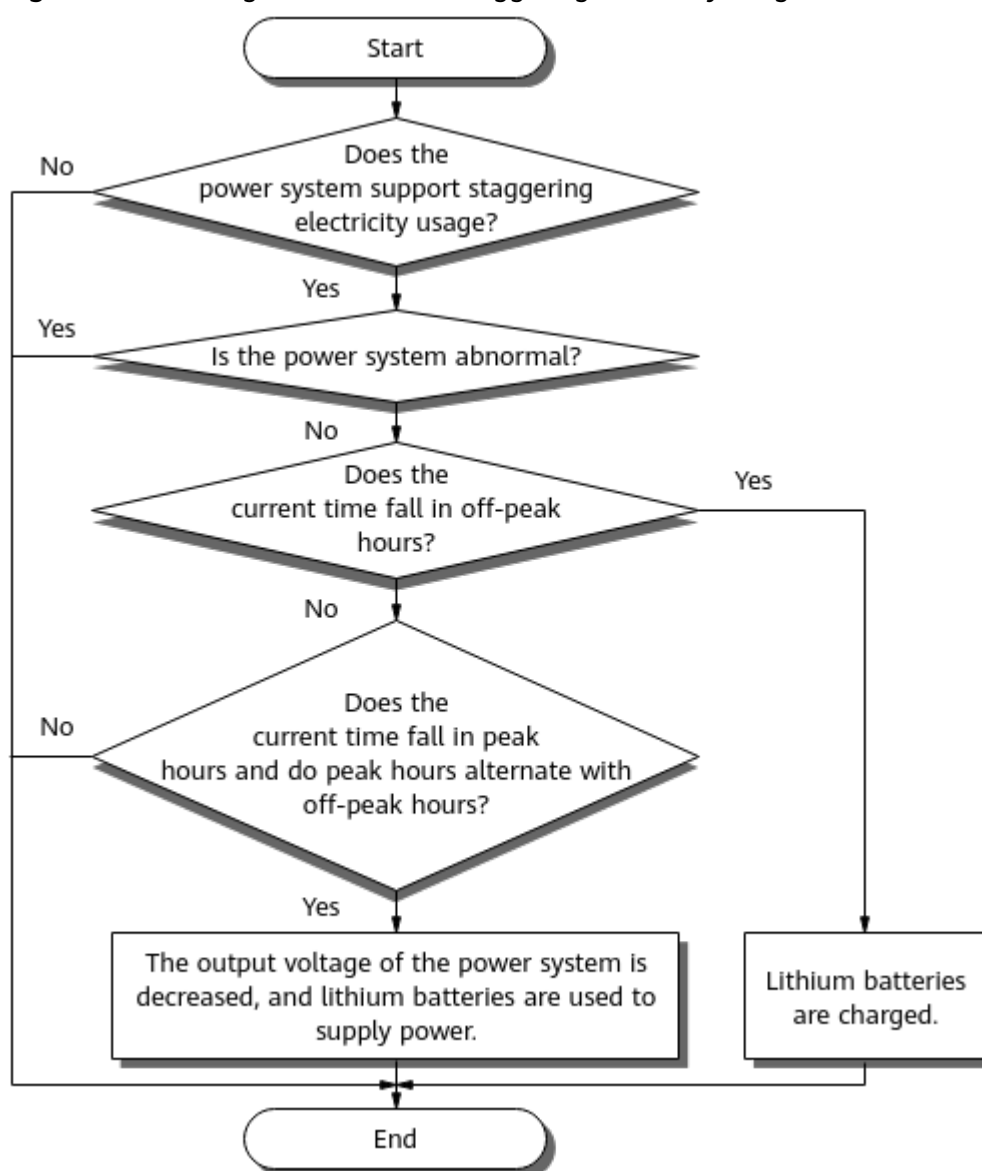
NOTE

- Before staggering electricity usage is enabled, the **ADD BATTERY** command needs to be executed to configure a battery with the battery type specified. If lead-acid and lithium batteries are used together but the battery type is specified as lithium battery, staggering electricity usage also takes effect, but the lead-acid battery lifespan is affected.
- For intelligent dynamic peak staggering, after the PMU resets or restarts, the power consumption data is cleared. In this case, it takes at least one day for the PMU to collect power consumption data again before intelligent dynamic peak staggering can take effect.

Working Mechanism of Staggering Electricity Usage

[Figure 8-1](#) shows how staggering electricity usage works on NEs after it is activated.

Figure 8-1 Working mechanism of staggering electricity usage



1. The PMU checks whether the power system supports staggering electricity usage. If it does, the process goes to the next step. If it does not, the process ends.

NOTE

Before staggering electricity usage is enabled, the **ADD BATTERY** command needs to be executed to configure a battery with the battery type specified. If lead-acid and lithium batteries are used together but the battery type is specified as lithium battery, staggering electricity usage takes effect, but the lead-acid battery lifespan is affected.

2. The PMU checks whether the power system is abnormal. If the power system is abnormal, the process ends. If it is normal, the process goes to the next step.
3. The PMU checks whether the current time falls in peak or off-peak hours.
 - If the current time falls in off-peak hours, lithium batteries are charged until they are fully charged or until off-peak hours end.

- If the current time falls in peak hours, and peak hours alternate with off-peak hours, the PMU decreases the output voltage of the PSU to a value greater than the DC voltage alarm lower threshold (specified by the **PMU.DCVLTHD (LTE eNodeB, 5G gNodeB)** parameter) and enables lithium batteries to supply power. The lithium batteries do not stop discharging until their remaining power becomes lower than the staggered discharge stop threshold (specified by the **EQUIPMENT.STAGGEREDDISCHARGESTOPTHLD (5G gNodeB, LTE eNodeB)** parameter) or until the peak hours end.

If the current time falls in neither peak nor off-peak hours, the process ends.

 NOTE

When lithium batteries are used to supply power and the difference between the value of the **PMU.DCVLTHD (LTE eNodeB, 5G gNodeB)** parameter and the current busbar voltage is less than 3 V, the discharging of lithium batteries is affected due to the PSU DC output. Therefore, when staggering electricity usage is enabled, the **PMU.DCVLTHD (LTE eNodeB, 5G gNodeB)** parameter cannot be set to a value greater than 500, that is, 50 V.

8.2 Network Analysis

8.2.1 Benefits

The method for calculating energy saving benefits produced by staggering electricity usage is as follows:

1. Assume that the electricity price in peak hours is P1 and that in off-peak hours is P2, the amount of electricity discharged by lithium batteries in peak hours is V, and lithium battery charging/discharging conversion efficiency is E (the charging/discharging conversion efficiency of Huawei lithium batteries is at least 92%). Then, the mains power consumption for charging lithium batteries during off-peak hours is V/E.
2. When staggering electricity usage is enabled, the mains electricity costs saved by using lithium batteries during peak hours is P1 x V, and the mains electricity costs for charging lithium batteries in off-peak hours is P2 x V/E. Therefore, the energy saving benefits can be calculated using the following formula: $G = P1 \times V - P2 \times V/E = (P1 - P2/E) \times V \times 0.95$.

Based on the benefit calculation method, the deployment suggestions for staggering electricity usage are as follows:

- To obtain more benefits, it is recommended that staggering electricity usage be enabled when the electricity price in peak hours is more than 1.1 times that in off-peak hours.
- Energy saving benefits are directly proportional to the differences in electricity prices between peak and off-peak hours and the charging/discharging conversion efficiency of lithium batteries.
- However, when the price difference between peak and off-peak hours cannot offset the losses caused by charging/discharging conversion of lithium batteries, there may even be losses from enabling staggering electricity usage.

 NOTE

The method for calculating benefits produced by intelligent dynamic peak staggering is the same as that produced by fixed staggering electricity usage. The difference between the two functions lies in that when intelligent dynamic peak staggering is enabled, the battery discharge capacity is calculated based on the target backup power use duration.

8.2.2 Impacts

The impacts between this function and other functions are described in [8.3.2.3 Function Impacts](#).

This function has no impact on the network.

8.3 Requirements

8.3.1 Licenses

Fixed Staggering Electricity Usage

None

Intelligent Dynamic Peak Staggering

RAT	Feature ID	Feature Name	Model	Sales Unit
GSM	MRFD-191202	Intelligent Peak Staggering (GSM)	LGB3IPS1	Per site
UMTS	MRFD-191212	Intelligent Peak Staggering (UMTS)	LQW9IPS01	Per NodeB
FDD	MRFD-191222	Intelligent Peak Staggering (LTE FDD)	LT1SNGPEST10	Per eNodeB
TDD	MRFD-191232	Intelligent Peak Staggering (LTE TDD)	LT4S0IPSTG00	Per eNodeB
NR	MRFD-191262	Intelligent Peak Staggering (NR)	NR0S00TPS000	Per gNodeB

 NOTE

In a multimode base station, intelligent dynamic peak staggering takes effect as long as one license for a RAT is available.

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.3.2.1 Prerequisite Functions

None

8.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
NR LTE FDD LTE TDD UMTS GSM	iLiBattScan	BATCTPA.ILIBATT SCANSW (5G gNodeB, LTE eNodeB)	<i>Power Supply Management</i>	None

8.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
NR LTE FDD LTE TDD UMTS GSM	PV deployment for base stations	PSU.PSUTYPE (5G gNodeB, LTE eNodeB)	<i>Green Site</i>	The benefits of intelligent dynamic peak staggering are affected when base stations use the PV deployment function. • When solar power supply is used, power discharge in off-peak hours is restricted, and the benefits provided by staggering electricity usage decrease. • When the solar power is greater than the load power and staggering electricity usage is enabled, lithium batteries do not discharge power in peak hours. Staggering electricity usage does not provide any benefits in this case.

8.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

Boards

- Main control boards: Only the UMPT supports this function.
- Baseband processing units: All baseband processing units support this function.

RF Modules

This function does not depend on RF modules.

8.3.4 Networking

Fixed Staggering Electricity Usage

None

Intelligent Dynamic Peak Staggering

In co-BBU separate-MPT scenarios, it is recommended that intelligent dynamic peak staggering be enabled only on the main control board with a license.

8.3.5 Others

Only Huawei AC power systems support staggering electricity usage. Requirements for the PMU hardware model, PMU software version, and batteries are as follows.

- Requirements of fixed staggering electricity usage are described in [Table 8-1](#) and [Table 8-3](#).
- Requirements of intelligent dynamic peak staggering are described in [Table 8-2](#) and [Table 8-3](#).

Table 8-1 Requirements of fixed staggering electricity usage for the PMU

Power System Type	PMU Hardware Model	Cabinet Type	PMU Software Version
ETP	CCUC	APM5930(AC) (Ver.A) APM30H (Ver.F_B~E)	If the hardware version is PMU.19, the PMU software version must be 255 or later. If the hardware version is PMU.22, the PMU software version must be 254 or later.
ETP	PMU 11C	APM5950H (AC)	If the hardware version is PMU.35, the PMU software version must be 2251 or later.
ETP	PMU 11C	APM5950H-L	If the hardware version is PMU.35, the PMU software version must be 2265 or later.

Table 8-2 Requirements of intelligent dynamic peak staggering for the PMU

Power System Type	PMU Hardware Model	Cabinet Type	PMU Software Version
ETP	CCUC	APM5930(AC) (Ver.A)	If the hardware version is PMU.19, the PMU software version must be 2350 or later.
ETP	PMU 11C	APM5950H (AC)	If the hardware version is PMU.35, the PMU software version must be 2350 or later.
ETP	PMU 11C	APM5950H-L	If the hardware version is PMU.35, the PMU software version must be 2350 or later.

Table 8-3 Requirements of staggering electricity usage for batteries

Battery Type	Battery Model	Support Staggering Electricity Usage or Not
Lithium batteries	ESM-48100B1/ ESM-48150B1	Yes
Lead-acid batteries	N/A	No
Hybrid use of lead-acid and lithium batteries	N/A	No

NOTE

- Before staggering electricity usage is enabled, the **ADD BATTERY** command needs to be executed to configure a battery with the battery type specified. If lead-acid and lithium batteries are used together but the battery type is specified as lithium battery, staggering electricity usage also takes effect, but the lead-acid battery lifespan is affected.
- For intelligent dynamic peak staggering, after the PMU resets or restarts, the power consumption data is cleared. In this case, it takes at least one day for the PMU to collect power consumption data again before intelligent dynamic peak staggering can take effect.

8.4 Operation and Maintenance

8.4.1 Data Configuration

8.4.1.1 Data Preparation

Table 8-4 and **Table 8-5** describe the parameters used for the activation of fixed staggering electricity usage and intelligent dynamic peak staggering, respectively.

Table 8-4 Parameters used for activation of fixed staggering electricity usage

Parameter Name	Parameter ID	Setting Notes
Staggered Power Switch	EQUIPMENT.STAGGEREDPOWERSWITCH (LTE eNodeB, 5G gNodeB)	Set this parameter to ON to enable staggering electricity usage.
Staggered Discharge Stop Threshold	EQUIPMENT.STAGGEREDREDDISCHARGESTOP THLD (5G gNodeB, LTE eNodeB)	Set this parameter to its recommended value.
Staggered Power Effective Mode	EQUIPMENT.SPCEM (LTE eNodeB, 5G gNodeB)	This parameter specifies the staggered power mode of the AC power supply managed by the PMU. Set this parameter to FIXED when configuring fixed staggering electricity usage.
Record ID	STAGGEREDPOWERRCORD.RECORDID (5G gNodeB, LTE eNodeB)	This parameter specifies the ID of a staggered power consumption period. It uniquely identifies a staggered power consumption period on a base station.
Record Type	STAGGEREDPOWERRCORD.RECORDTYPE (5G gNodeB, LTE eNodeB)	This parameter specifies the type of a staggered power consumption period record. Set this parameter to BY_MONTH if the local peak and valley electricity periods are changed by month. Set this parameter to BY_DAY if the local peak and valley electricity periods are changed by day.
Effective Month	STAGGEREDPOWERRCORD.EFFECTIVEMONTH (5G gNodeB, LTE eNodeB)	This parameter specifies the effective month of the period when the local peak and valley electricity periods are changed by month. Set this parameter to NULL if the local peak and valley electricity periods are changed by day.

Parameter Name	Parameter ID	Setting Notes
Effective Day	STAGGEREDPOWERRECORD.EFFECTIVEDAY (5G gNodeB, LTE eNodeB)	This parameter specifies the effective day of the period when the local peak and valley electricity periods are changed by day. Set this parameter to NULL if the local peak and valley electricity periods are changed by month.
Start Time	STAGGEREDPOWERRECORD.STARTTIME (5G gNodeB, LTE eNodeB)	This parameter specifies the start time of the staggered power consumption period.
End Time	STAGGEREDPOWERRECORD.ENTIME (5G gNodeB, LTE eNodeB)	This parameter specifies the end time of the staggered power consumption period.
Price Level	STAGGEREDPOWERRECORD.PRICELEVEL (5G gNodeB, LTE eNodeB)	This parameter specifies the electricity price level of the staggered power consumption period. • The PEAK value indicates the peak electricity price level. • The FLAT value indicates the normal electricity price level. • The LOW value indicates the valley electricity price level.

Table 8-5 Parameters used for activation of intelligent dynamic peak staggering

Parameter Name	Parameter ID	Setting Notes
Staggered Power Switch	EQUIPMENT.STAGGEREDPOWERSWITCH (LTE eNodeB, 5G gNodeB)	The value ON is recommended.

Parameter Name	Parameter ID	Setting Notes
Staggered Power Effective Mode	EQUIPMENT.SPCEM (LTE eNodeB, 5G gNodeB)	This parameter specifies the staggered power mode of the AC power supply managed by the PMU. Set this parameter to DYNAMIC when configuring intelligent dynamic peak staggering.
Target Backup Power Use Duration	EQUIPMENT.TGBKD (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan.
Record ID	STAGGEREDPOWERRECORD.RECORDID (5G gNodeB, LTE eNodeB)	This parameter specifies the ID of a staggered power consumption period. It uniquely identifies a staggered power consumption period on a base station.
Record Type	STAGGEREDPOWERRECORD.RECORDTYPE (5G gNodeB, LTE eNodeB)	This parameter specifies the type of a staggered power consumption period record. Set this parameter to BY_MONTH if the local peak and valley electricity periods are changed by month. Set this parameter to BY_DAY if the local peak and valley electricity periods are changed by day.
Effective Month	STAGGEREDPOWERRECORD.EFFECTIVEMONTH (5G gNodeB, LTE eNodeB)	This parameter specifies the effective month of the period when the local peak and valley electricity periods are changed by month.
Effective Day	STAGGEREDPOWERRECORD.EFFECTIVEDAY (5G gNodeB, LTE eNodeB)	This parameter specifies the effective day of the period when the local peak and valley electricity periods are changed by day.
Start Time	STAGGEREDPOWERRECORD.STARTTIME (5G gNodeB, LTE eNodeB)	This parameter specifies the start time of the staggered power consumption period.
End Time	STAGGEREDPOWERRECORD.ENDTIME (5G gNodeB, LTE eNodeB)	This parameter specifies the end time of the staggered power consumption period.

Parameter Name	Parameter ID	Setting Notes
Price Level	STAGGEREDPOWERRECORD.PRICELEVEL <i>(5G gNodeB, LTE eNodeB)</i>	This parameter specifies the electricity price level of the staggered power consumption period. • The PEAK value indicates the peak electricity price level. • The FLAT value indicates the normal electricity price level. • The LOW value indicates the valley electricity price level.

8.4.1.2 Using MML Commands

Staggering electricity usage needs to be activated on the MAE-Deployment and cannot be activated using MML commands on the base station side. For details, see [8.4.1.3 Using the MAE-Deployment](#).

Before using MML commands, refer to [8.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Fixed staggering electricity usage

```
//Adjusting the staggered discharge stop threshold for a single base station after fixed staggering electricity usage is enabled on the MAE-Deployment  
SET EQUIPMENT: STAGGEREDPOWERSWITCH=ON, SPCEM=FIXED, STAGGEREDDISCHARGESTOPHLD=80;
```

Intelligent dynamic peak staggering

```
//Adjusting the target backup power use duration for a single base station after intelligent dynamic peak staggering is enabled on the MAE-Deployment  
SET EQUIPMENT:STAGGEREDPOWERSWITCH=ON,SPCEM=DYNAMIC,TGBKD=60;
```

Deactivation Command Examples

```
//Disabling staggering electricity usage for a single base station after fixed staggering electricity usage or intelligent dynamic peak staggering is enabled on the MAE-Deployment  
SET EQUIPMENT:STAGGEREDPOWERSWITCH=OFF;
```

8.4.1.3 Using the MAE-Deployment

NOTICE

Before activating staggering electricity usage in batches, ensure that the following conditions are met:

- The electricity prices in peak, flat, and valley time are consistent among the involved sites, and the configuration information in the data plan table is also consistent among these sites.
- The configured number of batteries is consistent with the actual number of batteries. If they are inconsistent, staggering electricity usage may exit abnormally. You can run the **DSP PMU** command to check whether staggering electricity usage has exited abnormally.
- All sites meet the conditions for enabling intelligent dynamic peak staggering, as indicated by the evaluation results provided by the intelligent staggering evaluation tool. You can contact Huawei engineers to obtain the tool.

Step 1 On the MAE-Deployment, choose **Advanced > Feature Operation and Maintenance > Time-of-Use Power Configuration**.

Step 2 On the **Time-of-Use Power Configuration** tab page, set **Scenario** to **Config By Month** or **Config By Day**, and click **Export Default Template** to export the corresponding data plan table.

Step 3 Fill in the time segment information with the peak, flat, and low electricity price types specified in the data plan table as required, as shown in [Figure 8-2](#).

Figure 8-2 Time segment information

No.	Start Time	End Time	Rate
1	00:00	08:00	LOW
2	08:00	17:00	PEAK
3	17:00	21:00	FLAT
4	21:00	24:00	LOW

Step 4 On the **NE List** sheet of the data plan table:

1. To enable fixed staggering electricity usage, set **Staggered Power Effective Mode** to **Fixed** and set **Staggered Discharge Stop Threshold** as required.
2. To enable intelligent dynamic peak staggering, set **Staggered Power Effective Mode** to **Dynamic** and set **Target Backup Power Use Duration** based on the requirements of the customer, for example, 120 minutes.

Step 5 On the **Time-of-Use Power Configuration** tab page, select the prepared data plan table in the **Data Plan** area and click **Import** to import the table.

Step 6 Click **Execution** to generate incremental configuration and rollback scripts.

Step 7 Choose **Area Management > Planned Area > Export Incremental Scripts** to export the incremental configuration and rollback scripts.

- Step 8** Use the script executor of the MAE-Deployment to deliver incremental configuration scripts to activate staggering electricity usage, or deliver rollback scripts to deactivate this function.
- Step 9** (Optional) Use the parameter comparison function of the MAE-Deployment to check whether the parameter settings in the **STAGGEREDPOWERRECORD** MO are consistent among base stations that were once enabled with staggering electricity usage.
- If the parameter settings for some base stations are inconsistent, repeat steps [Step 1](#) to [Step 8](#) for these base stations based on the planned data.
 - If they are consistent, no further action is required.

 NOTE

For details about the parameter comparison function of the MAE-Deployment, see "Parameter Comparison" in *iMaster MAE Product Documentation*.

----End

8.4.2 Activation Verification

Using MML Commands

After enabling staggering electricity usage, run the following commands to check whether it has taken effect.

- Step 1** Run the **LST EQUIPMENT** command.
1. If the value of **Staggered Power Switch** and **Staggered Power Effective Mode** in the command output is **ON** and **FIXED**, respectively, fixed staggering electricity usage has been enabled.
 2. If the value of **Staggered Power Switch** and **Staggered Power Effective Mode** in the command output is **ON** and **DYNAMIC**, respectively, intelligent dynamic peak staggering has been enabled.
- Step 2** Run the **LST STAGGEREDPOWERRECORD** command to verify the existence of configuration information for staggering electricity usage.
- Step 3** Run the **DSP TIME** command. Check whether the system time is within the effective period of staggering electricity usage.
- If it is, go to [Step 4](#).
 - If it is not, wait until the start time of the effective period of staggering electricity usage arrives.
- Step 4** Run the **DSP PMU** command to query the current status of staggering electricity usage.
- If the value of **Staggered Power Effective Status** is **Enabled**, staggering electricity usage has taken effect.
 - If the value of **Staggered Power Effective Status** is **Disabled**, you can determine the possible cause of the ineffectiveness based on the value of **Staggered Power Current Disabled Cause**, as described in [Table 8-6](#) or [Table 8-7](#).

Table 8-6 Possible causes as to why fixed staggering electricity usage does not take effect

Cause	Description
Switch turned off	The function switch is not turned on.
Power capability not supported	The software or hardware version of the target PMU does not support staggering electricity usage.
Configuration failure	The BBU fails to query the effective status, or the queried effective status is inconsistent with the local configuration.
Power failure ^a	Staggering electricity usage does not take effect due to internal software or hardware faults in the power system. In this case, the PMU reports related alarms, which can be the following: – ALM-25621 Power Supply DC Output Out of Range – ALM-25622 Mains Input Out of Range – ALM-25624 Battery Power Unavailable – ALM-25626 Power Module Abnormal – ALM-25630 Power Module and Monitoring Module Communication Failure – ALM-25633 PMU Internal Interface Communication Failure – ALM-25634 Battery Not In Position – ALM-25635 Battery Failure – ALM-25654 Battery Temperature Unacceptable – ALM-25625 Battery Current Out of Range
Lithium battery not configured	Lithium batteries have been installed but not configured.
Battery type not supported ^b	Staggering electricity usage is not supported when lead-acid and lithium batteries are used together, or when only lead-acid batteries are used.
Low SOH of lithium battery	The state of health (SOH) of lithium batteries is lower than 70%.
Effect condition not met	The power system supports staggering electricity usage and is normal. However, the conditions for staggering electricity usage to take effect are not met, and this function does not take effect. For example, the remaining power of lithium batteries is insufficient, the number of discharging times within a day exceeds two, or the current time is not within the peak hours.

Cause	Description
	a: When the communication with the PMU is interrupted, the BBU reports ALM-25600 Monitoring Device Maintenance Link Failure. Staggering electricity usage does not take effect immediately after the PMU restarts. Instead, fixed staggering electricity usage is available only after the communication between the PMU and BBU recovers. b: If the battery type does not support fixed staggering electricity usage, this function does not take effect in the current staggered power consumption period. Whether it is supported in the following staggered power consumption period will be checked again based on the battery type.

Table 8-7 Possible causes as to why intelligent dynamic peak staggering does not take effect

Cause	Description
Switch turned off	The function switch is not turned on.
Power capability not supported	The software or hardware version of the target PMU does not support intelligent dynamic peak staggering.
Configuration failure	The BBU fails to query the effective status, or the queried effective status is inconsistent with the local configuration.
Power failure ^a	Intelligent dynamic peak staggering does not take effect due to internal software or hardware faults in the power system. In this case, the PMU reports related alarms, which can be the following: - ALM-25621 Power Supply DC Output Out of Range - ALM-25622 Mains Input Out of Range - ALM-25624 Battery Power Unavailable - ALM-25626 Power Module Abnormal - ALM-25630 Power Module and Monitoring Module Communication Failure - ALM-25633 PMU Internal Interface Communication Failure - ALM-25634 Battery Not In Position - ALM-25635 Battery Failure - ALM-25654 Battery Temperature Unacceptable - ALM-25625 Battery Current Out of Range
Lithium battery not configured	Lithium batteries have been installed but not configured.

Cause	Description
Battery type not supported ^b	Intelligent dynamic peak staggering is not supported when lead-acid and lithium batteries are used together, or when only lead-acid batteries are used.
Low SOH of lithium battery	The SOH of lithium batteries is lower than 70%.
Effect condition not met	The power system supports staggering electricity usage and is normal. However, the conditions for intelligent dynamic peak staggering to take effect are not met. For example, the remaining power of lithium batteries is insufficient, the number of discharging times within a day exceeds two, the current time is not within the peak hours, the backup time does not meet the backup power use duration requirements, or the PMU reset causes the recollection of the power consumption data.
Intelligent dynamic staggered power mode not supported by power capability	Power capability does not support the intelligent dynamic staggered power mode.
No available license	There is no available license.
NULL	Staggered discharge is in progress.
a: When the communication with the PMU is interrupted, the BBU reports ALM-25600 Monitoring Device Maintenance Link Failure. Staggering electricity usage does not take effect immediately after the PMU restarts. Instead, intelligent dynamic peak staggering is available only after the communication between the PMU and BBU recovers. b: If the battery type does not support staggering electricity usage, staggering electricity usage does not take effect in the current staggered power consumption period. Whether intelligent dynamic peak staggering is supported in the following staggered power consumption period will be checked again based on the battery type.	

----End

Observing Counters

Observe the value of the **VS.PMUBoard.StaggeredEnergy (5G gNodeB, LTE eNodeB)** counter on the MAE-Access. If the counter value is not 0, staggering electricity usage has taken effect.

8.4.3 Network Monitoring

Activating System KPIs

- Step 1** Log in to iMaster MAE, search for **KPI Management**, and click **KPI Management**.
- Step 2** Click **Show inactive KPIs**, enter the function subset keyword **PMU Power Saving Measurement**, select the system counter 1593835894 (**VS.PMUBoard.StaggeredEnergy (5G gNodeB, LTE eNodeB)**), and click **Activate/Deactivate**.
- Step 3** The **Activate/Deactivate KPI** dialog box is displayed. Select the KPI and click **Submit** to activate it. After the activation is submitted, click **Close** to close the current dialog box.
- Step 4** Wait until the activation progress bar reaches 100%, which indicates that the KPI is activated successfully.
- End

Creating a Custom KPI

- Step 1** Log in to iMaster MAE, search for **KPI Management**, and click **KPI Management**.
- Step 2** On the **KPI Management** page, click **New**. On the **New KPI** page, set **NE type** to **3GBTS** and **Object type** to **PMU Board**, enter the name of the KPI to be created **Peak Staggering Revenue (CNY)**, and click **New Formula**.
- Step 3** On the **New Formula** page, enter the following formula: Peak Staggering Revenue (CNY) = [PMU staggered discharge capacity] x ({Ppeak} – {Plow}/{E}) x {0.95}. In this formula:
- CNY refers to the Chinese yuan renminbi. You need to select a currency unit based on the site requirements.
 - PMU staggered discharge capacity is indicated by **VS.PMUBoard.StaggeredEnergy (5G gNodeB, LTE eNodeB)**.
 - Ppeak refers to peak electricity price, which needs to be manually entered.
 - Plow refers to valley electricity price, which needs to be manually entered.
 - E indicates the charge and discharge efficiency of the lithium battery. The value of E is 0.94.
 - The coefficient 0.95 is used in the formula because the proportion of the suppressed charge capacity to the total discharge capacity ranges from 3% to 7% and the percentage 5% is used for calculating the revenue.
- Step 4** Click **OK**, enter the name of the custom KPI on the **New KPI** page, and click **OK**. The custom KPI is created successfully.

----End

NOTE

The preceding steps take the creation of a 3G custom KPI as an example. The custom KPI belongs to power monitoring board in terms of the object type and depends only on the **VS.PMUBoard.StaggeredEnergy (5G gNodeB, LTE eNodeB)** counter. So you only need to create a custom KPI on any of the 3GBTS, 4GBTS, or 5GBTS and the custom KPI can be displayed for all the three RATs.

Creating a Whole Network Power Consumption Query Report

- Step 1** Log in to iMaster MAE, search for **Report Management**, and click **Report Management**.
- Step 2** Click **New**. On the report creation page, set **NE type** to **Whole Network** and click **Select Object** to select the type of the object to be queried. Then, select **Whole Network**.
- Step 3** Click **Select KPI**, enter a keyword to search for the newly created custom KPI **Peak Staggering Revenue (CNY)**, and select the KPI.
- Step 4** Click **Select Time**. You can set the report query time as required. Click **OK**, click **Save**, and enter the customized report name. A message is displayed, indicating that the report is saved successfully. The whole network power consumption query report is created successfully.

----End

Creating an NE Power Consumption Query Report

- Step 1** Log in to iMaster MAE, search for **Report Management**, and click **Report Management**.
- Step 2** Click **New**. On the report creation page, set **NE type** to **3GBTS** and **Object type** to **PMU Board**. Click **Select Object** to select the type of the object to be queried. The range can be customized.
- Step 3** Click **Select KPI** and select the newly created custom KPI **Peak Staggering Revenue (CNY)**.
- Step 4** Click **Select Time**. You can set the report query time as required. Click **OK**, click **Save**, and enter the customized report name. The NE power consumption query report is created successfully.

----End

NOTE

The preceding steps use the 3G NE power consumption query report as an example. For a 3G NE, 4G NE, and 5G NE, respectively set **NE type** to **3GBTS**, **4GBTS**, and **5GBTS**. For a multimode NE, select any of **3GBTS**, **4GBTS**, and **5GBTS**.

Creating an Object Group Power Consumption Query Report

- Step 1** Log in to iMaster MAE, search for **Object Group Management**, and click **Object Group Management**.
- Step 2** In the navigation tree on the left, click **3GBTS** and click **New**. The **New Object Group** tab page is displayed.
- Step 3** On the **New Object Group** tab page, set **NE type** to **3GBTS**, and set **Object type** to **PMU Board**. You can customize the NE range. Click **OK** to create an object group.
- Step 4** On the **Report Management** page, click **New**. On the report creation page, set **NE type** to **3GBTS** and **Object type** to **PMU Board Group**. Then, click **Select Object**. Select the new peak staggering revenue object group.

Step 5 Click **Select KPI** and select the newly created custom KPI **Peak Staggering Revenue (CNY)**.

Step 6 Click **Select Time**. You can set the report query time as required. Click **OK**, click **Save**, and enter the customized report name. The object group power consumption query report is created successfully.

----End

 NOTE

The preceding steps use the 3G object group power consumption query report as an example. For a 3G NE, 4G NE, and 5G NE, respectively set **NE type** to **3GBTS**, **4GBTS**, and **5GBTS**. For a multimode NE, select any of **3GBTS**, **4GBTS**, and **5GBTS**.

Viewing a Report

Step 1 Log in to iMaster MAE, search for **Report Management**, and click **Report Management**.

Step 2 Click the report name to go to a new page to view the revenue data.

Step 3 (Optional) Click **New Graph (Optional)**. In the displayed dialog box, select a chart as required and click **OK**. The chart is displayed based on the revenue data.

----End

9 PV Deployment for Base Stations

9.1 Principles

The APM5950H, APM5950H-L, and ICC cabinets can now use solar supply units (SSUs) in addition to the existing PSUs and batteries to provide power supply. With such a PV deployment function, the SSUs are preferentially used when there is sunlight. If the generated power is insufficient, the SSUs and PSUs work together to supply power when the mains supply is available, and the SSUs and batteries work together to supply power when the mains supply is unavailable. In short, when there is sunlight, solar power is preferred to save electricity fees and reduce carbon emissions.

The PV deployment function is controlled by the **PSU.PSUTYPE (5G gNodeB, LTE eNodeB)** parameter and can be enabled after an SSU is added. If there are multiple SSUs, add them one by one.

This function applies to GSM, UMTS, NR, and LTE. For GSM, only the eGBTS supports this function.

9.2 Network Analysis

9.2.1 Benefits

This function saves electricity fees and reduces carbon emissions.

9.2.2 Impacts

The impacts between this function and other functions are described in [9.3.2.3 Function Impacts](#).

Impact on the counter "Power consumption measured by the power system of *N*" (*N* refers to GSM, UMTS, LTE, NR, or RFA): In PV deployment scenarios, the power system measures the output power of PSUs and SSUs. The solar energy supplies DC power; as such, the power consumption measured at the AC measurement point may be less than that measured at the DC measurement point.

9.3 Requirements

9.3.1 Licenses

None

9.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

9.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
GSM UMTS LTE FDD LTE TDD NR	Power System Type	PMU.PTYPE (5G gNodeB , LTE eNodeB) set to ETP	None	If PSU.PSUTYPE (5G gNodeB , LTE eNodeB) is set to SSU for a PSU, PMU.PTYPE (5G gNodeB , LTE eNodeB) must be set to ETP for the corresponding PMU.

9.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
GSM UMTS LTE FDD LTE TDD NR	Adaptive Power Backup	N/A	15 Adaptive Power Backup	When the Adaptive Power Backup feature takes effect, PV deployment for base stations cannot be enabled. This is because the PV deployment function will cause the calculation results related to the Adaptive Power Backup feature to be inaccurate.

9.3.2.3 Function Impacts

After the PV deployment function is enabled, the power system can use mains and solar power to provide power supply. The solar power supply is used preferentially. Since an extra power supply is introduced, the following functions are affected.

RAT	Function Name	Function Switch	Reference	Description
NR LTE FDD LTE TDD UMTS GSM	Staggering electricity usage (fixed)/ Intelligent dynamic peak staggering	EQUIPMENT.STAGGEREDPOWERSWITCH (LTE eNodeB, 5G gNodeB)	<i>Green Site</i>	The benefits of staggering electricity usage are affected when base stations use the PV deployment function. • When solar power supply is used, power discharge in off-peak hours is restricted, and the benefits provided by staggering electricity usage decrease. • When the solar power is greater than the load power and staggering electricity usage is enabled, lithium batteries do not discharge power in peak hours. Staggering electricity usage does not provide any benefits in this case.

9.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

DBS5900 (macro base station)

Boards

No requirements

RF Modules

No requirements

Power Supply

- The cabinet must be an APM5950H, APM5950H-L, or ICC series cabinet. For details about ICC series cabinets and the mapping between PMUs and cabinets, see "Mapping Between Physical Cabinets and PTYPE Values" in "Hardware" in section "Basic Power Supply Management Functions" in *Power Supply Management*.
- SSUs, PSUs, and lead-acid or lithium batteries are installed at the site. The solar power supply is used only as an auxiliary power supply. If there is no solar power supply, the PSU and battery configurations must meet the normal power supply requirements of the site load. It is recommended that the configured SSU power be less than the typical power of the site load. The restrictions on the number of SSUs configured in different cabinets are as follows:
 - For an APM5950H, at most two SSUs can be configured.
 - For an APM5950H-L, at most one SSU can be configured.
 - For an ICC series cabinet, at most nine SSUs can be configured.
- You are advised to install PSUs before installing SSUs.
 - For an APM5950H, it is recommended that the SSU be installed in slot 7 or 6. Do not install two SSUs in the same phase. Otherwise, AC phase loss may occur. If PSU current limiting protection in case of AC circuit breaker capability insufficiency is enabled, PSU current limiting protection may take effect earlier than expected when AC phase loss occurs. In this case, power supply to the site may be disconnected.
 - For an APM5950H-L, it is recommended that the SSU be installed in slot 4.
 - For an ICC series cabinet, the configured slot number is only a logical slot number. It is recommended that the SSU slot numbers be configured one by one after the PSU slot number is configured.
- Other components related to solar power are also required, including PV modules, PV module supports, and solar junction boxes (SJBs). For an APM5950H, the passive circuit breaker unit (PVDU or PVBU) is required. For an APM5950H-L, the passive circuit breaker unit (PVBU) is required.

9.3.4 Others

- Base station software version
The base station software of RAN24.1/eRAN18.1/SRAN18.1/5G RAN6.1 or later is required.

9.4 Operation and Maintenance

9.4.1 Data Configuration

9.4.1.1 Data Preparation

Table 9-1 describes the required parameters for the PV deployment function.

Table 9-1 Parameters for the PV deployment function

Parameter Name	Parameter ID	Setting Notes
PSU Type	PSU.PSUTYPE (5G gNodeB, LTE eNodeB)	At least one SSU must be configured for the PV deployment function. SSU configuration requirements vary according to cabinet types. For details, see 9.3.4 Others .

9.4.1.2 Using MML Commands

Before using MML commands, refer to [9.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- Step 1** (Optional) Activate the PMU software. (The base station software version must meet the requirements of this function. For details, see [9.3.4 Others](#).)

```
//Activating the PMU software
ACT SOFTWARE: OT=BOARD, CN=0, SRN=7, SN=0;
//Checking whether the PMU software has been upgraded to the target version
DSP BRDVER: CN=0,SRN=7,SN=0;
```

- Step 2** Run the **DSP BATTERY** command to check whether the system is configured with batteries. If batteries are not configured, perform step 3 to configure batteries based on the type and number of batteries installed.

```
DSP BATTERY: CN=0;
```

- Step 3** (Optional) Run the **ADD BATTERY** command with parameters set based on the type and number of batteries installed.

```
//Using lithium batteries as an example
ADD BATTERY: CN=0, SRN=7, SN=0, BTYPE=LI_BAT, HTSDF=ENABLE, LVSD=ENABLE;
```

- Step 4** Run the **ADD PSU** command with **PSUTYPE** set to **SSU**. If there are multiple SSUs, add them one by one. The following provides an example to add an SSU in slot 7 of the APM5950H:

```
ADD PSU:CN=0,SRN=7,SN=7,PSUTYPE=SSU;
```

----End

Deactivation Command Examples

Run the **RMV PSU** command to remove the SSU from the corresponding slot. If there are multiple SSUs, remove them one by one.

9.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

9.4.2 Activation Verification

Using MML Commands

Run the **DSP PSU** command to check whether the running status of the SSU is normal. If there are multiple SSUs, run this command for each SSU. The function is successfully activated when the effective result and query result are consistent.

When there is sunlight and the corresponding upper-level circuit breaker is ON, you can run the **DSP PMU** command to query the total SSU output current. If the output current is not zero, the function has been activated successfully.

NOTE

When solar power input overvoltage occurs due to improper PV module connection, the **DSP PSU** command output shows that the SSU running status is **Protect**. The SSU running status queried by running the **DSP PSU** command is **Shut down** in the following scenarios:

- The power system is started within 5 minutes.
- The solar energy is insufficient or the solar power device is not connected.
- No battery is configured or the battery circuit is disconnected.
- The battery is powered off because the voltage of the power system is too low or the temperature of the power system is too high.
- A large number of lithium batteries are faulty, and the backup time of the power system is less than 30 minutes.
- The mains supply is unavailable and the backup time of the power system is less than 30 minutes when **Load Shutdown Flag** is set to **DISABLE**.

Using Counters

When there is sunlight and the corresponding upper-level circuit breaker is ON, check the **VS.PMUBoard.SolarOutputEnergy (LTE eNodeB, 5G gNodeB)** counter value on the MAE-Access. If the value is not 0, the PV deployment function has taken effect.

9.5 Network Monitoring

View the **VS.PMUBoard.SolarOutputEnergy (LTE eNodeB, 5G gNodeB)** counter on the MAE-Access to observe the total output current of SSUs when the PV deployment function is enabled.

10 Third-Party Green Power Access Measurement and O&M Management

10.1 Principles

The green power input unit (GIU) can be used to implement measurement and O&M management of third-party green power supplies. When the third-party green power source is photovoltaic (PV) power, the GIU can be used to perform intelligent photovoltaics (IPV) management, including PV-related energy yield measurement, basic alarm management, and O&M management.

The GIU can be used in the following scenarios where:

- A Huawei cabinet is used.
- The iSDU06D is used with a third-party power system. iSDU is short for intelligent site digitalized unit.

The GIU supports the following functions:

- Checks the input current and voltage of green power (such as PV power) and measures its energy yields.
- Provides basic alarm management and O&M management functions for the green power (such as PV power) system.

This function applies to GSM, UMTS, NR, and LTE. For GSM, only the eGBTS supports this function.

10.2 Network Analysis

10.2.1 Benefits

Using third-party green power supplies can reduce mains electricity fees and carbon emissions.

10.2.2 Impacts

The impacts between this function and other functions are described in [10.3.2.3 Function Impacts](#).

Impact on the counter "Power consumption measured by the power system of N " (N refers to GSM, UMTS, LTE, NR, or RFA): When a third-party green power supply and mains supply are used together, the power consumption measured by the power system includes the output power of PSUs and GIUs. The third-party green power supply provides DC power. Therefore, the power consumption at the AC measurement point may be less than that at the DC measurement point.

10.3 Requirements

10.3.1 Licenses

None

10.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

10.3.2.1 Prerequisite Functions

None

10.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
GSM UMTS LTE FDD LTE TDD NR	Adaptive Power Backup	N/A	15 Adaptive Power Backup	When the Adaptive Power Backup feature takes effect, third-party green power access measurement and O&M management cannot be enabled. This is because the PV deployment function will cause the calculation results related to the Adaptive Power Backup feature to be inaccurate.

10.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
NR LTE FDD LTE TD D UM TS GS M	Staggering electricity usage (fixed)/ Intelligent dynamic peak staggering	EQUIPMENT.STAGGEREDPOWER_SWITCH (LTE eNodeB, 5G gNodeB)	<i>Green Site</i>	The benefits of staggering electricity usage are affected by third-party green power access measurement and O&M management. • When solar power supply is used, power discharge in off-peak hours is restricted, and the benefits provided by staggering electricity usage decrease. • When the solar power is greater than the load power and staggering electricity usage is enabled, lithium batteries do not discharge power in peak hours. Staggering electricity usage does not provide any benefits in this case.

RAT	Function Name	Function Switch	Reference	Description
NR LTE FDD LTE TDD UMTS GSM	PV deployment for base stations	PSU.PSUTYPE (5G gNodeB, LTE eNodeB)	<i>Green Site</i>	The benefits of PV deployment for base stations are affected by third-party green power access measurement and O&M management. If both third-party green power (such as PV power) and SSUs are used to supply power, a conflict arises in the statistical source of the total output current of the SSUs and the solar energy yield. As a result, the statistics are inaccurate. Therefore, you are not advised to enable SSU-based PV deployment when a third-party green power supply is used.

10.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

Equipment

To enable third-party green power access measurement and O&M management, the base station must be configured with a GIU. The following table lists the cabinets and devices that support the GIU.

Table 10-1 Cabinets and devices that support the GIU

Cabinet or Equipment Type	Description
APM5950H cabinet	N/A

Cabinet or Equipment Type	Description
APM5950H-L cabinet	N/A
APM5930 cabinet	N/A
APM30H Ver.E cabinet	When used with an APM30H Ver.E cabinet, this function supports only lead-acid batteries. To support lithium batteries, a cabinet upgrade package is required.
iSDU06D	- The GIU can be used as an optional component of the iSDU06D in PV deployment scenarios. - The iSDU06D can be used with the GIU to implement green power access on condition that the third-party power system is configured with backup batteries and the iSDU06D can obtain the mains status.

Boards

The PMU 11B does not support this function.

Power Supply Systems

- A third-party green power (such as PV power) system, such as the PVDU-150A1 and PVPU-72N1, must be installed outside the APM cabinet or iSDU06D.
- The output power of the third-party green power (such as PV power) must be less than the typical power of the site load.
- To ensure that the green power (such as PV power) is preferentially used to supply power to site loads, it is recommended that the voltage of the third-party green power supply be 0.5 V to 1.0 V higher than the busbar voltage of the APM cabinet.
- Hierarchical power-off takes effect on condition that the iSDU06D can obtain the mains status. The mains status can be obtained through Simple Network Management Protocol (SNMP) or dry contacts.

10.3.4 Others

The GIU can be used when a Huawei cabinet is used and when the iSDU06D is used with a third-party power system. The base station software version requirements are as follows:

- When a Huawei cabinet is used, the software of SRAN19.1 or later is required.
- When the iSDU06D is used with a third-party power system, the software of SRAN20.0 or later is required.

10.4 Operation and Maintenance

10.4.1 Data Configuration

10.4.1.1 Data Preparation

The following table lists the parameter settings for third-party green power access measurement and O&M management.

Table 10-2 Parameter settings for third-party green power access measurement and O&M management

Parameter Name	Parameter ID	Setting Notes
PDU Type	PDU.PDUTYPE <i>(LTE eNodeB, 5G gNodeB)</i>	This function requires at least one GIU. For an APM5950H cabinet, it is recommended that a maximum of two GIUs be configured. For an APM5950H-L/APM5930/APM30H (Ver.E) cabinet, it is recommended that a maximum of one GIU be configured. • For the APM5950H, the GIU can be installed in slots 8 to 12. • For the APM5950H-L, the GIU can be installed in slots 5 to 7. • For the APM5930, the GIU can be installed in slots 6 to 10. • For the APM30H Ver.E, the GIU can be installed in slots 6 to 9.
Green Power Startup Mode	PDU.GPSM <i>(LTE eNodeB, 5G gNodeB)</i>	This parameter indicates the startup mode of the green power module connected to the GIU. It can be set to SELFSTART (default value) or NON-SELFSTART . When this parameter is set to SELFSTART , the green power module starts and outputs power as long as there is sunlight, requiring no power supply from the base station power system. When this parameter is set to NON-SELFSTART , the green power module starts only after it is power supplied by the base station power system.
Switch	ALMPORT.SW <i>(5G gNodeB, LTE eNodeB)</i>	It is recommended that this parameter be set to ON for the port through which alarms are required to be detected and reported.

Parameter Name	Parameter ID	Setting Notes
Port Type	ALMPORT.PT (5G gNodeB, LTE eNodeB)	Set this parameter to the actual port type based on the network plan.
Alarm Voltage	ALMPORT.AVOL (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.

10.4.1.2 Using MML Commands

Before using MML commands, refer to [10.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Basic Configuration Examples

If the iSDU06D is used with a third-party power system, first obtain the status of the mains supply to the third-party power system.

- If the iSDU06D is connected to the third-party power system through the SNMP, see [Data Configuration](#) in section "iSDU06D Used with a Third-Party Power System."
- The following is a configuration example when the iSDU06D is connected to the third-party power system through a dry contact.

```
//Setting the dry contact signal input port, for example, setting CN, SRN, SN, PN, PT, and AVOL as required, SW to ON, and AID to 65033
SET ALMPORT: CN=0, SRN=17, SN=0, PN=0, SW=ON, AID=65033, PT=BOOL, AVOL=LOW;
//Modifying the port binding relationship for the mains supply outage alarm, for example, setting ISDSWITCH to ENABLE and NMSACN, NMSASRN, NMSASN, and NMSAPN as required
MOD NMSABIND: ISDSWITCH=ENABLE, NMSACN=0, NMSASRN=17, NMSASN=0, NMSAPN=0;
```

Activation Command Examples

- Step 1** (Optional) Activate the PMU software. (The base station software version must meet the requirements of this function. For details, see [10.3.4 Others](#).)

```
//Activating the PMU software
ACT SOFTWARE: OT=BOARD, CN=0, SRN=7, SN=0;
//Checking whether the PMU software has been upgraded to the target version
DSP BRDVER: CN=0, SRN=7, SN=0;
```

- Step 2** Run the **DSP BATTERY** command to check whether the system is configured with batteries. If batteries are not configured, perform step 3 to configure batteries based on the type and number of batteries installed.

```
DSP BATTERY: CN=0;
```

Step 3 (Optional) Run the **ADD BATTERY** command with parameters set based on the type and number of batteries installed.

//Batteries can be configured only when Huawei cabinets are used. Using lithium batteries as an example
ADD BATTERY: CN=0, SRN=7, SN=0, BTYPE=LI_BAT, HTSDF=ENABLE, LVSDf=ENABLE;

Step 4 Run the **ADD PDU** command with **PDUTYPE** set to **GIU**. To add multiple GIUs, run **ADD PDU** multiple times. The following provides an example to add a GIU in slot 8 of the APM5950H:

ADD PDU: CN=0, SRN=7, SN=8, PDUTYPE=GIU, GPSM=SELFSTART;

Step 5 (Optional) Activate the GIU software. (The base station software version must meet the requirements of this function. For details, see [10.3.4 Others](#).)

//Activating the GIU software
ACT SOFTWARE: OT=BOARD, CN=0, SRN=7, SN=8;
//Checking whether the GIU software has been upgraded to the target version
DSP BRDVER: CN=0, SRN=7, SN=8;

----End

Deactivation Command Examples

Run the **RMV PDU** command to remove the GIU in the corresponding slot. To remove multiple GIUs, run **RMV PDU** multiple times.

10.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

10.4.2 Activation Verification

Using MML Commands

- Run the **DSP PDU** command to query the running status of all GIUs. If the running status of all GIUs is normal, this function has taken effect.
- Assume that the third-party green power is PV power. If there is sunlight and the corresponding circuit breaker is ON, you can run the **DSP PMU** command to query the total output current of the solar supply units (SSUs). If the output current is not zero, the function has taken effect.
- Run the **LST PDU** command to query the **Green Power Startup Mode** parameter value for all GIUs. If the value is the same as the configured value, the configuration has taken effect.

NOTE

When the GIU experiences overtemperature or input overvoltage, the running status of the GIU queried by running the **DSP PDU** command will be displayed as **Protected**, and ALM-25626 Power Module Abnormal will be reported with **Specific Problem** being **Power Distribution Module Protection**. You can try to remove and then insert the GIU locally. If the fault persists, replace the GIU.

In certain scenarios, the GIU running status queried by running the **DSP PDU** command is **Shut down** and the PV deployment function is disabled. The possible causes are as follows:

- When the GIU is used in scenarios where Huawei cabinets are used:
 - The DIP switch is turned off.
 - The circuit breaker on the panel is turned off.
 - The power system was started less than 5 minutes ago.
 - No battery is configured or the battery circuit is disconnected.
 - The battery is powered off because the voltage of the power system is too low or the temperature of the power system is too high.
 - A large number of lithium batteries are faulty, and the backup time of the power system is less than 30 minutes.
 - The mains supply is unavailable and the backup time of the power system is less than 30 minutes when **Low Voltage Shutdown Flag** is set to **DISABLE**.
- When the GIU is used in scenarios where the iSDU06D is used with a third-party power system:
 - The DIP switch is turned off.
 - The circuit breaker on the panel is turned off.
 - The system was started less than 5 minutes ago.
 - 57 V power distribution scenario: The mains supply is disconnected (the mains status has been obtained) or the mains status cannot be obtained.
 - 53 V power distribution scenario: The mains is disconnected (the mains status has been obtained) or the power distribution voltage drops to below 49 V.

Using Counters

Check the **VS.PMUBoard.SolarOutputEnergy (LTE eNodeB, 5G gNodeB)** counter value on the MAE-Access. If the value is not zero, the third-party green power supply has output power.

10.5 Network Monitoring

View the **VS.PMUBoard.SolarOutputEnergy (LTE eNodeB, 5G gNodeB)** counter on the MAE-Access to observe the total output current of SSUs when a third-party green power supply is used.

11 Low Power Consumption Mode (LTE)

11.1 Principles

When a power insufficiency or power failure alarm is reported or user intervention is involved, cells enter low power consumption mode to reduce the power consumption of the RF modules and extend the service time of the base station. This function is controlled by the **CellLowPower.LowPwrSwitch** parameter.

Low power consumption mode uses open-loop control. That is, the backup power duration increases by an unknown amount. This function aims to extend the backup power duration. When it takes effect, coverage and performance may be affected.

11.1.1 Application Scenario

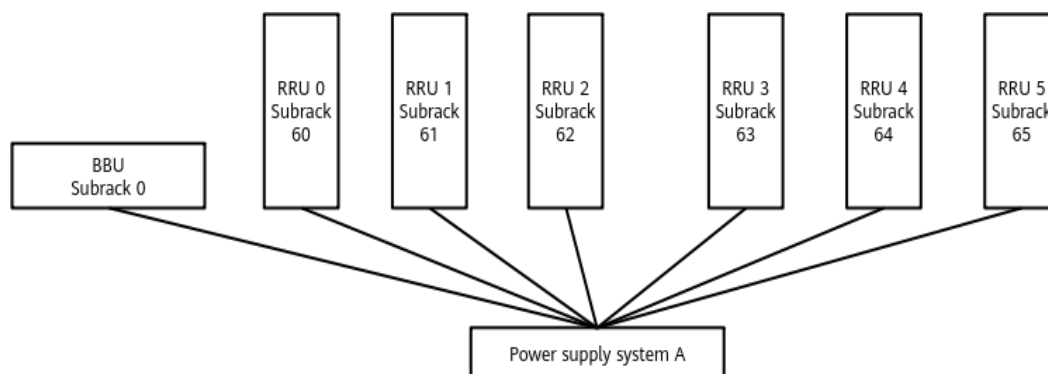
The low power consumption mode can be used in centralized and distributed power supply scenarios.

Centralized Power Supply Scenario

A centralized power supply scenario (as shown in [Figure1 Centralized power supply scenario](#)) is when main base station devices (including the BBU and AAUs/RRUs) use the same power supply system and backup power supply system. If the mains supply fails, the same backup power supply system provides power for all main devices, and the same power alarms are reported if any.

In this scenario, all cells enabled with this function will enter low power consumption mode if the mains supply fails for a period longer than the transitional period.

Figure 11-1 Centralized power supply scenario

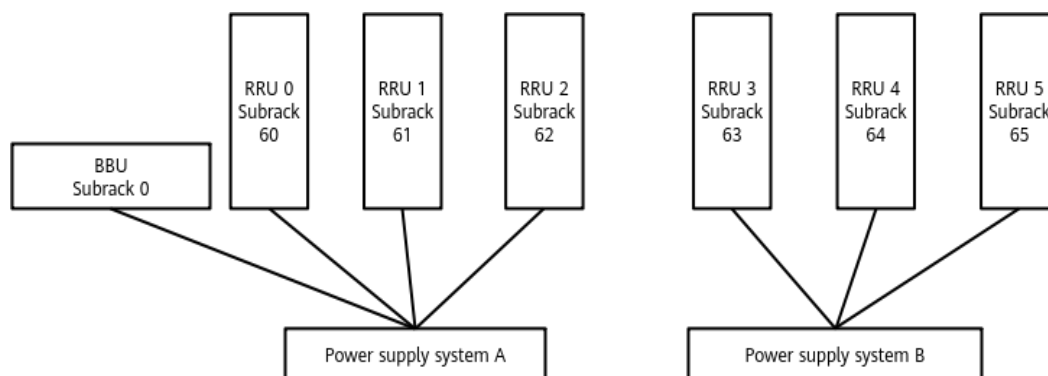


Distributed Power Supply Scenario

A distributed power supply scenario (as shown in [Figure2 Distributed power supply scenario](#)) is when main base station devices (including the BBU and AAUs/RRUs) use multiple power supply systems and backup power supply systems. If the mains supply of a power supply system fails, the corresponding backup power supply system provides power for the related main devices, but the power supply systems of other devices remain unchanged, and different power alarms may be reported for different devices.

In this scenario, only cells that are enabled with this function and served by AAUs/RRUs experiencing power supply failures will enter low power consumption mode if the mains supply fails for a period longer than the transitional period.

Figure 11-2 Distributed power supply scenario



To ensure that only cells served by AAUs/RRUs experiencing power supply failures enter low power consumption mode in the distributed power supply scenario, power supply group IDs must be correctly configured for AAUs/RRUs before low power consumption mode is enabled. A power supply group ID identifies a group of AAUs/RRUs (a single AAU/RRU may exist in the group) using the same power supply system, and is specified by the **RRU.PSGID** parameter. When configuring this parameter, comply with the following principles:

- One power supply group ID is configured for AAUs/RRUs using the same power supply system, whereas different power supply group IDs are configured for AAUs/RRUs using different power supply systems.

- The power supply group ID of the AAUs/RRUs using the same power supply system as the BBU is **0**.

As shown in **Figure2 Distributed power supply scenario**, the power supply group ID of the RRUs in subbracks 60, 61, and 62 must be set to **0**, and that of the RRUs in subbracks 63, 64, and 65 can be set to **1**.

If the mains supply fails for a period longer than the transitional period for the power supply system whose monitoring is managed by the BBU, the NR cells that are enabled with low power consumption mode and served by AAUs/RRUs with the **RRU.PSGID** parameter set to **0** will enter the low power consumption mode.

If the mains supply fails for a period longer than the transitional period for the power supply system whose monitoring is managed by an AAU/RRU, the NR cells that are enabled with low power consumption mode and served by the AAU/RRU or AAUs/RRUs having the same **RRU.PSGID** setting as the preceding AAU/RRU will enter the low power consumption mode.

NOTE

- In the distributed power supply scenario, multiple dry contact ports may report alarms indicating mains supply failures that occur in non-Huawei power supply systems. If this is the case, the same alarm ID (specified by the **ALMPORT.AID** parameter) must be set for all these dry contact ports.
- If a BBU monitors multiple power supplies that use different AC input sources or are controlled by separate circuit breakers, the mains input exception status of these power supplies may be inconsistent if the mains supply is abnormal. If the inconsistency lasts for more than the value of the **NRDUCellPowerSaving.TimeToTrigger** parameter, low power consumption mode cannot work properly.
- If the mains supply is abnormal, the base station reports ALM-25622 Mains Input Out of Range. In this case, rectify the fault onsite as soon as possible (for example, by restoring the mains supply or resetting the circuit breaker).
- If the **RRU.PSGID** parameter value is inconsistent with actual power supply relationships, AAUs/RRUs with normal power supplies may enter low power consumption mode when this function is triggered. As a result, the cells served by these AAUs/RRUs may experience capability degradation or become unavailable due to shutdown.

11.1.2 Triggering

An eNodeB can enter low power consumption mode in either of the following ways:

- Automatic triggering mode
After the low power consumption mode switch (specified by **CellLowPower.LowPwrSwitch**) is turned on, the eNodeB reports ALM-25622 Mains Input Out of Range or detects a user-defined mains supply interruption alarm within a specified period.
The period during which the eNodeB remains in the low power consumption mode is specified by **CellLowPower.StartTime** and **CellLowPower.StopTime**. The user-defined mains supply interruption alarm can be set through the **MAINSALARMBIND** MO:
- Manual intervention mode
After the low power consumption mode switch (specified by **CellLowPower.LowPwrSwitch**) is turned on, the forcible entering switch of

low power consumption mode (specified by **CellLowPower.LowPwrOrder**) also needs to be turned on. Then, the cell enters low power consumption mode regardless of whether it is within the specified period of low power consumption or an alarm is reported.

11.1.3 eNodeB Actions

When low power consumption mode is triggered, the eNodeB does not enter this mode immediately but works normally for a period of time. This period is called a transitional period.

- If the ALM-25622 Mains Input Out of Range or a user-defined mains supply interruption alarm is cleared, or the user cancels manual intervention within this period, the eNodeB exits low power consumption mode.
- If the ALM-25622 Mains Input Out of Range or a user-defined mains supply interruption alarm persists, or manual intervention continues within this period, the eNodeB takes different actions depending on the specific backup power saving policy after the period ends. The backup power saving policy is specified by the **CellLowPower.BakPwrSavPolicy** parameter.

The eNodeB triggers low power consumption mode and takes power saving measures based on the backup power saving policy.

11.1.3.1 Power Saving Policy

Low power consumption mode measures are specified by the **CellLowPower.BakPwrSavPolicy** parameter.

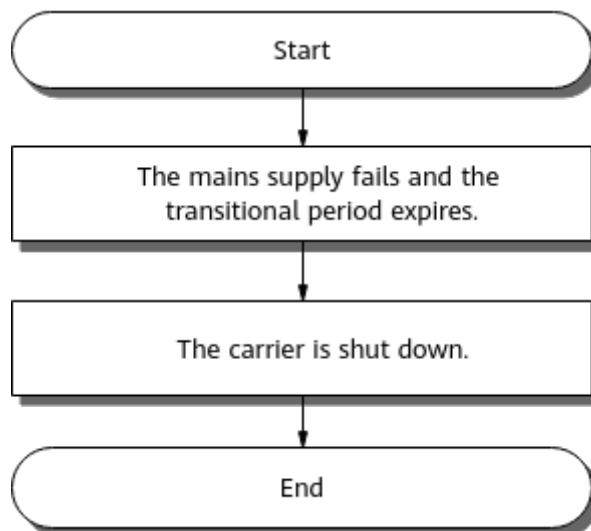
Backup power saving policies in low power consumption mode include low power consumption mode measures and precise backup power saving policies. If the **CellLowPower.LowPwrOrder** parameter is set to **ON** to enable manual intervention mode, only low power consumption mode measures are supported while precise backup power saving policies are not.

Low Power Consumption Mode Measure

When the **CellLowPower.BakPwrSavPolicy** parameter is set to **POLICY1**, **POLICY2**, **POLICY3**, **CUSTOMIZED**, or **PRIORITY_CUSTOMIZED**, the eNodeB executes low power consumption mode measures.

- Policy 1 (indicated by **POLICY1**): If the mains supply fails and the transitional period exceeds 5 minutes, the eNodeB shuts down the carrier.

Figure 11-3 Policy 1



- Policy 2 (indicated by **POLICY2**), policy 3 (indicated by **POLICY3**), and customized policy (indicated by **CUSTOMIZED**): The eNodeB executes power saving measures in sequence based on the mains supply failure duration, as shown in [Figure 11-4](#). The measures are as follows: lowering the transmit power for services, lowering the transmit power for reference signals, shutting down some transmit channels, and shutting down the carrier.

Figure 11-4 Execution of low power consumption mode measures in sequence

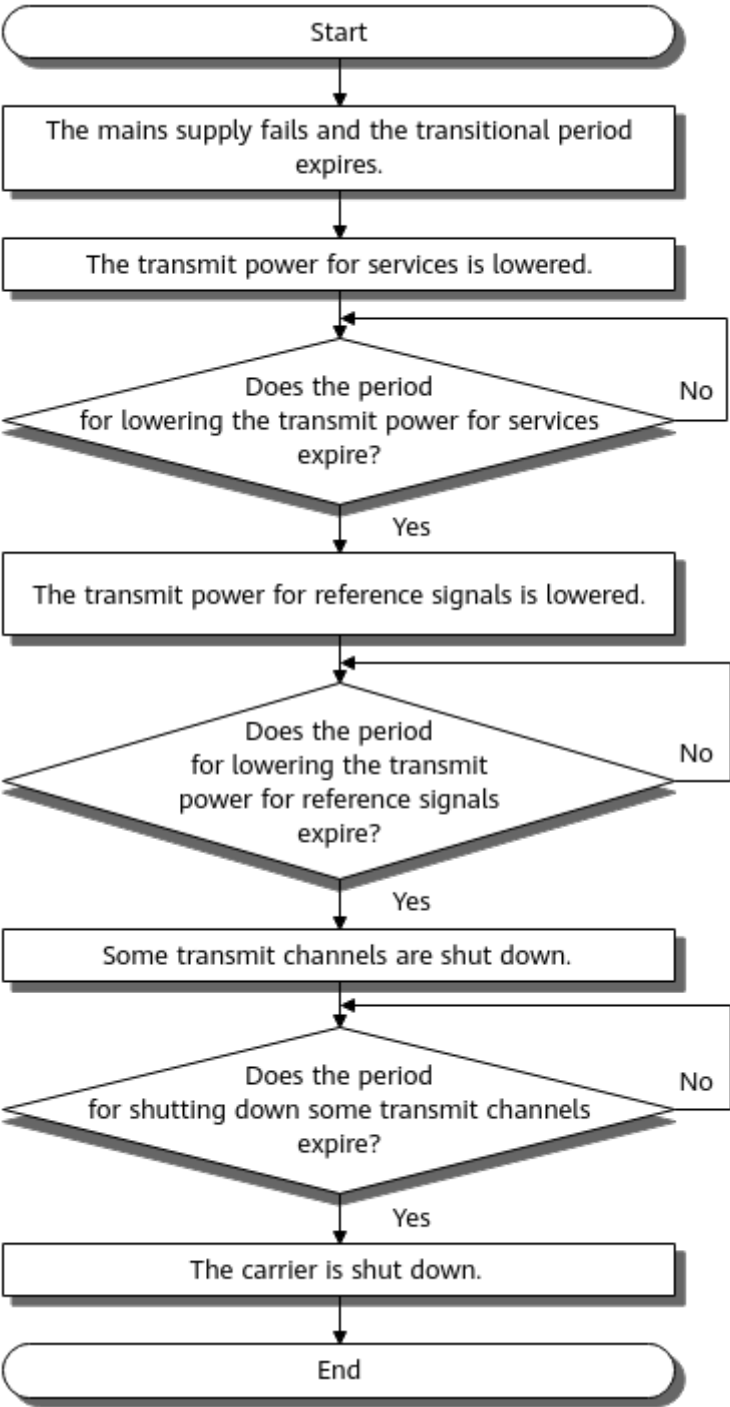


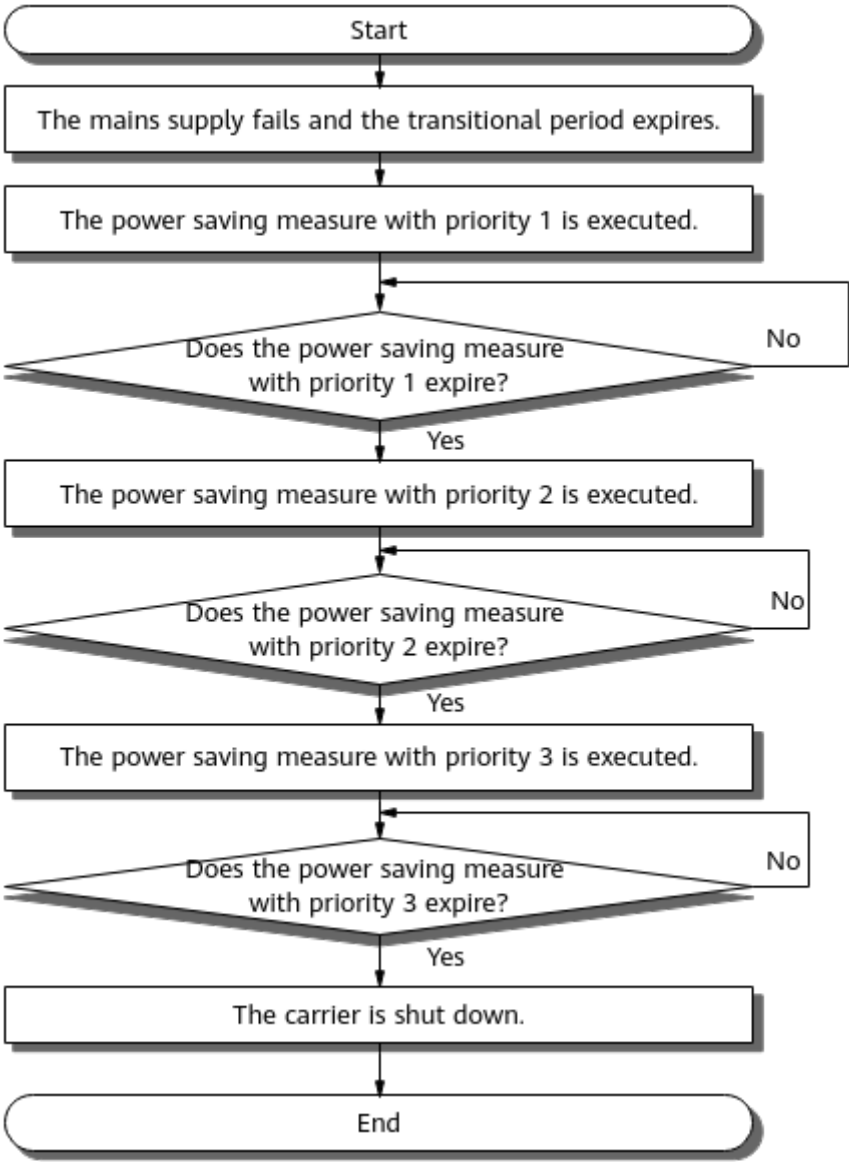
Table 11-1 lists the settings of the transitional period and periods for executing some power saving measures.

Table 11-1 Settings of the periods for a customized policy, policy 2, and policy 3

Back power save policy Parameter Value	Transitional Period (min)	Period for Lowering the Transmit Power for Services (min)	Period for Lowering the Power for Reference Signals (min)	Period for Shutting Down Some Transmit Channels (min)
POLICY2	5	120	120	120
POLICY3	5	120	120	1440
CUSTOMIZED	CellLowPower.EnterTimeLen	CellLowPower.CellUsedPowerReduceTimeLen	CellLowPower.RsPowerReduceTimeLen	CellLowPower.RfShutDownTimeLen

- Priority-based customized policy (indicated by **PRIORITY_CUSTOMIZED**): The eNodeB executes power saving measures in sequence based on the mains supply failure duration and customized measure priorities, as shown in [Figure3 Priority-based customized policy](#).

Figure 11-5 Priority-based customized policy



[Table2 Configurations of the priority-based customized policy](#) lists the parameters specifying the periods and priorities for some power saving measures.

Table 11-2 Configurations of the priority-based customized policy

Item	Transition Period	Lowering the Transmit Power for Services	Lowering the Power for Reference Signals	Shutting Down Some Transmit Channels
Duration	CellLowPower.EnterTimeLen	CellLowPower.CellUsedPowerReduceTimeLen	CellLowPower.RsPwrReduceTimeLen	CellLowPower.RfShutDownTimeLen

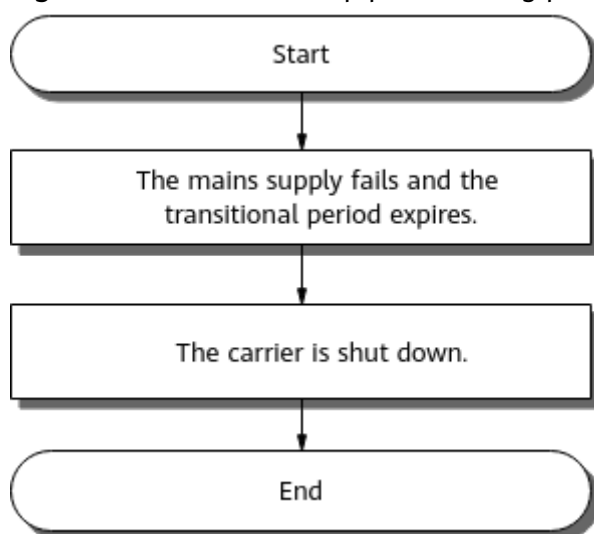
Item	Transition Period	Lowering the Transmit Power for Services	Lowering the Power for Reference Signals	Shutting Down Some Transmit Channels
Priority	N/A	CellLowPower.CellUsedPwrReduceStartPri	CellLowPower.RsPwrReduceStartPri	CellLowPower.RfShutDownStartPri

Precise Backup Power Saving Policy

When the **CellLowPower.BakPwrSavPolicy** parameter is set to **PRECISE_BACKUP_POLICY1**, **PRECISE_BACKUP_POLICY2**, or **PRECISE_BACKUP_POLICY3**, the eNodeB chooses a precise backup power saving policy.

- Precise backup power saving policy 1 (**CellLowPower.BakPwrSavPolicy** set to **PRECISE_BACKUP_POLICY1**): When the eNodeB performs precise backup power saving policy 1, it shuts down the carrier without any determination procedure in the case that the transitional period exceeds 5 minutes, as shown in [Figure 11-6](#).

Figure 11-6 Precise backup power saving policy 1



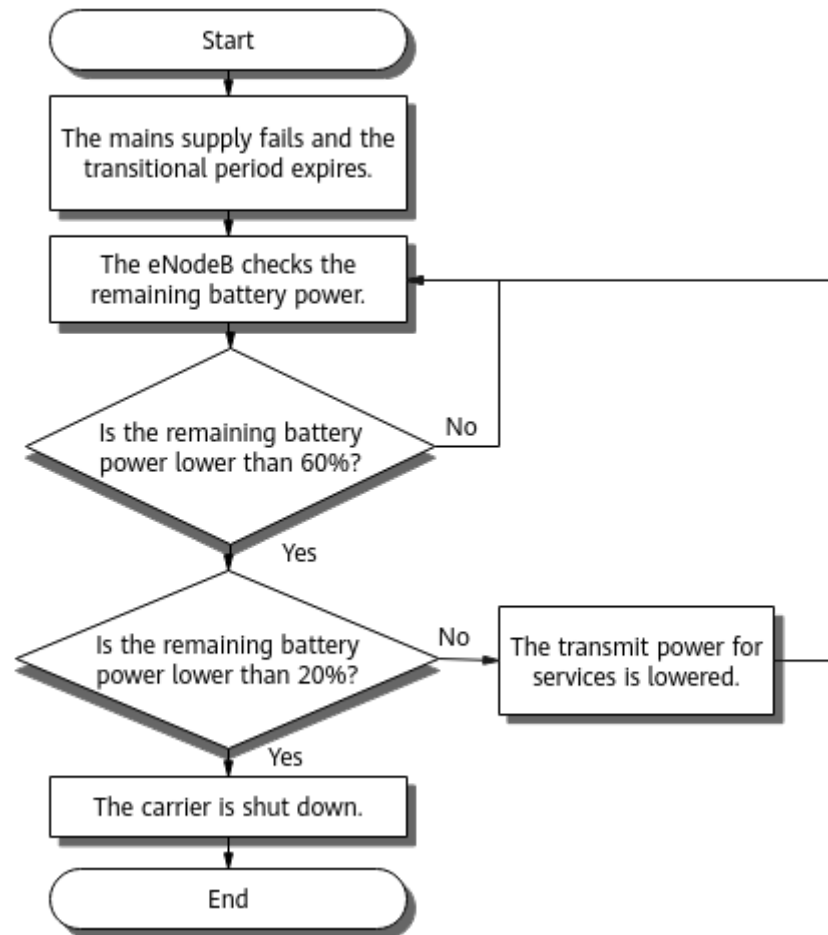
NOTE

When the **CellLowPower.BakPwrSavPolicy** parameter is set to **PRECISE_BACKUP_POLICY1** for a cell and the maximum backup power duration is defined and takes effect on the MAE-Deployment side, the MAE-Deployment identifies the cell as a non-reserved cell. The cell will be shut down if the maximum backup power duration requirement is not met. When the low power consumption mode switch (specified by the **CellLowPower.LowPwrSwitch** parameter) is off, the cell, however, is identified as a reserved cell and will not be shut down within the maximum backup power duration.

- Precise backup power saving policy 2 (**CellLowPower.BakPwrSavPolicy** set to **PRECISE_BACKUP_POLICY2**): When the eNodeB performs precise backup

power saving policy 2, it takes different low power consumption mode measures based on the remaining battery power. **Figure 11-7** shows the execution of precise backup power saving policy 2 when a Huawei power system is used. If a third-party power system is used, a PMU 15A(Ver.B) must be configured. Otherwise, precise backup power saving policy 2 is not supported.

Figure 11-7 Precise backup power saving policy 2

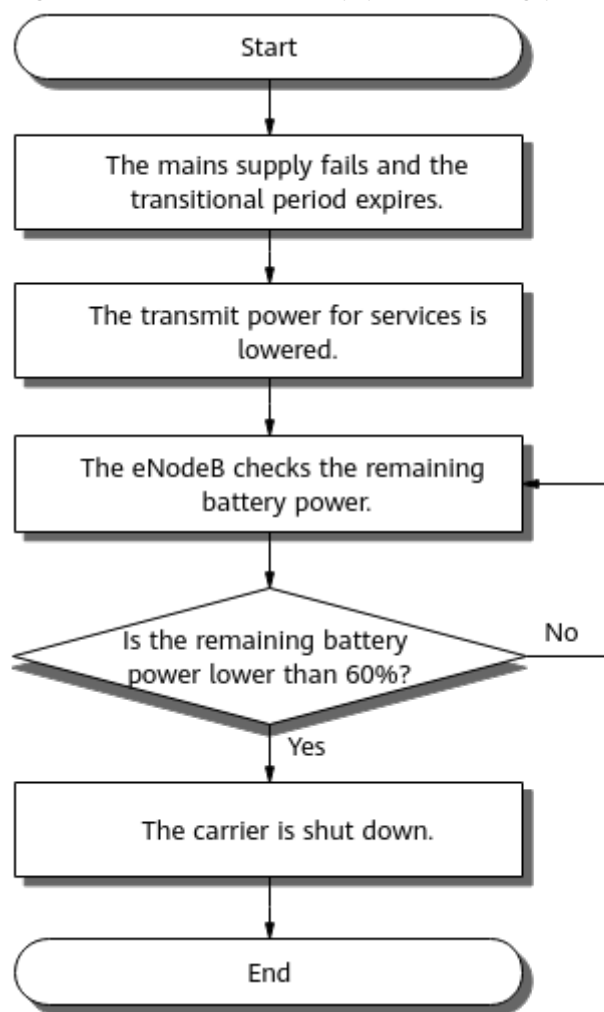


For the precise backup power saving policies, the remaining battery power equals the remaining lithium battery capacity multiplied by the lithium battery efficiency. The remaining capacity and efficiency of the lithium battery are indicated by the **Remaining Capacity(%)** and **Lithium Battery Efficiency** parameters in the command output of **DSP BATTERY**.

- Precise backup power saving policy 3 (**CellLowPower.BakPwrSavPolicy** set to **PRECISE_BACKUP_POLICY3**): When the eNodeB performs precise backup power saving policy 3, it takes different low power consumption mode measures based on the remaining battery power. **Figure 11-8** shows the execution of precise backup power saving policy 3 when a Huawei power system is used. The execution of precise backup power saving policy 3 depends on whether a PMU 15A(Ver.B) is configured when a third-party power system is used.

- If a PMU 15A(Ver.B) is configured and the mains supply failure duration exceeds the transitional period, the eNodeB executes the same procedure as that in scenarios where a Huawei power system is used.
- If no PMU 15A(Ver.B) is configured in the preceding scenario, the eNodeB only lowers the available service power for cells until the mains supply recovers or battery discharging stops.

Figure 11-8 Precise backup power saving policy 3



11.1.3.2 Power Saving Measure

In low power consumption mode, the following power saving measures can be used in a cell: lowering the transmit power for services in the cell, lowering the transmit power for reference signals, shutting down some transmit channels, and shutting down the carrier of the cell.

Table 11-3 lists the power saving measures that are supported by low power consumption mode and precise backup power saving policies. The priority and effective period of some power saving measures can be specified. For details, see [11.1.3.1 Power Saving Policy](#).

Table 11-3 Power saving measures of different backup power saving policies

Lo w Po wer Co nsu mp tio n Mo de	Back power save policy Parameter Value	Lowering Transmit Power for Services in the Cell	Lowering Transmit Power for Reference Signals	Shutting Down Some Transmit Channels	Shutting Down the Carrier
Lo w po wer con su mp tion mo de me asu res	POLICY1	No	No	No	Yes
	POLICY2	Yes	Yes	Yes	Yes
	POLICY3	Yes	Yes	Yes	Yes
	CUSTOMIZED	Yes	Yes	Yes	Yes
	PRIORITY_CUS TOMIZED	Yes	Yes	Yes	Yes
Pre cise bac kup po wer savi ng poli cies	PRECISE_BACK UP_POLICY1	No	No	No	Yes

Low Power Consumption Mode	Back power save policy Parameter Value	Lowering Transmit Power for Services in the Cell	Lowering Transmit Power for Reference Signals	Shutting Down Some Transmit Channels	Shutting Down the Carrier
	PRECISE_BACKUP_POLICY2	- Yes: This policy is supported only if Huawei lithium batteries (used to provide backup power) are used with either of the following power systems: (1) a third-party power system configured with a PMU 15A(Ver.B); (2) a Huawei power system. - No: This policy is not supported in other	No	No	- Yes: This policy is supported only if Huawei lithium batteries (used to provide backup power) are used with either of the following power systems: (1) a third-party power system configured with a PMU 15A(Ver.B); (2) a Huawei power system. - No: This policy is not supported in other scenarios.

Low Power Consumption Mode	Back power save policy Parameter Value	Lowering Transmit Power for Services in the Cell	Lowering Transmit Power for Reference Signals	Shutting Down Some Transmit Channels	Shutting Down the Carrier
		scenarios.			

Lo w Po wer Co nsu mp tio n Mo de	Back power save policy Parameter Value	Lowering Transmit Power for Services in the Cell	Lowering Transmit Power for Reference Signals	Shutting Down Some Transmit Channels	Shutting Down the Carrier
	PRECISE_BACK UP_POLICY3	Yes	No	No	• Yes: This policy is supported only if Huawei lithium batteries (used to provide backup power) are used with either of the following power systems: (1) a third-party power system configured with a PMU 15A(Ver.B); (2) a Huawei power system. • No: If other power systems are used, the eNodeB only lowers

Low Power Consumption Mode	Back power save policy Parameter Value	Lowering Transmit Power for Services in the Cell	Lowering Transmit Power for Reference Signals	Shutting Down Some Transmit Channels	Shutting Down the Carrier
					the available service power for cells until the mains supply recovers or battery discharging stops.
Note: If Back power save policy is set to POLICY1 , POLICY2 , POLICY3 , CUSTOMIZED , or PRIORITY_CUSTOMIZED , the same power saving measure is used regardless of the power system type and backup power type.					

Lowering Transmit Power for Services in the Cell

If this measure is taken, the eNodeB lowers the transmit power of the traffic channels in the cell and prohibits incoming UE handovers from neighboring cells.

If the eNodeB takes low power consumption mode measures in sequence based on the mains supply failure duration and the **CellLowPower.BakPwrSavPolicy** parameter is set to **CUSTOMIZED** or **PRIORITY_CUSTOMIZED**, the values of the **CellLowPower.CellUsedPwrReduceTimeLen** and **CellLowPower.CellUsedPwrRatio** parameters can be customized. The latter two parameters specify the duration in which the transmit power of the traffic channels is lowered and the percentage of the total transmit power normally available that can be used within this duration, respectively.

Lowering Transmit Power for Reference Signals

When the eNodeB takes low power consumption mode measures in sequence based on the mains supply failure duration:

- The eNodeB lowers the transmit power of the traffic channels and reference signals simultaneously.
- If the **CellLowPower.BakPwrSavPolicy** parameter is set to **CUSTOMIZED** or **PRIORITY_CUSTOMIZED**, the duration in which the reference signal transmit power is lowered and the decrease in the reference signal power can be

customized by using the **CellLowPower.RsPwrReduceTimeLen** and **CellLowPower.RsPwrAdjOffset** parameters, respectively.

- When the period for lowering the power of reference signals expires, the eNodeB shuts down some of the transmit channels too.

Shutting Down Some Transmit Channels

When the eNodeB takes low power consumption mode measures in sequence based on the mains supply failure duration:

- The eNodeB lowers the transmit power of the traffic channels and reference signals and shuts down some transmit channels.
- If the **CellLowPower.BakPwrSavPolicy** parameter is set to **CUSTOMIZED** or **PRIORITY_CUSTOMIZED**, the duration in which the channels are shut down can be customized by using the **CellLowPower.RfShutDownTimeLen** parameter.
- When the period for channel shutdown expires, the eNodeB also shuts down the carrier.

The procedure for shutting down some transmit channels is similar to that for RF channel intelligent shutdown. For details, see "eNodeB Actions" in section "RF Channel Intelligent Shutdown" in *Energy Conservation and Emission Reduction* in the feature documentation of the corresponding RAT. The differences between them are as follows:

- Shutting down some transmit channels does not take the UEs in the cell and load of the cell into consideration. The transmit channels will be shut down even if there are UEs in the cell.
- After the channels are shut down, the transmit power of other channels in the cell does not increase.

Shutting Down the Carrier

The eNodeB shuts down the carrier of the cell.

If the eNodeB chooses a precise backup power saving policy, the eNodeB needs to transfer UEs out of the cell before shutting down the carrier. The procedure for UE transfer is the same as that for automatic cell shutdown. For details, see "eNodeB Actions" in section "Adaptive Power Consumption" in *Energy Conservation and Emission Reduction* in the feature documentation of the corresponding RAT.

11.1.4 Leaving

The condition for exiting low power consumption mode depends on whether the eNodeB enters this mode automatically or forcibly:

- If the eNodeB automatically enters the low power consumption mode due to an alarm, the eNodeB exits this mode when any of the following conditions is met:
 - The **CellLowPower.LowPwrSwitch** parameter is set to **OFF**.
 - The period configured for the low power consumption mode elapses. This period is specified by the **CellLowPower.StartTime** and **CellLowPower.StopTime** parameters.

- The alarm that caused the eNodeB to enter the low power consumption mode is cleared.
- If the eNodeB forcibly enters the low power consumption mode, the eNodeB exits this mode when either of the following conditions is met:
 - The **CellLowPower.LowPwrOrder** parameter is set to **OFF**.
 - The **CellLowPower.LowPwrSwitch** parameter is set to **OFF**.

11.2 Network Analysis

11.2.1 Benefits

Low power consumption mode is recommended when the mains supply is unstable in centralized power supply scenarios.

If Huawei power supply system and lithium batteries are used for power supply, precise backup power saving policies are recommended for low power consumption mode. That is, it is recommended that the **CellLowPower.BakPwrSavPolicy** parameter be set to **PRECISE_BACKUP_POLICY1**, **PRECISE_BACKUP_POLICY2**, or **PRECISE_BACKUP_POLICY3**.

Power saving efficiency may decrease when two modes share the same PA. This is because the PA can be shut down only when carriers for both modes are shut down. For example, when the GSM and LTE modes use the same PA, the PA cannot be shut down if the LTE carrier is shut down but the GSM carrier is not.

11.2.2 Impacts

The impacts between this function and other functions are described in [11.3.2.3 Function Impacts](#).

In addition, low power consumption mode has the following impacts:

- This function has a negative impact on cell coverage. If the carrier of a cell is shut down, it cannot provide services.
- This function enables a base station to automatically enter low power consumption mode when ALM-25622 Mains Input Out of Range or a user-defined mains supply interruption alarm is generated, extending the base station service time and the duration that the base station is maintainable. With this function, the transmit power of a cell may be lowered, or the cell may become unavailable, affecting system capacity and network performance.
- If carrier shutdown is performed, it has the same network impacts as those of automatic cell shutdown. For details, see "Impacts" in "Adaptive Power Consumption" in *Energy Conservation and Emission Reduction in eRAN Feature Documentation*.
- If a third-party power system is used with Huawei lithium batteries, and intelligent anti-theft (except the lithium battery software lock function) is enabled for the lithium batteries, the lithium batteries need to reserve some power for anti-theft monitoring and cannot be completely discharged. As a result, the backup power use duration is affected. For details about the

lithium battery software lock function, see *Power Supply Management*. For details about other intelligent anti-theft functions for Huawei lithium batteries, contact Huawei engineers.

11.3 Requirements

11.3.1 Licenses

RAT	Feature ID	Feature Name	Model	Sales Unit
LTE FDD	LOFD-001040	Low Power Consumption Mode	LT1S00LP CM00	Per cell
LTE TDD	TDLOFD-001040	Low Power Consumption Mode	LT1ST0LP CM00	Per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation.

11.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

11.3.2.1 Prerequisite Functions

None

11.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Inter-eNodeB SFN based on eNodeB coordination	Cell.EuCellStandbyMode	<i>SFN</i>	None

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Inter-eNodeB adaptive SFN/SDMA based on eNodeB coordination	CellAlgoSwitch.SfnULSchSwitch CellAlgoSwitch.SfnDLSchSwitch Cell.EuCellStandbyMode	<i>SFN</i>	None
LTE FDD	Intelligent power-off of carriers in the same coverage as UMTS networks	InterRatCellsShutdown.ForceShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	None
NB-IoT	NB-IoT cell	Cell.NbCellFlag set to TRUE	<i>NB-IoT Basics (FDD)</i>	Low power consumption mode is not supported in NB-IoT cells.
LTE FDD	RB reservation for NB-IoT deployment	CellRbReserve.RbRsvMode set to NB_DEPLOYMENT	None	When an NB-IoT network is deployed in LTE in-band mode, RBs in LTE FDD cells must be reserved for NB-IoT. Enabling low power consumption mode in such FDD cells affects NB-IoT carriers, and affects services of some UEs in NB-IoT cells. Therefore, low power consumption mode cannot work with RB reservation for NB-IoT deployment.
LTE FDD LTE TDD	Zero Guard Band Between Contiguous Intra-Band Carriers	CONTIG_INTRA_BAND_CARRIER_SWITCH option of the ContigIntraBandCarrierSwitch parameter	<i>Seamless Intra-Band Carrier Joining (FDD)</i> <i>Seamless Intra-Band Carrier Joining (TDD)</i>	None

RAT	Function Name	Function Switch	Reference	Description
LTE TDD	Inter-eNodeB SFN based on eNodeB coordination	Cell.EuCellStandbyMode	<i>SFN</i>	None
LTE TDD	Inter-eNodeB adaptive SFN/SDMA based on eNodeB coordination	CellAlgoSwitch.SfnULSchSwitch CellAlgoSwitch.SfnDLSchSwitch Cell.EuCellStandbyMode	<i>SFN</i>	None
LTE TDD	Assisted massive MIMO	Cell.MultiRruCellMode set to ASSISTED_MASSIVE_MIMO_CELL	None	None
LTE TDD	Intelligent Multi-Beam of 8T8R	None	<i>Intelligent Multi-Beam of 8T8R (TDD)</i>	None
LTE FDD LTE TDD	High-speed cell sleeping	HSCellSleepInterfAvoidSwitch option of the HighSpdAdaptionPara.InterfAvoidMode parameter	<i>High Speed Mobility</i>	Low power consumption mode cannot work with high-speed cell sleeping.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	RuralLink antenna solution (LTE)	• SectorSplitGroup.BeamTemplateId set to 0 • SectorSplitGroup.BeamDeploymentMode set to MODE_5 • SectorSplitCell.CellBeamIndex set to 0, 1, or 2	<i>RuralLink Solution</i>	None
LTE FDD	LumiLink solution (LTE)	• SectorSplitGroup.BeamTemplateId set to 0 • SectorSplitGroup.BeamDeploymentMode set to MODE_5 • SectorSplitCell.CellBeamIndex set to 0, 1, or 2	<i>LumiLink Solution</i>	Low power consumption mode cannot work with the LTE LumiLink solution.
LTE FDD	GUL joint carrier shutdown	InterRatCellsShutdown.ForcedShutdownSwitch	<i>Multi-RAT Coordinated Energy Saving</i>	GUL joint carrier shutdown cannot work with low power consumption mode.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	SOC level-based low power consumption mode	• IPWRM.SOCCLVLADJ SW • SOCLVL_BASED_LOW_PWR_SW option of the CellLowPower.LowPwrConsumeModeExtSw parameter	<i>Green Site</i>	• LTE SOC level-based low power consumption mode (controlled by the SOCLVL_BASED_LOW_PWR_SW option of the CellLowPower.LowPwrConsumeModeExtSw parameter) cannot work with LTE low power consumption mode. • When the CellLowPower.BakPwrSavPolicy parameter is set to LOW_PWR_AT_SOCLVL1, LOW_PWR_AT_SOCLVL2, or LOW_PWR_AT_SOCLVL3, LTE low power consumption mode cannot be enabled.
LTE FDD LTE TDD	Base-station-level low power consumption mode	LowPwrConsumptionSw option of the ENodeBAlgoSwitch.PowerSaveSwitch parameter	None	If CellLowPower.BakPwrSavPolicy is set to PRIORITY_CUSTOMIZED , base-station-level low power consumption mode cannot be enabled.
LTE TDD	Beam Deployment Mode	SectorSplitGroup.BeamDeploymentMode	None	For an 8T8R cell, if the SectorSplitGroup.BeamDeploymentMode parameter is set to NOT_CFG or MODE_3 , low power consumption mode cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Voltage adjustment during simulated loading	CellSimuLoad.SimuLoadAdjVoltTime set to a value other than 0	None	Voltage adjustment during simulated loading cannot work with low power consumption mode.

11.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Scheduling	None	<i>Downlink Scheduling</i>	When an eNodeB in low power consumption mode lowers transmit power for services in the cells, services are scheduled based on the configured maximum transmit power for services.
LTE FDD LTE TDD	Power control	CellAlgoSwitch.DlPcAlgoSwitch CellAlgoSwitch.UlPcAlgoSwitch	<i>Power Control</i>	When an eNodeB in low power consumption mode lowers transmit power for reference signals, the eNodeB lowers the transmit power of reference signals for the entire cell based on the adjustment amount configured for the reference signal transmit power.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Intelligent power-off of carriers in the same coverage	CellShutDown.CellShutDownSwitch	<i>Energy Conservation and Emission Reduction</i>	When a capacity cell in the intelligent power-off of carriers in the same coverage state meets the conditions for entering low power consumption mode, the cell enters low power consumption mode only after it exits the intelligent power-off of carriers in the same coverage state and the carrier is turned on again. If any co-coverage neighboring basic cell of a capacity cell is in low power consumption mode, the capacity cell does not enter the intelligent power-off of carriers in the same coverage state.
LTE FDD LTE TDD	Dynamic carrier shutdown	CARRIER_DYNAMIC_SHUTDOWN_SW option of the CellShutDown.CarrierShutdownEnhancedSw parameter	<i>Energy Conservation and Emission Reduction</i>	When the local cell in the dynamic carrier shutdown state meets the conditions for entering low power consumption mode, the cell exits the dynamic carrier shutdown state and enters low power consumption mode.
LTE FDD LTE TDD	RF channel intelligent shutdown	CellRfShutdown.RfShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	Low power consumption mode takes effect when it is enabled together with RF channel intelligent shutdown.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	RF channel dynamic muting	RF_CHN_DYN_MUTING_SW option of the CellRfChnDy nMuting.RfChnDynMutingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	RF channel dynamic muting does not take effect when low power consumption mode is in effect.
LTE FDD	RF channel dynamic muting in the presence of TM9 UEs	RF_CHN_DYN_MUTING_TM9_SW option of the CellRfChnDy nMuting.RfChnDynMutingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	RF channel dynamic muting in the presence of TM9 UEs does not take effect when low power consumption mode is in effect.
LTE FDD	RF channel dynamic muting in TM9	TM9_TTI_CHN_DYN_MUTING_SW option of the CellRfChnDy nMuting.RfChnDynMutingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	RF channel dynamic muting in TM9 does not take effect when low power consumption mode is in effect.
LTE FDD LTE TDD	CRS dynamic muting	CRS_DYNAMIC_MUTING_SW option of the SymbolPwrSaving.SymbolShutdownEnhancedSw parameter	<i>Energy Conservation and Emission Reduction</i>	CRS dynamic muting does not take effect when low power consumption mode is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Dynamic voltage adjustment	CellAlgoSwitch.DynAdjVoltageSwitch	<i>Energy Conservation and Emission Reduction</i>	During the transitional period of low power consumption mode, dynamic voltage adjustment can still take effect. When the transitional period elapses and the eNodeB takes further actions to lower the power consumption, dynamic voltage adjustment no longer takes effect.
LTE FDD LTE TDD	MIMO	Cell.TxRxMode	<i>MIMO</i>	When an eNodeB in low power consumption mode shuts down some transmit channels, the eNodeB cannot work in MIMO mode.
LTE FDD	Downlink cell coverage enhancement	CellAlgoSwitch.CellDLCoverageEnhancementSwitch	None	Downlink cell coverage enhancement does not take effect when an eNodeB enters low power consumption mode.
LTE FDD LTE TDD	Dynamic Power Sharing Between LTE Carriers	CellDynamicPowerSharing.DynamicPowerSharingSwitch	<i>Dynamic Power Sharing Between LTE Carriers</i>	Power sharing does not work for cells in low power consumption mode.
LTE FDD LTE TDD	Smart carrier selection based on virtual grids	SMART_CARRIER_SELECTION_SW option of the MultiCarrierUnifiedSch.MultiCarrierUnifiedSchSwitch parameter	<i>Multi-carrier Unified Scheduling</i>	Entering low power consumption mode affects the progress of virtual grid model building.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Downlink 2CC aggregation	CaDl2CCExtSwitch option of the CaMgtCfg.CellCaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Cells that are shut down due to low power consumption mode do not participate in carrier aggregation (CA). In CA-group-based configuration mode, after such a cell in a CA group exits this power saving state, this cell cannot be configured as an SCell for an existing CA UE until the UE accesses the network again.
LTE TDD	Inter-cell downlink D-MIMO	InterCellDmimoJTSwitch option of the CellAlgoSwitch.DMIMOAlgoSwitch parameter	<i>D-MIMO (TDD)</i>	A cell within an inter-cell D-MIMO cluster cannot serve as a coordinating cell for joint transmission (JT) and inter-RRU channel calibration may fail when the cell enters low power consumption mode. As a result, the inter-cell D-MIMO cluster is disabled.
LTE FDD	CRS port mapping detection and reconfiguration	AutoRecfgSwitch parameter in the STR CRSPORTOPT DET command	<i>MIMO</i>	CRS port mapping detection and reconfiguration are not supported in low power consumption mode, and the detection result indicates that the scenario is not supported. If the triggering conditions for low power consumption mode are met during CRS port mapping detection, this mode takes effect after the detection is complete.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	SFN cell	Cell.MultiRruCellMode	<i>SFN</i>	For an SFN cell, if some physical cells initiate cell reestablishment when the eNodeB in low power consumption mode shuts down the corresponding carrier, the reestablishment of the physical cells will fail with "RF Out of Service" alarms reported.
LTE TDD	SFN cell	Cell.MultiRruCellMode	<i>SFN</i>	For an SFN cell, if some physical cells initiate cell reestablishment when the eNodeB in low power consumption mode shuts down the corresponding carrier, the reestablishment of the physical cells will fail with "RF Out of Service" alarms reported.
LTE FDD LTE TDD	Automatic cell shutdown	CellAutoShutdown.CellAutoShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	When the triggering conditions for both low power consumption mode and automatic cell shutdown are met, automatic cell shutdown will take effect but low power consumption mode will not.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Channel calibration	Joint channel calibration: RruJointCalP araCfg.TxChn CalSwitch Downlink-only module channel calibration: RruJointCalP araCfg.AauPa ssivePortCali bPeriod	<i>MIMO</i>	If channel calibration is being performed when PAs are shut down after low power consumption mode takes effect, channel calibration fails. You can run the DSP CELLCALIBRATION command to check whether channel calibration is successful.
LTE TDD	Intra-frequency split	None	<i>Soft Split Resource Duplex (TDD)</i>	Shutting down some transmit channels and the subsequent operations in low power consumption mode can take effect only in the primary cell, and the transmit channels or carriers of the split cell will be shut down as those of the primary cell are shut down.
LTE FDD	UL coordinated channel shutdown	CellRfShutdo wn.MultiRatJ ointChnShut downSw	<i>Multi-RAT Coordinated Energy Saving</i>	A cell in the UL coordinated channel shutdown state cannot enter low power consumption mode.
LTE FDD LTE TDD	RF module deep dormancy	RRU.DORMA NCYSW (LTE eNodeB, 5G gNodeB) set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	RF module deep dormancy takes precedence over low power consumption mode when both functions are enabled. Low power consumption mode does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	RF module super dormancy	RRU.DORMANCY set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	RF module super dormancy takes precedence over low power consumption mode and low power consumption mode does not take effect when both functions are enabled.
LTE FDD LTE TDD	LNR coordinated channel shutdown	CellRfShutdown.MultiRatJointChnShutdownSw	<i>Multi-RAT Coordinated Energy Saving</i>	Low power consumption mode takes precedence over LNR coordinated channel shutdown when both functions are enabled.
LTE FDD LTE TDD	Cell Switch-off Based on Automatic Co-coverage Identification (LTE)	CellShutdown.CellShutdownSwitch set to ON_CO_COVERAGE_AUTO_IDENT	<i>Energy Conservation and Emission Reduction</i>	When the local cell in the Cell Switch-off Based on Automatic Co-coverage Identification state meets the conditions for entering low power consumption mode, the cell enters low power consumption mode only after exiting the Cell Switch-off Based on Automatic Co-coverage Identification state.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Multi-carrier coordinated energy saving	CellShutdown n.CellShutdownSwitch set to ON_MULTI_CARRIER_HIER_SHUTDOWN or ON_MULTICARRIER_HIERMANUAL_SHUTDOWN	<i>Energy Conservation and Emission Reduction</i>	When the local cell in the multi-carrier coordinated energy saving state meets the conditions for entering low power consumption mode, the cell enters low power consumption mode only after it exits the multi-carrier coordinated energy saving state and the carrier is turned on again. When all co-coverage neighboring basic cells of the local cell are in low power consumption mode, the local cell does not enter the multi-carrier coordinated energy saving state.
LTE FDD	Extreme power sharing	LTE_EXTREME_POWER_SHARING_SW option of the CellDynamicPowerSharing.DynamicPowerSharingSwitch parameter	<i>Dynamic Power Sharing Between LTE Carriers</i>	Cells that have entered the energy saving state cannot participate in power sharing. These cells will attempt to participate in power sharing every 10 minutes.
LTE FDD	Centralized power sharing	LTE_CENTRAL_POWER_SHARING_SW option of the CellDynamicPowerSharing.DynamicPowerSharingSwitch parameter	<i>Smart 8T8R (FDD)</i>	Cells that have entered the energy saving state cannot participate in power sharing. These cells will attempt to participate in power sharing every 10 minutes.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Downlink intelligent joint power control	DL_INTEL_JOINT_PC_SW option of the CellDlSchExtParam.DlSchExtSwitch parameter	<i>Smart 8T8R (FDD)</i>	Downlink intelligent joint power control does not take effect when a cell enters low power consumption mode.
LTE FDD	Downlink intelligent joint AMC and power control	DL_INTEL_JOINT_AMC_AND_PC_SW option of the CellDlSchExtParam.DlSchExtSwitch parameter	<i>Smart 8T8R (FDD)</i>	Downlink intelligent joint AMC and power control does not take effect when a cell enters low power consumption mode.
LTE FDD	Enhanced extreme power sharing	LTE_ENH_EXTREME_POWER_SHARING_SW option of the CellDynPowerSharing.DynPowerSharingSwitch parameter	<i>Smart 8T8R (FDD)</i>	Cells that have entered the energy saving state cannot participate in power sharing. These cells will attempt to participate in power sharing every 10 minutes.
LTE FDD	Hybrid power sharing	LTE_HYBRID_POWER_SHARING_SW option of the CellDynPowerSharing.DynPowerSharingSwitch parameter	<i>Smart 8T8R (FDD)</i>	Cells that have entered the energy saving state cannot participate in power sharing. These cells will attempt to participate in power sharing every 10 minutes.
LTE FDD	Channel isolation for PIM interference avoidance	PIM_INTERFERENCE_CHANNEL_SHUTDOWN_SW option of the PimInterfSmartAvoid.PimAvoidSwitch parameter	<i>Interference Detection and Suppression</i>	When low power consumption mode takes effect, channel isolation for PIM interference avoidance is terminated.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Intelligent PIM interference avoidance	PIM_INTERF_SMART_AVOID_SW option of the PimInterfSmartAvoid.PimAlgoSwitch parameter	<i>Interference Detection and Suppression</i>	Low power consumption mode is terminated during PIM interference detection.
LTE FDD	PIM interference cancellation	PIM_INTERF_CANCELLATION_SW option of the PimInterfSmartAvoid.PimAlgoSwitch parameter	<i>Interference Detection and Suppression</i>	After low power consumption mode takes effect in an interfering cell or a victim cell, the eNodeB re-evaluates whether to continue the implementation of PIM interference cancellation in each cell. • If PIM interference persists after low power consumption mode takes effect, PIM interference cancellation continues. • If PIM interference disappears after low power consumption mode takes effect, PIM interference cancellation is stopped to reduce resource consumption caused by periodic PIM interference characteristic training and interference characteristic extraction.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	RF PIM interference cancellation	RF_PIM_CAN_SW option of the RRUEXTINFO.RruOptlFeatSw parameter	<i>Interference Detection and Suppression</i>	PIM interference characteristics change after carrier shutdown takes effect in an interfering or victim cell. If this is the case, the PIM interference cancellation process is restarted, which causes fluctuations in PIM interference cancellation performance.
LTE FDD LTE TDD	Intelligent power control energy saving	INTEL_INTRA_CELL_PC_SW option of the CellPwrSavingsAlgo.PwrSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Intelligent power control energy saving does not take effect when a cell enters low power consumption mode.
LTE FDD LTE TDD	Dynamic carrier shutdown phase 2	CellPwrSavingsAlgo.CarrierDynShutdownPh2Sw set to CAPACITY_CELL	<i>Energy Conservation and Emission Reduction</i>	When the local cell in the dynamic carrier shutdown phase 2 state meets the conditions for entering low power consumption mode, the cell exits the dynamic carrier shutdown phase 2 state and enters low power consumption mode.
LTE TDD	Clock synchronization detection (TDD)	ClkDetect.ClkAsyncDetectSwitch	<i>Synchronization</i>	Clock out-of-synchronization may fail to be detected in scenarios where carriers are shut down in low power consumption mode.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	LTE FDD and NR Flash Dynamic Power Sharing	LTE_NR_DYN_POWER_SHARING_SW option of the CellDynamicPowerSharing parameter	<i>LTE and NR Power Allocation</i>	Cells that have entered the energy saving state cannot participate in LTE FDD and NR Flash Dynamic Power Sharing. These cells will attempt to participate in power sharing every 10 minutes.
LTE FDD	Downlink coordinated power control	DL_COORD_POWER_SWITCH option of the CellMimoParaCfg parameter	<i>Low-Band Booster</i>	Downlink coordinated power control does not take effect when a cell enters low power consumption mode.
LTE FDD	Massive MIMO downlink coordinated power control	MM_DL_COORD_PC_SWITCH option of the SectorSplitGroup parameter	<i>Massive MIMO Enhancements (FDD)</i>	Massive MIMO downlink coordinated power control does not take effect when a cell enters low power consumption mode.
LTE FDD	Optimization-based scheduling	TIME_FREQ_2D_UNIFIED_OPT_SCH_SWITCH and TIME_FREQ_UNIFIED_OPT_SCH_ENH_SWITCH options of the CellMultiCarrierUnifiedScheduling parameter	<i>Multi-carrier Unified Scheduling</i>	When low power consumption mode is enabled, the accuracy of calculating the resources available to the relevant cell is compromised. As a result, the benefits of optimization-based scheduling are affected.

RAT	Function Name	Function Switch	Reference	Description
LTE TDD	Dynamic cell-level threshold adjustment for switching between non-antenna-selection dual-stream beamforming and TM9 hybrid precoding	NONASBF_AND_TM9HY_THLD_ADP_SW option of the CellBfMimoP araCfg.BfMimoAlgoOptSwitch parameter	<i>TMA (TDD)</i>	Low power consumption mode decreases the probability of activating dynamic cell-level threshold adjustment for switching between non-antenna-selection dual-stream beamforming and TM9 hybrid precoding, while the dynamic cell-level threshold adjustment increases the probability of entering the low power consumption mode.
LTE FDD LTE TDD	Adaptive Power Backup	None	<i>Green Site</i>	- Low power consumption mode cannot take effect when the Adaptive Power Backup feature is in effect. If low power consumption mode has been enabled in some cells, the Adaptive Power Backup feature cannot take effect. - When the MAE is disconnected from the base station, this feature rolls back to low power consumption mode.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	SOC level-based low power consumption mode	• IPWRM.SOCLVLADJSW • SOCLVL_BASED_LOW_PWR_SW option of the CellLowPower.LowPowerConsumptionModeExtSw parameter	<i>Green Site</i>	LTE low power consumption mode does not take effect as long as the IPWRM.SOCLVLADJSW parameter is set to ON .
LTE FDD	LTE-NR multi-dimensional scheduling optimization	LTE: LNR_MULTI_D_OPT_SCH_SW option of the GigaBandCell.GigaBandCellAlgoSw parameter NR: LNR_MULTI_D_OPT_SCH_SW option of the NRDUCellGigaBand.GigaBandCellAlgoSw parameter	<i>LTE FDD and NR Inter-RAT Unified Scheduling</i>	When low power consumption mode is enabled, the accuracy of calculating the resources available to the relevant cell is compromised. As a result, the benefits of LTE-NR multi-dimensional scheduling optimization are affected.
LTE FDD	LNR CRS muting	LNR_CRSMUTE_SW option of the CellAlgoExtSwitch.LnrZeroBufferzoneSw parameter	<i>LTE and NR Zero Bufferzone</i>	LNR CRS muting does not take effect in a cell that has entered low power consumption mode.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	CMAS	CellAlgoSwitch.CmasSwitch	<i>CMAS</i>	The following applies when low power consumption mode is in effect: • When PWS message-triggered exit of cells from energy saving is disabled (by deselecting the PWS_RESTART_IN_D_ENH_SW option of the EnodebAlgoExtSwitch.PwsIndicationSwitch parameter), CMAS warning notifications cannot be properly sent because low power consumption mode shuts down carriers. • With PWS message-triggered exit of cells from energy saving enabled, cells exit low power consumption mode upon receipt of a PWS message.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	ETWS	None	<i>ETWS</i>	The following applies when low power consumption mode is in effect: - When PWS message-triggered exit of cells from energy saving is disabled (by deselecting the PWS_RESTART_IN_D_ENH_SW option of the EnodebAlgoExtSwitch.PwsIndicationSwitch parameter), ETWS warning notifications cannot be sent because low power consumption mode shuts down carriers. - With PWS message-triggered exit of cells from energy saving enabled, cells exit low power consumption mode upon receipt of a PWS message.
LTE FDD	RF channel dynamic muting in TM4	TM4_TTI_CHN_DYN_MUTING_SW option of the CellRfChnDynamicMuting.RfChnDynMutingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	RF channel dynamic muting in TM4 does not take effect when low power consumption mode is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Port sparseness	PORT_SPARSENESS_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Precise backup power saving policies in low power consumption mode take precedence over port sparseness when they are both enabled.
LTE FDD LTE TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATSW parameter	<i>Multi-RAT Coordinated Energy Saving</i>	RF module fast deep dormancy takes precedence over low power consumption mode when both functions are enabled. Low power consumption mode does not take effect when RF module fast deep dormancy is in effect.
LTE FDD	Cell RX channel interference and noise query	This function is controlled by the STRCELLRXINTRFNOIQUERY command.	<i>MIMO</i>	If low power consumption mode or precise backup power saving policy is in effect in a cell and reporting of invalid uplink interference counter values is enabled (controlled by the UL_IF_INVALID_IN_CARRROFF_SW option of the ENodebCounterParaGrp.ENodebCounterAlgoSwitch parameter), then RX channel interference and noise query for the cell will not produce any results.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Inter-cell cable sequence detection	This function is controlled by the STR INTERCELLCR OSSFEEDERT ST command.	<i>MIMO</i>	If low power consumption mode is in effect for a cell and the DSP CELLLOWPOWER command output shows RF Shutdown Status = ENABLED , the cell is not selected for inter-cell cable sequence detection.

11.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

Manual intervention mode: 3900 and 5900 series base stations (macro base stations)

Automatic triggering mode: 3900 and 5900 series base stations (macro base stations)

Boards

- There are no requirements if the eNodeB takes low power consumption mode measures in sequence.
- There are no requirements if the eNodeB uses a precise backup power saving policy.

RF Modules

None

Power Supply

- Power cabinet: The mains failure alarm can be reported.
- Available backup power at the site: If the eNodeB enables low power consumption mode, batteries are required to provide backup power. This function cannot be enabled at sites without backup power. You are advised to precheck the battery health of base stations equipped with lead-acid batteries and not to enable this function when batteries are severely aged (without discharge capability or with extremely short discharge time). Otherwise, the gains in terms of the extended backup power use duration may decrease.

- The power supply system can be provided by Huawei or another vendor.
 - The Huawei power supply system can be connected to a Huawei base station through the two RS485 signal input ports on the universal power and environment interface unit (UPEU) or universal environment interface unit (UEIU). They can also be connected through the RS485 signal input port or the power cable on the RRU.
 - A third-party power system can be connected to a Huawei base station through the Boolean signal input ports on the UPEU or UEIU. That is, they are connected through a dry contact.

In this case, you need to configure the **ALMPORT** MO to set information about the dry contact port and configure the **MAINSALARMBIND** MO to define the alarms generated by this port as mains supply failure alarms. Under this setting, a mains supply failure alarm will be reported to the Huawei base station through the port if the third-party power system experiences a mains supply failure. Upon detection of this alarm, the Huawei base station can start low power consumption mode to extend the duration of basic coverage service provisioning.

 NOTE

The name and severity of the alarms reported by the dry contact port can be customized by running the **SET ENVALMPARA** command.

- **Table 11-4** lists the requirements of low power consumption mode measures and precise backup power saving policies on power supplies and backup power.

Table 11-4 Requirements of different policies on different power supplies

Po we r Su ppl y Ty pe	Type of Batteries Used to Provide Backup Power	Low Power Consumption Mode Measure	Precise Backup Power Saving Policy
Hu aw ei po we r sys te m ^a	Huawei lithium batteries	Supported	Supported
	Non-Huawei lithium batteries	Supported	Not supported

Power Supply Type	Type of Batteries Used to Provide Backup Power	Low Power Consumption Mode Measure	Precise Backup Power Saving Policy
Third-party power system	Huawei lithium batteries	Supported	Supported only when a PMU 15A(Ver.B) is configured. ^b
	Non-Huawei lithium batteries used to provide backup power	Supported	Not supported
a: When low power consumption mode is used with a Huawei power system, the power system must meet the requirements listed in Table 11-5. b: In this scenario, precise backup power saving policies are not supported in C-RAN networking.			

Table 11-5 Huawei power system requirements of low power consumption mode

Power System Type	Power System	PMU Model	PMU Software Version
ETP	APM5930	CCUC	When the hardware version is PMU.19, the PMU software version must be 219 or later. When the hardware version is PMU.22, the PMU software version must be 213 or later.
ETP	OPM200 series	PMU 16A	The PMU software version must be 177 or later.
ETP	APM5950 H-L	PMU 11C	If the hardware version is PMU.35, the PMU software version must be 2265 or later.
ETP	APM5950 H	PMU 11C	If the hardware version is PMU.35, the PMU software version must be 2251 or later.

 NOTE

- These measures and policies can be used for a base station only when **Battery Type** is **Li-Ion Battery** and **Number Of Lithium Batteries** is at least 1 in the command output of **DSP BATTERY** used to query the status of Huawei lithium batteries. For details about the Huawei power system configurations, lithium battery configurations, and mapping between power systems and lithium batteries, see *Power Supply Management*.
- The software version of the PMU can be queried by running the **DSP BRDVER** command.
- If the communication between the Huawei lithium batteries and the PMU is faulty, the low power consumption mode function cannot run properly. In this case, the base station maintains the current operation taken in low power consumption mode until the communication recovers.
- The power cabinet where an SMU-type power system (of which the type is specified by **PMU.PTYPE (5G gNodeB, LTE eNodeB)**) is located does not support the precise backup power saving policy.
- Precise backup power saving policies do not apply to scenarios where multiple power systems are used and lithium batteries in each power system are managed by an individual PMU.

11.3.4 Cells

When the **CellLowPower.LowPwrSwitch** parameter is set to **ON** to enable low power consumption mode and the **eUCellSectorEqm.CellBeamMode** is set to **MASSIVE_MIMO_Ph1** or **MASSIVE_MIMO_Ph2**, **CellLowPower.BakPwrSavPolicy** specifying the backup power saving policy must be set to **PRECISE_BACKUP_POLICY1**, **PRECISE_BACKUP_POLICY2**, or **PRECISE_BACKUP_POLICY3**.

11.3.5 Others

None

11.4 Operation and Maintenance

11.4.1 Data Configuration

11.4.1.1 Data Preparation

[Table 11-6](#) describes the parameters used for function activation.

Table 11-6 Parameters used for activation

RAT	Parameter Name	Parameter ID	Setting Notes
LTE FDD LTE TDD	Local cell ID	CellLowPower.LocalCellId	None

RAT	Parameter Name	Parameter ID	Setting Notes
LTE FDD LTE TDD	Low power consumption mode switch	CellLowPower.LowPwrSwitch	It is recommended that this switch be turned on when low power consumption mode is required.
LTE FDD LTE TDD	Low power consumption mode order	CellLowPower.LowPwrOrder	Use the recommended value.
LTE FDD LTE TDD	Start time	CellLowPower.StartTime	It is recommended that this parameter be set to 00:00:00 when low power consumption mode is enabled. This parameter must be set to the same value for all split cells in which low power consumption mode is enabled in a split group.
LTE FDD LTE TDD	Stop time	CellLowPower.StopTime	It is recommended that this parameter be set to 23:59:59 when low power consumption mode is enabled. This parameter must be set to the same value for all split cells in which low power consumption mode is enabled in a split group.
LTE FDD LTE TDD	Cell used power reduce time length	CellLowPower.CellUsedPowerReduceTimeLen	Use the recommended value.
LTE FDD LTE TDD	Reference signal power reduce time length	CellLowPower.RsPwrReduceTimeLen	Use the recommended value.
LTE FDD LTE TDD	RF shutdown time length	CellLowPower.RfShutdownTimeLen	Use the recommended value.

RAT	Parameter Name	Parameter ID	Setting Notes
LTE FDD LTE TDD	Cell used power ratio	CellLowPower.CellUsedPowerRatio	Use the recommended value. When a cell is heavily loaded, setting this parameter to a larger value saves less energy but decreases the peak throughput by a smaller amount. When a cell is lightly loaded, setting this parameter to a smaller value saves more energy but decreases the peak throughput by a larger amount.
LTE FDD LTE TDD	Reference signal power adjust offset	CellLowPower.RsPwrAdjOffset	Use the recommended value. - Setting this parameter to a smaller value, for example, -5, results in a greater reduction of reference signal power. This saves more energy, but greatly reduces system coverage. - Setting this parameter to a larger value, for example, -1, results in a modest reduction of reference signal power. This saves less energy, but reduces system coverage only slightly. Therefore: If there are many cell center users (CCUs), a smaller value is recommended to save more energy. If there are many cell edge users (CEUs), a larger value is recommended to provide better coverage.
LTE FDD LTE TDD	Cell Used Power Reduce Startup Priority	CellLowPower.CellUsedPowerReduceStartPri	Set this parameter based on the network plan.
LTE FDD LTE TDD	RF Shutdown Startup Priority	CellLowPower.RfShutdownStartPri	Set this parameter based on the network plan.
LTE FDD LTE TDD	RS Power Reduce Startup Priority	CellLowPower.RsPwrReduceStartPri	Set this parameter based on the network plan.

RAT	Parameter Name	Parameter ID	Setting Notes
LTE FDD LTE TDD	Time length for entering the low power consumption mode	CellLowPower.EnterTimeLen	Use the recommended value.

RAT	Parameter Name	Parameter ID	Setting Notes
LTE FDD LTE TDD	Back power save policy	CellLowPower.BakPwrSavePolicy	If a multimode base station uses the same set of storage batteries to support services of all RATs: • The value POLICY3 is recommended if the services of the local cell need to be given the highest priority in the case of mains supply failures. • The value POLICY2 is recommended if the services of the local cell need to be given the medium priority in the case of mains supply failures. • The value POLICY1 is recommended if the services of the local cell need to be given the lowest priority and the rated power of this cell needs to be decreased to save power for cells of other RATs in the case of mains supply failures. • The value CUSTOMIZED is used if the duration-related parameters need to be customized for the local cell. • The value PRIORITY_CUSTOMIZED is used if the duration-related parameters and the sequence for power saving measures to take effect need to be customized for the local cell. When Huawei power supply system and lithium batteries are used for power supply: • The value PRECISE_BACKUP_POLICY1 is recommended if the services of the local cell need to be given the lowest priority and the rated power of this cell needs to be decreased to save power for cells of other RATs. • The value PRECISE_BACKUP_POLICY2 is recommended if the services of the local cell need to be given the highest priority in the case of mains supply failures.

RAT	Parameter Name	Parameter ID	Setting Notes
			The value PRECISE_BACKUP_POLICY3 is recommended if the services of the local cell need to be given the medium priority in the case of mains supply failures. This parameter must be set to the same value for all split cells in which low power consumption mode is enabled in a split group.

 NOTE

For an LTE FDD SFN cell, the entering and exiting of low power consumption mode are triggered on a physical cell basis in a distributed power supply scenario only when the durations for lowering transmit power for services and reference signals are both set to **0** or when the precise backup power saving policy 1 is used.

If the power equipment for the base station is not provided by Huawei, you must bind a user-defined mains supply outage alarm to a specified port, as described in [Table 11-7](#).

Table 11-7 Data preparation

RAT	Parameter Name	Parameter ID	Setting Notes
LTE FDD LTE TDD	Switch	ALMPORT.SW	Set this parameter to ON if a non-Huawei power supply system is connected to the base station through a dry contact.
LTE FDD LTE TDD	Cabinet No.	ALMPORT.CN	Set this parameter based on the network plan.
LTE FDD LTE TDD	Subrack No.	ALMPORT.SRN	Set this parameter based on the network plan.
LTE FDD LTE TDD	Slot No.	ALMPORT.SN	Set this parameter based on the network plan.

RAT	Parameter Name	Parameter ID	Setting Notes
LTE FDD LTE TDD	Port No.	ALMPORT.PN	Set this parameter based on the network plan.
LTE FDD LTE TDD	Alarm ID	ALMPORT.AI/D	Set this parameter based on the network plan.
LTE FDD LTE TDD	Port Type	ALMPORT.PT	Set this parameter based on the analog or Boolean value output by the non-Huawei power supply system when the mains supply fails.
LTE FDD LTE TDD	Alarm Voltage	ALMPORT.AV/OL	Set this parameter based on the Boolean value output by the non-Huawei power supply system when the mains supply fails.
LTE FDD LTE TDD	Base Station Almpport Switch	MAINSALAR MBIND.ISDS WITCH	Set this parameter to ENABLE if the alarms generated by the dry contact port need to be defined as mains supply outage alarms.
LTE FDD LTE TDD	Almpport Cabinet No.	MAINSALAR MBIND.NMS ACN	Specifies the cabinet number of the dry contact that reports a mains supply outage alarm.
LTE FDD LTE TDD	Almpport Subrack No.	MAINSALAR MBIND.NMS ASRN	Specifies the subrack number of the dry contact that reports a mains supply outage alarm.
LTE FDD LTE TDD	Almpport Slot No.	MAINSALAR MBIND.NMS ASN	Specifies the slot number of the dry contact that reports a mains supply outage alarm.
LTE FDD LTE TDD	Almpport Port No.	MAINSALAR MBIND.NMS APN	Specifies the port number of the dry contact that reports a mains supply outage alarm.

Table 11-8 describes the parameters that are required to configure the power supply relationships between RF units and power systems in distributed power supply scenarios in LTE FDD.

Table 11-8 Data preparation

RAT	Parameter Name	Parameter ID	Setting Notes
LTE FDD	Power Supply Group ID	RRU.PSGID	Specifies the ID of a power supply group, which is formed by one or more RRUs or RFUs that are served by the same power system. RRUs and RFUs served by the same power system correspond to the same group ID, and RRUs and RFUs served by different power systems must be configured with different group IDs. The default value of this parameter is 0. Use the default value for RRUs that share the same power system as the BBU.

 NOTE

If the **RRU.PSGID** parameter value is inconsistent with the actual power supply relationships in LTE FDD, triggering low power consumption mode may cause the RRU whose power supply is normal to enter this function. In such a case, the capacity of the corresponding cell may degrade, or this cell may be shut down and become unavailable.

11.4.1.2 Using MML Commands

Before using MML commands, refer to [11.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//(Optional) Running the SET ALMPORT command to turn on the port switch and configure the
information about the dry contact port and running the MOD NMSABIND command to define the alarms
generated by the dry contact port as mains supply outage alarms, if low power consumption mode is used
with a non-Huawei power supply system
SET ALMPORT: CN=0, SRN=0, SN=19, PN=0, SW=ON, AID=65033, PT=BOOL, AVOL=LOW;
MOD NMSABIND: ISDSWITCH=ENABLE, NMSACN=0, NMSASRN=0, NMSASN=19, NMSAPN=0;
//(Optional, required in distributed power supply scenarios) Configuring a power supply group ID
MOD RRU: CN=0, SRN=60, SN=0, PSGID=1;
//Configuring the low power consumption mode (assuming that the local cell has a medium priority)
MOD CELLLOWPOWER: LocalCellId=0, LowPwrSwitch=ON, LowPwrOrder=OFF, StartTime=00&00&00,
StopTime=23&59&59, BakPwrSavPolicy=POLICY2;
//Configuring the low power consumption mode based on user-defined priorities (if the mains supply is
faulty, carrier shutdown takes effect after the following periods separately last for 5 minutes in sequence:
smooth transition period, reference signal power reduction period, available power reduction period, and RF
shutdown period)
```

MOD CELLLOWPOWER: LocalCellId=0, LowPwrSwitch=ON, LowPwrOrder=OFF, StartTime=00&00&00, StopTime=23&59&59, BakPwrSavPolicy=PRIORITY_CUSTOMIZED, CellUsedPwrReduceTimeLen=5, RsPwrReduceTimeLen=5, RfShutDownTimeLen=5, CellUsedPwrRatio=50, RsPwrAdjOffset=-1, EnterTimeLen=5, CellUsedPwrReduceStartPri=START_PRIORITY_SECOND, RsPwrReduceStartPri=START_PRIORITY_FIRST, RfShutdownStartPri=START_PRIORITY_THIRD;

NOTE

An eNodeB processes alarms reported by a dry contact port only after the eNodeB is bound to the dry contact port. A user-defined dry contact port is specified by running the **SET ALMPORT** command. The name, severity, and type of the alarm reported by the user-defined dry contact are specified by running the **SET ENVALMPARA** command.

Deactivation Command Examples

Deactivating the low power consumption mode

MOD CELLLOWPOWER: LocalCellId=0, LowPwrSwitch=OFF;

11.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

In LTE TDD, this function can be activated in batches using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance > MAE-Deployment Guidelines > Enhanced Feature Management > Feature Operation and Maintenance**.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

11.4.2 Activation Verification

- Entering the Low Power Consumption Mode Because of Abnormal Mains Supply

- Step 1** Run the **LST CELLLOWPOWER** command to verify that the value of the **Low power consumption mode switch** parameter is **On** and the value of the **Low power consumption mode order** parameter is **Off**, and to verify **Start time** and **Stop time**.
- Step 2** Run the **DSP TIME** command to verify that the system time falls within the period specified by the **Start time** and **Stop time** parameters.
- Step 3** Use batteries instead of AC power to provide power. Verify that ALM-25622 Mains Input Out of Range has been reported.
- Step 4** After the transitional period, run the **DSP CELL** command to view the value of the **Cell power save state** parameter. For details about the command execution interval, see the duration-related parameter settings in [11.1.3 eNodeB Actions](#). If the value is **Low Power Consumption Mode**, this feature has been activated.
- Step 5** Run the **DSP CELLLOWPOWER** command to check whether the smooth transition period or a power-saving measure is in effect. If the relevant status is **Enabled** for a power-saving measure, this measure has taken effect for the cell.

- Smooth Transition Status
- Cell Used Power Reduce Status
- RS Power Reduce Status
- RF Shutdown Status
- Physical Cell Shutdown Status
- Cell Shutdown Status

In the distributed power supply scenario, if the power supply of the RRUs serving some physical cells in the SFN cells is abnormal and all of the following conditions are met:

- **CellLowPower.CellUsedPwrReduceTimeLen** is set to **0**, or **CellLowPower.CellUsedPwrReduceStartPri** is set to **NOT_WORK**.
- **CellLowPower.RsPwrReduceTimeLen** is set to **0**, or **CellLowPower.RsPwrReduceStartPri** is set to **NOT_WORK**.

Then, the value of **Physical Cell Shutdown Status** is **Enabled**. Otherwise, the value of **Cell Shutdown Status** is **Enabled**.

----End

- Forcibly Entering the Low Power Consumption Mode

Step 1 Run the **LST CELLLOWPOWER** command to verify that the value of the **Low power consumption mode switch** parameter is **On** and the value of the **Low power consumption mode order** parameter is **On**.

Step 2 After the transitional period, run the **DSP CELL** command to view the value of the **Cell power save state** parameter. For details about the command execution interval, see the duration-related parameter settings in [11.1.3 eNodeB Actions](#). If the value is **Low Power Consumption Mode**, this feature has been activated.

Step 3 Run the **DSP CELLLOWPOWER** command to check whether the smooth transition period or a power-saving measure is in effect. If the relevant status is **Enabled** for a power-saving measure, this measure has taken effect for the cell.

- Smooth Transition Status
- Cell Used Power Reduce Status
- RS Power Reduce Status
- RF Shutdown Status
- Physical Cell Shutdown Status
- Cell Shutdown Status

In the distributed power supply scenario, if the power supply of the RRUs serving some physical cells in the SFN cells is abnormal and all of the following conditions are met:

- **CellLowPower.CellUsedPwrReduceTimeLen** is set to **0**, or **CellLowPower.CellUsedPwrReduceStartPri** is set to **NOT_WORK**.
- **CellLowPower.RsPwrReduceTimeLen** is set to **0**, or **CellLowPower.RsPwrReduceStartPri** is set to **NOT_WORK**.

Then, the value of **Physical Cell Shutdown Status** is **Enabled**. Otherwise, the value of **Cell Shutdown Status** is **Enabled**.

----End

11.4.3 Network Monitoring

Benefit Monitoring

To monitor and evaluate the performance of this feature, operators can view the following performance indicators on the MAE-Access:

- eNodeB power consumption
It is indicated by the **VS.EnergyCons.BTS.Adding.Service.LTE** or **VS.EnergyCons.BTS.Adding.LTE** counter value on the MAE-Access.
Operators can evaluate the performance of the low power consumption mode by comparing the power consumed by an eNodeB before and after this function is enabled. The decrease in eNodeB power consumption indicates how much energy is saved.
- Power consumption of an RF module
It is indicated by the **VS.EnergyCons.BTSBoard** counter on the MAE-Access. For the meaning of this counter, see *eNodeB Performance Counter Reference*.
If low power consumption mode is triggered due to abnormal mains supply or is forcibly triggered, operators can evaluate the performance of low power consumption mode by comparing the power consumed by an RF module before and after this function is enabled. The decrease in RF module power consumption indicates how much energy is saved.
- Duration of cell unavailability due to energy saving
It is indicated by the **L.Cell.Unavail.Dur.EnergySaving** counter on the MAE-Access. For the meaning of this counter, see *Performance Counter Reference of eNodeBs*.

On the live network, you can compare one-week eNodeB power consumption before and after the function is enabled to evaluate its energy saving gains.

Possible Issues

If a fault occurs, take the steps described in this section.

- Fault 1
Low power consumption mode switch is set to **ON**, **Low power consumption mode order** is set to **OFF**, and the cell cannot enter the low power consumption mode as expected after the transitional period has elapsed.

Perform the following operations:

- Step 1** Check whether the eNodeB system time falls within the low power consumption period.
- Step 2** Check whether batteries are correctly configured for the eNodeB. For detailed operations about how to configure batteries, see *Power Supply Management*.
- Step 3** Check whether ALM-25622 Mains Input Out of Range has been reported with the cause "AC Failure." The low power consumption mode is triggered only after the

eNodeB power source is switched from AC power to batteries due to an abnormal mains power supply.

----End

12 Low Power Consumption Mode (NR)

12.1 Principles

In the scenario where both a power supply system and batteries are configured, the available duration of backup power for basic coverage services will be impacted if the power supply system becomes faulty while the deployed batteries are insufficient. Low power consumption mode can be used to extend the duration of basic coverage service provisioning in this scenario. When an alarm indicating power supply insufficiency or failures is reported, cells with high power consumption enter low power consumption mode to reduce the power consumed by RF modules. The application scenarios of low power consumption mode (NR) are the same as those of low power consumption mode (LTE). For details, see [11.1.1 Application Scenario](#).

Low power consumption mode uses open-loop control. That is, the backup power duration increases by an unknown amount. This function aims to extend the backup power use duration. When it takes effect, coverage and performance may be affected.

Low power consumption mode can be enabled by selecting the **LOW_PWR_MODE_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter. The energy saving type, backup power saving policy, and other information under an energy saving policy are specified by the parameters in the **NRDUCellPowerSaving** MO, as listed in [Table 12-1](#).

Table 12-1 NRDUCellPowerSaving MO configuration

Parameter	Meaning
NRDUCellPowerSaving.PowerSavingPolicyIndex	This parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed for the local cell.

Parameter	Meaning
NRDUCellPowerSaving.PowerSavingType	This parameter specifies the energy saving type indicated by an energy saving policy index. Low power consumption mode requires that this parameter be set to LOW_PWR_MODE , meaning that under the corresponding energy saving policy, the local cell will enter low power consumption mode based on the backup power saving policy.
NRDUCellPowerSaving.BackupPwrSavingPolicy	This parameter specifies the backup power saving policy. Table 12-2 lists the parameter configurations and implementation policies.

Table 12-2 Configurations that can be defined by users for different backup power saving policies

Backup Power Saving Policy	Parameter Configuration	Policy Execution
PRECISE_BAC KUP_POLICY1 (Precise backup power saving policy 1)	NRDUCellPowerSaving.TimeToTrigger , which specifies the transitional period, during which services are normal and the cell does not enter low power consumption mode	The carrier of the cell is shut down when the mains supply fails for a period longer than the transitional period. The shutdown lasts until the mains supply recovers or battery discharging stops.
PRECISE_BAC KUP_POLICY2 (Precise backup power saving policy 2)	NRDUCellPowerSaving.TimeToTrigger , which specifies the transitional period, during which services are normal and the cell does not enter low power consumption mode	Symbol power saving is performed for the cell when the mains supply fails for a period longer than the transitional period. Symbol power saving lasts until the mains supply recovers or battery discharging stops.

Backup Power Saving Policy	Parameter Configuration	Policy Execution
PRECISE_BACKUP_POLICY3 (Precise backup power saving policy 3)	NRDUCellPowerSaving.TimeToTrigger , which specifies the transitional period, during which services are normal and the cell does not enter low power consumption mode	- When a Huawei APM cabinet (APM5930/APM5950H/APM5950H-L) is used, symbol power saving is first performed for the cell when the mains supply fails for a period longer than the transitional period. When the battery power drops below 60%, carrier shutdown is then performed for the cell and lasts until the mains supply recovers or battery discharging stops. - When an MTS series power system (whose power monitoring module is an SMU02C) is connected with the MAE for O&M management, symbol power saving is performed for the cell when the mains supply fails for a period longer than the transitional period. Symbol power saving lasts until the mains supply recovers or battery discharging stops. - Scenarios using a third-party power system: - If no PMU 15A(Ver.B) is used, the execution process is the same as that in the scenario where an MTS series power system (whose power monitoring module is an SMU02C) is connected with the MAE for O&M management. - If a PMU 15A(Ver.B) is used, the execution process is the same as that in the scenario where an APM cabinet is used.

Backup Power Saving Policy	Parameter Configuration	Policy Execution
CUSTOMIZED_BACKUP_POLICY (Customized backup power saving policy)	• NRDUCellPowerSaving.TimeToTrigger, which specifies the transitional period, during which services are normal and the cell does not enter low power consumption mode • NRDUCellPowerSaving.RfChnShutdownDuration, which specifies the duration during which channel shutdown is performed for the cell in low power consumption mode	Channel shutdown is performed for the cell when the mains supply fails for a period longer than the transitional period. By default, half of the channels are shut down (whether channel shutdown takes effect in single-module multi-carrier scenarios depends on the negotiation result). The channel shutdown duration can be specified. After the specified duration elapses, carrier shutdown is performed for the cell together with channel shutdown. The shutdown lasts until the mains supply recovers or battery discharging stops.
Note: If precise backup power saving policy 1, precise backup power saving policy 2, or customized backup power saving policy is used, the policy execution process when an APM cabinet and non-Huawei lithium batteries (used to provide power backup) are used or a third-party power system is used is the same as that when an APM cabinet and Huawei lithium batteries (used to provide power backup) are used.		

The following requires attention during the execution process of backup power saving policies for low power consumption mode:

- Before carrier shutdown is performed for the cell, UEs need to be transferred out of the cell. For details, see section "Energy Saving Procedure" in "Timing Carrier Shutdown" in *Energy Conservation and Emission Reduction*.
- Different from channel shutdown for power saving, channel shutdown in low power consumption mode is performed regardless of the number of UEs in the current cell and the cell load and also takes effect after the transitional period.
- Channel shutdown in low power consumption mode is performed in accordance with the following rules:
 - Half of the RF channels are shut down by default.
 - The **RF Channel Shutdown Mode** parameter is set to **DL_CHANNEL_SHUTDOWN** by default.
- In multi-carrier scenarios, negotiation among multiple carriers is performed after the transitional period. During the negotiation, this function brings no gains. In single-carrier scenarios, this function takes effect immediately after the transitional period without negotiation.
- In multi-carrier scenarios, channel shutdown (that is, transmit channel blocking) takes effect only when the following conditions are met:

- The requirements on the **Backup Power Saving Policy** parameter settings: The **Backup Power Saving Policy** parameter is successfully set to **CUSTOMIZED_BACKUP_POLICY** for all cells served by the same RF module, and the negotiation for the combinations of channels to be shut down is successful. Channel shutdown may not take effect due to the following causes: (1) The **Backup Power Saving Policy** parameter is not set to **CUSTOMIZED_BACKUP_POLICY** for some of the cells served by the same RF module. (2) Not the periods specified for RF channel shutdown in all cells overlap.
- The requirements on the **RF Channel Shutdown Duration** parameter settings: After the transitional period specified by the **Time to Trigger** parameter, there is 10 minutes for negotiation. It is recommended that **RF Channel Shutdown Duration** be set to a value greater than 15 minutes. If the duration specified by the configured **RF Channel Shutdown Duration** parameter value is not long enough for negotiation, channel shutdown does not take effect. If the negotiation succeeds, the actual RF channel shutdown duration is the configured **RF Channel Shutdown Duration** parameter value minus the negotiation duration.

12.2 Network Analysis

12.2.1 Benefits

Low power consumption mode prolongs the backup power use duration of basic services when the mains supply fails. In LTE and NR co-RF module scenarios, it is recommended that this function be enabled for both LTE and NR cells and the same backup power saving policy be configured for these cells.

Energy saving gains may decrease when the temperature differential of RF modules is excessively large. This is to prevent energy saving-triggered shutdown from affecting the lifespan of RF modules.

12.2.2 Impacts

The impacts between this function and other functions are described in [12.3.2.3 Function Impacts](#).

If the backup power saving policy configured by the operator includes carrier shutdown, the following will occur after low power consumption mode takes effect:

- Existing UEs in the cells whose carriers are shut down cannot perform services, slightly increasing the service drop rate.
- New UEs within the coverage of the cells whose carriers are shut down cannot access the network.
- During NR carrier shutdown, new UEs are prohibited from accessing an NR cell. If UEs attempt to be handed over to the NR cell, the values of the counters related to handover preparation failures (such as **N.HO.IntraFreq.FailOut.PrepFailure**) slightly increase.
- During NR carrier shutdown, if UEs fail to be handed over to target cells, the values of counters related to the number of intra-NR incoming handover

failures (including **N.HO.FailIn.HOCancel**, **N.HO.FailIn.AdmitFail**, **N.HO.FailIn.AdmitFail.TNL**, **N.HO.FailIn.AdmitFail.UserLic**, and **N.HO.FailIn.AdmitFail.Load**), or the values of the LTE counters related to the number of inter-RAT incoming handover failures for different causes (including **L.IRATHO.N2E.PreInFail**, **L.IRATHO.N2E.PreInFail.AdmitFail**, **L.IRATHO.N2E.PreInFail.TgtNotAllowed**, and **L.IRATHO.N2E.PreInFail.HOCancel**) slightly increase.

- Channel shutdown in low power consumption mode does not check the existing UEs in the cell and cell load rate. After channel shutdown takes effect, the cell coverage may be affected. (The number of handovers and the handover success rate may change, the access success rate may decrease, and the service drop rate may increase.) In addition, the power compensation for common channels may increase the common-channel interference between base stations when the inter-base-station distance is small.
- After channel shutdown in low power consumption mode takes effect, power compensation is performed for common channels based on hardware capabilities. However, coverage may be affected due to insufficient power compensation.
- For a 2T or 4T NR FDD/NR SDL RRU, as the RRU has a small number of channels, shutting down half of the channels decreases the rank used, causing a significant increase in the PRB usage and affecting the energy saving gains.
- The processing delay increases, the scheduling capability deteriorates, and the scheduling time is extended. The decrease in the downlink cell and UE throughputs is described in the following table. The longer the distance between a UE and a base station, the larger the decrease in the uplink and downlink throughput.

Scenario	Decrease in the Downlink Cell and UE Throughputs
Single base station networking	0% to 70%
Continuous networking	0% to 50%

- The exact time when carrier shutdown actually starts taking effect may not be synchronized with that when related performance counters are reported. Consequently, if carrier shutdown starts taking effect in a measurement period, there will be a measurement error of less than 5 minutes between the reported results and the actual results for the shut-down cell in the measurement period, as listed in [Table 12-3](#).

Table 12-3 Related performance counters and KPIs

Function Subset	Performance Counter and KPI	Description
- PRB Measurement - Hyper Cell TRP PRB Measurement - DU Cell Operator PRB Measurement	- Counters related to the number of available PRBs - Counters related to the number of available PDCCH CCEs - PRB usage	None
DU Cell Algorithm Measurement	- Counters related to the total number of physical uplink shared channel (PUSCH) transmission time intervals (TTIs) - Counters related to the total number of PDSCH TTIs	None
DU Cell SI Message Measurement	Number of SI transmissions	None

Function Subset	Performance Counter and KPI	Description
- Channel Quality Measurement - DU Cell TRP Channel Quality Measurement - Uplink Interference Measurement on PRB 0 to PRB 99 - Uplink Interference Measurement on PRB 100 to PRB 199 - Uplink Interference Measurement on PRB 200 to PRB 272	- Counters related to interference and noise received in the uplink - Counters related to uplink received signal strength indicator (RSSI) - Counters related to uplink power - Counters related to the average interference and noise detected on a single PRB - Counters related to the maximum interference and noise detected on a single PRB	The PWR_SAVING_COUNTER_CORR_SW option of the gNBOamParam.StatisticsStrategySwitch parameter specifies whether to report invalid uplink interference counter values during carrier shutdown. - If this option is selected, invalid uplink interference counter values are reported for carriers that have been shut down. - If this option is deselected, measured uplink interference counter values are reported for such carriers.

- In the case of false physical random access channel (PRACH) detection where the base station mistakenly regards interference or noise as the random access preambles of UEs, measurement for the following counters and KPIs will continue:
 - Counters related to the number of UEs scheduled in the uplink and downlink in a cell
 - Counters related to uplink data transmission duration
 - Counters related to the number of transport blocks (TBs) used for uplink data transmission
 - Counters related to MCS index distribution for scheduling on the PUSCH
 - Counters related to the number of RBs used for scheduling on the PUSCH
 - PUSCH RB usage
- As the cell in the carrier shutdown state does not send downlink paging messages, the counter listed in [Table 12-4](#) is no longer measured.

Table 12-4 Counter related to paging

Function Subset	Counter
DU Cell Paging Measurement	Number of paged UEs over the Uu interface in a cell

- If a third-party power system is used with Huawei lithium batteries, and intelligent anti-theft (except the lithium battery software lock function) is enabled for the lithium batteries, the lithium batteries need to reserve some power for anti-theft monitoring and cannot be completely discharged. As a result, the backup power use duration is affected. For details about the lithium battery software lock function, see *Power Supply Management*. For details about other intelligent anti-theft functions for Huawei lithium batteries, contact Huawei engineers.

12.3 Requirements

12.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-021203	gNodeB Power Saving	NR0S0GNBPS00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists* in 5G RAN feature documentation.

12.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

12.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	LTE and NR coordinated symbol power saving	LTE: SymbolPwrSaving.MultiRatSymbolShutdownSw NR: MULTI_RAT_SYMBOL_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	Low power consumption mode depends on LTE and NR coordinated symbol power saving when the NRDUCellPowerSaving.BackupPwrSavingPolicy parameter is set to PRECISE_BACKUP_POLICY2 or PRECISE_BACKUP_POLICY3 and the multimode RF module serving the NR cell serves both LTE and NR carriers.
NR FDD	MU-MIMO CSI-RS Beam Sweeping Mode	NRDUCellCsirs.MuMimoCsirsBeamSweepMode	None	When a customized backup power saving policy is used in low power consumption mode, the MU-MIMO CSI-RS Beam Sweeping Mode parameter must be set to NOT_CONFIG to maintain channel shutdown.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	3D coverage pattern	NRDUCCELLTRPBEAM.CoverageScenario	<i>Beam Management</i>	- When a customized backup power saving policy is used in low power consumption mode, scenario-based beams supported by channel shutdown are used. - The NRDUCCELLTRPBEAM.CoverageScenario parameter can only be set to DEFAULT, SCENARIO_1, SCENARIO_2, SCENARIO_3, SCENARIO_4, SCENARIO_6, SCENARIO_7, SCENARIO_8, SCENARIO_9, SCENARIO_12, SCENARIO_13, EXPAND_SCENARIO_1, EXPAND_SCENARIO_5, EXPAND_SCENARIO_7, EXPAND_SCENARIO_8, SCENARIO_27, SCENARIO_28, SCENARIO_30, SCENARIO_31, SCENARIO_36, or CUSTOMIZED_BEAM_SCENARIO.
NR FDD	3D coverage pattern	NRDuFddCellTrpBeam.CoverageScenario	<i>Beam Management</i>	When the NRDUCell.LampSiteCellFlag parameter is set to NO, the NRDUCellPowerSaving.BackupPwrSavingPolicy parameter is set to CUSTOMIZED_BACKUP_POLICY, the NRDUCell.DuplexMode parameter is set to CELL_FDD, and the NRDUCellTrp.TxRxMode parameter is set to 32T32R, the CoverageScenario parameter in the NRDuFddCellTrpBeam MO corresponding to the NRDUCellTrp MO must be set to DEFAULT, SCENARIO_210, SCENARIO_220, SCENARIO_240, or SCENARIO_242.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	CSI-RS resource management	• NRDUCell.DuplexMode • NRDUCellCsirs.UserCsirsBeamType	<i>Channel Management</i>	• The customized backup power mode retains the restrictions on FDD channel shutdown and CSI-RS beam types. • When the NRDUCell.DuplexMode parameter is set to CELL_FDD and the NRDUCellCsirs.UserCsirsBeamType parameter is set to TYPE8 or TYPE10 for a cell, the BackupPwrSavingPolicy parameter in the NRDUCellPowerSaving MO of the cell cannot be set to CUSTOMIZED_BACKUP_POLICY.

12.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	3D coverage pattern	NRDUCellTrpBeam.ConnModeCoverageScenario	None	When a customized backup power saving policy is used in low power consumption mode, the NRDUCellTrpBeam.ConnModeCoverageScenario parameter must be set to DEFAULT .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	NR DU cell TRP beam configuration	NRDUCellTrpBeam.TrpAntType	None	- The weight for DRS shutdown is not available, so channel shutdown cannot be enabled. - When the NRDUCellTrpBeam.TrpAntType parameter is set to 8T_2222, 8T_44, or 8T_422, the Backup Power Saving Policy parameter cannot be set to CUSTOMIZED_BACKUP_POLICY for low power consumption mode.
NR FDD Low-frequency NR TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	After low power consumption mode is enabled, it takes effect for cells based on actual conditions, affecting the performance of high-reliability services for UEs.
NR FDD	Dual-plane function in intra-frequency dual-network redundancy networking	DUAL_PLANE_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	When a customized backup power saving policy is used in low power consumption mode, dual-plane function in intra-frequency dual-network redundancy networking cannot be enabled.
NR FDD Low-frequency NR TDD	Primary/Backup function in intra-frequency dual-network redundancy networking	PRIMARY_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	When a customized backup power saving policy is used in low power consumption mode, primary/backup function in intra-frequency dual-network redundancy networking cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Dual-BBU backup in dual-network redundancy networking	DUAL_BBU_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	When a customized backup power saving policy is used in low power consumption mode, dual-BBU backup in dual-network redundancy networking cannot be enabled.
NR FDD	3D coverage pattern	BEAM_POWER_REDUCTION_ALGO_SW option of the NRDUfddCellTrpBeam.ScenarioBeamAlgoSw parameter	<i>Beam Management</i>	When the Backup Power Saving Policy parameter is set to CUSTOMIZED_BACKUP_POLICY for low power consumption mode, FDD channel shutdown and beam power reduction cannot be enabled.
Low-frequency NR TDD	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	When a customized backup power saving policy is used in low power consumption mode, distributed massive MIMO cannot be enabled.
NR TDD	LTE TDD and NR spectrum sharing	LTE_NR_TDD_SPECTRUM_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter	None	When a customized backup power saving policy is used in low power consumption mode, LTE TDD and NR spectrum sharing cannot be enabled.
NR FDD	Downlink power aggregation	DL_POWER_AGG_SW option of the NRDUCellChnPwr.LteNrSpctShrPwrAggSwitch parameter	None	When a customized backup power saving policy is used in low power consumption mode, downlink power aggregation cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	Hybrid DSS Based on Asymmetric Bandwidth	- LTE: SpectrumCloud.SpectrumCloudSwitch parameter set to LTE_NR_SPECTRUM_SHR, and the LNR_SPECTRUM_SHR_ASYM_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter selected - NR: LTE_NR_FDD_SPECTRUM_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter and LTE_NR_FDD_SPECTRUM_SHR_ASYM_SW option of the NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch parameter selected	<i>LTE FDD and NR Hybrid Dynamic Spectrum Sharing</i>	When a customized backup power saving policy is used in low power consumption mode, Hybrid DSS Based on Asymmetric Bandwidth cannot be enabled.
NR TDD NR FDD	PDCCH rate matching	PDCCH_RATEMATCHING_SW option of the NRDUCELLPDSCH.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	There are reliability risks when PDCCH rate matching and channel shutdown take effect at the same time. Therefore, PDCCH rate matching and channel shutdown cannot be enabled at the same time.

RAT	Function Name	Function Switch	Reference	Description
NR TDD NR FDD	Adaptive Power Backup	LOW_PWR_MODE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Green Site</i>	Both Adaptive Power Backup and low power consumption mode involve control on cell services in power backup scenarios. To avoid affecting algorithm accuracy, Adaptive Power Backup and low power consumption mode cannot be enabled at the same time, and Adaptive Power Backup takes precedence over low power consumption mode.
NR FDD	RuralLink antenna solution (NR)	EASY_OMNI_SITE_MODE_SW option of the NRDUfddCellTrpBeam.ScenarioModeSw parameter	<i>RuralLink Solution</i>	Low power consumption mode and the RuralLink antenna solution (NR) cannot be enabled at the same time.
NR FDD Low-frequency NR TDD	Multi-carrier joint carrier shutdown	MULTI_CARR_JOINT_CARR_SHDN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction Multi-RAT Coordinated Energy Saving</i>	Multi-carrier joint carrier shutdown does not work with low power consumption mode.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	SSB coverage pattern for emergency and normal services	NRDUCellTrpBeam.CoverageScenario	<i>Emergency Communications Solution (Low-Frequency TDD)</i>	- If a cell uses a customized backup power saving policy for low power consumption mode, the NRDUCellTrpBeam.CoverageScenario parameter cannot be set to SCENARIO_33, SCENARIO_34, or SCENARIO_35 for the cell. - If a cell uses a customized backup power saving policy for low power consumption mode, the NRDUCellTrpBeam.ConnModeCovScenarioExt parameter must be set to DEFAULT for the cell.
Low-frequency NR TDD	Connected mode coverage pattern for emergency and normal services	NRDUCellTrpBeam.ConnModeCovScenarioExt	<i>Emergency Communications Solution (Low-Frequency TDD)</i>	

12.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Intelligent carrier shutdown	INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When intelligent carrier shutdown takes effect for a carrier: - If the RF module is not serving any other NR carrier, symbol power saving in low power consumption mode does not take effect. - If the RF module is serving other NR carriers, the status of symbol power saving in low power consumption mode for these carriers is not affected.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When timing carrier shutdown takes effect for a carrier: - If the RF module is not serving any other NR carrier, symbol power saving in low power consumption mode does not take effect. - If the RF module is serving other NR carriers, the status of symbol power saving in low power consumption mode for these carriers is not affected.
NR FDD Low-frequency NR TDD	RF module deep dormancy	RRU.DORMANCY_SW (LTE eNodeB, 5G gNodeB) set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	RF module deep dormancy takes precedence over low power consumption mode when both functions are enabled. Low power consumption mode does not take effect when RF module deep dormancy is in effect.
NR FDD Low-frequency NR TDD	RF module super dormancy	RRU.DORMANCY_SW (LTE eNodeB, 5G gNodeB) set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	RF module super dormancy takes precedence over low power consumption mode when both functions are enabled. Low power consumption mode does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Cell-level MRO	MRO_SCENARIO_IDENT_SW option of the gNB Mro.ScenarioIdentifySwitch parameter and the INTRA_FREQ_MRO_SW option of the NRCellMro.MroSwitch parameter	<i>MRO</i>	When both low power consumption mode and intra-frequency MRO are enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, MRO may take effect later than expected.
NR FDD Low-frequency NR TDD	UE-level MRO	UE_MRO_SWITCH option of the NRCellMro.MroSwitch parameter	<i>MRO</i>	When both low power consumption mode and UE-level MRO are enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, the penalty mechanism accuracy of UE-level MRO may be affected.
NR FDD	LTE FDD and NR Flash Dynamic Power Sharing	LTE_NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	Cells that have entered the energy saving state cannot participate in LTE FDD and NR Flash Dynamic Power Sharing. These cells will attempt to participate in power sharing after exiting the energy saving state.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE and NR multi-band power adaptive scheduling	• LTE: LNR_MULTI_BAND_PWR_ADPT_SCH_SW option of the CellDynPowerSharing.DynPowerSharingSwitch parameter • NR: LNR_MULTI_BAND_PWR_ADPT_SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	Only low power consumption mode takes effect when low power consumption mode and LTE and NR multi-band power adaptive scheduling are both enabled and a customized backup power saving policy is used in low power consumption mode.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Remote interference adaptive avoidance	RIM_ADAPT_AVOID_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	When both low power consumption mode and remote interference adaptive avoidance are enabled and a customized backup power saving policy is used in low power consumption mode: • If the coverage compensation policy in NR low power consumption mode is disabled (by deselecting the COV_COMPENSATION_POLICY_SW option of the NRDUCellPowerSaving.LowPwrModeAlgoSwitch parameter), channel shutdown in low power consumption mode does not take effect. • If the coverage compensation policy in NR low power consumption mode is enabled, low power consumption mode and remote interference adaptive avoidance can both take effect. In this case, coverage performance of low power consumption mode may deteriorate.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Enhanced remote interference adaptive avoidance	RIM_ADAPT_AVOID_ENH_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	When both low power consumption mode and enhanced remote interference adaptive avoidance are enabled and a customized backup power saving policy is used in low power consumption mode: • If the coverage compensation policy in NR low power consumption mode is disabled (by deselecting the COV_COMPENSATION_POLICY_SW option of the NRDUCellPowerSaving.LowPwrModeAlgoSwitch parameter), channel shutdown in low power consumption mode does not take effect. • If the coverage compensation policy in NR low power consumption mode is enabled, low power consumption mode and enhanced remote interference adaptive avoidance can both take effect. In this case, coverage performance of low power consumption mode may deteriorate.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Remote interference adaptive suppression	RIM_ADAPT_SUPPRESSION_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	When both low power consumption mode and remote interference adaptive suppression are enabled and a customized backup power saving policy is used in low power consumption mode: • If the coverage compensation policy in NR low power consumption mode is disabled (by deselecting the COV_COMPENSATION_POLICY_SW option of the NRDUCellPowerSaving.LowPwrModeAlgoSwitch parameter), channel shutdown in low power consumption mode does not take effect. • If the coverage compensation policy in NR low power consumption mode is enabled, low power consumption mode and remote interference adaptive suppression can both take effect. In this case, coverage performance of low power consumption mode may deteriorate.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Remote interference resistance for random access	RA_REMOTE_INTERFERESW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	When both low power consumption mode and remote interference resistance for random access are enabled and a customized backup power saving policy is used in low power consumption mode: • If the coverage compensation policy in NR low power consumption mode is disabled (by deselecting the COV_COMPENSATION_POLICY_SW option of the NRDUCellPowerSaving.LowPwrModeAlgoSwitch parameter), channel shutdown in low power consumption mode does not take effect. • If the coverage compensation policy in NR low power consumption mode is enabled, low power consumption mode and remote interference resistance for random access can both take effect. In this case, coverage performance of low power consumption mode may deteriorate.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Remote interference resistance for PUSCH demodulation	RIM_PUSCH_DEMOD_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	When both low power consumption mode and remote interference resistance for PUSCH demodulation are enabled and a customized backup power saving policy is used in low power consumption mode: • If the coverage compensation policy in NR low power consumption mode is disabled (by deselecting the COV_COMPENSATION_POLICY_SW option of the NRDUCellPowerSaving.LowPwrModeAlgorithmSwitch parameter), channel shutdown in low power consumption mode does not take effect. • If the coverage compensation policy in NR low power consumption mode is enabled, low power consumption mode and remote interference resistance for PUSCH demodulation can both take effect. In this case, coverage performance of low power consumption mode may deteriorate.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Precoding enhancement for non-antenna-selection UEs	DL_NAS_PRECODE_ENH_SW option of the NRDUCellPdschPreambleSw parameter	<i>MIMO (TDD)</i>	Coverage performance of low power consumption mode may deteriorate when both low power consumption mode and precoding enhancement for non-antenna-selection UEs are enabled and a customized backup power saving policy is used in low power consumption mode.
NR FDD	LTE FDD and NR Flash Dynamic Power Sharing	LTE_NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch parameter	<i>LTE and NR Power Allocation</i>	Only low power consumption mode takes effect when both low power consumption mode and LTE FDD and NR Flash Dynamic Power Sharing are enabled and a customized backup power saving policy is used in low power consumption mode.
Low-frequency NR TDD	LTE TDD and NR Flash Dynamic Power Sharing	LTE_NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch parameter	<i>LTE and NR Power Allocation</i>	Only low power consumption mode takes effect when both low power consumption mode and LTE TDD and NR Flash Dynamic Power Sharing are enabled and a customized backup power saving policy is used in low power consumption mode.
NR FDD Low-frequency NR TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	Only low power consumption mode takes effect when both low power consumption mode and NR inter-carrier dynamic power sharing are enabled and a customized backup power saving policy is used in low power consumption mode.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SC_H_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	When both low power consumption mode (NR) and NR multi-band power adaptive scheduling are enabled and a customized backup power saving policy is used, only low power consumption mode takes effect.
NR FDD Low-frequency NR TDD	Power saving scheduling	POWER_SAVING_SCHEDULE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Only low power consumption mode takes effect when both low power consumption mode and power saving scheduling are enabled and a customized backup power saving policy is used in low power consumption mode.
Low-frequency NR TDD	Broadcast beam densification	SCENARIO_BEAM_DENSIFY_ALGO_SW option of the NRDUCellTrpBeam.ScenarioBeamAlgoSw parameter	<i>Beam Management</i>	Changes in beams will cause termination of low power consumption mode when both low power consumption mode and broadcast beam densification are enabled and a customized backup power saving policy is used in low power consumption mode.
FDD Low-frequency NR TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	Only low power consumption mode takes effect when both low power consumption mode and intra-base-station DL CoMP are enabled and a customized backup power saving policy is used in low power consumption mode.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	High static PSD ratios between LTE TDD and NR	SPEC_ENH_LOOSE_PSD_LIMIT option of the RRU.CUSTOMED_RFSPECSW parameter	<i>LTE and NR Power Allocation</i>	Coverage performance of low power consumption mode may deteriorate when both low power consumption mode and high static PSD ratios between LTE TDD and NR are enabled and a customized backup power saving policy is used in low power consumption mode.
NR FDD Low-frequency NR TDD	Rate-matching-pattern-configuration-based PDCCH rate matching	PDCCH_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	Channel shutdown in low power consumption mode does not take effect when both low power consumption mode and rate-matching-pattern-configuration-based PDCCH rate matching are enabled and a customized backup power saving policy is used in low power consumption mode.
NR FDD Low-frequency NR TDD	Adaptive Power Backup	None	<i>Green Site</i>	- Precise backup power saving policy 2 or 3 cannot take effect when the Adaptive Power Backup feature is in effect. If precise backup power saving policy 2 or 3 has been enabled in some cells, the Adaptive Power Backup feature cannot take effect. - When the MAE is disconnected from the base station, the feature rolls back to precise backup power saving policy 1.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	Intelligent MCS optimization for uplink SU-MIMO	UL_SU_MCS_INTELL_OPT_SW option of the NRDUCellAiAlgo.AiAmcAlgoSwitch parameter	<i>Smart 8T8R (FDD)</i>	When intelligent MCS optimization for uplink SU-MIMO and low power consumption mode are both enabled, the former function does not take effect or its gains decrease.
LTE FDD LTE TDD UMTS GSM	SOC level-based low power consumption mode	• IPWRM.SOCLVLADJSW • LTE: SOCLVL_BASED_LOW_PWR_SW option of the CellLowPower.LowPwrConsumptionModeExtSw parameter - UMTS: SOCLVL_BASED_LOW_PWR_SW option of the NODEBALGPARRA.LOWPWRCONSUMEMODEEXTSW parameter - GSM: GBTSENERGYMGTPARA.SOC LVLLLOWPWR MODESW	<i>Green Site</i>	NR low power consumption mode does not take effect as long as the IPWRM.SOCLVLADJSW parameter is set to ON .
NR FDD	Intelligent PIM interference avoidance	PIM_INTERF_ART_AVOID_SW option of the NRDUCellPimInterfAvd.PimAlgoSwitch parameter	<i>Interference Avoidance</i>	The low power consumption mode is terminated during PIM interference detection.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	Channel isolation for PIM interference avoidance	PIM_INTERF_CHN_SHUTDOWN_SW option of the NRDUCellPimInterfAvd.PimAlgoSwitch parameter	<i>Interference Avoidance</i>	- Channel isolation for PIM interference avoidance is terminated when the carrier shutdown function takes effect in the low power consumption mode. - Only channel isolation for PIM interference avoidance takes effect when both low power consumption mode and channel isolation for PIM interference avoidance are enabled and a customized backup power saving policy is used in low power consumption mode.
Low-frequency NR TDD	L/NR Inter-band Deep Coordination	- LTE: LNR_INTER_BAND_DEEP_COORD_SW option of the CellDynPowerSharing.DynmicPowerSharingSwitch parameter - NR: LNR_INTER_BAND_DEEP_COORD_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter	<i>LTE and NR Power Allocation</i>	Only low power consumption mode takes effect when low power consumption mode and L/NR Inter-band Deep Coordination are both enabled and a customized backup power saving policy is used in low power consumption mode.

12.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable AAUs/RRUs that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Power Supply

- Power cabinet: The mains failure alarm can be reported.
- Available backup power at the site: If the eNodeB enables low power consumption mode, batteries are required to provide backup power. This function cannot be enabled at sites without backup power. You are advised to precheck the battery health of base stations equipped with lead-acid batteries and not to enable this function when batteries are severely aged (without discharge capability or with extremely short discharge time). Otherwise, the gains in terms of the extended backup power use duration may decrease.
- The power supply system can be provided by Huawei or another vendor.
 - The Huawei power supply system can be connected to a Huawei base station through the two RS485 signal input ports on the universal power and environment interface unit (UPEU) or universal environment interface unit (UEIU). They can also be connected through the RS485 signal input port or the power cable on the RRU.
 - A third-party power system can be connected to a Huawei base station through the Boolean signal input ports on the UPEU or UEIU. That is, they are connected through a dry contact.

In this case, you need to configure the **ALMPORT** MO to set information about the dry contact port and configure the **MAINSALARMBIND** MO to define the alarms generated by this port as mains supply failure alarms. Under this setting, a mains supply failure alarm will be reported to the Huawei base station through the port if the third-party power system experiences a mains supply failure. Upon detection of this alarm, the Huawei base station can start low power consumption mode to extend the duration of basic coverage service provisioning.

NOTE

The name and severity of the alarms reported by the dry contact port can be customized by running the **SET ENVALMPARA** command.

- When low power consumption mode is used with a Huawei power system and Huawei lithium batteries, a PMU must be configured on the base station. [Table 12-5](#) lists the requirements for the PMU.

Table 12-5 PMU requirements of low power consumption mode

Power System Type	Cabinet Type	PMU Model	PMU Software Version
ETP	APM5930	APM5930(CCUC)	If the hardware version is PMU.19, the PMU software version must be 219 or later. If the hardware version is PMU.22, the PMU software version must be 213 or later.
ETP	None	OPM200	The PMU software version must be 177 or later.
ETP	APM5950H-L	PMU 11C	If the hardware version is PMU.35, the PMU software version must be 2265 or later.
ETP	APM5950H	PMU 11C	If the hardware version is PMU.35, the PMU software version must be 2251 or later.
SMU	MTS series power system ^a	SMU02C (02312M ML-009)	If the hardware version is PMU.101, the PMU software version must be 2300 or later.
a: The MTS series power cabinet in use must meet the capability requirements for its connection with the MAE for O&M management, which is supported from SRAN21.1 onwards. For details about the MTS series power system model and version requirements, contact Huawei engineers.			

 NOTE

- Low power consumption mode can be used for a base station (with precise backup power saving policy 1/2/3 implemented) only when **Battery Type** is **Li-Ion Battery** and **Number Of Lithium Batteries** is at least 1 in the command output of **DSP BATTERY**. For details about the power supply configurations, lithium battery configurations, and mapping between power supplies and lithium batteries, see *Power Supply Management*. If a customized backup power saving policy is used, there is no requirement for the **Battery Type** parameter settings.
- The software version of the PMU can be queried by running the **DSP BRDVER** command.
- If the communication between the lithium batteries and the PMU is faulty, the low power consumption mode function cannot run properly. In this case, the base station maintains the current operation taken in low power consumption mode until the communication recovers.
- Precise backup power saving policy 1, precise backup power saving policy 2, and precise backup power saving policy 3 do not apply to scenarios where multiple power systems are used and lithium batteries in each power system are managed by an individual PMU.

- When low power consumption mode is enabled in a scenario with a third-party power system and Huawei lithium batteries (used to provide power backup), the third-party power system must support connection through a dry contact and a PMU 15A(Ver.B) must be configured. In this scenario, low power consumption mode is not supported in C-RAN networking.

12.3.4 Cells

- **NRDUCell.DeployPosition** cannot be set to **DEPLOY_IAB**.
- **NRDUCell.SlotAssignment** cannot be set to **1_1_DSDSDSDSDS**.
- **NRDUCell.FrequencyBand** cannot be set to **N257, N258, N260, or N261**.
- If **NRDUCellPowerSaving.BackupPwrSavingPolicy** is set to **PRECISE_BACKUP_POLICY1** or **PRECISE_BACKUP_POLICY3**, **NRDUCell.DuplexMode** must be set to **CELL_TDD, CELL_FDD, or CELL_SDL**.
- If **NRDUCellPowerSaving.BackupPwrSavingPolicy** is set to **CUSTOMIZED_BACKUP_POLICY**, **NRDUCell.FrequencyBand** must be set to **N1, N2, N3, N5, N7, N8, N12, N13, N20, N25, N28, N38, N39, N40, N41, N66, N71, N75, N77, N78, or N79**.
- For NR TDD cells, if **NRDUCellPowerSaving.BackupPwrSavingPolicy** is set to **CUSTOMIZED_BACKUP_POLICY**, **NRDUCellTrp.TxRxMode** must be set to **2T2R, 4T4R, 8T8R, 32T32R, or 64T64R**.
- For NR FDD and NR SDL cells, if **NRDUCellPowerSaving.BackupPwrSavingPolicy** is set to **CUSTOMIZED_BACKUP_POLICY**, **NRDUCellTrp.TxRxMode** must be set to **2T2R, 2T, 4T4R, 4T, 2T4R, 8T8R, or 32T32R**.

12.3.5 Others

None

12.4 Operation and Maintenance

12.4.1 Data Configuration

12.4.1.1 Data Preparation

[Table 12-6](#) and [Table 12-7](#) describe the parameters used for function activation and optimization, respectively. The tables do not describe parameters related to cell establishment.

Table 12-6 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Power Supply Group ID	RRU.PSGID	One power supply group ID is configured for AAUs/RRUs using the same power supply system, whereas different power supply group IDs are configured for AAUs/RRUs using different power supply systems. The power supply group ID of the AAUs/RRUs using the same power supply system as the BBU is 0 .
NR DU Cell ID	NRDUCellAlgoSwitch.NrDuCellId	Set this parameter based on the network plan.
Power Saving Switch	NRDUCellAlgoSwitch.PowerSavingSwitch	Select the LOW_PWR_MODE_SW option when low power consumption mode is required.
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	Set this parameter based on the network plan.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	Set this parameter to LOW_PWR_MODE when low power consumption mode is required.

Parameter Name	Parameter ID	Setting Notes
Backup Power Saving Policy	NRDUCellPowerSaving.BackupPwrSavingPolicy	If a multimode base station uses the same set of storage batteries to support the services of all RATs: • The value PRECISE_BACKUP_POLICY1 is recommended if the services of the local cell require a low priority and the rated power of this cell needs to be decreased to save power for cells of other RATs in the case of mains supply failures. • The value PRECISE_BACKUP_POLICY2 is recommended if the services of the local cell require a high priority in the case of mains supply failures. • The value PRECISE_BACKUP_POLICY3 or CUSTOMIZED_BACKUP_POLICY is recommended if the services of the local cell require a medium priority in the case of mains supply failures. • When a target backup power use duration is specified using the MAE-Deployment plug-in, the plug-in defines the cells whose backup policy is PRECISE_BACKUP_POLICY1 and with the LOW_PWR_MODE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter selected as non-reserved cells and cells with this option deselected as reserved cells.

Parameter Name	Parameter ID	Setting Notes
Switch	ALMPORT.SW	Set this parameter to ON if a non-Huawei power supply system is connected to the base station through a dry contact port.
Cabinet No.	ALMPORT.CN	Set this parameter based on the network plan.
Subrack No.	ALMPORT.SRN	Set this parameter based on the network plan.
Slot No.	ALMPORT.SN	Set this parameter based on the network plan.
Port No.	ALMPORT.PN	Set this parameter based on the network plan.
Alarm ID	ALMPORT.AID	Set this parameter based on the network plan.
Port Type	ALMPORT.PT	Set this parameter based on the analog or Boolean value output by the non-Huawei power supply system when the mains supply fails.
Alarm Voltage	ALMPORT.AVOL	Set this parameter based on the Boolean value output by the non-Huawei power supply system when the mains supply fails.
Base Station Almport Switch	MAINSALARMBIND.ISDSWITCH	Set this parameter to ENABLE if the alarms generated by the dry contact port are defined as mains supply failure alarms.
Almport Cabinet No.	MAINSALARMBIND.NMSACN	Set this parameter based on the network plan.
Almport Subrack No.	MAINSALARMBIND.NMSASRN	Set this parameter based on the network plan.
Almport Slot No.	MAINSALARMBIND.NMSASN	Set this parameter based on the network plan.
Almport Port No.	MAINSALARMBIND.NMSAPN	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Time to Trigger	NRDUCellPowerSaving.TimeToTrigger	You are advised to set this parameter to its recommended value.
Low Power Mode Algorithm Switch	NRDUCellPowerSaving.LowPwrModeAlgoSwitch	Select the COV_COMPENSATION_POLICY_SW option when the coverage compensation policy in NR low power consumption mode is required.

 NOTE

The parameters in the **ALMPORT** MO must be set based on the analog or Boolean value output by the non-Huawei power supply system when the mains supply fails.

Table 12-7 Parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
Alarm ID	ALMCURCFG.AID	Set this parameter based on the network plan.
Alarm Name	ALMCURCFG.ANM	Set this parameter based on the network plan.
Alarm Level	ALMCURCFG.ALVL	Set this parameter based on the network plan.
Event Type	ALMCURCFG.ASS	Set this parameter based on the network plan.

12.4.1.2 Using MML Commands

Before using MML commands, refer to [12.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
 //(Optional) Running the SET ALMPORT command to turn on the port switch and configure the information about the dry contact port and running the MOD NMSABIND command to define the alarms
```

generated by the dry contact port as mains supply failure alarms, if low power consumption mode is used with a third-party power system
 SET ALMPORT: CN=0, SRN=0, SN=19, PN=0, SW=ON, AID=65033, PT=BOOL, AVOL=LOW;
 MOD NMSABIND: ISDSWITCH=ENABLE, NMSACN=0, NMSASRN=0, NMSASN=19, NMSAPN=0;
 //(Optional; required in distributed power supply scenarios) Setting a power supply group ID
 MOD RRU: CN=0,SRN=60,SN=0,PSGID=0;
 //Enabling low power consumption mode
 MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=LOW_PWR_MODE_SW-1;
 //Setting an energy saving policy of low power consumption mode
 ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1,
 PowerSavingType=LOW_PWR_MODE, BackupPwrSavingPolicy=PRECISE_BACKUP_POLICY1, TimeToTrigger=5;
 //Enabling the coverage compensation policy in NR low power consumption mode
 MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1,
 PowerSavingType=LOW_PWR_MODE, LowPwrModeAlgoSwitch=COV_COMPENSATION_POLICY_SW-1;

NOTE

- If a third-party power system experiences a mains supply failure, the following must be performed before defining the alarm generated by the dry contact port:
 - Check whether the alarm reported for the third-party power system is triggered in low level or high level.
 - Ensure that the cable connections of the dry contact port are correct when there is no mains supply to the third-party power system.
- If an MTS power cabinet is regarded as a third-party power system for management and the mains fault is obtained through dry contacts, the alarm is triggered in low level, and there is no mains supply to the ALM1 port on the interface board in the cabinet.

Optimization Command Examples

//(Optional) Customizing the name, severity, and other information of the alarms reported by the dry contact port, if low power consumption mode is used with a third-party power system
 SET ENVALMPARA: ALMID=65033, ANM="Non-Huawei power supply system mains fault", ALVL=MAJOR, ASS=HARDWARE;

Deactivation Command Examples

//(Optional) Running the SET ALMPORT command to turn off the port switch and running the MOD NMSABIND command to set ISDSWITCH to DISABLE, if low power consumption mode does not need to be used with a third-party power system
 SET ALMPORT: CN=0, SRN=0, SN=19, PN=0, SW=OFF;
 MOD NMSABIND: ISDSWITCH=DISABLE;
 //Disabling the coverage compensation policy in NR low power consumption mode
 MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1,
 PowerSavingType=LOW_PWR_MODE, LowPwrModeAlgoSwitch=COV_COMPENSATION_POLICY_SW-0;
 //Disabling low power consumption mode
 MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=LOW_PWR_MODE_SW-0;
 //Removing the energy saving policy of low power consumption mode
 RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1;

12.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

12.4.2 Activation Verification

After the low power consumption mode function is activated, use either of the following methods to check whether the function has taken effect:

- Method 1:

When low power consumption mode is used with the Huawei power supply system:

- a. Run the **LST NRDUCELLALGOSWITCH** command. If the value of the **Power Saving Switch** parameter is **Low Power Consumption Mode Switch:On** in the command output, low power consumption mode has been enabled.
- b. Run the **LST NRDUCELLPOWERSAVING** command. Verify that there is at least one record with **Power Saving Type** being **Low Power Consumption Mode**.
- c. Wait for a period of time specified by the **NRDUCellPowerSaving.TimeToTrigger** parameter after a mains fault occurs. Then, run the **DSP NRDUCELL** command. Verify that the value of the **Power Saving State** parameter is **Low Power Consumption Mode:Enabled**, which indicates that low power consumption mode has taken effect.

When low power consumption mode is used with a non-Huawei power supply system:

- a. Run the **LST ALMPORT** command. Verify that the parameter settings in the **ALMPORT** MO are consistent with the analog or Boolean value output by the non-Huawei power supply system when the mains supply fails.
- b. Run the **LST NMSABIND** command. Verify that the value of the **Base Station Almpor Switch** parameter is **ENABLE**.
- c. Run the **LST NRDUCELLALGOSWITCH** command. Verify that the value of the **Power Saving Switch** parameter is **Low Power Consumption Mode Switch:On** in the command output.
- d. Run the **LST NRDUCELLPOWERSAVING** command. Verify that there is at least one record with **Power Saving Type** being **Low Power Consumption Mode**.
- e. Wait for a period of time specified by the **NRDUCellPowerSaving.TimeToTrigger** parameter after a mains fault occurs. Then, run the **DSP NRDUCELL** command. Verify that the value of the **Power Saving State** parameter is **Low Power Consumption Mode:Enabled**, which indicates that low power consumption mode has taken effect.

- Method 2:

Run the **DSP NRDUCELLPOWERSAVING** command to query the status of low power consumption mode.

- If the value of **Power Saving Type** is **Low Power Consumption Mode** and the value of **Current State** is **Symbol Power Saving Enabled in LPC Mode, Channel Shutdown Enabled in LPC Mode, or Carrier Shutdown Enabled in LPC Mode**, low power consumption mode has taken effect.
- If the value of **Power Saving Type** is **Low Power Consumption Mode** and the value of **Current State** is **Symbol Power Saving Disabled in LPC Mode, Channel Shutdown Disabled in LPC Mode, or Carrier Shutdown Disabled in LPC Mode**, then the cell should have entered low power consumption mode with symbol power saving or carrier shutdown

performed, but for some reason has not. You can check the **Current Disabled Cause** parameter value to determine the reason why low power consumption mode does not take effect. [Table 12-8](#) lists possible reasons.

- If the value of **Power Saving Type** is **Low Power Consumption Mode** and the value of **Current State** is **Disabled**, check the **Current Disabled Cause** parameter value to determine the reason why low power consumption mode does not take effect. [Table 12-8](#) lists possible reasons.

Table 12-8 Possible reasons why low power consumption mode does not take effect

Reason	Description
Insufficient License	No license for this function is available.
Switch Turned Off	None
Cell Unavailable	The cell is blocked or deactivated, fails to be set up, or is unavailable due to other reasons.
Pwr Normal/In Time-to-Trig/ EPMU Incompat	• If low power consumption mode is used with the Huawei power supply system, the mains supply is normal or a transitional period is ongoing. • If low power consumption mode is used with a non-Huawei power supply system, the mains supply is normal, a transitional period is ongoing, or the MOD NMSABIND command has not been executed to define the alarms generated by the dry contact port as mains supply failure alarms. • If low power consumption mode is used with an external power supply system interconnected through a protocol, low power consumption mode is not triggered when the mains supply fails.
Impacted Feature Taken Effect	Impacted features have taken effect.

Reason	Description
Others	• A multimode RF module serves both LTE and NR carriers, and multi-RAT coordinated symbol power saving is not enabled for the LTE and NR cells. • A multimode RF module is in the negotiation state or the negotiation fails, and channel shutdown does not take effect. • Intelligent carrier shutdown or timing carrier shutdown takes effect for a carrier, and the RF module does not serve any other NR carrier. • Internal software or other reasons cause the function to fail. • Low power consumption mode does not take effect in SUL cells.

 NOTE

Resetting the PMU affects the energy saving gains brought by low power consumption mode but does not affect the function reliability. The function status during the PMU reset is as follows:

1. No mains supply: The PMU reports an alarm indicating no mains supply. Then the BBU delivers energy saving level-1 configurations, and the related NR cells configured with low power consumption mode enter the energy saving state based on the configurations.
2. PMU reset: When all PMU faults are rectified, the BBU delivers energy saving level-0 configurations.
3. The related NR cells configured with low power consumption mode receive the energy saving level-0 configurations and the function is disabled.
4. The mains supply still fails and the alarm indicating no mains supply is still not cleared. The BBU delivers energy saving level-1 configurations. The related NR cells configured with low power consumption mode re-enter the energy saving state based on the configurations.

12.4.3 Network Monitoring

Benefit Monitoring

Performance counters on the MAE-Access are used to monitor and evaluate the performance of this function.

- Site power consumption
If the iSDU06D/iSDU03D is configured and interconnected with a third-party power system through a protocol, the **VS.EPMU.DCEnergy** counter can be used for observation.

- gNodeB power consumption
It is indicated by the **VS.EnergyCons.BTS.Adding.Service.NR** or **VS.EnergyCons.BTS.Adding.NR** counter on the MAE-Access.
- RF module power consumption
It is indicated by the **VS.EnergyCons.BTSBoard** counter on the MAE-Access. If low power consumption mode is triggered due to abnormal mains supply or is forcibly triggered, operators can evaluate the performance of low power consumption mode by comparing the power consumed by an RF module before and after this function is enabled. The decrease in RF module power consumption indicates how much energy is saved.
- Main equipment power consumption
It is indicated by the **VS.EnergyCons.BTS.Adding.Service.GSM**, **VS.EnergyCons.BTS.Adding.Service.UMTS**, **VS.EnergyCons.BTS.Adding.Service.LTE**, **VS.EnergyCons.BTS.Adding.Service.NR**, **VS.EnergyCons.BTS.Adding.GSM**, **VS.EnergyCons.BTS.Adding.UMTS**, **VS.EnergyCons.BTS.Adding.LTE**, and **VS.EnergyCons.BTS.Adding.NR** counters on the MAE-Access. By monitoring the power consumption of the main equipment before and after the function takes effect, you can obtain the increase in the backup power use duration. (Pay attention to the mapping between physical sites and logical sites.)
- Duration of cell unavailability due to energy saving
It is indicated by the **N.Cell.Unavail.Dur.EnergySaving** counter on the MAE-Access. On the live network, you can compare one-week gNodeB power consumption before and after the function is enabled to evaluate its energy saving gains.
- Channel shutdown
It is indicated by the **VS.RADIOEQM.ChannelShutdown.Dur** counter on the MAE-Access. The result can be observed only after transmit channel blocking is enabled. Determine the gains in the effective duration of channel shutdown in low power consumption mode based on the function switch, mains supply alarms, and counter subscription.

13 SOC Level-based Low Power Consumption Mode

13.1 Principles

When the remaining backup power is insufficient, SOC level-based low power consumption mode enables cells to shut down carriers and enter the low power consumption mode. This reduces the power consumption of RF modules and prolongs the service time of the base station.

This function applies to GSM, UMTS, and LTE.

Parameter Configuration

SOC level-based low power consumption mode can be enabled only when all of the following conditions are met:

- **IPWRM.SOCLVLADJSW** is set to **ON**, and the upper thresholds of SOC levels specified by **IPWRM.SOCLVL1UTHD**, **IPWRM.SOCLVL2UTHD**, and **IPWRM.SOCLVL3UTHD** are set.
- Switches for this function can be turned on separately for one RAT or for multiple RATs and the corresponding backup power saving policies are configured.

- On the LTE side, this function is controlled by the **SOCLVL_BASED_LOW_PWR_SW** option of the **CellLowPower.LowPwrConsumeModeExtSw** parameter, and the **CellLowPower.BakPwrSavPolicy** parameter is set to **LOW_PWR_AT_SOCLVL1**, **LOW_PWR_AT_SOCLVL2**, or **LOW_PWR_AT_SOCLVL3**.

After LTE SOC level-based low power consumption mode takes effect, all carriers of cells are shut down.

- On the UMTS side, this function is controlled by the **SOCLVL_BASED_LOW_PWR_SW** option of the **NODEBALGPARA.LOWPWRCONSUMEMODEEXTSW** parameter, and the **NODEBALGPARA.BAKPWRSAPOLICY** parameter is set to

LOW_PWR_AT_SOCLVL1, LOW_PWR_AT_SOCLVL2, or LOW_PWR_AT_SOCLVL3.

After UMTS SOC level-based low power consumption mode takes effect, carriers of reserved cells are not shut down while all carriers of non-reserved cells are shut down. Reserved cells are configured using the **ULOCCELL.RSV** parameter.

- On the GSM side, this function is controlled by the **GBTSENERGYMGTPARA.SOCLVLOWPWRMODESW** parameter, and the **GBTSENERGYMGTPARA.SOCLVLPWRSVAPOLICY** parameter is set to **LOW_PWR_AT_SOCLVL1, LOW_PWR_AT_SOCLVL2, or LOW_PWR_AT_SOCLVL3.**

After GSM SOC level-based low power consumption mode takes effect, some carriers of cells are shut down. The carriers that are to be shut down are specified by the **GTRXDEV.SDFLAG** and **BTSLOCKBCCH.BTSPOWEROFFLOCKBCCH** parameters on the BSC side.

After SOC level-based low power consumption mode is enabled, cells shut down carriers based on the SOC level of the remaining backup power and enter the low power consumption mode.

For example, if **CellLowPower.BakPwrSavPolicy** is set to **LOW_PWR_AT_SOCLVL1** on the LTE side, **NODEBALGPARA.BAKPWRSVAPOLICY** is set to **LOW_PWR_AT_SOCLVL2** on the UMTS side, and **GBTSENERGYMGTPARA.SOCLVLPWRSVAPOLICY** is set to **LOW_PWR_AT_SOCLVL3** on the GSM side, the following processing is performed for each RAT:

- If the remaining backup power is less than or equal to the **IPWRM.SOCLVL1UTHD** parameter value, LTE SOC level-based low power consumption mode takes effect and all carriers of LTE cells are shut down.
- If the remaining backup power is less than or equal to the **IPWRM.SOCLVL2UTHD** parameter value, UMTS SOC level-based low power consumption mode takes effect and all carriers of non-reserved UMTS cells are shut down.
- If the remaining backup power is less than or equal to the **IPWRM.SOCLVL3UTHD** parameter value, GSM SOC level-based low power consumption mode takes effect and the selected carriers of GSM cells are shut down.

Application Scenario

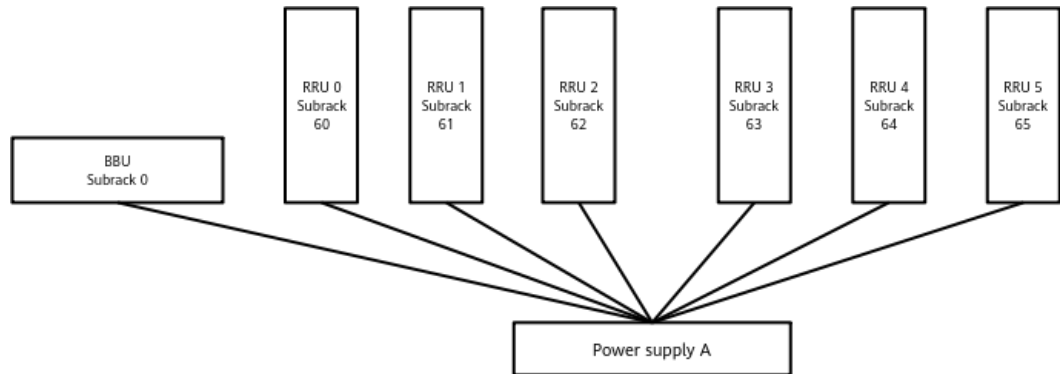
SOC level-based low power consumption mode applies only to centralized power supply scenarios. All cells enabled with this function enter this mode at the same time.

The differences between centralized and distributed power supply scenarios are as follows:

- Centralized power supply scenario
A centralized power supply scenario is when main base station devices (including the BBU and AAUs/RRUs) use the same power supply system and backup power supply system. If the mains supply fails, the same backup power supply system provides power for all main devices, and the same

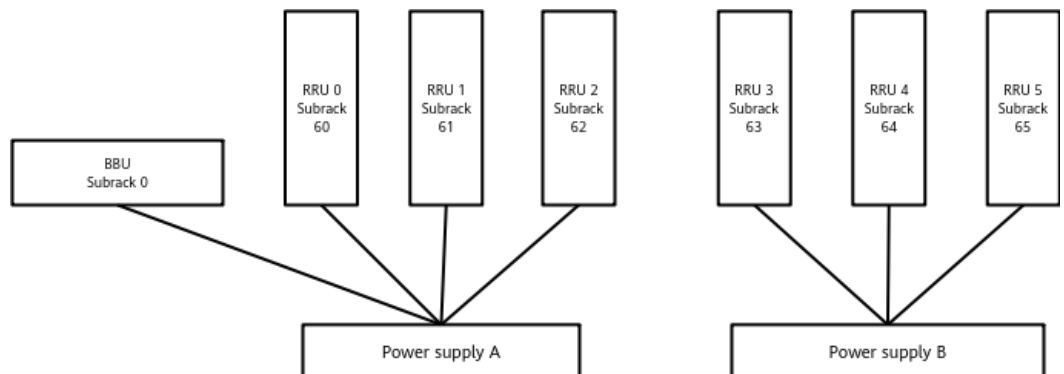
power alarms are reported if any. [Figure 13-1](#) illustrates a centralized power supply scenario.

Figure 13-1 Centralized power supply scenario



- Distributed power supply scenario (this function is not applicable)
A distributed power supply scenario is when main base station devices (including the BBU and AAUs/RRUs) use multiple power supply systems and backup power supply systems. If the mains supply of a power supply system fails, the corresponding backup power supply system provides power for the related main devices, but the power supply systems of other devices remain unchanged, and different power alarms may be reported for different devices. [Figure 13-2](#) illustrates a distributed power supply scenario.

Figure 13-2 Distributed power supply scenario



13.2 Network Analysis

13.2.1 Benefits

In scenarios where the backup power is insufficient and the power supply is unstable (for example, at solar-only rural sites during rainy seasons), the service time of the base stations can be prolonged. Due to the power configuration of each RAT, service fluctuations, and PV environment fluctuations, how long the service time of the base stations can be prolonged may fluctuate.

13.2.2 Impacts

The impacts between this function and other functions are described in [13.3.2.3 Function Impacts](#).

This function has the following network impacts:

LTE side:

This function has a negative impact on cell coverage. If carriers of a cell are shut down, the cell cannot provide services.

After carrier shutdown, UEs may be handed over to neighboring cells. As a result, the values of counters related to handovers (listed in [Table 13-1](#)) fluctuate.

Table 13-1 Counters related to handovers

Counter Description	Counter Name
Number of intra-eNodeB inter-frequency outgoing handover attempts in a cell	L.HHO.IntraeNB.InterFreq.PrepAttOut
Number of intra-eNodeB inter-frequency outgoing handover executions in a cell	L.HHO.IntraeNB.InterFreq.ExecAttOut
Number of successful intra-eNodeB inter-frequency outgoing handovers in a cell	L.HHO.IntraeNB.InterFreq.ExecSuccOut
Number of inter-eNodeB inter-frequency outgoing handover attempts in a cell	L.HHO.IntereNB.InterFreq.PrepAttOut
Number of inter-eNodeB inter-frequency outgoing handover executions in a cell	L.HHO.IntereNB.InterFreq.ExecAttOut
Number of successful inter-eNodeB inter-frequency outgoing handovers in a cell	L.HHO.IntereNB.InterFreq.ExecSuccOut
Number of intra-eNodeB intra-duplex-mode incoming handover attempts in a cell	L.HHO.IntraeNB.PrepAttIn
Number of intra-eNodeB intra-duplex-mode incoming handover executions in a cell	L.HHO.IntraeNB.ExecAttIn
Number of successful intra-eNodeB intra-duplex-mode incoming handovers in a cell	L.HHO.IntraeNB.ExecSuccIn

Counter Description	Counter Name
Number of inter-eNodeB intra-duplex-mode incoming handover attempts in a cell	L.HHO.IntereNB.PrepAttIn
Number of inter-eNodeB intra-duplex-mode incoming handover executions in a cell	L.HHO.IntereNB.ExecAttIn
Number of successful inter-eNodeB intra-duplex-mode incoming handovers in a cell	L.HHO.IntereNB.ExecSucIn

UMTS side:

After the function takes effect, some cells under the NodeB are shut down, diminishing the uplink and downlink power resources of the NodeB.

There will be no carrier signals in the coverage areas of the cells that have been shut down. If all cells are shut down, WCDMA signals are not available in the area covered by the NodeB.

The maximum CPU usage of the control-plane subsystem may experience an intermittent increase if multiple users access the carrier at the same time after the carrier is recovered from the shutdown state.

GSM side:

After this function takes effect, some carriers in a cell are shut down, reducing the site capacity.

Capacity reduction may cause network congestion. As a result, the access success rate and handover success rate will decrease, and services may even be interrupted.

13.3 Requirements

13.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
LOFD-210202	LTE SOC Level-Based Low Power Consumption Mode	LT1SLSLLPCM0	per Cell
TDLOFD-210612	LTE SOC Level-Based Low Power Consumption Mode	LT4S00LBLP00	per Cell

Feature ID	Feature Name	Model	Sales Unit
WRFD-270201	UMTS SOC Level-Based Low Power Consumption Mode	LQW9USLPCM01	per NodeB
GBFD-270201	GSM SOC Level-Based Low Power Consumption Mode	LGB3SOCLLPM	per BTS

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation.

13.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

13.3.2.1 Prerequisite Functions

None

13.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	LTE low power consumption mode	CellLowPower.LowPwrSwitch	<i>Green Site</i>	- LTE SOC level-based low power consumption mode (controlled by the SOCLVL_BASED_LOW_PWR_SW option of the CellLowPower.LowPwrConsumeModeExtSw parameter) cannot work with LTE low power consumption mode. - When the CellLowPower.BakPwrSavPolicy parameter is set to LOW_PWR_AT_SOCLVL 1, LOW_PWR_AT_SOCLVL 2, or LOW_PWR_AT_SOCLVL 3, LTE low power consumption mode cannot be enabled.
LTE FDD LTE TDD	High-speed cell sleeping	HSCellSleepInterfAvoidSwitch option of the HighSpdAdaption-Para.InterfAvoidMode parameter	<i>High Speed Mobility</i>	LTE SOC level-based low power consumption mode cannot work with high-speed cell sleeping.
LTE FDD LTE TDD	Massive MIMO assistance cell	Cell.MultiRruCellFlag set to BOOLEAN_TRUE Cell.MultiRruCellMode set to ASSISTED_MASSIVE_MIMO_CELL	None	LTE SOC level-based low power consumption mode cannot be enabled in massive MIMO assistance cells.
LTE FDD LTE TDD	Base-station-level low power consumption mode	LowPwrConsumptionSw option of the ENodeBAlgoSwitch.PowerSaveSwitch parameter	None	LTE SOC level-based low power consumption mode cannot work with base-station-level low power consumption mode.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Zero Guard Band Between Contiguous Intra-Band Carriers	CONTIG_INTRA_BAND_CARR_SW option of the ContigIntraBand-Carr.ContigIntraBandCarrSw parameter	<i>Seamless Intra-Band Carrier Joining (FDD)</i> <i>Seamless Intra-Band Carrier Joining (TDD)</i>	LTE SOC level-based low power consumption mode cannot work with Zero Guard Band Between Contiguous Intra-Band Carriers.
NB-IoT	NB-IoT cell	Cell.NbCellFlag set to TRUE	<i>NB-IoT Basics (FDD)</i>	LTE SOC level-based low power consumption mode is not supported by NB-IoT cells.
NB-IoT	RB Reserve Mode	CellRbReserve.RbRsrvMode set to NB_DEPLOYMENT	None	RB reservation for NB-IoT deployment cannot work with LTE SOC level-based low power consumption mode.
LTE TDD	Beam Deployment Mode	SectorSplitGroup.BeamDeploymentMode	None	For an 8T8R cell, if the SectorSplitGroup.BeamDeploymentMode parameter is set to NOT_CFG or MODE_3 , LTE SOC level-based low power consumption mode cannot be enabled.
LTE TDD	LTE TDD and NR Dynamic Spectrum Sharing	SpectrumCloud.SpectrumCloudSwitch set to LTE_NR_SPECTRUM_SHR	None	LTE SOC level-based low power consumption mode cannot work with LTE TDD and NR Dynamic Spectrum Sharing.
LTE TDD	LTE and NR coordinated spectrum sharing	SpectrumCloud.SpectrumCloudSwitch set to LTE_NR_COORD_SPECTRUM_SHR	None	LTE SOC level-based low power consumption mode cannot work with LTE and NR coordinated spectrum sharing.

RAT	Function Name	Function Switch	Reference	Description
UMTS	UMTS low power consumption mode	NODEBALGPARA.LOWPWRCONSUMPMODESW	<i>Green BTS</i>	If the SOCVLV_BASED_LOW_PWR_SW option of the NODEBALGPARA.LOWPWRCONSUMEMODEEXTSW parameter is selected, UMTS low power consumption mode cannot be enabled.
UMTS	Multi-Carrier Switch off Based on Power Backup	NODEBPOWEROUTAGE.ISDSW	<i>Green BTS</i>	If the SOCVLV_BASED_LOW_PWR_SW option of the NODEBALGPARA.LOWPWRCONSUMEMODEEXTSW parameter is selected, Multi-Carrier Switch off Based on Power Backup cannot be enabled.
UMTS	Mixed multi-RRU cells	ULOCCELL.LOCELITYPE set to MIXED_MULTIRRU_CELL	None	UMTS SOC level-based low power consumption mode cannot be enabled in mixed multi-RRU cells.
GSM	GSM low power consumption mode	GBTSENERGYMGTPARA.LOWPWRCONSUMPMODESW	<i>Intelligent Power Consumption Decrease</i>	If the GBTSENERGYMGTPARA.SOCVLVLOWPWARMODESW parameter is set to ON , GSM low power consumption mode cannot be enabled.
NR LTE FDD LTE TDD UMTS GSM	iLiBattScan	BATCTPA.ILIBATTSCANSW (5G gNodeB, LTE eNodeB)	<i>Power Supply Management</i>	None

13.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	NR low power consumption mode	LOW_PWR_MODE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Green Site</i>	NR low power consumption mode does not take effect as long as the IPWRM.SOCLVLADJSW parameter is set to ON .
LTE FDD LTE TDD	LTE low power consumption mode	CellLowPower.LowPwrSwitch	<i>Green Site</i>	LTE low power consumption mode does not take effect as long as the IPWRM.SOCLVLADJSW parameter is set to ON .
LTE FDD LTE TDD	Intelligent power-off of carriers in the same coverage	CellShutdown.CellShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	- When a capacity cell in the intelligent power-off of carriers in the same coverage state meets the conditions for entering SOC level-based low power consumption mode, the cell enters SOC level-based low power consumption mode only after it exits the intelligent power-off of carriers in the same coverage state and the carrier is turned on again. - If any co-coverage neighboring basic cell of a capacity cell is in SOC level-based low power consumption mode, the capacity cell does not enter the intelligent power-off of carriers in the same coverage state.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Dynamic carrier shutdown	CARRIER_DYN_SHUTDOWN_SW option of the CellShutdown.CarrierShutdownEnhancedSw parameter	<i>Energy Conservation and Emission Reduction</i>	When the local cell in the dynamic carrier shutdown state meets the conditions for entering SOC level-based low power consumption mode, the cell exits the dynamic carrier shutdown state and enters SOC level-based low power consumption mode.
LTE FDD LTE TDD	RF channel intelligent shutdown	CellRfShutdown.RfShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	A cell in SOC level-based low power consumption mode cannot enter the RF channel intelligent shutdown state.
LTE FDD LTE TDD	RF channel dynamic muting	RF_CHN_DYN_MUTING_SW option of the CellRfChnDynMuting.RfChnDynMutingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	RF channel dynamic muting does not take effect when SOC level-based low power consumption mode is in effect.
LTE FDD	RF channel dynamic muting in TM9	TM9_TTI_CHN_DYN_MUTING_SW option of the CellRfChnDynMuting.RfChnDynMutingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	RF channel dynamic muting in TM9 does not take effect when SOC level-based low power consumption mode is in effect.
LTE FDD LTE TDD	CRS dynamic muting	CRS_DYNAMIC_MUTING_SW option of the SymbolPwrSavings.SymbolShutdownEnhancedSw parameter	<i>Energy Conservation and Emission Reduction</i>	CRS dynamic muting does not take effect when SOC level-based low power consumption mode is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Dynamic voltage adjustment	CellAlgoSwitch.DynAdjVoltSwitch	<i>Energy Conservation and Emission Reduction</i>	During the transitional period of SOC level-based low power consumption mode, dynamic voltage adjustment can still take effect. When the transitional period elapses and the eNodeB takes further actions to lower the power consumption, dynamic voltage adjustment no longer takes effect.
LTE FDD LTE TDD	MIMO	Cell.TxRxMode	<i>MIMO</i>	When an eNodeB in SOC level-based low power consumption mode shuts down some transmit channels, the eNodeB cannot work in MIMO mode.
LTE FDD	Downlink cell coverage enhancement	CellAlgoSwitch.CellDLCoverEnhanceSwitch	None	Downlink cell coverage enhancement does not take effect when an eNodeB enters SOC level-based low power consumption mode.
LTE FDD LTE TDD	Dynamic Power Sharing Between LTE Carriers	CellDynPowerSharing.DynamicPowerSharingSwitch	<i>Dynamic Power Sharing Between LTE Carriers</i>	Power sharing does not work for cells in SOC level-based low power consumption mode.
LTE FDD LTE TDD	Smart carrier selection based on virtual grids	SMART_CARRIER_SELECTION_SW option of the MultiCarrUnifiedSch.MultiCarrierUnifiedSchSw parameter	<i>Multi-carrier Unified Scheduling</i>	Entering SOC level-based low power consumption mode affects the progress of virtual grid model building.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Downlink 2CC aggregation	CaDL2CCExtSwitch option of the CaMgtCfg.CellCaAlgoSwitch parameter	<i>Carrier Aggregation</i>	- Cells that are shut down due to SOC level-based low power consumption mode do not participate in CA. - In CA-group-based configuration mode, after such a cell in a CA group exits this power saving state, this cell cannot be configured as an SCell for an existing CA UE until the UE accesses the network again.
LTE TDD	Inter-cell downlink D-MIMO	InterCellDmimoJTSwitch option of the CellAlgoSwitch.DMIMOAlgoSwitch parameter	<i>D-MIMO (TDD)</i>	A cell within an inter-cell D-MIMO cluster cannot serve as a coordinating cell for joint transmission (JT) and inter-RRU channel calibration may fail when the cell enters SOC level-based low power consumption mode. As a result, the inter-cell D-MIMO cluster is disabled.
LTE FDD	CRS port mapping detection and reconfiguration	AutoRecfgSwitch parameter in the STR CRSPORTOPTDET command	<i>MIMO</i>	- CRS port mapping detection and reconfiguration are not supported in SOC level-based low power consumption mode, and the detection result indicates that the scenario is not supported. - If the triggering conditions for SOC level-based low power consumption mode are met during CRS port mapping detection, this mode takes effect after the detection is complete.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Automatic cell shutdown	CellAutoShutdown.CellAutoShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	When both SOC level-based low power consumption mode and automatic cell shutdown meet their triggering conditions, automatic cell shutdown will take effect but SOC level-based low power consumption mode will not.
LTE FDD	Channel calibration	Joint channel calibration: RruJointCalParaCfg.TxChnCalSwitch Downlink-only module channel calibration: RruJointCalParaCfg.AauPassivePortCalibPeriod	<i>MIMO</i>	If channel calibration is being performed when PAs are shut down after SOC level-based low power consumption mode takes effect, channel calibration fails. You can run the DSP CELLCALIBRATION command to check whether channel calibration is successful.
LTE FDD	UL coordinated channel shutdown	CellRfShutdown.MultiRatJointChnShutdownSw	<i>Multi-RAT Coordinated Energy Saving</i>	A cell in the UL coordinated channel shutdown state cannot enter SOC level-based low power consumption mode.
LTE FDD LTE TDD	RF module deep dormancy	RRU.DORMANCY SW (LTE eNodeB, 5G gNodeB) set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	RF module deep dormancy takes precedence over SOC level-based low power consumption mode when both functions are enabled. SOC level-based low power consumption mode does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	RF module super dormancy	RRU.DORMANCY SW set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	RF module super dormancy takes precedence over SOC level-based low power consumption mode and SOC level-based low power consumption mode does not take effect when both functions are enabled.
LTE FDD LTE TDD	Cell Switch-off Based on Automatic Co-coverage Identification (LTE)	CellShutdown.CellShutdownSwitch set to ON_CO_COV_AUTTO_IDENT	<i>Energy Conservation and Emission Reduction</i>	When the local cell in the Cell Switch-off Based on Automatic Co-coverage Identification state meets the conditions for entering SOC level-based low power consumption mode, the cell enters SOC level-based low power consumption mode only after exiting the Cell Switch-off Based on Automatic Co-coverage Identification state.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Multi-carrier coordinated energy saving	CellShutdown.CeIlShutdownSwitch set to ON_MULTI_CARRIER_HIER_SHUTDOWN or ON_MULTICARRIER_MANUAL_SHUTDOWN	<i>Energy Conservation and Emission Reduction</i>	- When the local cell in the multi-carrier coordinated energy saving state meets the conditions for entering SOC level-based low power consumption mode, the cell enters SOC level-based low power consumption mode only after it exits the multi-carrier coordinated energy saving state and the carrier is turned on again. - When all co-coverage neighboring basic cells of the local cell are in SOC level-based low power consumption mode, the local cell does not enter the multi-carrier coordinated energy saving state.
LTE FDD	Centralized power sharing	LTE_CENTRAL_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>Smart 8T8R (FDD)</i>	Cells that have entered the energy saving state cannot participate in power sharing. These cells will attempt to participate in power sharing every 10 minutes.
LTE FDD	Downlink intelligent joint power control	DL_INTEL_JOINT_PC_SW option of the CellDLSchExtParam.DLSchExtSwitch parameter	<i>Smart 8T8R (FDD)</i>	Downlink intelligent joint power control does not take effect in a cell that has entered SOC level-based low power consumption mode.
LTE FDD	Downlink intelligent joint AMC and power control	DL_INTEL_JOINT_AMC_AND_PC_SW option of the CellDLSchExtParam.DLSchExtSwitch parameter	<i>Smart 8T8R (FDD)</i>	Downlink intelligent joint AMC and power control does not take effect in a cell that has entered SOC level-based low power consumption mode.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Enhanced extreme power sharing	LTE_ENH_EXT_POWER_SHARING_SW option of the CellDynPowerSharing.DynPowerSharingSwitch parameter	<i>Smart 8T8R (FDD)</i>	Cells that have entered the energy saving state cannot participate in power sharing. These cells will attempt to participate in power sharing every 10 minutes.
LTE FDD	Hybrid power sharing	LTE_HYBRID_POWER_SHARING_SW option of the CellDynPowerSharing.DynPowerSharingSwitch parameter	<i>Smart 8T8R (FDD)</i>	Cells that have entered the energy saving state cannot participate in power sharing. These cells will attempt to participate in power sharing every 10 minutes.
LTE FDD	Channel isolation for PIM interference avoidance	PIM_INTERF_CHN_SHUTDOWN_SW option of the PimInterfSmartAvoid.PimAlgoSwitch parameter	<i>Interference Detection and Suppression</i>	When SOC level-based low power consumption mode takes effect, channel isolation for PIM interference avoidance is terminated.
LTE FDD	Intelligent PIM interference avoidance	PIM_INTERF_SMART_AVOID_SW option of the PimInterfSmartAvoid.PimAlgoSwitch parameter	<i>Interference Detection and Suppression</i>	SOC level-based low power consumption mode is terminated during PIM interference detection.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	PIM interference cancellation	PIM_INTERF_CANCELLATION_SW option of the PimInterfSmartAvoid.PimAlgoSwitch parameter	<i>Interference Detection and Suppression</i>	After SOC level-based low power consumption mode takes effect in an interfering cell or a victim cell, the eNodeB re-evaluates whether to continue the implementation of PIM interference cancellation in each cell. • If PIM interference persists after SOC level-based low power consumption mode takes effect, PIM interference cancellation continues. • If PIM interference disappears after SOC level-based low power consumption mode takes effect, PIM interference cancellation is stopped to reduce resource consumption caused by periodic PIM interference characteristic training and interference characteristic extraction.
LTE FDD LTE TDD	Intelligent power control energy saving	INTEL_INTRA_CELL_PC_SW option of the CellPwrSavingAlgorithm.PwrSavingAlgorithmSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Intelligent power control energy saving does not take effect in a cell that has entered SOC level-based low power consumption mode.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Dynamic carrier shutdown phase 2	CellPwrSavingAlgo.CarrierDynShutdownPh2Sw set to CAPACITY_CELL	<i>Energy Conservation and Emission Reduction</i>	When the local cell in the dynamic carrier shutdown phase 2 state meets the conditions for entering SOC level-based low power consumption mode, the cell exits the dynamic carrier shutdown phase 2 state and enters SOC level-based low power consumption mode.
LTE TDD	Clock synchronization detection (TDD)	ClkDetect.ClkAsyncDetectSwitch	<i>Synchronization</i>	Clock out-of-synchronization may fail to be detected in scenarios where carriers are shut down in SOC level-based low power consumption mode.
LTE FDD	LTE FDD and NR Flash Dynamic Power Sharing	LTE_NR_DYN_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	Cells that have entered the energy saving state cannot participate in LTE FDD and NR Flash Dynamic Power Sharing. These cells will attempt to participate in power sharing every 10 minutes.
LTE FDD	Downlink coordinated power control	DL_COORD_PC_SW option of the CellMimoParaCfg.MimoSwitch parameter	<i>Low-Band Booster</i>	Downlink coordinated power control does not take effect in a cell that has entered SOC level-based low power consumption mode.
LTE FDD	Massive MIMO downlink coordinated power control	MM_DL_COORD_PC_SW option of the SectorSplitGroup.IntelligentBeamschSwitch parameter	<i>Massive MIMO Enhancements (FDD)</i>	Massive MIMO downlink coordinated power control does not take effect in a cell that has entered SOC level-based low power consumption mode.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Optimization-based scheduling	TIME_FREQ_2D_UNIF_OPT_SCH_SW and TIME_FREQ_UNIF_OPT_SCH_ENH_SW options of the CellMultiCarrUniSch.CellMultiCarrUnifiedSchSw parameter	<i>Multi-carrier Unified Scheduling</i>	When SOC level-based low power consumption mode is enabled, the accuracy of calculating the resources available to the relevant cell is compromised. As a result, the benefits of optimization-based scheduling are affected.
LTE TDD	Dynamic cell-level threshold adjustment for switching between non-antenna-selection dual-stream beamforming and TM9 hybrid precoding	NONASBF_AND_TM9HY_THLD_ADP_SW option of the CellBfMimoParaCfg.BfMimoAlgoOptSwitch parameter	<i>TMA (TDD)</i>	SOC level-based low power consumption mode decreases the probability of activating dynamic cell-level threshold adjustment for switching between non-antenna-selection dual-stream beamforming and TM9 hybrid precoding, while the dynamic cell-level threshold adjustment increases the probability of entering SOC level-based low power consumption mode.
LTE TDD	Intra-frequency split	Cell.MultiCellsShareMode set to SPLIT SsrdCellGroup.PrimaryLocalCellID	<i>Soft Split Resource Duplex (TDD)</i>	SOC level-based low power consumption mode does not take effect in intra-frequency split cells.
LTE FDD LTE TDD	Inter-eNodeB SFN based on eNodeB coordination	Cell.MultiRruCellMode	<i>SFN</i>	SOC level-based low power consumption mode does not take effect in inter-eNodeB SFN cells based on eNodeB coordination.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD UMTS GSM	Intelligent shutdown of carriers due to PSU failure	- LTE/UMTS/GSM eGBTS: EQUIPMENT.PSUFP (LTE eNodeB, 5G gNodeB) - GSM GBTS: BTSPSUFP.CFG FLAG	<i>Base Station Equipment Reliability</i>	Intelligent shutdown of carriers due to PSU failure does not take effect as long as the IPWRM.SOCLVLADJSW parameter is set to ON .
LTE FDD UMTS	UMTS and LTE Dynamic Power Sharing	- UMTS: ULOCCELL.ULPOWERSHARE - LTE: UMTS_LTE_DYNAMIC_POWER_SHARING_SW and LTE_DYN_POWER_SHARING_SW options of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>UMTS and LTE Dynamic Power Sharing</i>	When SOC level-based low power consumption mode is in effect, UMTS and LTE Dynamic Power Sharing does not take effect.
UMTS	UMTS low power consumption mode	NODEBALGPARA.LOWPWRCONSUMPMODESW	<i>Green BTS</i>	UMTS low power consumption mode does not take effect as long as the IPWRM.SOCLVLADJSW parameter is set to ON .
UMTS	Coverage Compensation for Site Outage	UNBMPPHYPARA.CovCompensationDataColSw UNODEBALGPARA.SiteOutageCovCompensationSw	<i>Coverage Compensation for Site Outage</i>	Coverage Compensation for Site Outage does not take effect as long as the IPWRM.SOCLVLADJSW parameter is set to ON .

RAT	Function Name	Function Switch	Reference	Description
UMTS	Multi-Carrier Switch off Based on Power Backup	NODEBPOWEROUTAGE.ISDSW	<i>Green BTS</i>	Multi-Carrier Switch off Based on Power Backup does not take effect as long as the IPWRM.SOCLVLADJSW parameter is set to ON .
UMTS	Narrowband Interference Suppression	ULOCCELL.NBIS	<i>Narrowband Interference Suppression</i>	Entering SOC level-based low power consumption mode affects the accuracy of the detection of narrowband interference based on the RTWP in a cell.
UMTS GSM	GSM and UMTS Intelligent Shutdown Based on RAT Priority	- eGBTS: GBTSENERGYMGTPARA.PWROFFPROTECTSTARTTIME - BSC: BTSBAKPWR.PWROFFPROTECTSTARTTIME	<i>Intelligent Power Consumption Decrease</i>	GSM and UMTS Intelligent Shutdown Based on RAT Priority does not take effect as long as the IPWRM.SOCLVLADJSW parameter is set to ON .
GSM	Active Backup Power Control	- eGBTS: GBTSENERGYMGTPARA.BAKPWRSVMMETHOD set to BYPWR - BSC: BTSBAKPWR.BAKPWRSVMMETHOD set to BYPWR	<i>Intelligent Power Consumption Decrease</i>	Active Backup Power Control does not take effect as long as the IPWRM.SOCLVLADJSW parameter is set to ON .
GSM	Shutdown of TRXs for Batteries	- eGBTS: GBTSENERGYMGTPARA.BAKPWRSVMMETHOD set to BYPWR - BSC: BTSBAKPWR.BAKPWRSVMMETHOD set to BYPWR	<i>Power Supply Management</i>	Shutdown of TRXs for Batteries does not take effect as long as the IPWRM.SOCLVLADJSW parameter is set to ON .

RAT	Function Name	Function Switch	Reference	Description
GSM	GSM low power consumption mode	GBTSENERGYMGTPARA.LOWPWRCONSUMPMODESW	<i>Intelligent Power Consumption Decrease</i>	GSM low power consumption mode does not take effect as long as the IPWRM.SOCLVLADJSW parameter is set to ON .
GSM	TRX Power Amplifier Intelligent Shutdown	• BSCDSTPA.BSCDYN SWITCHTRXPAALLOW • GCELLBASICPARA.DYNOpenTrxPower • GTRXDEV.CPS	<i>Intelligent Power Consumption Decrease</i>	The TRXs that are shut down in SOC level-based low power consumption mode cannot be activated through TRX Power Amplifier Intelligent Shutdown even if the network load increases.
GSM	Dynamic Cell Power Off	GCELLDYNTURN OFF.TURNOFFENABLE	<i>Dynamic Cell Power Off</i>	The TRXs that are shut down in SOC level-based low power consumption mode cannot be activated through Dynamic Cell Power Off even if the network load increases.
LTE FDD	LTE-NR multi-dimensional scheduling optimization	LTE: LNR_MULTI_D_OPT_SCH_SW option of the GigaBandCell.GigaBandCellAlgoSw parameter NR: LNR_MULTI_D_OPT_SCH_SW option of the NRDUCellGigaBand.GigaBandCellAlgoSw parameter	<i>LTE FDD and NR Inter-RAT Unified Scheduling</i>	When SOC level-based low power consumption mode is enabled, the accuracy of calculating the resources available to the relevant cell is compromised. As a result, the benefits of LTE-NR multi-dimensional scheduling optimization are affected.
LTE FDD	LNR CRS muting	LNR_CRS_MUTE_SW option of the CellAlgoExtSwitch.LnrZeroBufferzoneSw parameter	<i>LTE and NR Zero Buffer zone</i>	LNR CRS muting does not take effect in a cell that has entered SOC level-based low power consumption mode.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	CMAS	CellAlgoSwitch.CmasSwitch	<i>CMAS</i>	The following applies when SOC level-based low power consumption mode is in effect: • When PWS message-triggered exit of cells from energy saving is disabled (by deselecting the PWS_RESTART_IND_ENH_SW option of the EnodebAlgoExtSwitch.PwsIndicationSwitch parameter), CMAS warning notifications cannot be properly sent in LTE cells since carriers in those cells are shut down. However, UEs can still receive CMAS warning notifications from other carriers. • With PWS message-triggered exit of cells from energy saving enabled, cells exit the energy saving state upon receipt of a PWS message.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	ETWS	None	<i>ETWS</i>	The following applies when SOC level-based low power consumption mode is in effect: - When PWS message-triggered exit of cells from energy saving is disabled (by deselecting the PWS_RESTART_IND_ENH_SW option of the EnodebAlgoExtSwitch.PwsIndicationSwitch parameter), ETWS warning notifications cannot be sent in LTE cells since carriers in those cells are shut down. However, UEs can still receive ETWS warning notifications from other carriers. - With PWS message-triggered exit of cells from energy saving enabled, cells exit the energy saving state upon receipt of a PWS message.
UMTS	NCR cells	ULOCCELL.NCRCELL set to TRUE	<i>RuralC ow Soluti on</i>	The ULOCCELL.RSV parameter must be set to 1 for NCR cells, which serve as reserved cells where carriers are not deactivated.

RAT	Function Name	Function Switch	Reference	Description
UMTS	RF module deep dormancy	RRU.DORMANCY <i>SW</i> set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	RF module deep dormancy takes precedence over UMTS SOC level-based low power consumption mode when both functions are enabled. UMTS SOC level-based low power consumption mode does not take effect when RF module deep dormancy is in effect.
LTE FDD LTE TDD UMTS	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATS <i>W</i> parameter	<i>Multi-RAT Coordinated Energy Saving</i>	RF module fast deep dormancy takes precedence over SOC level-based low power consumption mode when both functions are enabled. SOC level-based low power consumption mode does not take effect when RF module fast deep dormancy is in effect.
UMTS	RF module super dormancy	RRU.DORMANCY <i>SW</i> set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	RF module super dormancy takes precedence over UMTS SOC level-based low power consumption mode when both functions are enabled. UMTS SOC level-based low power consumption mode does not take effect when RF module super dormancy is in effect.
GSM	RF module super dormancy	RRU.DORMANCY <i>SW</i> set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	The RF modules whose carriers are shut down cannot be woken up from the super dormancy state.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Port sparseness	PORT_SPARSENESS_SW option of the CellPwrSavingAlgorithm.PwrSavingAlgorithmSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	SOC level-based low power consumption mode takes precedence over port sparseness when both functions are enabled.
UMTS	RuralLink antenna solution (UMTS)	• SECTOR.ANTNUM set to 0 • SECTOREQM.ANTCFGMODE set to BEAM • ULOCCELL.LOCELLTYPE set to NORMAL_CELL • BASEBANDEQM.UMTSDEMMODE set to DEM_4_CHANNEL or DEM_ADPT_CHANNEL • ULOCCELL.BFMODE set to FOURTXBEAMOPT • ULOCCELL.FOURTXDEPLOYMENTMODE set to MODE_2 • BEAMCELLGROUP.BEAMCELLGROUPTYPE set to FOURTXBEAMOPT	<i>Green Site</i>	In UMTS three-sector split cell scenarios, SOC level-based low power consumption mode can shut down the carriers of the three cells in a split group only if the cells are all non-reserved cells.
UMTS	UMTS Static Split Beam	ULOCCELL.BFMODE	<i>Green BTS</i>	In low-frequency massive MIMO scenarios, SOC level-based low power consumption mode can shut down all the carriers of the two cells in a split group only if the cells are both non-reserved cells.

13.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

Boards

No requirements

RF Modules

No requirements

Power Supply

Power cabinets: Only Huawei power cabinets that can be managed by a PMU are supported. Third-party power cabinets are not supported.

Backup power: Before enabling this function, ensure that backup power is available at the site and the backup battery type is lithium battery.

Others:

- A maximum of one power monitoring module can be configured under a node.
- If a site monitoring unit (SMU) or PMU is used, it must be connected to the BBU through an RS485 cable.

13.3.4 Cells

- The **CellLowPower.BakPwrSavPolicy** parameter must be set to the same value for all cells in the same sector split group.
- For LTE TDD, the TX/RX mode of cells cannot be 32T32R or 64T64R.
- For LTE FDD, this function is not supported when the TX/RX mode of cells is not 32T32R or 16T16R and the **SECTOREQM.ANTCFGMODE** parameter is set to **BEAM**.

13.3.5 Others

When this function is enabled, it is only possible for a power monitoring unit to report to the MAE through BBU in-band networking.

13.4 Operation and Maintenance

13.4.1 Data Configuration

13.4.1.1 Data Preparation

Table 13-2 describes the parameters used for function activation.

Table 13-2 Parameters used for activation

RAT	Parameter Name	Parameter ID	Setting Notes
LTE FDD LTE TDD UMTS GSM	Power Supply Group ID	RRU.PSGID	Set this parameter to 0 if SOC level-based low power consumption mode is required.
LTE FDD LTE TDD UMTS GSM	SOC Level Adjust Switch	IPWRM.SOCLVLADJSW	Set this parameter to ON if SOC level-based low power consumption mode is required.
LTE FDD LTE TDD UMTS GSM	SOC Level 1 Upper Threshold	IPWRM.SOCLVL1UTHD	Set this parameter based on the network plan. IPWRM.SOCLVL1UTHD must be set to a value greater than or equal to the IPWRM.SOCLVL2UTHD parameter value, and IPWRM.SOCLVL2UTHD must be set to a value greater than or equal to the IPWRM.SOCLVL3UTHD parameter value.
LTE FDD LTE TDD UMTS GSM	SOC Level 2 Upper Threshold	IPWRM.SOCLVL2UTHD	
LTE FDD LTE TDD UMTS GSM	SOC Level 3 Upper Threshold	IPWRM.SOCLVL3UTHD	

RAT	Parameter Name	Parameter ID	Setting Notes
LTE FDD LTE TDD	Low Power Consumption Mode Extension Sw	CellLowPower.LowPwrConsumptionModeExtSw	Select the SOCLVL_BASED_LOW_PWR_SW option of this parameter if LTE SOC level-based low power consumption mode is required and IPWRM.SOCLVLADISW is set to ON .
LTE FDD LTE TDD	Back power save policy	CellLowPower.BakPwrSavPolicy	Set this parameter based on the network plan. This parameter must be set to the same value for all split cells in which SOC level-based low power consumption mode is enabled in a split group.
UMTS	Low Power Consumption Mode Extension Switch	NODEBALGPARA.LOWPWRCONSUMEMODEEXTSW on the UMTS side	Select the SOCLVL_BASED_LOW_PWR_SW option of this parameter if UMTS SOC level-based low power consumption mode is required and IPWRM.SOCLVLADISW is set to ON .
UMTS	Backup Power Saving Policy	NODEBALGPARA.BAKPWRSAPOLICY	Set this parameter based on the network plan.
GSM	SOC Level-based Low Power Consumption Mode Switch	GBTSENERGYMGTPARA.SOCLVLLOWPWRRMODESW	Set this parameter to ON if GSM SOC level-based low power consumption mode is required and IPWRM.SOCLVLADISW is set to ON .
GSM	SOC Level-based Power Saving Policy	GBTSENERGYMGTPARA.SOCLVLPWRSAPOLICY	Set this parameter based on the network plan.

13.4.1.2 Using MML Commands

Before using MML commands, refer to [13.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//(Optional) Configuring a monitoring unit and batteries by running the ADD PMU and ADD BATTERY
commands, respectively, if a Huawei digital power system is used
ADD CABINET: CN=0, TYPE=VIRTUAL;
ADD PMU: CN=0, SRN=7, PTYPE=ETP, MPN=0;
ADD BATTERY: CN=0, SRN=7, BTYPE=LI_BAT;
//(Optional) Configuring a power supply group ID
MOD RRU: CN=0, SRN=60, SN=0, PSGID=0;
//Turning on the SOC level adjustment switch and configuring the upper thresholds of SOC levels
SET IPWRM: SOCLVLADJSW=ON, SOCLVL1UTHD=40, SOCLVL2UTHD=30, SOCLVL3UTHD=20;
//Enabling LTE SOC level-based low power consumption mode
MOD CELLLOWPOWER: LocalCellId=0, LowPwrConsumeModeExtSw=SOCLVL_BASED_LOW_PWR_SW-1,
BakPwrSavPolicy=LOW_PWR_AT_SOCLVL1;
//Configuring reserved and non-reserved UMTS cells
MOD ULOCELL: ULOCELLID=0, LOCELLTYPE=NORMAL_CELL, RSV=FALSE;
MOD ULOCELL: ULOCELLID=1, LOCELLTYPE=NORMAL_CELL, RSV=TRUE;
//Enabling UMTS SOC level-based low power consumption mode
SET NODEBALGPARA: LOWPWRCONSUMEMODEEXTSW=SOCLVL_BASED_LOW_PWR_SW-1,
BAKPWRSAVPOLICY=LOW_PWR_AT_SOCLVL2;
//On the BSC side, configuring carriers that can be shut down
SET GTRXDEV: IDTYPE=BYID, TRXID=0, SDFLAG=ENABLE;
//On the BSC side, configuring whether to allow BCCH TRX shutdown
SET BTSLOCKBCCH: IDTYPE=BYID, BTSID=0, BTSPowerOFFLOCKBCCH=YES;
//Enabling GSM SOC level-based low power consumption mode
SET GBTSENERGYMGTPARA: SOCLVLLOWPWRMODESW=ON,
SOCLVLPWRSAPOLICY=LOW_PWR_AT_SOCLVL3;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling GSM SOC level-based low power consumption mode
SET GBTSENERGYMGTPARA: SOCLVLLOWPWRMODESW=OFF;
//Disabling UMTS SOC level-based low power consumption mode
SET NODEBALGPARA: LOWPWRCONSUMEMODEEXTSW=SOCLVL_BASED_LOW_PWR_SW-0;
//Disabling LTE SOC level-based low power consumption mode
MOD CELLLOWPOWER: LocalCellId=0, LowPwrConsumeModeExtSw=SOCLVL_BASED_LOW_PWR_SW-0;
//Turning off the SOC level adjustment switch
SET IPWRM: SOCLVLADJSW=OFF;
```

13.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

13.4.2 Activation Verification

After SOC level-based low power consumption mode is enabled, perform the following operations to check whether it has taken effect:

LTE SOC level-based low power consumption mode

- Step 1** Run the **LST IPWRM** command. If the value of **SOC Level Adjust Switch** is **ON** in the command output, the SOC level adjustment switch has been turned on.
- Step 2** Run the **LST CELLLOWPOWER** command. If the value of **Low Power Consumption Mode Extension Sw** is **SOCLVL_BASED_LOW_PWR_SW:On** and that of **Back power save policy** is **LOW_PWR_AT_SOCLVLX** in the command output, LTE SOC level-based low power consumption mode has been enabled.
- Step 3** Perform the following operations to check whether LTE SOC level-based low power consumption mode has taken effect:
- Run the **DSP CELL** command. If the value of **Cell power save state** is **Low Power Consumption Mode-Cell Shutdown** in the command output, LTE SOC level-based low power consumption mode has taken effect.
 - Run the **DSP CELLLOWPOWER** command. If the value of **Cell Shutdown Status** is **ENABLED** in the command output, LTE SOC level-based low power consumption mode has taken effect.

----End

UMTS SOC level-based low power consumption mode

- Step 1** Run the **LST IPWRM** command. If the value of **SOC Level Adjust Switch** is **ON** in the command output, the SOC level adjustment switch has been turned on.
- Step 2** Run the **LST NODEBALGPARA** command. If the value of **Low Power Consumption Mode Extension Switch** is **SOC Level-Based Low Power Consumption Mode Switch:ON** and that of **Backup Power Saving Policy** is **Low Power Consumption At SOC Level X** in the command output, UMTS SOC level-based low power consumption mode has been enabled.
- Step 3** Run the **DSP ULOCELLPWRSAV** command. If the value of **SOC Level-Based Low Power Consumption Mode Power Saving State** is **Enable** in the command output, UMTS SOC level-based low power consumption mode has taken effect.

----End

GSM SOC level-based low power consumption mode

- Step 1** Run the **LST IPWRM** command. If the value of **SOC Level Adjust Switch** is **ON** in the command output, the SOC level adjustment switch has been turned on.
- Step 2** Run the **LST GBTSENERGYMGTPARA** command. If the value of **SOC Level-based Low Power Consumption Mode Switch** is **On** and that of **SOC Level-based Power Saving Policy** is **Low Power Consumption At SOC Level X** in the command output, GSM SOC level-based low power consumption mode has been enabled.

Step 3 Five minutes after GSM SOC level-based low power consumption mode is enabled, check whether the corresponding event is reported or check the corresponding performance counter.

- EVT-26290 Backup Power Saving is reported on the eGBTS side and the value of **Power Saving Status** is **Enter Common Power Saving Shutdown TRX Status**, indicating that GSM SOC level-based low power consumption mode has taken effect.
- If the value of the BSC performance counter **TRX.PWR.ADJUST.OFF.DUR** is greater than 0, GSM SOC level-based low power consumption mode has taken effect.

----End

You can run the **DSP BACKUPPWRSAVINGLVL** command to query the backup power saving level and reason of a base station. The values of **Base Station Backup Power Saving Level** include **MINOR**, **MAJOR**, and **CRITICAL**, which correspond to the SOC levels 1, 2, and 3, respectively. If SOC level-based low power consumption mode takes effect, the value of **Base Station Backup Power Saving Reason** is **SOCLEVEL_CHANGE:YES**.

13.4.3 Network Monitoring

- RF module power consumption
It is indicated by the **VS.EnergyCons.BTSBoard** counter on the MAE-Access. Operators can evaluate the performance of SOC level-based low power consumption mode by comparing the power consumed by an RF module before and after this function is enabled. The decrease in RF module power consumption indicates how much energy is saved.
- Main equipment power consumption
It is indicated by the **VS.EnergyCons.BTS.Adding.Service.GSM**, **VS.EnergyCons.BTS.Adding.Service.UMTS**, **VS.EnergyCons.BTS.Adding.Service.LTE**, **VS.EnergyCons.BTS.Adding.GSM**, **VS.EnergyCons.BTS.Adding.UMTS**, and **VS.EnergyCons.BTS.Adding.LTE** counters on the MAE-Access. By monitoring the power consumption of the main equipment before and after the function takes effect, you can obtain the increase in the backup power duration.

14 Base Station Hierarchical Power-Off

14.1 Principles

When mains supply is disconnected, a base station can be powered by storage batteries if it is configured with a power backup system. If base station energy consumption is high, the power of backup batteries may not be sufficient enough for the base station to provide services for a target duration.

Against this backdrop, base station hierarchical power-off is introduced. Users can set different load power-off voltage or time values to enable the hierarchical power-off of RF modules when the mains supply is faulty. This extends storage batteries' service duration and life. Base station hierarchical power-off applies to the following scenarios:

- The Huawei power system is used.
- The iSDU06D/iSDU03D/iSDU03S/iSDU03T is used with a third-party power system.

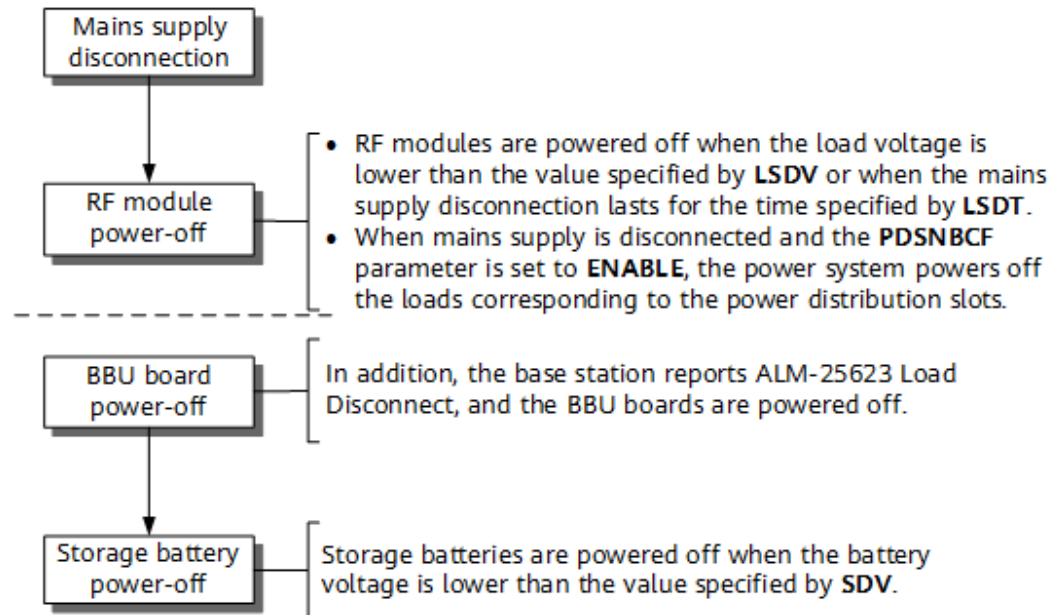
This function is especially suitable for base station cascading scenarios. When mains supply to the upper-level base station is disconnected while that to the lower-level base station is connected, the hierarchical power-off function can be used to extend the working period of transmission equipment in the upper-level base station as much as possible to mitigate the impact on the lower-level base station.

14.1.1 Working Principles

Scenario 1: Huawei Power System

A base station powers off RF modules, BBU boards, and storage batteries level by level when mains supply is disconnected, as shown in the following figure.

Figure 14-1 Principle for base station hierarchical power-off



- Time- or voltage-based subrack-level power-off: When mains supply is disconnected and the **LSDF (LTE eNodeB, 5G gNodeB)** and **PDSNBCF (LTE eNodeB, 5G gNodeB)** parameters are set to **ENABLE**, the loads corresponding to the power distribution slots of the power modules managed by the PMU are powered off. When the base station voltage decreases to the value specified by the **LSDV (LTE eNodeB, 5G gNodeB)** parameter or when the mains supply disconnection lasts for the time specified by the **LSDT (LTE eNodeB, 5G gNodeB)** parameter, the base station shuts down RF modules powered by the power modules managed by the PMU, but maintains power supply to BBU boards and transmission equipment.
- Time-based slot-level power-off: When the **PDSATTR.CSDTS (LTE eNodeB, 5G gNodeB)** parameter is set to **ON** and the mains supply disconnection lasts for the time specified by the **PDSATTR.CLSDT (LTE eNodeB, 5G gNodeB)** parameter, the base station shuts down the RF modules powered by the corresponding power distribution modules.
- Time-based port-level power-off: For a PMU whose power system type is SMU (that is, a PMU for which **PMU.PTYPE (LTE eNodeB, 5G gNodeB)** is set to **SMU**), when the **PDPATTR.CSDTS (LTE eNodeB, 5G gNodeB)** parameter is set to **ON** and the mains supply disconnection lasts for the time specified by the **PDPATTR.CLSDT (LTE eNodeB, 5G gNodeB)** parameter, power-off is triggered for the corresponding port.
- Voltage-based port-level power-off: For a PMU whose power system type is SMU (that is, a PMU for which **PMU.PTYPE (LTE eNodeB, 5G gNodeB)** is set to **SMU**), when the **PDPATTR.CUSLOADSHUTDOWNMODE (5G gNodeB, LTE eNodeB)** parameter is set to **ON_VOL** and the base station voltage drops to the threshold specified by the **PDPATTR.CUSLOADSHUTDOWNVOLTAGE (5G gNodeB, LTE eNodeB)** parameter after the mains supply is disconnected, power-off is triggered for the corresponding port.

When a base station is being powered off, ALM-25623 Load Disconnect is reported. The reporting procedure varies depending on the power-off mode.

- In voltage-based power-off mode, the base station reports ALM-25623 Load Disconnect when its voltage decreases to the threshold specified by the **LSDV (LTE eNodeB, 5G gNodeB)** or **PDPATTR.CUSLOADSHUTDOWNVOLTAGE (5G gNodeB, LTE eNodeB)** parameter. After the OSS receives the alarm, it instructs the base station to power off baseband processing units (BBPs) and the universal baseband radio interface unit (UBRI). RF modules will be powered off 1 minute after this alarm is reported.
- In time-based power-off mode:
 - If the **LSDT (LTE eNodeB, 5G gNodeB)** or **CLSDT (LTE eNodeB, 5G gNodeB)** parameter is set to **0**, the base station is powered off immediately after reporting ALM-25623 Load Disconnect.
 - If the **LSDT (LTE eNodeB, 5G gNodeB)** or **CLSDT (LTE eNodeB, 5G gNodeB)** parameter is set to a value greater than **0**, the base station reports ALM-25623 Load Disconnect 1 minute before being powered off. After the OSS receives the alarm, it instructs the base station to power off BBPs and the UBRI. RF modules will be powered off 1 minute after this alarm is reported.

When the voltage falls below the value specified by the **SDV (5G gNodeB, LTE eNodeB)** parameter, storage batteries are disconnected. In other words, the base station and all modules are disconnected.

NOTE

- In BBU board power-off, only the BBPs and the UBRI are powered off and other boards are not powered off.
- This alarm is triggered only by the power supply system connected to the BBU. If the BBU is powered by multiple power supply systems, BBU board power-off is triggered as long as one power supply system reports the alarm.
- If the PMU is managed by the CCU that is connected to the BBU, BBU board power-off is triggered upon the reporting of the alarm.
- A PMU whose power system type is SMU supports the customized port-level power-off time only if it is installed in an MTS or ICC series power subrack with intelligent circuit breakers; and it supports voltage-based subrack-level power-off only if it is installed in an MTS or ICC series power subrack without intelligent circuit breakers. Contact Huawei engineers to check whether there are intelligent circuit breakers installed in the power subrack.
- In terms of voltage-based port-level power-off, if the power-off voltage is higher than the power-on voltage specified for the SMU, this function does not take effect.

Scenario 2: iSDU06D/iSDU03D/iSDU03S/iSDU03T Used with a Third-Party Power System

When mains supply is disconnected, subrack-level or port-level power-off is triggered for the iSDU06D/iSDU03D/iSDU03S/iSDU03T, which is specified by the following parameters: For subrack-level power-off, **PMU.PTYPE (LTE eNodeB, 5G gNodeB)** needs to be set to **ISDU** for the iSDU06D/iSDU03D/iSDU03S/iSDU03T.

- Time-based subrack-level power-off: It is implemented by running the **MOD PMU** command with the **PMU.LSDF (LTE eNodeB, 5G gNodeB)**, **PMU.LSDTF (LTE eNodeB, 5G gNodeB)**, and **PMU.LSDT (LTE eNodeB, 5G gNodeB)** parameters specified.

When the mains supply disconnection lasts for the time specified by the **PMU.LSDT (LTE eNodeB, 5G gNodeB)** parameter, subrack-level power-off is triggered.

- Voltage-based subrack-level power-off: It is implemented by running the **MOD PMU** command with the **PMU.LSDF (LTE eNodeB, 5G gNodeB)** and **PMU.LSDV (LTE eNodeB, 5G gNodeB)** parameters specified.

When the mains supply is unavailable and the input voltage is lower than the **PMU.LSDV (LTE eNodeB, 5G gNodeB)** parameter value, subrack-level power-off is triggered.

- Time-based port-level power-off: It is implemented by running the **MOD PDPATTR** command with the **PDPATTR.CSDTS (LTE eNodeB, 5G gNodeB)** parameter set to **ON** by port and the **PDPATTR.CLSDT (LTE eNodeB, 5G gNodeB)** parameter specified. When the mains supply disconnection lasts for the time specified by the **PDPATTR.CLSDT (LTE eNodeB, 5G gNodeB)** parameter, power-off is triggered for the corresponding port.

When the mains supply disconnection lasts for the time specified by the **PDPATTR.CLSDT (LTE eNodeB, 5G gNodeB)** parameter, power-off is triggered for the corresponding port.

Table 14-1 lists the hierarchical power-off modes supported by iSDUs. After hierarchical power-off is triggered, the iSDU06D/iSDU03D/iSDU03S/iSDU03T reports ALM-25623 Load Disconnect.

Table 14-1 Hierarchical power-off modes supported by different iSDUs

iSDU Model	Number of Output Ports on the iSDU	Ports Supporting Base Station Hierarchical Power-Off	Ports Not Supporting Base Station Hierarchical Power-Off	Time- or Voltage-based Subrack-Level Power-Off	Time-based Port-Level Power-Off
iSDU06D	10	Six 50 A ports	Four 30 A ports	Six 50 A ports powered off at the same time	Six 50 A ports powered off separately
iSDU03D	5	Three 50 A ports	Two 30 A ports	Three 50 A ports powered off at the same time	Three 50 A ports powered off separately
iSDU03S	7	Three 40 A ports	Four 30 A ports	Three 40 A ports powered off at the same time	Three 40 A ports powered off separately

iSDU Model	Number of Output Ports on the iSDU	Ports Supporting Base Station Hierarchical Power-Off	Ports Not Supporting Base Station Hierarchical Power-Off	Time- or Voltage-based Subrack-Level Power-Off	Time-based Port-Level Power-Off
iSDU03T	3	Three 50 A ports	N/A	Three 50 A ports powered off at the same time	Three 50 A ports powered off separately

14.1.2 Parameters

Parameters Related to RF Module Power-Off

The *LSDF (LTE eNodeB, 5G gNodeB)*, *PDSNBCF (LTE eNodeB, 5G gNodeB)*, and *LSDTF (LTE eNodeB, 5G gNodeB)* parameters apply to the power modules corresponding to the PMU, and the *CSDTS (LTE eNodeB, 5G gNodeB)* parameter applies to a power distribution slot. The following figure shows the applications of four load power-off parameters.

Figure 14-2 Power-module-level parameter application

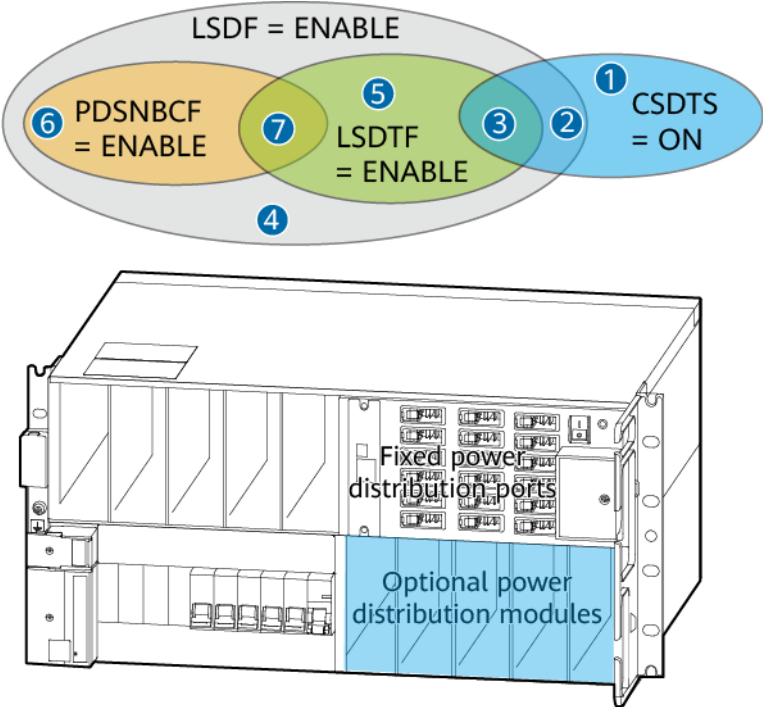


Table 14-2 Power-module-level parameter application analysis

No.	Load Shutdown Flag (<i>LSDF</i> (LTE <i>eNodeB</i> , 5G <i>gNodeB</i>))	Load Shutdown Time Flag (<i>LSDTF</i> (LTE <i>eNodeB</i> , 5G <i>gNodeB</i>))	Power Distribution Slot No-Backup Flag (<i>PDSNB CF</i> (LTE <i>eNodeB</i> , 5G <i>gNodeB</i>))	Customized Shutdown Time Switch (<i>CSDTS</i> (LTE <i>eNodeB</i> , 5G <i>gNodeB</i>))	Power-Off Scenario Without Mains
1	DISABLE	DISABLE	DISABLE	ON	Time-based slot power-off: When the mains supply disconnection lasts for the time specified by the <i>CLSDT</i> (LTE <i>eNodeB</i> , 5G <i>gNodeB</i>) parameter, the RF module powered by the optional power distribution module in the slot whose <i>CSDTS</i> (LTE <i>eNodeB</i> , 5G <i>gNodeB</i>) is set to ON is powered off.
2	ENABLE	DISABLE	DISABLE	ON	• Low voltage power-off: When the load voltage is lower than the value of <i> LSDV</i> (LTE <i>eNodeB</i>, 5G <i>gNodeB</i>), all RF modules corresponding to the PMU are powered off. • If low voltage power-off requirements are not met, time-based slot power-off is used.

No.	Load Shutdown Flag (<i>LSDF</i> (LTE eNodeB, 5G gNodeB))	Load Shutdown Time Flag (<i>LSDTF</i> (LTE eNodeB, 5G gNodeB))	Power Distribution Slot No-Backup Flag (<i>PDSNB CF</i> (LTE eNodeB, 5G gNodeB))	Customized Shutdown Time Switch (<i>CSDTS</i> (LTE eNodeB, 5G gNodeB))	Power-Off Scenario Without Mains
3	ENABLE	ENABLE	DISABLE	ON	- Low voltage power-off - If low voltage power-off requirements are not met, time-based slot power-off is used and is not affected by scheduled power-off. - Scheduled power-off: When the mains supply disconnection lasts for the time specified by the <i>LSDT (LTE eNodeB, 5G gNodeB)</i> parameter, all RF modules corresponding to the PMU are powered off.
4	ENABLE	DISABLE	DISABLE	OFF	Low voltage power-off
5	ENABLE	ENABLE	DISABLE	OFF	- Low voltage power-off - Scheduled power-off
6	ENABLE	DISABLE	ENABLE	OFF	- RF modules powered by the power distribution modules in the optional power distribution slot are powered off immediately when the mains supply is unavailable. - Low voltage power-off

No.	Load Shutdown Flag (<i>LSDF</i> (LTE <i>eNodeB</i> , 5G <i>gNodeB</i>))	Load Shutdown Time Flag (<i>LSDTF</i> (LTE <i>eNodeB</i> , 5G <i>gNodeB</i>))	Power Distribution Slot No-Backup Flag (<i>PDSNB CF</i> (LTE <i>eNodeB</i> , 5G <i>gNodeB</i>))	Customized Shutdown Time Switch (<i>CSDTS</i> (LTE <i>eNodeB</i> , 5G <i>gNodeB</i>))	Power-Off Scenario Without Mains
7	ENABLE	ENABLE	ENABLE	OFF	• RF modules powered by the power distribution modules in the optional power distribution slot are powered off immediately when the mains supply is unavailable. • Low voltage power-off • Scheduled power-off

A PMU whose power system type is SMU installed in an MTS or ICC series power subrack supports either of the following functions:

- Customized port-level power-off time and voltage-based port-level power-off (only if the power subrack is configured with intelligent circuit breakers)
- Voltage-based subrack-level power-off (if the power subrack is not configured with intelligent circuit breakers)

The following table lists the application scenarios of hierarchical power-off of a PMU whose power system type is SMU.

Table 14-3 SMU hierarchical power-off application analysis

No.	Load Shutdown Flag (<i>LSDF (LTE eNodeB, 5G gNodeB)</i>)	Customized Shutdown Time Switch (<i>CSDTS (LTE eNodeB, 5G gNodeB)</i>)	Customized Load Shutdown Optional Mode (<i>PDPATTR.CU SLOADSHUT DOWNMODE (5G gNodeB, LTE eNodeB)</i>)	Power-Off Scenario Without Mains
1	ENABLE	OFF	OFF	Low voltage power-off: When the load voltage is lower than the value of <i>LSDV (LTE eNodeB, 5G gNodeB)</i>, all RF modules corresponding to the PMU are powered off.
2	DISABLE	ON	OFF	Time-based port-level power-off: When the mains supply disconnection lasts for the time specified by the <i>CLSDT (LTE eNodeB, 5G gNodeB)</i> parameter, the RF module on the corresponding port is powered off.
3	DISABLE	OFF	ON_VOL	Voltage-based port-level power-off: When the base station voltage drops to the <i>CUSLOADSHUTDOWN-VOLTAGE (5G gNodeB, LTE eNodeB)</i> parameter value, the RF module on the corresponding port is powered off.

Parameters Related to BBU Board Power-Off

BBU board power-off flag: *SDBBLSD (5G gNodeB, LTE eNodeB)*

BBU board power-off is triggered when the base station reports ALM-25623 Load Disconnect and *SDBBLSD (5G gNodeB, LTE eNodeB)* is set to **ON** (eGBTS/NodeB/eNodeB/gNodeB) or **Yes** (GBTS).

Parameters Related to Storage Battery Power-Off

- Storage battery power-off flag: *LVSDF (5G gNodeB, LTE eNodeB)*
- Voltage threshold for triggering storage battery power-off: *SDV (5G gNodeB, LTE eNodeB)*
- Storage batteries are automatically powered off when the following conditions are met:
- *LVSDF (5G gNodeB, LTE eNodeB)* is set to **ENABLE**.
 - Battery voltage is lower than the value specified by *SDV (5G gNodeB, LTE eNodeB)*.

Parameters Related to iSDU06D/iSDU03D/iSDU03S/iSDU03T Power-Off

The following table lists the applications of three load power-off parameters in different combinations.

Table 14-4 Parameter application analysis

No.	Voltage-based Subrack-Level Power-Off (<i>PMU.LSDF (LTE eNodeB, 5G gNodeB)</i>)	Time-based Subrack-Level Power-Off (<i>PMU.LSDT F (LTE eNodeB, 5G gNodeB)</i>)	Time-based Port-Level Power-Off (<i>PDSATTR.CSDTS (LTE eNodeB, 5G gNodeB)</i>)	Power-Off Scenario
1	DISABLE	DISABLE	ON	Time-based port-level power-off
2	ENABLE	DISABLE	ON	Voltage-based subrack-level power-off and time-based port-level power-off are both set, and the one for which the triggering conditions are met first takes effect.
3	ENABLE	ENABLE	ON	Voltage-based subrack-level power-off and time-based port-level power-off are both set, and the one for which the triggering conditions are met first takes effect.
4	ENABLE	DISABLE	OFF	Voltage-based subrack-level power-off

No.	Voltage-based Subrack-Level Power-Off (PMU.LSDF (LTE eNodeB, 5G gNodeB))	Time-based Subrack-Level Power-Off (PMU.LSDT F (LTE eNodeB, 5G gNodeB))	Time-based Port-Level Power-Off (PDSATTR.CSDTS (LTE eNodeB, 5G gNodeB))	Power-Off Scenario
5	ENABLE	ENABLE	OFF	Voltage-based subrack-level power-off and time-based subrack-level power-off are both set, and the one for which the triggering conditions are met first takes effect.

14.2 Network Analysis

14.2.1 Benefits

After base station hierarchical power-off takes effect, the service time of batteries is prolonged.

14.2.2 Impacts

The impacts between this function and other functions are described in [14.3.2.3 Function Impacts](#).

After base station hierarchical power-off takes effect, the services of the cells served by the RF modules that are powered off are unavailable.

14.3 Requirements

14.3.1 Licenses

No requirements

14.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

14.3.2.1 Prerequisite Functions

None

14.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
GSM UMTS LTE FDD LTE TDD NR	Base station hierarchical power-off	For a PMU whose power system type is SMU (that is, a PMU for which PMU.PTYPE (LTE eNodeB, 5G gNodeB) is set to SMU): - Time-based port-level power-off: PDPATTR.CSDTS (5G gNodeB, LTE eNodeB) - Voltage-based port-level power-off: PDPATTR.CUSLOADS HUTDOWNMODE (5G gNodeB, LTE eNodeB)	14 Base Station Hierarchical Power-Off	For a PMU whose power system type is SMU (that is, a PMU for which PMU.PTYPE (LTE eNodeB, 5G gNodeB) is set to SMU), if PDPATTR.CSDTS (5G gNodeB, LTE eNodeB) is set to ON , PDPATTR.CUSLOADS HUTDOWNMODE (5G gNodeB, LTE eNodeB) cannot be set to ON_VOL .

14.3.2.3 Function Impacts

None

14.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- Scenario 1: Huawei power system
 - 3900 and 5900 series base stations (macro base stations)
 - DBS3900 LampSite and DBS5900 LampSite
- Scenario 2: iSDU06D/iSDU03D/iSDU03S/iSDU03T used with a third-party power system
 - 3900 and 5900 series base stations (macro base stations)

Power Supply

For details, see "Power Supply Systems" and "Mapping Between Physical Cabinets and PTYPE Values" in "Hardware" in section "Basic Power Supply Management Functions" in *Power Supply Management*.

Only the SMU02C (BOM number: 02312MML-009) supports voltage-based port-level power-off.

Boards

No requirements

14.3.4 Others

14.3.4.1 Scenario 1: Huawei Power System

Any version is permitted for RF module power-off and storage battery power-off under low input power voltage.

The base station software of GBSS16.0/RAN16.0/eRAN7.0/SRAN9.0/5G RAN1.0 or later is required for scheduled RF module power-off.

The base station software of GBSS17.1/RAN17.1/eRAN8.1/SRAN10.1/5G RAN1.0 or later is required for BBU board power-off.

The base station software of RAN23.0/eRAN17.0/SRAN17.0/5G RAN5.0 or later is required for the non-backup power capability for power distribution slots.

The base station software of GBSS28.1/RAN28.1/eRAN22.1/SRAN22.1/5G RAN10.1 or later is required for voltage-based port-level power-off.

- Power supply system type

[14.3.3 Hardware](#) describes requirements of this function for the power supply system type.

NOTE

In the BTS3900 or BTS3900L using an IMS06 cabinet, RF modules and the BBU are powered by the same DCDC. If hierarchical power-off is used and RF modules are powered off, the entire BBU will be powered off. Therefore, this function is not recommended in this scenario.

- PMU software version

The software version of the PMU delivered with a Huawei wireless base station must be 126 or later.

14.3.4.2 Scenario 2: iSDU06D/iSDU03D/iSDU03S/iSDU03T Used with a Third-Party Power System

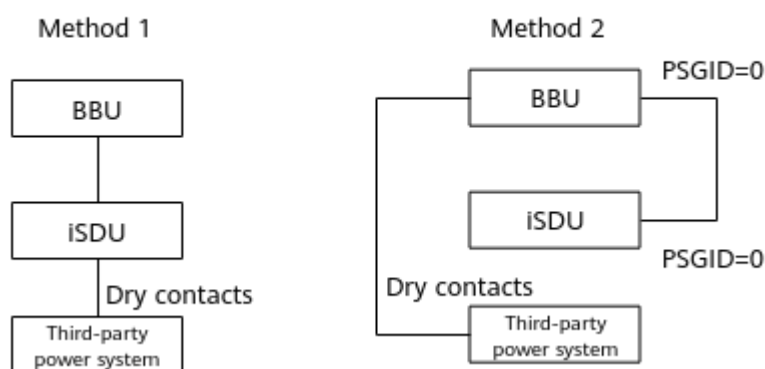
Third-Party Power Systems

Hierarchical power-off takes effect on condition that the iSDU06D/iSDU03D/iSDU03S/iSDU03T can obtain the mains status. All of them can obtain the mains status through dry contacts; in addition, the iSDU06D/iSDU03D can obtain the mains status through the SNMP.

There are two methods for obtaining the mains status through dry contacts, as shown in **14.3.4.2 Scenario 2: iSDU06D/iSDU03D/iSDU03S/iSDU03T Used with a Third-Party Power System**.

- Method 1: The status of the mains supply to a third-party power system is obtained through iSDU06D/iSDU03D/iSDU03S/iSDU03T dry contacts.
- Method 2: When the power supply group ID (specified by **PMU.PSGID (LTE eNodeB, 5G gNodeB)**) configured for the iSDU06D/iSDU03D/iSDU03S/iSDU03T is the same as that configured for BBU dry contacts (which is fixed to 0), the status of the mains supply to a third-party power system can be obtained through these dry contacts and then forwarded to the iSDU06D/iSDU03D/iSDU03S/iSDU03T.

Figure 14-3 Methods for obtaining the mains status through dry contacts



NOTE

If the iSDU is interconnected with a third-party power system using the SNMP and bound to a dry contact at the same time, the mains status obtained through the SNMP-based interconnection is used.

Deployment Requirements for Third-Party Power Systems

- iSDU06D/iSDU03D: If a third-party power system supports the SNMP, it is recommended that the mains status be obtained through the SNMP. If the third-party power system does not support the SNMP, it is recommended that BBU dry contacts be used to forward the mains status. If the BBU dry contacts are fully occupied, iSDU06D/iSDU03D dry contacts are used to obtain the mains status.
- iSDU03S/iSDU03T: It is recommended that BBU dry contacts be used to forward the mains status. If the BBU dry contacts are fully occupied, iSDU03S/iSDU03T dry contacts are used to obtain the mains status.

Manager

The following table lists the restrictions that the manager of the iSDU06D/iSDU03D/iSDU03S/iSDU03T imposes on enabling hierarchical power-off:

Table 14-5 Restrictions that the manager of the iSDU06D/iSDU03D/iSDU03S/iSDU03T imposes on enabling hierarchical power-off

Manager of the iSDU06D/iSDU03D/iSDU03S/iSDU03T	Subrack-Level Power-Off	Port-Level Power-Off
BBU	Supported	Supported
RRU or AAU	Not supported	Supported

 NOTE

- When the iSDU06D/iSDU03D/iSDU03S/iSDU03T is managed by an RRU or AAU (MSRN: 60–254), port-level hierarchical power-off is supported. However, the **CSDTS** parameter cannot be set to **ON** for the power supply port on the RRU or AAU corresponding to the iSDU06D/iSDU03D/iSDU03S/iSDU03T. This port is called the manager port.
- Ports specified by the parameters in the **PDPATTR** MO are disabled by default. Do not enable the manager port and configure other circuit breaker output ports based on site requirements.

14.4 Operation and Maintenance (eGBTS/NodeB/eNodeB/gNodeB)

It is recommended that this function be enabled in areas where the mains supply fails frequently or the transmission equipment has high requirements on power supply continuity.

14.4.1 Activation and Optimization

14.4.1.1 Data Preparation

Hierarchical power-off can be implemented when the Huawei power system is used or when the iSDU06D/iSDU03D/iSDU03S/iSDU03T is used with a third-party power system. Data preparation in different scenarios is as follows:

- [Table 14-6](#) and [Table 14-7](#) list the parameter settings when the Huawei power system is used.
- [Table 14-9](#) and [Table 14-10](#) list the parameter settings when the iSDU06D/iSDU03D/iSDU03S/iSDU03T is used with a third-party power system.

Table 14-6 Parameter settings for base station hierarchical power-off

Parameter Name	Parameter ID	Setting Notes
Load Shutdown Flag	PMU.LSDF (5G gNodeB, LTE eNodeB)	• ENABLE is recommended for the SC48200 PMU. • DISABLE is recommended for other PMUs. (Note: This parameter must be set to DISABLE in an IBC10.)
Load Shutdown Voltage	PMU.LSDV (5G gNodeB, LTE eNodeB)	Configuration restrictions: • For an SC48200 power system: – PMU.LSDV (5G gNodeB, LTE eNodeB) must be set to a value greater than or equal to BATTERY.SDV (5G gNodeB, LTE eNodeB) plus 5. – PMU.LSDV (5G gNodeB, LTE eNodeB) must be set to a value less than or equal to PMU.DCVLTHD (5G gNodeB, LTE eNodeB) minus 5. • For any other power system: – PMU.LSDV (5G gNodeB, LTE eNodeB) must be set to a value greater than or equal to BATTERY.SDV (5G gNodeB, LTE eNodeB). – PMU.LSDV (5G gNodeB, LTE eNodeB) must be set to a value less than PMU.DCVLTHD (5G gNodeB, LTE eNodeB). The recommended values are as follows: • If lead-acid batteries are used to provide power backup: – 465 is recommended for the SC48200 power supply system. – 440 is recommended for other power supply systems. • If lithium batteries are used to provide power backup^[1]: – 495 is recommended for a power supply system using the IBBS20L to provide power backup. – 476 is recommended for a power supply system using the ESM-48100B1/ESM-48150B1/ESM-48100U2 to provide power backup. – 460 is recommended when the APM30H (Ver.B)/APM30H (Ver.C)/APM30H (Ver.D)/APM30H (Ver.E) cabinet uses lithium batteries for power backup.
Load Shutdown Time Flag	PMU.LSDTF (5G gNodeB, LTE eNodeB)	Set this parameter based on site requirements.

Parameter Name	Parameter ID	Setting Notes
Load Shutdown Time	PMU.LSDT (5G gNodeB, LTE eNodeB)	Set this parameter based on site requirements.
Shut Down BBU Boards in Load Power Off	EQUIPMENT.SDBBLS (5G gNodeB, LTE eNodeB)	Set this parameter based on site requirements.
Low Voltage Shutdown Flag	BATTERY.LVSDF (5G gNodeB, LTE eNodeB)	ENABLE is recommended.
Shutdown Voltage	BATTERY.SDV (5G gNodeB, LTE eNodeB)	Configuration restrictions: - For an SC48200 power system: PMU.LSDV (5G gNodeB, LTE eNodeB) must be set to a value greater than or equal to BATTERY.SDV (5G gNodeB, LTE eNodeB) plus 5. - For any other power system: PMU.LSDV (5G gNodeB, LTE eNodeB) must be set to a value greater than or equal to BATTERY.SDV (5G gNodeB, LTE eNodeB). - If the power system is an SMU, PDPATTR.CUSLOADSHUTDOWNVOLTAGE (5G gNodeB, LTE eNodeB) must be set to a value greater than BATTERY.SDV (5G gNodeB, LTE eNodeB). The recommended values are as follows: - If lead-acid batteries are used to provide power backup: 460 is recommended for the SC48200 PMU. 430 is recommended for other PMUs. - If lithium batteries are used to provide power backup^[1]: 482 is recommended for a power supply system using the IBBS20L to provide power backup. 462 is recommended for a power supply system using the ESM-48100B1/ESM-48150B1/ESM-48100U2 to provide power backup. 430 is recommended when the APM30H (Ver.B)/APM30H (Ver.C)/APM30H (Ver.D)/APM30H (Ver.E) cabinet uses lithium batteries for power backup.

Parameter Name	Parameter ID	Setting Notes
Power Distribution Slot No-Backup Flag	PMU.PDSNBCF (LTE eNodeB, 5G gNodeB)	The PMU.LSDF (5G gNodeB, LTE eNodeB) parameter must be set to DISABLE or ENABLE. • If the PMU.LSDF (LTE eNodeB, 5G gNodeB) parameter is set to DISABLE, the PMU.PDSNBCF (LTE eNodeB, 5G gNodeB) parameter cannot be set. • If the PMU.LSDF (LTE eNodeB, 5G gNodeB) parameter is set to ENABLE, the PMU.PDSNBCF (LTE eNodeB, 5G gNodeB) parameter can be set. Its default value is DISABLE.

 NOTE

[1]: The end-of-discharge voltage of the lithium battery is higher than that of the lead-acid battery. If the default value for the lead-acid battery is used, discharge undervoltage protection will be performed on the lithium battery first. Then, storage battery power-off under low voltage and load power-off will not take effect.

Table 14-7 Parameter settings for customized time-based load power-off function

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	PDSATTR.CN (LTE eNodeB, 5G gNodeB)	Set this parameter to the cabinet number of the PMU that manages the power distribution slot.
Subrack No.	PDSATTR.SRN (LTE eNodeB, 5G gNodeB)	Set this parameter to the subrack number of the PMU that manages the power distribution slot.
Slot No.	PDSATTR.SN (LTE eNodeB, 5G gNodeB)	Set this parameter to the number of the power distribution slot.
Customized Shutdown Time Switch	PDSATTR.CSDTS (LTE eNodeB, 5G gNodeB)	When this switch is turned on, the power output of the corresponding power distribution slot is shut down when the mains failure duration exceeds the user-defined power-off time. When this switch is turned off, the customized power-off function is disabled for the corresponding power distribution slot.
Customized Load Shutdown Time	PDSATTR.CLSDT (LTE eNodeB, 5G gNodeB)	After the customized power-off function switch is turned on for a power distribution slot, the RF module powered by the power distribution slot is powered off when the AC power failure duration of the power distribution slot exceeds the specified time threshold. The value of this parameter is displayed as NULL in the DSP PDSATTR command output when the power output is not switched off in the case of a mains failure.

Parameter Name	Parameter ID	Setting Notes
Power Distribution Slot Description	PDSATTR.PDSDESC <i>(LTE eNodeB, 5G gNodeB)</i>	It is recommended that this parameter be set to the location information of the RF module that is powered by the power distribution slot. The location information includes the name of the NE to which the RF module belongs and the cabinet number, subrack number, and slot number of the RF module.

Table 14-8 Parameter settings for SMU port-level power-off

Parameter Name	Parameter ID	Setting Notes
Port No.	PDPATTR.PORTNO <i>(5G gNodeB, LTE eNodeB)</i>	Set this parameter based on the network plan.
Customized Shutdown Time Switch	PDPATTR.CSDTS <i>(5G gNodeB, LTE eNodeB)</i>	It is recommended that this parameter be set to ON if the power output of the corresponding power distribution port is shut down when the mains failure duration exceeds the user-defined power-off time.
Customized Load Shutdown Time	PDPATTR.CLSDT <i>(5G gNodeB, LTE eNodeB)</i>	Set this parameter based on the network plan.
Customized Load Shutdown Optional Mode	PDPATTR.CUSLOADSHUTDOWNMODE <i>(5G gNodeB, LTE eNodeB)</i>	It is recommended that this parameter be set to ON_VOL if the power output of the corresponding power distribution port is shut down when the port voltage is lower than the customized load power-off voltage.
Customized Load Shutdown Voltage	PDPATTR.CUSLOADSHUTDOWNVOLTAGE <i>(5G gNodeB, LTE eNodeB)</i>	Set this parameter based on the network plan. Configuration restrictions: • PDPATTR.CUSLOADSHUTDOWNVOLTAGE <i>(5G gNodeB, LTE eNodeB)</i> must be set to a value greater than or equal to BATTERY.SDV <i>(5G gNodeB, LTE eNodeB)</i>. • PDPATTR.CUSLOADSHUTDOWNVOLTAGE <i>(5G gNodeB, LTE eNodeB)</i> must be set to a value less than PMU.DCVLTHD <i>(5G gNodeB, LTE eNodeB)</i>.

Table 14-9 Parameter settings for iSDU06D/iSDU03D/iSDU03S/iSDU03T port-level power-off

Parameter Name	Parameter ID	Setting Notes
Port No.	PDPATTR.PORTNO (LTE eNodeB, 5G gNodeB)	A value in the range of 0 to 5 indicates one of six 50 A ports.
Customized Shutdown Time Switch	PDPATTR.CSDTS (LTE eNodeB, 5G gNodeB)	Set this parameter based on the site requirements.
Customized Load Shutdown Time	PDPATTR.CLSDT (LTE eNodeB, 5G gNodeB)	Set the time based on the total battery backup power and duration allocated to the device. Value range: 0 to 1440 (unit: minute)

Table 14-10 Parameter settings for iSDU06D/iSDU03D/iSDU03S/iSDU03T subrack-level power-off

Parameter Name	Parameter ID	Setting Notes
Load Shutdown Flag	PMU.LSDF (LTE eNodeB, 5G gNodeB)	Set this parameter based on the site requirements.
Load Shutdown Voltage	PMU.LSDV (LTE eNodeB, 5G gNodeB)	Set the voltage based on the battery type and specifications. Value range: 350 to 556 (unit: 0.1 V), default value: 440 Note: If you do not know the battery specifications, you are advised not to set the voltage parameter.
Load Shutdown Time Flag	PMU.LSDTF (LTE eNodeB, 5G gNodeB)	Set this parameter based on the site requirements.
Load Shutdown Time	PMU.LSDT (LTE eNodeB, 5G gNodeB)	Set the time based on the total battery backup power and duration allocated to the device. Value range: 0 to 1440 (unit: minute)

Parameter Name	Parameter ID	Setting Notes
Power System Type	PMU.PTYPE (LTE eNodeB, 5G gNodeB)	Set this parameter to ISDU if the iSDU06D/iSDU03D/iSDU03S/iSDU03T is to be configured.

14.4.1.2 Using MML Commands

Before using MML commands, refer to [14.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

14.4.1.2.1 Scenario 1: Huawei Power System

Activation Command Examples

To enable the function of powering off modules when the mains supply is faulty, perform the following steps:

- Step 1** Run the **MOD PMU** command. In this step, set parameters as follows:
- **LSDF (5G gNodeB, LTE eNodeB)** is set to **ENABLE**.
 - **LSDV (5G gNodeB, LTE eNodeB)** is set to a value based on site requirements, such as **465**.
 - **LSDTF (5G gNodeB, LTE eNodeB)** is set to **ENABLE**.
 - **LSDT (5G gNodeB, LTE eNodeB)** is set to a value based on site requirements, such as **30**.
 - **PDSNBCF** is set to a value based on site requirements, such as **ENABLE**.
- Step 2** Run the **SET EQUIPMENT** command with **SDBBLSD (5G gNodeB, LTE eNodeB)** set to **YES**.
- Step 3** Run the **MOD BATTERY** command with parameters set as follows:
- **LVSDF (5G gNodeB, LTE eNodeB)** is set to **ENABLE**.
 - **SDV (5G gNodeB, LTE eNodeB)** is set to a value based on site requirements, such as **430**.

----End

To enable the function of powering off RF modules based on the power distribution slots when the mains supply is faulty, perform the following step:

- Step 1** Run the **ADD PDSATTR** command with parameters set as follows:

- ***CSDTS (LTE eNodeB, 5G gNodeB)*** is set to **ON**.
- ***CLSDT (LTE eNodeB, 5G gNodeB)*** is set to a value based on site requirements, such as **10**.

----End

To enable the function of powering off RF modules by SMU port based on the time when the mains supply is faulty, perform the following step:

Step 1 Run the **MOD PDPATTR** command. In this step, set parameters as follows:

- ***CSDTS (LTE eNodeB, 5G gNodeB)*** is set to **ON**.
- ***CLSDT (LTE eNodeB, 5G gNodeB)*** is set to a value based on site requirements, such as **10**.

----End

To enable the function of powering off RF modules by SMU port based on the voltage when the mains supply is faulty, perform the following step:

Step 1 Run the **MOD PDPATTR** command. In this step, set parameters as follows:

- ***CUSLOADSHUTDOWNMODE (5G gNodeB, LTE eNodeB)*** is set to **ON_VOL**.
- ***CUSLOADSHUTDOWNVOLTAGE (5G gNodeB, LTE eNodeB)*** is set to a value based on site requirements, such as **440**.

----End

Deactivation Command Examples

To deactivate RF module power-off, run the **MOD PMU** command with **PMU.LSDF** set to **DISABLE**.

To deactivate BBU board power-off, run the **SET EQUIPMENT** command with **EQUIPMENT.SDBBLSD** set to **OFF**.

To deactivate storage battery power-off, run the **MOD BATTERY** command with **BATTERY.LVSDF** set to **DISABLE**.

To deactivate the function of powering off RF modules based on the power distribution slots when the mains supply is faulty, run the **MOD PDSATTR** command with ***CSDTS (LTE eNodeB, 5G gNodeB)*** set to **OFF**.

To deactivate the function of powering off RF modules by SMU port based on the time when the mains supply is faulty, run the **MOD PDPATTR** command with ***CSDTS (LTE eNodeB, 5G gNodeB)*** set to **OFF**.

To deactivate the function of powering off RF modules by SMU port based on the voltage when the mains supply is faulty, run the **MOD PDPATTR** command with ***CUSLOADSHUTDOWNMODE (5G gNodeB, LTE eNodeB)*** set to **OFF**.

14.4.1.2.2 Scenario 2: iSDU06D/iSDU03D/iSDU03S/iSDU03T Used with a Third-Party Power System

Basic Configuration Examples

If the iSDU06D/iSDU03D/iSDU03S/iSDU03T is used with a third-party power system, first obtain the status of the mains supply to the third-party power system.

- If the iSDU06D/iSDU03D is connected to the third-party power system through the SNMP, see [Data Configuration](#) in section "iSDU06D/iSDU03D Used with a Third-Party Power System."
- The following is a configuration example when the iSDU06D/iSDU03D/iSDU03S/iSDU03T is connected to the third-party power system through dry contacts.

Scenario 1: The status of the mains supply to a third-party power system is obtained through iSDU06D/iSDU03D/iSDU03S/iSDU03T dry contacts.

```
//Setting the dry contact signal input port, for example, setting CN, SRN, SN, PN, PT, and AVOL as required, SW to ON, and AID to 65033
SET ALMPORT: CN=0, SRN=17, SN=0, PN=0, SW=ON, AID=65033, PT=BOOL, AVOL=LOW;
//Modifying the port binding relationship for the mains supply outage alarm, for example, setting ISDSWITCH to ENABLE and NMSACN, NMSASRN, NMSASN, and NMSAPN as required
MOD NMSABIND: ISDSWITCH=ENABLE, NMSACN=0, NMSASRN=17, NMSASN=0, NMSAPN=0;
```

Scenario 2: When the power supply group ID configured for the iSDU06D/iSDU03D/iSDU03S/iSDU03T is the same as that configured for BBU dry contacts (which is fixed to 0), the status of the mains supply to a third-party power system is obtained through these dry contacts and then forwarded to the iSDU06D/iSDU03D/iSDU03S/iSDU03T.

```
//Setting the dry contact signal input port, for example, setting CN, SRN, SN, PN, PT, and AVOL as required, SW to ON, and AID to 65033
SET ALMPORT: CN=0, SRN=0, SN=18, PN=0, SW=ON, AID=65033, PT=BOOL, AVOL=LOW;
//Modifying the port binding relationship for the mains supply outage alarm, for example, setting ISDSWITCH to ENABLE and NMSACN, NMSASRN, NMSASN, and NMSAPN as required
MOD NMSABIND: ISDSWITCH=ENABLE, NMSACN=0, NMSASRN=0, NMSASN=18, NMSAPN=0;
//Setting the power supply group ID for the iSDU06D/iSDU03D/iSDU03S/iSDU03T
MOD PMU: CN=0, SRN=17, PTYPE=ISDU, PSGID=0;
```

Activation Command Examples

Step 1 (Optional) To enable time-based port-level power-off when the mains supply is faulty, perform the following steps: Run the **MOD PDPATTR** command. In this step, set parameters as follows:

- **CN** and **SRN** are set to the cabinet number and subrack number of the PMU, respectively. **SN** is set to the number of the power distribution slot.
- **PORTNO** is set as required. For example, the value 0 indicates the first port.
- **CSDTS** is set to **ON**.
- **CLSDT** is set as required, for example, to **120** (unit: minute).

```
//Modifying the load power-off configuration of a power distribution port, for example, setting the customized load power-off time of the first power distribution port to 120 minutes
MOD PDPATTR: CN=0, SRN=17, SN=0, PORTNO=0, CSDTS=ON, CLSDT=120;
```

Step 2 (Optional) To enable time-based subrack-level power-off when the mains supply is faulty, perform the following steps: Run the **MOD PMU** command. In this step, set parameters as follows:

- **LSDF** is set to **ENABLE**.
- **LSDTF** is set to **ENABLE**.
- **LSDT** is set as required, for example, to **120** (unit: minute).
- **PTYPE** is set to **ISDU** if the iSDU06D/iSDU03D/iSDU03S/iSDU03T is to be configured.

//Modifying the load power-off configuration of the entire subrack, for example, setting the customized subrack-level load power-off time to 120 minutes
MOD PMU: CN=0, SRN=17, PTYPE=ISDU, LSDF=ENABLE, LSDTF=ENABLE, LSDT=120;

Step 3 (Optional) To enable voltage-based subrack-level power-off when the mains supply is faulty, perform the following steps: Run the **MOD PMU** command. In this step, set parameters as follows:

- **LSDF** is set to **ENABLE**.
- **LSDV** is set as required, for example, to **440** (unit: 0.1 V).
- **PTYPE** is set to **ISDU** if the iSDU06D/iSDU03D/iSDU03S/iSDU03T is to be configured.

//Modifying the load power-off configuration of the entire subrack, for example, setting the customized subrack-level load power-off voltage to 44 V
MOD PMU: CN=0, SRN=17, PTYPE=ISDU, LSDF=ENABLE, LSDV=440;

----End

Deactivation Command Examples

Step 1 (Optional) To deactivate time-based port-level power-off, run the **MOD PDPATTR** command with **CSDTS** set to **OFF**.

Step 2 (Optional) To deactivate time-based subrack-level power-off, run the **MOD PMU** command with **LSDTF** set to **DISABLE**.

Step 3 (Optional) To deactivate voltage-based subrack-level power-off, run the **MOD PMU** command with **LSDF** set to **DISABLE**.

----End

14.4.1.3 Using the MAE-Deployment

For detailed operations, see .

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

14.4.2 Activation Verification

14.4.2.1 Scenario 1: Huawei Power System

No Backup for Power Distribution Slots

Step 1 Run the **MOD PMU** command with **PDSNBCF** set to **ENABLE**.

Step 2 Turn off the circuit breaker for the mains supply so that the storage batteries provide power for the base station. If secondary loads powered by power

distribution slots are powered off in this case, no backup for power distribution slots has been activated.

----End

RF Module Power-Off Under Low Voltage

To check whether RF module power-off under low voltage has been activated, perform the following steps:

- Step 1** Run the **MOD PMU** command with **PMU.LSDV** and **PMU.LSDT** set to their maximum values.
- Step 2** When the voltage of loads is lower than the value of **PMU.LSDV**, check whether RF modules are powered off. If all RF modules are powered off in this case, this function has been activated.

----End

Scheduled RF Module Power-Off

To check whether scheduled RF module power-off has been activated, perform the following steps:

- Step 1** Run the **MOD PMU** command with **PMU.LSDT** set to a small value (for example, one minute) and **PMU.LSDV** set to its minimum value.
- Step 2** Turn off the circuit breaker for the mains supply so that the storage batteries provide power for the base station.
- Step 3** When the mains supply disconnection lasts for the time specified by **PMU.LSDT**, check whether the RF modules are powered off. If all RF modules are powered off in this case, this function has been activated.

----End

BBU Board Power-Off

To check whether BBU board power-off has been activated, perform the following steps:

- Step 1** Perform related steps in sections "RF Module Power-Off Under Low Voltage" and "Scheduled RF Module Power-Off" to trigger RF module power-off.
- Step 2** When the base station reports ALM-25623 Load Disconnect, check whether the BBPs and the UBRI board are powered off. If they are powered off in this case, BBU board power-off has been activated.

----End

Storage Battery Power-Off Under Low Voltage

To check whether storage battery power-off under low voltage has been activated, perform the following steps:

- Step 1** Run the **MOD BATTERY** command with **BATTERY.SDV** set to its maximum value.

- Step 2** Turn off the circuit breaker for the mains supply so that the storage batteries provide power for the base station.
- Step 3** When the voltage of storage batteries is lower than the value specified by **BATTERY.SDV**, check whether the storage batteries are disconnected from the base station. If the base station is powered off in this case, this function has been activated.
- End

RF Module Power-Off Based on Power Distribution Slots

To check whether RF module power-off based on power distribution slots is activated, perform the following steps:

- Step 1** Run the **MOD PDSATTR** command with **CLSDT (LTE eNodeB, 5G gNodeB)** set to a small value (for example, 1).
- Step 2** Run the **DSP PDSATTR** command to check the setting result.
- Step 3** Turn off the circuit breaker for the mains supply so that the storage batteries provide power for the base station.
- Step 4** When the mains supply disconnection lasts for the time specified by **LSDT (LTE eNodeB, 5G gNodeB)**, check whether the storage batteries are disconnected from RF modules powered by the corresponding power distribution module. If all RF modules powered by the corresponding power distribution module are powered off in this case, RF module power-off based on power distribution slots has been activated.
- End

SMU Time-based Port-Level RF Module Power-Off

To check whether SMU time-based port-level RF module power-off is activated, perform the following steps:

- Step 1** Run the **MOD PDPATTR** command with **CLSDT (LTE eNodeB, 5G gNodeB)** set to a small value (for example, 1).
- Step 2** Run the **DSP PDPATTR** command to check the setting result.
- Step 3** Turn off the circuit breaker for the mains supply so that the storage batteries provide power for the base station.
- Step 4** When the mains supply disconnection lasts for the time specified by **CLSDT (LTE eNodeB, 5G gNodeB)**, check whether the storage batteries are disconnected from RF modules powered by the corresponding port. If all RF modules powered by the corresponding port are powered off in this case, time-based port-level RF module power-off has been activated.
- End

SMU Voltage-based Port-Level RF Module Power-Off

To check whether SMU voltage-based port-level RF module power-off is activated, perform the following steps:

- Step 1** Run the **MOD PDPATTR** command with **CUSLOADSHUTDOWNMODE (5G gNodeB, LTE eNodeB)** set to its maximum value.
- Step 2** Run the **DSP PDPATTR** command to check the setting result.
- Step 3** Turn off the circuit breaker for the mains supply so that the storage batteries provide power for the base station.
- Step 4** When the port voltage is less than the **CUSLOADSHUTDOWNVOLTAGE (5G gNodeB, LTE eNodeB)** parameter value, check whether the storage batteries are disconnected from RF modules powered by the corresponding port. If all RF modules powered by the corresponding port are powered off in this case, voltage-based port-level RF module power-off has been activated.

----End

14.4.2.2 Scenario 2: iSDU06D/iSDU03D/iSDU03S/iSDU03T Used with a Third-Party Power System

Time-based Port-Level Power-Off

- Step 1** Run the **MOD PDPATTR** command with **CSDTS** set to **ON** and **CLSDT** set as required.
- Step 2** Turn off the circuit breaker for the mains supply so that the storage batteries provide power for the base station.
- Step 3** When the mains supply disconnection lasts for the time specified by **CLSDT**, check whether the modules corresponding to the ports are powered off. If they are powered off in this case, time-based port-level power-off has been activated.

----End

Time-based Subrack-Level Power-Off

- Step 1** Run the **MOD PMU** command with **LSDTF** set to **ENABLE** and **LSDT** set as required.
- Step 2** Turn off the circuit breaker for the mains supply so that the storage batteries provide power for the base station.
- Step 3** When the mains supply disconnection lasts for the time specified by **LSDT**, check whether all the modules corresponding to the subrack are powered off. If they are powered off in this case, time-based subrack-level power-off has been activated.

----End

Voltage-based Subrack-Level Power-Off

- Step 1** Run the **MOD PMU** command with **LSDF** set to **ENABLE** and **LSDV** set as required.
- Step 2** Turn off the circuit breaker for the mains supply so that the storage batteries provide power for the base station.
- Step 3** When the input voltage of the iSDU06D/iSDU03D is lower than the value of **LSDV**, check whether all the modules corresponding to the subrack are powered off. If

they are powered off in this case, voltage-based subrack-level power-off has been activated.

----End

14.5 Operation and Maintenance (GBTS)

It is recommended that this function be enabled in areas where the mains supply fails frequently or the transmission equipment has high requirements on power supply continuity.

14.5.1 Activation and Optimization

14.5.1.1 Data Preparation

[Table 14-11](#) lists the parameter settings for base station hierarchical power-off.

Table 14-11 Parameter settings for base station hierarchical power-off

Parameter Name	Parameter ID	Setting Notes
Load Shutdown Flag	BTSAPMUBP.LSDF	Recommended value for the SC48200 power supply system: ENABLE Recommended value for other power supply systems: DISABLE
Load Shutdown Voltage	BTSAPMUBP.LSDV	• If lead-acid batteries are used to provide power backup: – 465 is recommended for the SC48200 power supply system. – 440 is recommended for other power supply systems. • If lithium batteries are used to provide power backup^[1]: – 495 is recommended for a power supply system using the IBBS20L to provide power backup. – 476 is recommended for a power supply system using the ESM-48100B1/ESM-48150B1/ESM-48100U2 to provide power backup. – 460 is recommended when the APM30H (Ver.B)/APM30H (Ver.C)/APM30H (Ver.D)/APM30H (Ver.E) cabinet uses lithium batteries for power backup.
Load Shutdown Time Flag	BTSAPMUBP.LSDTF	Set this parameter based on site requirements.
Load Shutdown Time	BTSAPMUBP.LSDT	Set this parameter based on site requirements.

Parameter Name	Parameter ID	Setting Notes
Load Shutdown Close BBU board	BTSOTHPARA.SDB BLSD	Set this parameter based on site requirements.
Low Voltage Shutdown Flag	BTSAPMUBP.LVSDF	Recommended value: ENABLE
Shutdown Voltage	BTSAPMUBP.SDV	• If lead-acid batteries are used to provide power backup: – 460 is recommended for the SC48200 power system. – 430 is recommended for other power systems. • If lithium batteries are used to provide power backup^[1]: – 482 is recommended for a power supply system using the IBBS20L to provide power backup. – 462 is recommended for a power supply system using the ESM-48100B1/ESM-48150B1/ESM-48100U2 to provide power backup. – 430 is recommended when the APM30H (Ver.B)/APM30H (Ver.C)/APM30H (Ver.D)/APM30H (Ver.E) cabinet uses lithium batteries for power backup.

 NOTE

[1]: The end-of-discharge voltage of the lithium battery is higher than that of the lead-acid battery. If the default value for the lead-acid battery is used, discharge undervoltage protection will be performed on the lithium battery first. Then, storage battery power-off under low voltage and load power-off will not take effect.

14.5.1.2 Using MML Commands

Before using MML commands, refer to [14.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

 NOTE

For details about the MML command examples for enabling base station hierarchical power-off for a GBTS when the iSDU06D/iSDU03D/iSDU03S/iSDU03T is used, see [14.4.1.2 Using MML Commands](#).

Activation Command Examples

Step 1 Run the **SET BTSAPMUBP** command with parameters set as follows:

- **BTSAPMUBP.LSDF** set to **ENABLE**
- **BTSAPMUBP.LSDV** set to a value based on site requirements, such as **465**
- **BTSAPMUBP.LSDTF** set to **ENABLE**
- **BTSAPMUBP.LSDT** set to a value based on site requirements, such as **30**
- **BTSAPMUBP.LVSDF** set to **ENABLE**
- **BTSAPMUBP.SDV** set to a value based on site requirements, such as **430**

Step 2 Run the **SET BTSOTHPARA** command with **BTSOTHPARA.SDBBLSD** set to **YES**.

----End

Deactivation Command Examples

Run the **SET BTSAPMUBP** command with the following parameters set as required:

- To deactivate automatic battery power-off under low voltage, set **BTSAPMUBP.LVSDF** to **DISABLE**.
- To deactivate automatic battery power-off under high temperature, set **BTSAPMUBP.HTSDF** to **DISABLE**.
- To deactivate automatic load power-off under low voltage, set **BTSAPMUBP.LSDF** to **DISABLE**.

14.5.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

14.5.2 Activation Verification

NOTE

For details about the activation verification of base station hierarchical power-off for a GBTS when the iSDU06D/iSDU03D/iSDU03S/iSDU03T is used, see [14.4.1.3 Using the MAE-Deployment](#).

RF Module Power-Off Under Low Voltage

To check whether RF module power-off under low voltage has been activated, perform the following steps:

Step 1 Run the **SET BTSAPMUBP** command with **LSDV** and **LSDT** set to their maximum values.

Step 2 Turn off the circuit breaker for the mains supply so that the storage batteries provide power for the base station.

- Step 3** When the voltage of loads is lower than the value of **LSDV**, check whether RF modules are powered off. If all RF modules are powered off in this case, RF module power-off under low voltage has been activated.

----End

Scheduled RF Module Power-Off

To check whether scheduled RF module power-off has been activated, perform the following steps:

- Step 1** Run the **SET BTSAPMUBP** command with **LSDT** set to a small value (for example, one minute) and **LSDV** set to its minimum value.
- Step 2** Turn off the circuit breaker for the mains supply so that the storage batteries provide power for the base station.
- Step 3** When the mains supply disconnection lasts for the time specified by **LSDT**, check whether RF modules are powered off. If all RF modules are powered off in this case, scheduled RF module power-off has been activated.

----End

BBU Board Power-Off

To check whether BBU board power-off has been activated, perform the following steps:

- Step 1** Perform related steps in sections "RF Module Power-Off Under Low Voltage" and "Scheduled RF Module Power-Off" to trigger RF module power-off.
- Step 2** When the base station reports ALM-25623 Load Disconnect, check whether the BBPs, UBRI, and standby main control board are powered off. If they are powered off in this case, BBU board power-off has been activated.

----End

Storage Battery Power-Off Under Low Voltage

To check whether storage battery power-off under low voltage has been activated, perform the following steps:

- Step 1** Run the **SET BTSAPMUBP** command with **SDV** set to its maximum value.
- Step 2** Turn off the circuit breaker for the mains supply so that the storage batteries provide power for the base station.
- Step 3** When the voltage of storage batteries is lower than the value of **SDV**, storage batteries are automatically powered off. If the base station is powered off in this case, storage battery power-off under low voltage has been activated.

----End

Load Power-Off Under Low Voltage

To check whether load power-off under low voltage has been activated, perform the following steps:

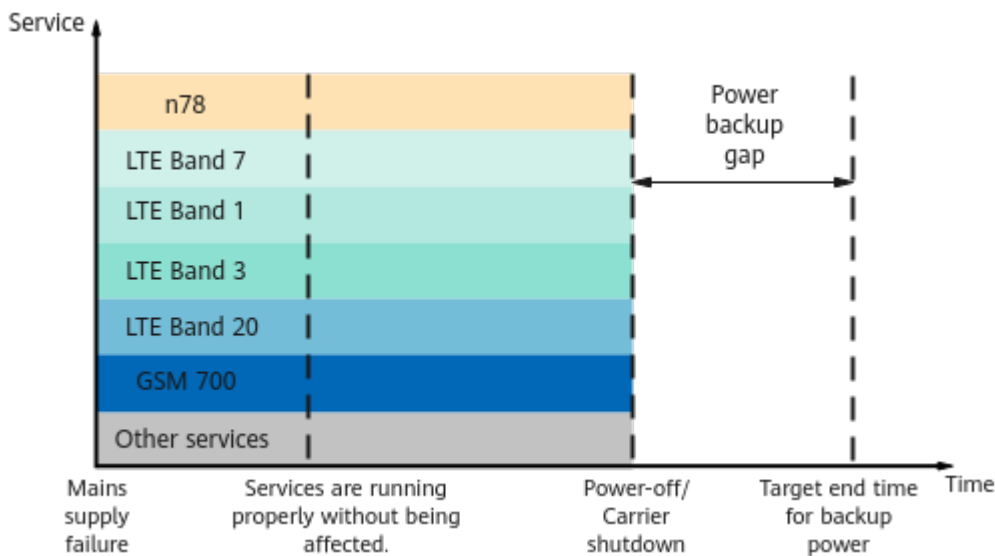
- Step 1** Run the **SET BTSAPMUBP** command with **LSDV** set to its maximum value.
- Step 2** Turn off the circuit breaker for the mains supply so that the storage batteries provide power for the base station.
- Step 3** When the voltage of loads is lower than the value specified by **LSDV**, check whether the storage batteries are powered off. If all RF modules are powered off in this case, load power-off under low voltage has been activated.
- End

15 Adaptive Power Backup

15.1 Principles

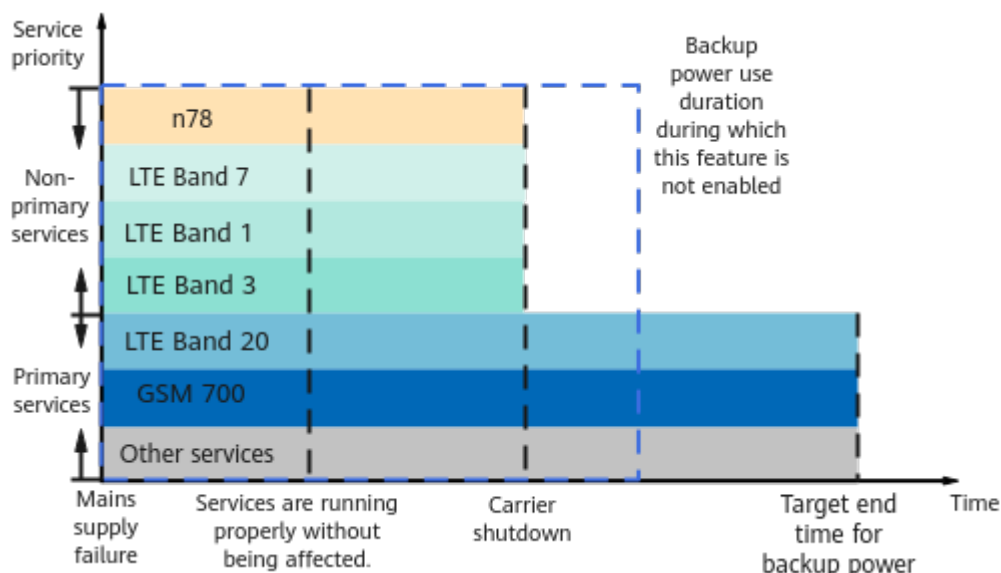
The Adaptive Power Backup feature is a power backup solution tailored for the entire site for the sake of the lossless running duration and target backup power use duration. Traditional service power backup, as shown in [Figure 15-1](#), may cause a power backup gap. As a result, the target backup power use duration cannot be achieved. When the backup power configuration cannot meet the target backup power use duration, the Adaptive Power Backup feature shuts down non-primary cell-level services at a proper time based on the primary cell-level services set by the operator, historical power consumption, and service load trend. In this way, the target backup power use duration can be ensured for primary services.

Figure 15-1 Traditional service power backup



The Adaptive Power Backup feature is introduced to extend the in-service time of the site with a small amount of or no additional backup power investment, as shown in [Figure 15-2](#). The principles of this feature are as follows:

Figure 15-2 Principles of Adaptive Power Backup



- When the mains supply is normal, primary and non-primary services are running at the same time.
- When the mains supply is disconnected:
 - Primary and non-primary services operate at the same time within the lossless running duration.
 - Non-primary services may be shut down beyond the lossless running duration.

NOTE

When microwave services need to be considered as the most primary services of a site, the Adaptive Power Backup feature allows all cells to be set as non-reserved cells and backup power resources to be preferentially used for microwave services.

The Adaptive Power Backup feature can be configured and activated for NEs only on the MAE-Deployment.

Terms Related to the Adaptive Power Backup Feature

- Primary services: services that must not be affected in the case of a power failure, including BBU services, reserved-cell services, and services on other devices
 - BBU: a module that functions as the root node of RF modules and must run properly during the operation of reserved cells
 - Reserved cells: cells that are preferentially guaranteed when the mains supply is disconnected, for example, GSM, UMTS, or LTE 900 MHz cells
 - Other devices: devices except Huawei wireless base stations, such as transmission devices

- Non-primary services: services that can be affected in the case of a power failure, that is, the services in non-reserved cells
 - Non-reserved cells: cells that need to be shut down when the mains supply is disconnected and the remaining battery capacity is not sufficient for the site to run for the target backup power use duration. Generally, capacity-layer cells of each RAT are defined as non-reserved cells, such as UMTS 2.1 GHz, LTE 1.8 GHz, LTE 2.6 GHz, and NR 3.5 GHz cells. Non-reserved cells can be configured as planned.
- Remaining battery capacity: total energy that batteries can discharge after the mains supply is disconnected. The Adaptive Power Backup feature is applicable to lithium batteries only. The remaining capacity is essential for determining whether primary loads can operate for the target backup power use duration.

15.2 Network Analysis

15.2.1 Benefits

When the mains supply is disconnected and the backup power configuration cannot meet the target power backup duration, the Adaptive Power Backup feature shuts down non-primary cell-level services at a proper time based on the primary cell-level services set by the operator, historical power consumption, and service load trend. In this way, primary services can operate for the target power backup duration.

15.2.2 Impacts

The impacts between this function and other functions are described in [15.3.2.3 Function Impacts](#).

When the mains supply is disconnected, non-primary services may be shut down beyond the lossless running duration. As a result, the corresponding services are unavailable.

15.3 Requirements

15.3.1 Licenses

None

15.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

15.3.2.1 Prerequisite Functions

RAT	Function ID	Function Name	Reference	Description
GSM	GBFD-241201	Low Power Consumption Mode	<i>Intelligent Power Consumption Decrease</i>	N/A
UMTS	WRFD-241203	Low Power Consumption Mode	<i>Green BTS</i>	N/A
LTE FDD	LOFD-001040	Low Power Consumption Mode	11 Low Power Consumption Mode (LTE)	N/A
LTE TDD	TDLOFD-001040	Low Power Consumption Mode		N/A
NR	FOFD-021203	gNodeB Power Saving	12 Low Power Consumption Mode (NR)	N/A

15.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
GSM UMTS LTE FDD LTE TDD NR	PV deployment for base stations Third-party green power access measurement and O&M management	N/A	9 PV Deployment for Base Stations 10 Third-Party Green Power Access Measurement and O&M Management	When the Adaptive Power Backup feature takes effect, PV deployment for base stations and third-party green power access measurement and O&M management cannot be enabled. This is because the PV deployment function will cause the calculation results related to the Adaptive Power Backup feature to be inaccurate.

15.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
GSM UMTS LTE FDD LTE TDD NR	Staggering electricity usage	N/A	8 Staggering Electricity Usage	- If staggered discharge is performed first, the remaining battery capacity decreases. In this case, the time for performing the power consumption reduction measure is advanced when the Adaptive Power Backup feature is executed. - If the Adaptive Power Backup feature is executed first, the gains of staggering electricity usage decrease.
NR	Precise backup power saving policies in low power consumption mode	NRDUCellPowerSaving.BackupPwrSavingPolicy set to PRECISE_BACKUP_POLICY1 , PRECISE_BACKUP_POLICY2 , or PRECISE_BACKUP_POLICY3	12 Low Power Consumption Mode (NR)	- Precise backup power saving policy 2 or precise backup power saving policy 3 cannot take effect when the Adaptive Power Backup feature is in effect. If precise backup power saving policy 2 or precise backup power saving policy 3 has been enabled in some cells, the Adaptive Power Backup feature cannot take effect. - When the MAE is disconnected from the base station, the feature rolls back to precise backup power saving policy 1.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Low power consumption mode	CellLowPower .LowPwrSwitc h	11 Low Power Consumption Mode (LTE)	- Low power consumption mode cannot take effect when the Adaptive Power Backup feature is in effect. If low power consumption mode has been enabled in some cells, the Adaptive Power Backup feature cannot take effect. - When the MAE is disconnected from the base station, this feature rolls back to low power consumption mode.

15.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

Hardware

The Adaptive Power Backup feature applies to scenarios where a Huawei APM cabinet is used and where the iSDU06D/iSDU03D is used with a third-party power system. Requirements are as follows:

- Scenarios where a Huawei APM cabinet is used: Only the APM5930(AC) (Ver.A), APM5950H, and APM5950H-L supports the feature. One APM is used and provides power supply at a time. A PMU must be configured. [Table 15-1](#) and [Table 15-2](#) list the requirements for the PMU and batteries, respectively.
- Scenarios where the iSDU06D/iSDU03D is used with a third-party power system: [Table 15-3](#) lists the requirements of the Adaptive Power Backup feature for a third-party power system and iSDU06D/iSDU03D. [Table 15-2](#) lists the requirements for batteries.

NOTE

For details about the requirements for a third-party power system when it is used with the iSDU06D/iSDU03D, see [17.1 iSDU06D/iSDU03D Support for Connection with a Third-Party Power System \(Using the SNMP\)](#).

Table 15-1 Requirements of Adaptive Power Backup for scenarios where a Huawei APM cabinet is used

Power System Type	PMU Hardware Model	Cabinet Type	PMU Software Version
ETP	CCUC	APM5930(AC) (Ver.A)	If the hardware version is PMU.19, the PMU software version must be 2300 or later. If the hardware version is PMU.22, the PMU software version must be 2301 or later. If the hardware version is PMU.34, the PMU software version must be 2301 or later.
ETP	PMU 11C	APM5950H	If the hardware version is PMU.35, the PMU software version must be 2301 or later.
ETP	PMU 11C	APM5950H-L	If the hardware version is PMU.35, the PMU software version must be 2301 or later.

Table 15-2 Requirements of Adaptive Power Backup for batteries

Battery Type	Battery Model	Whether Adaptive Power Backup Is Supported
Lithium batteries	ESM-48100B1 ESM-48150B1 ESM48100U2	Supported
Lead-acid batteries	N/A	Not supported
Hybrid use of lead-acid and lithium batteries	N/A	Not supported

Table 15-3 Requirements of Adaptive Power Backup for scenarios where the iSDU06D/iSDU03D is used with a third-party power system

Third-Party Power System	Hardware Model of the PMU in the iSDU06D/iSDU03D	Software Version of the PMU in the iSDU06D/iSDU03D
- Supports SNMP-based interconnection and works as the only power supply - Must be an external power monitoring unit (EPMU)	PMU 12B (Only for the iSDU06D)	All versions

Boards

None

15.3.4 Others

MAE Version

The Adaptive Power Backup feature applies to scenarios where a Huawei APM cabinet is used and where the iSDU06D/iSDU03D is used with a third-party power system. Requirements are as follows:

- Scenarios where a Huawei APM cabinet is used: The MAE version must be iMaster MAE V100R023C00 or later.
- Scenarios where the iSDU06D/iSDU03D is used with a third-party power system: The MAE version must be iMaster MAE V100R024C00 or later.

NOTE

The Adaptive Power Backup plug-in is released with the MAE-Deployment component. The plug-in version must be consistent with the MAE version.

Base Station Software Version

The Adaptive Power Backup feature applies to scenarios where a Huawei APM cabinet is used and where the iSDU06D/iSDU03D is used with a third-party power system. Requirements are as follows:

- Scenarios where a Huawei APM cabinet is used: The base station version must be BTS3900 V100R019C00 or later.
- Scenarios where the iSDU06D/iSDU03D is used with a third-party power system: The base station version must be BTS3900 V100R020C00 or later.

Scenario

Only centralized power supply is supported. In the centralized power supply scenario, the BBU and RRU use the same power supply and backup system. When

the mains supply is disconnected, main devices are supplied by the same set of backup power system with consistent power alarms.

Device

- Lithium batteries must be configured. Lead-acid battery only or hybrid use of lead-acid and lithium batteries is not supported.
- NEs in the same power supply and backup group must be managed by the same MAE. Multiple OSSs are not supported. When NEs in the same group are managed by different MAEs, the target power backup duration will be incorrectly calculated.

15.4 Operation and Maintenance

15.4.1 Data Configuration

15.4.1.1 Data Preparation

1. Subscribe to counters.

The Adaptive Power Backup feature applies to scenarios where a Huawei APM cabinet is used and where the iSDU06D/iSDU03D is used with a third-party power system. Requirements are as follows:

- Scenarios where a Huawei APM cabinet is used: Before enabling the Adaptive Power Backup feature, check whether the counters listed in the following table have been subscribed to during a 60-minute measurement period.

Counter ID	Counter Name	Description
1593835640	VS.EnergyCons.BTS.Measuring.GSM	Power consumption measured by the power system of the GSM
1593835629	VS.EnergyCons.BTS.Measuring.UMTS	Power consumption measured by the power system of UMTS
1593835638	VS.EnergyCons.BTS.Measuring.LTE	Power consumption measured by the power system of the LTE
1593835805	VS.EnergyCons.BTS.Measuring.NR	Power consumption measured by the power system of NR
1593835639	VS.EnergyCons.BTS.Adding.GSM	The sum of power consumption of all boards in the GSM
1593835628	VS.EnergyCons.BTS.Adding.UMTS	The sum of power consumption of all boards in UMTS

Counter ID	Counter Name	Description
1593835637	VS.EnergyCons.BTS.Adding.LTE	The sum of power consumption of all boards in the LTE
1593835804	VS.EnergyCons.BTS.Adding.NR	The sum of power consumption of all boards in NR
1593835957	VS.EnergyCons.BTS.Adding.Service.GSM	Sum of power consumption of all boards (GSM services)
1593835958	VS.EnergyCons.BTS.Adding.Service.UMTS	Sum of power consumption of all boards (UMTS services)
1593835959	VS.EnergyCons.BTS.Adding.Service.LTE	Sum of power consumption of all boards (LTE services)
1593835961	VS.EnergyCons.BTS.Adding.Service.NR	Sum of power consumption of all boards (NR services)
1593835835	VS.EnergyCons.BU	Power consumption of baseband unit
73435300	VS.TCP.Usage.Mean	Average TCP Usage for Cell
1526726740	L.ChMeas.PR.B.DL.Used.Avg	Average number of used PDSCH PRBs
1526728433	L.ChMeas.PR.B.DL.Avail	Number of available downlink PRBs
1911816540	N.PR.B.DL.Used.Avg	Average number of used downlink PRBs
1911816679	N.PR.B.DL.Avail.Avg	Average number of available downlink PRBs

- Scenarios where the iSDU06D/iSDU03D is used with a third-party power system: Before enabling the Adaptive Power Backup feature, check whether the counters listed in the following table have been subscribed to during a 60-minute measurement period.

Counter ID	Counter Name	Description
1593835639	VS.EnergyCons.BTS.Adding.GSM	The sum of power consumption of all boards in the GSM
1593835628	VS.EnergyCons.BTS.Adding.UMTS	The sum of power consumption of all boards in UMTS

Counter ID	Counter Name	Description
1593835637	VS.EnergyCons.BTS.Adding.LTE	The sum of power consumption of all boards in the LTE
1593835804	VS.EnergyCons.BTS.Adding.NR	The sum of power consumption of all boards in NR
1593835957	VS.EnergyCons.BTS.Adding.Service.GSM	Sum of power consumption of all boards (GSM services)
1593835958	VS.EnergyCons.BTS.Adding.Service.UMTS	Sum of power consumption of all boards (UMTS services)
1593835959	VS.EnergyCons.BTS.Adding.Service.LTE	Sum of power consumption of all boards (LTE services)
1593835961	VS.EnergyCons.BTS.Adding.Service.NR	Sum of power consumption of all boards (NR services)
1593835835	VS.EnergyCons.BU	Power consumption of baseband unit
73435300	VS.TCP.Usage.Mean	Average TCP Usage for Cell
1526726740	L.ChMeas.PRBDL.Used.Avg	Average number of used PDSCH PRBs
1526728433	L.ChMeas.PRBDL.Avail	Number of available downlink PRBs
1911816540	N.PRBDL.Used.Avg	Average number of used downlink PRBs
1911816679	N.PRBDL.Avail.Avg	Average number of available downlink PRBs
1593835836	VS.EnergyCons.BTSBoard	Power consumption of board
1593835942	VS.EPMU.DCEnergy	DC energy of the EPMU

- Obtain information about NEs at the site and identify the mapping between physical sites and logical sites.

 NOTE

For example, a site is configured with two BBUs, which are managed as NE 01 and NE 02 on the MAE and powered by one power system. You need to manually identify that NE 01 and NE 02 are deployed at the same physical site.

- [Table 15-4](#), [Table 15-5](#), [Table 15-6](#), and [Table 15-7](#) describe the parameters used for function activation.

Table 15-4 GSM parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Low Power Consumption Mode Switch	GBTSENERGYMGTP ARA.LOWPWRCONS UMPMODESW	Set this parameter to ON when low power consumption mode is required.
Shutdown TRX Optimize Switch	GBTSENERGYMGTP ARA.SHUTDOWNTR XOPTSW	The value ON is recommended.
Shut Down Enabled	GTRXDEV.SDFLAG	The value ENABLE is recommended.
BTS Power-off Lock BCCH	BTSLOCKBCCH.BTSP OWEROFFLOCKBCC H	Set this parameter based on the network plan. • If this parameter is set to YES, the BSC is allowed to shut down all TRXs. • If this parameter is set to NO, the BSC can only shut down the non-BCCH TRXs for which Shut Down Enabled is set to ENABLE.

Table 15-5 UMTS parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Low Power Consumption Mode Switch	NODEBALGPARA.LO WPWRCONSUMPM ODESW	Set this parameter to ON when low power consumption mode is required.
Smooth Power Change Function Switch	NODEBSMTHPWR.S MTHPWRSWITCH	Set this parameter to OPEN when the smooth power change function is required.

Table 15-6 LTE parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Low power consumption mode switch	CELLLOWPOWER.Lo wPwrSwitch	Set this parameter to ON when low power consumption mode is required.

Parameter Name	Parameter ID	Setting Notes
Low power consumption mode order	CELLLOWPOWER.LowPwrOrder	Set this parameter to ON when the function of forcibly entering the low power consumption mode is required.
Start time	CELLLOWPOWER.StartTime	Set this parameter based on the network plan.
Stop time	CELLLOWPOWER.StopTime	Set this parameter based on the network plan.
Back power save policy	CELLLOWPOWER.BackPwrSavPolicy	Set this parameter based on the network plan.

Table 15-7 NR parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Power Saving Type	NRDUCELLPOWERSAVING.PowerSavingType	Set this parameter based on the network plan.
Low Power Consumption Mode Switch	LOW_PWR_MODE_SW option of the NRDUCELLALGOSWITCH.PowerSavingSwitch parameter	Select this option if low power consumption mode is required.
Backup Power Saving Policy	NRDUCELLPOWERSAVING.BackupPwrSavingPolicy	Set this parameter based on the network plan.
Time to Trigger	NRDUCELLPOWERSAVING.TimeToTrigger	Set this parameter based on the network plan.

15.4.1.2 Using MML Commands

This function needs to be activated on the MAE-Deployment and cannot be activated using MML commands on the base station side. For details, see [15.4.1.3 Using the MAE-Deployment](#).

Before using MML commands, refer to [15.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After

Modification" and "Caution" for the corresponding parameters in the parameter reference.

GSM

Activation command examples

1. Run the **LST RRU** command to check whether the value of **Power Supply Group ID** is **0**. If yes, the centralized power supply scenario is used and this solution can be implemented. Otherwise, do not implement this solution.
2. Run the **SET GBTSENERGYMGTPARA** command with **Low Power Consumption Mode Switch** of eGBTS set to **ON** and **Shutdown TRX Optimize Switch** set to **ON**.
3. Run the **SET GTRXDEV** command with **Shut Down Enabled** set to **ENABLE** to specify the carriers that can be shut down.
4. (Optional) In the dual-band network scenario or when the GSM mode can be shut down, run the BSC MML command **SET BTSLOCKBCCH** with **BTS Power-off Lock BCCH** set to **YES** to allow shutdown of BCCH carriers for which **Shut Down Enabled** was set to **ENABLE** in step 2. That is, the GSM cells corresponding to these carriers can be shut down.

```
//Turning on the eGBTS low power consumption mode switch
SET GBTSENERGYMGTPARA: LOWPWRCONSUMPMODESW=ON, SHUTDOWNTRXOPTSW=ON;
//Configuring the carriers that can be shut down
SET GTRXDEV: IDTYPE=BYID, TRXID=0, SDFLAG=ENABLE;
//Optional) Allowing the base station to shut down BCCH carriers when low power consumption mode is
enabled for the base station
SET BTSLOCKBCCH: IDTYPE=BYID, BTSID=0, BTSPWEROFFLOCKBCCH=YES;
```

Deactivation command examples

Run the **SET GBTSENERGYMGTPARA** command with **Low Power Consumption Mode Switch** set to **OFF**.

```
//Turning off the eGBTS low power consumption mode switch
SET GBTSENERGYMGTPARA: LOWPWRCONSUMPMODESW=OFF;
```

UMTS

Activation command examples

1. Run the **LST RRU** command to check whether the value of **Power Supply Group ID** is **0**. If yes, the centralized power supply scenario is used and this solution can be implemented. Otherwise, do not implement this solution.
2. Run the **MOD ULOCELL** command to configure reserved cells. In this step, set the **Reserved Cell** parameter to **TRUE** for the cell to be reserved.
3. Run the **SET NODEBALGPARA** command to configure the UMTS low power consumption mode. In this step, set **Low Power Consumption Mode Switch** to **ON**.
4. (Optional) To enable smooth power reduction, run the **SET NODEBSMTHPWRSW** command with **Smooth Power Change Function Switch** set to **OPEN**.

```
//Configuring reserved cells
MOD ULOCELL: ULOCELLID=1, LOCELLTYPE=NORMAL_CELL, RSV=TRUE;
//Turning on the UMTS low power consumption mode switch
SET NODEBALGPARA: LOWPWRCONSUMPMODESW=ON;
```

```
//(Optional) Enabling smooth power reduction  
SET NODEBSMTHPWRSW: SMTHPWRSWITCH=OPEN;
```

Deactivation command examples

Run the **SET NODEBALGPARA** command with **Low Power Consumption Mode Switch** set to **OFF**.

```
//Disabling UMTS low power consumption mode  
SET NODEBALGPARA:LOWPWRCONSUMPMODESW=OFF;
```

LTE

Activation command examples

1. Run the **LST RRU** command to check whether the value of **Power Supply Group ID** is **0**. If yes, the centralized power supply scenario is used and this solution can be implemented. Otherwise, do not implement this solution.
2. Run the **MOD CELLLOWPOWER** command to configure the low power consumption mode for LTE. If **Low Power Consumption Mode Switch** is set to **ON** and **Precise Backup Policy 1** is configured, the cell is a non-reserved cell. If **Low Power Consumption Mode Switch** is set to **OFF**, the cell is a reserved cell.

```
//(Optional, required in centralized power supply scenarios) Setting a power supply group ID  
MOD RRU: CN=0, SRN=60, SN=0, PSGID=0;  
//Defining non-reserved cells by enabling low power consumption mode and setting precise backup power saving policy 1  
MOD CELLLOWPOWER: LocalCellId=0, LowPwrSwitch=ON, LowPwrOrder=OFF, StartTime=00&00&00, StopTime=23&59&59, BakPwrSavPolicy= PRECISE_BACKUP_POLICY1;  
//Defining reserved cells by disabling low power consumption mode  
MOD CELLLOWPOWER: LocalCellId=0, LowPwrSwitch=OFF;
```

Deactivation command examples

1. For non-reserved cells, run the **MOD CELLLOWPOWER** command with **Low Power Consumption Mode Switch** set to **OFF**.
2. For reserved cells, restore the value to the initial one.

```
//Deactivating low power consumption mode for non-reserved cells  
MOD CELLLOWPOWER: LocalCellId=0, LowPwrSwitch=OFF;  
//For reserved cells, restoring the original value if LowPwrSwitch was set to ON  
MOD CELLLOWPOWER: LocalCellId=0, LowPwrSwitch=ON;  
//For reserved cells, restoring the original value if LowPwrSwitch was set to OFF  
MOD CELLLOWPOWER: LocalCellId=0, LowPwrSwitch=OFF;
```

NR

Activation command examples

1. Run the **LST RRU** command to check whether the value of **Power Supply Group ID** is **0**. If yes, the centralized power supply scenario is used and this solution can be implemented. Otherwise, do not implement this solution.
2. Run the **MOD NRDUCELLALGOSWITCH** command to configure the low power consumption mode for NR. If **Low Power Consumption Mode Switch** is set to **ON** and **Precise Backup Policy 1** is configured, the cell is a non-reserved cell. If **Low Power Consumption Mode Switch** is set to **OFF**, the cell is a reserved cell.

```
//(Optional, required in centralized power supply scenarios) Setting a power supply group ID  
MOD RRU: CN=0,SRN=60,SN=0,PSGID=0;
```

```
//Defining non-reserved cells by enabling low power consumption mode and setting an energy saving  
policy of low power consumption mode, that is, precise backup power saving policy 1  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=LOW_PWR_MODE_SW-1;  
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1,  
PowerSavingType=LOW_PWR_MODE, BackupPwrSavingPolicy=PRECISE_BACKUP_POLICY1, TimeToTrigger=5;  
//Defining reserved cells by disabling low power consumption mode  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=LOW_PWR_MODE_SW-0;
```

Deactivation command examples

1. For non-reserved cells, run the **MOD NRDUCELLALGOSWITCH** command with **Low Power Consumption Mode Switch** set to **OFF**.
2. For reserved cells, restore the value to the initial one.

```
//For non-reserved cells, turning off the low power consumption mode switch, and deleting the energy  
saving policy of low power consumption mode  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=LOW_PWR_MODE_SW-0;  
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=1;  
//(Optional) Turning off the low power consumption mode switch for reserved cells  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=LOW_PWR_MODE_SW-0;  
//(Optional) Turning on the low power consumption mode switch for reserved cells  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=LOW_PWR_MODE_SW-1;
```

15.4.1.3 Using the MAE-Deployment

Adaptive Power Backup can be configured using the MAE-Deployment. For details, contact Huawei engineers.

15.4.2 Activation Verification

After the configurations in [15.4.1 Data Configuration](#) are complete, you can query the state of a task on the MAE. The state of a task can be **Running**, **Blocked**, or **Stopped**. The state of a group in the task can be **Running**, **Blocked**, **Stopped**, **Pre-Calculating**, or **Pre-Calculation Failed**.

15.4.3 Network Monitoring

Adaptive Power Backup automatically generates a **Report.csv** file. The following table describes the parameters in the report. The **Report.csv** file is generated only when the mains supply is disconnected.

Table 15-8 Parameters in the Report.csv file

Parameter	Description
Group ID	Indicates the group ID. For the multimode base station (MBTS), the group ID is automatically planned.
NE Name	Indicates all NE names under the group ID.
MBTS name	Indicates the name of the separate-MPT MBTS.
Failed Mains Supply Time	Indicates the time when the mains supply is disconnected.

Parameter	Description
Target Backup Duration (Unit: hour)	Indicates the target power backup duration, which is the value entered when a task is created.
Battery Configured(Unit:Ah) ^a	Indicates the rated capacity of a lithium battery (unit: Ah).
Target Battery Configured(Unit:Ah) ^b	Indicates the power required for lossless running at the target power backup duration (unit: Ah).
Backup Time When Feature Disabled(Unit:hour) ^c	Indicates the estimated maximum power backup duration when the feature is disabled (unit: hour).
Backup Time When Feature Enable(Unit:hour)	Indicates the estimated maximum power backup duration when the feature is enabled (unit: hour).
Duration Of Lossless Running(Unit:hour)	Indicates the lossless running duration (unit: hour).
Cells Shutdown Time	Indicates the shutdown time of non-reserved cells.
Recovered Mains Supply Time	Indicates the mains supply recovery time.
a: Indicates the rated power of lithium batteries. The health of lithium batteries has been considered. b: Indicates the power required for lossless running for the target power backup duration (unit: Ah), that is, the site-level power in the past 24 hours. The average of the sum of the maximum values of the VS.EnergyCons.BTS.Measuring.GSM, VS.EnergyCons.BTS.Measuring.UMTS, VS.EnergyCons.BTS.Measuring.LTE, and VS.EnergyCons.BTS.Measuring.NR counters is used as the input parameter for calculating the battery capacity. If the calculated battery capacity is inconsistent with the configured battery capacity, the possible cause is that cells are added, base stations are newly deployed, or related counter data is incomplete. c: Indicates the estimated power backup duration based on the rated power of lithium batteries and historical power when services (non-primary loads + primary loads) are running without losses.	

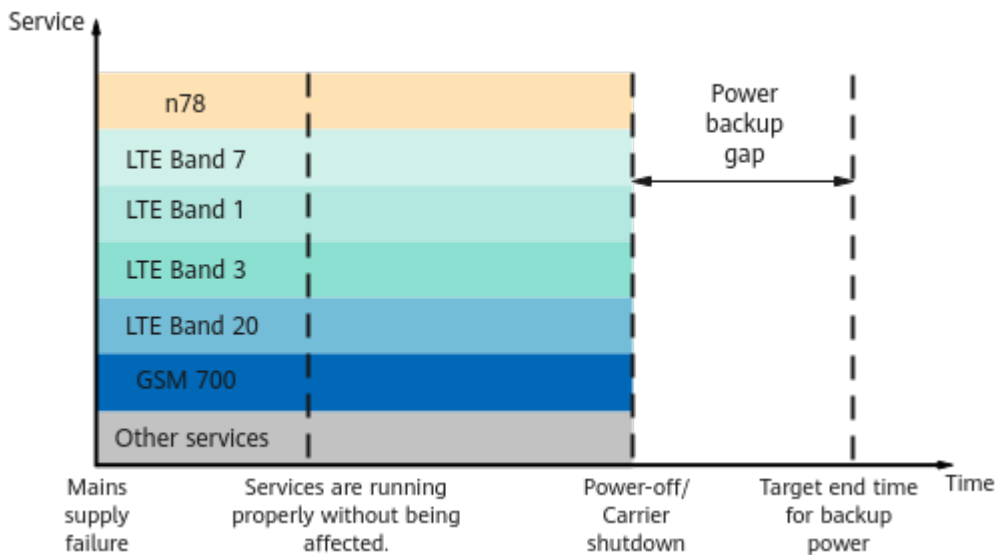
16 SitePower Backup Policy Management

16.1 Principles

Working Principles

With capacity expansion of wireless sites, there is an increase in the site-level power consumption, posing new challenges to the site power backup capability. Traditional power backup, as shown in [Figure 16-1](#), may cause a power backup gap. As a result, the target backup power use duration cannot be achieved.

Figure 16-1 Traditional power backup



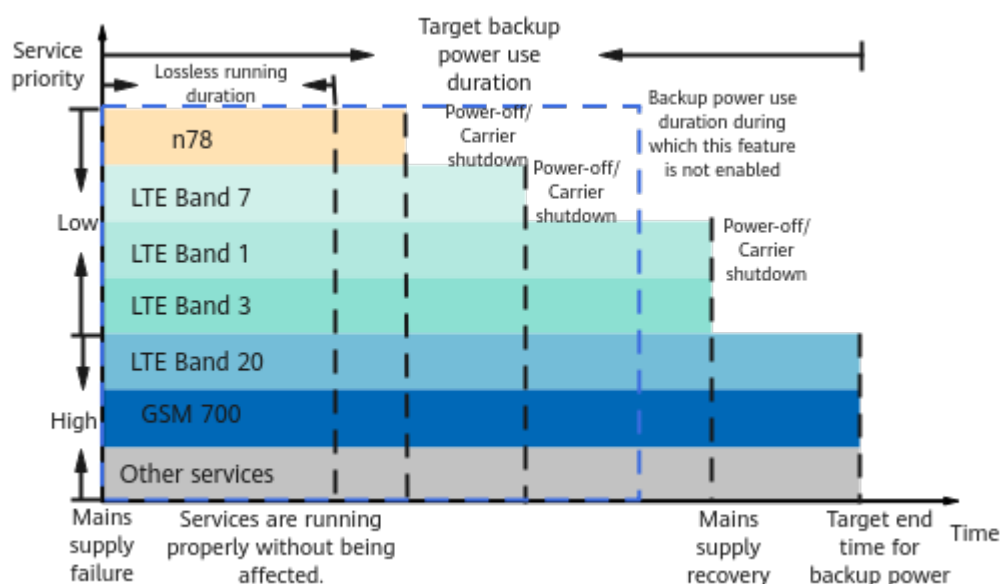
SitePower Backup Policy Management provides an intelligent power backup solution. When the backup power configuration cannot meet the target backup power use duration, this solution shuts down low-priority services at a proper time through module power-off or carrier shutdown based on the service priorities

defined by the operator, historical power consumption, service load trend, and remaining battery capacity. In this way, the service time of sites is extended and high-priority services can operate for the target backup power use duration with a small amount of or no additional backup power investment.

- When the mains supply is normal, high- and low-priority services both operate.
- When the mains supply is disconnected:
 - High- and low-priority services both operate within the lossless running duration.
 - Low-priority services may be preferentially shut down beyond the lossless running duration.

Figure 16-2 shows the principles of SitePower Backup Policy Management.

Figure 16-2 SitePower Backup Policy Management



Service priorities involved in SitePower Backup Policy Management are defined as follows:

- High-priority services: services that must not be affected in the case of a power failure, including BBU services, operator-defined high-priority services, and other related services
 - BBU services: Service operations must be ensured when SitePower Backup Policy Management is in effect.
 - Operator-defined high-priority services: services whose priority is defined as "high"
 - Other related services: services of devices other than Huawei wireless base stations, such as transmission devices and switches
- Low-priority services: services that can be affected in the case of a power failure and whose priority is defined as "low". Low-priority services can be distinguished by priority. A lower energy saving priority results in a shorter

backup power use duration and earlier power-off. A higher backup power use duration is ensured for high-priority services.

 NOTE

When microwave services need to be considered as the most primary services of a site, SitePower Backup Policy Management allows all services to be set as low-priority services and backup power resources to be preferentially used for microwave services.

Restrictions on configuring service priorities are as follows:

- The sites to be deployed are selected for a task and comply with the same priority rule.
- All frequency bands can be set to low-priority to guarantee transmission services and can also be set to high-priority (they are set to high-priority by default in GSM).
- For high-priority services, the energy saving priority is set to "Not involved" by default.
- Services with different priorities can have the same energy saving priority.
- Energy saving priorities are distinguished by value. The smaller the value, the earlier the power-off or carrier shutdown is performed. The larger the value, the later the power-off or carrier shutdown is performed. SitePower Backup Policy Management may perform power-off or carrier shutdown for multiple services with different priorities at the same time.
- The values corresponding to the energy saving priorities can be discontinuous.
- The frequency band parameters configured for the sites must match the frequency band information listed in the tool table.
- If no cell (for example, SUL cell) or service is matched, the service is processed as a high-priority service.

The task creation, service priority, lossless running duration, and target backup power use duration for SitePower Backup Policy Management are configured on the MAE-Access. [Table 16-1](#) and [Table 16-2](#) describe the configurations of the service priority, lossless running duration, and target backup power use duration mentioned in [Figure 16-2](#).

Table 16-1 Service priority configurations in SitePower Backup Policy Management

No.	RAT	Frequency Band	Priority	Energy Saving Priority
1	NR	n78	Low	1
2	LTE	7	Low	2
3	LTE	1	Low	3
4	LTE	3	Low	3
5	LTE	20	High	Not involved
6	GSM	GSM900	High	Not involved

Table 16-2 Related configurations in SitePower Backup Policy Management

Definition	Description	Configuration Mode	Unit	Value Range
Lossless running duration	Duration within which all services are not affected	Manual	Hour (h)	[0, 12]; < target backup power use duration
Target backup power use duration	Customer-expected backup power use duration for key services	Manual	Hour (h)	[0.5, 12]
Expected mains recovery time	Assumed mains recovery time, which is calculated by this feature to reduce traffic loss during the period when the mains supply fails and is not the actual power outage period	Feature-defined: Mains supply failure time + 0.8 x Target backup power use duration	None	None

Application Scenario

SitePower Backup Policy Management applies to the scenarios where:

- The backup power use duration is insufficient because capacity expansion is enabled for wireless networks but not for backup power.
- The backup power use duration is insufficient due to battery aging.
- The backup power use duration is insufficient due to mains supply deterioration.

16.2 Network Analysis

16.2.1 Benefits

When the mains supply is disconnected and the backup power configuration cannot meet the target backup power use duration, SitePower Backup Policy Management shuts down low-priority services (at the cell level) at a proper time

based on the service priorities defined by the operator, historical power consumption, service load trend, and remaining battery capacity. In this way, the service time of sites is extended and high-priority services can operate for the target backup power use duration with a small amount of or no additional backup power investment. After SitePower Backup Policy Management takes effect, the backup power use duration for high-priority services can be extended up to twice the standard length.

16.2.2 Impacts

The impacts between this function and other functions are described in [15.3.2.3 Function Impacts](#).

When the mains supply is disconnected, non-primary services may be shut down beyond the lossless running duration. As a result, the corresponding services are unavailable.

16.3 Requirements

16.3.1 Licenses

RAT	Feature ID	Feature Name	Model	Sales Unit
LTE TDD	TDLOFD-001040	Low Power Consumption Mode	LT1ST0LPCM00	Per Cell
NR	FOFD-021203	gNodeB Power Saving	NR0S0GNBPS00	per Cell
UMTS	WRFD-241203	UMTS Low Power Consumption Mode	LQW9ULPCM01	per NodeB

MAE licenses are required. For details, contact Huawei engineers.

16.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

16.3.2.1 Prerequisite Functions

None

16.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
GS M UM TS LTE FDD LTE TDD NR	PV deployment for base stations Third-party green power access measurement and O&M management	None	9 PV Deployment for Base Stations 10 Third-Party Green Power Access Measurement and O&M Management	When SitePower Backup Policy Management is in effect, PV deployment for base stations and third-party green power access measurement and O&M management cannot be enabled. This is because the PV deployment function will cause the calculation results related to the Adaptive Power Backup feature to be inaccurate.
GS M UM TS LTE FDD LTE TDD NR	Base station hierarchical power-off	(1) Scenarios where an APM power system is used: PDSATTR.CS DTS (LTE eNodeB, 5G gNodeB) (2) Scenarios where an MTS series power system is used or an iSDU06D/iSDU03D interconnected with an MTS series power system using SNMP is used: PDPATTR.CSD TS (LTE eNodeB, 5G gNodeB)	14 Base Station Hierarchical Power-Off	SitePower Backup Policy Management requires base station hierarchical power-off to take effect. When SitePower Backup Policy Management is in effect, you are not advised to modify the configurations related to base station hierarchical power-off.

RAT	Function Name	Function Switch	Reference	Description
GS M UM TS LTE FDD LTE TDD NR	Power Backup Perception	None	19 Power Backup Perception (Trial)	After SitePower Backup Policy Management takes effect, the backup power use duration perceived by Power Backup Perception may be inaccurate.
GS M UM TS LTE FDD LTE TDD	SOC level-based low power consumption mode	• LTE: SOCLVL_B ASED_LO W_PWR_S W option of the CellLowPower.LowPowerConsumptionModeExtSw parameter • UMTS: SOCLVL_B ASED_LO W_PWR_S W option of the NODEBALGPARA.LOWPWRCONSUMEMODEEXTSW parameter • GSM: GBTSENERGYMGTPARA.SOCLVLLOWPWRMODESW	13 SOC Level-based Low Power Consumption Mode	SitePower Backup Policy Management does not take effect after SOC level-based low power consumption mode takes effect.

RAT	Function Name	Function Switch	Reference	Description
GS M UM TS LTE FDD LTE TDD NR	Staggering electricity usage	EQUIPMENT. STAGGEREDP OWERSWITC H (LTE eNodeB, 5G gNodeB)	8 Staggering Electricity Usage	After staggering electricity usage takes effect, power backup capacity decreases. As a result, the site may fail to run for the target backup power use duration. Staggering electricity usage also affects the historical power consumption statistics of the site. As a result, the statistics of the entire site do not match the traffic statistics of the main equipment. If staggering electricity usage takes effect after SitePower Backup Policy Management, this affects power saving benefits of staggering electricity usage.

RAT	Function Name	Function Switch	Reference	Description
UM TS LTE FDD LTE TDD NR	Low power consumption mode	- NR: LOW_PWR_MODE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter - LTE: CellLowPower.LowPowerSwitch - UMTS: NODEBALGPARA.LOWPWRCO NSUMPMODESW	11 Low Power Consumption Mode (LTE) 12 Low Power Consumption Mode (NR) <i>Green BTS</i>	SitePower Backup Policy Management requires low power consumption mode to take effect. When SitePower Backup Policy Management is in effect, you are not advised to modify the configurations related to low power consumption mode.

16.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description (Optional)
GS M UM TS LTE FDD LTE TDD NR	Load low voltage disconnection (LLVD)	None	None	LLVD is usually performed at the end of battery discharge. SitePower Backup Policy Management does not affect the LLVD function. If LLVD takes effect before SitePower Backup Policy Management, the policy of SitePower Backup Policy Management will be updated again.

RAT	Function Name	Function Switch	Reference	Description (Optional)
GS M UM TS LTE FDD LTE TDD NR	Energy saving	None	<i>Energy Conservation and Emission Reduction</i> <i>Multi-RAT Coordinated Energy Saving</i>	If SitePower Backup Policy Management and a certain energy saving function are both enabled, the function that is enabled first takes effect. That is, if SitePower Backup Policy Management is in effect, the energy saving function cannot take effect. If the energy saving function is in effect, the policy of SitePower Backup Policy Management may be updated again.

16.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

Boards

Main control boards:

- Scenarios where a Huawei APM cabinet, an MTS series power system, or an iSDU06D/iSDU03D interconnected with an MTS series power system using SNMP is used: Only the main control boards that are upgraded to V100R021C10 or later supports SitePower Backup Policy Management.
- Scenarios where an MTS series power system is used with a DCPU in an intelligent subrack: Only the main control boards that are upgraded to V100R022C10 or later supports SitePower Backup Policy Management.

Baseband processing units: All baseband processing units support this function.

RF Modules

No requirements

Power Supply

Only Huawei lithium batteries support this function. The use of only lead-acid batteries and hybrid use of lead-acid and lithium batteries are both inapplicable. The requirements on power systems are as follows:

- Scenarios where an APM power system is used: Only the APM5930, APM5950H, and APM5950H-L support this function.
- Scenarios where an MTS series power system is used: For details, contact Huawei engineers.

NOTE

1. In scenarios where an MTS series power system is used, power-off management is supported only when the ports on the MTS series power system in an intelligent subrack are in the customized load or BLVD load state.
 2. Power-off is supported if an MTS series power system is used with a DCDCU in an intelligent subrack.
- Scenarios where an iSDU06D/iSDU03D interconnected with an MTS series power system using SNMP is used: The software and hardware requirements of SNMP-based interconnection must be met. For details, see [17.1 iSDU06D/iSDU03D Support for Connection with a Third-Party Power System \(Using the SNMP\)](#).

NOTE

In scenarios where an iSDU06D/iSDU03D interconnected with an MTS series power system using SNMP is used, the MTS series power system can be connected to the NetEco through BBU in-band networking.

Only centralized power supply is supported. In the centralized power supply scenario, the BBU and RRU use the same power supply and backup system.

16.3.4 Others

SitePower Backup Policy Management requires that all main equipment be provided by Huawei.

MAE version:

- Scenarios where a Huawei APM cabinet or an MTS series power system is used: The MAE version must be iMaster MAE V100R025C10 or later.
- Scenarios where an MTS series power system is used with a DCDCU in an intelligent subrack: The MAE version must be iMaster MAE V100R026C10 or later.
- Scenarios where an iSDU06D/iSDU03D interconnected with an MTS series power system using SNMP is used:
 - Scenarios where an MTS series power system is not connected to the NetEco: The MAE version must be iMaster MAE V100R025C10 or later.
 - Scenarios where an MTS series power system is connected to the NetEco through BBU in-band networking: The MAE version must be iMaster MAE V100R026C10 or later.

NEs in the same power supply and backup group must be managed by the same MAE. If NEs in the same group are managed by different MAEs, the calculated target backup power use duration will be incorrect.

16.4 Operation and Maintenance

16.4.1 Data Configuration

16.4.1.1 Data Preparation

1. Subscribe to counters.

SitePower Backup Policy Management is applicable to scenarios where an APM or MTS series power system is used and scenarios where an iSDU06D/iSDU03D interconnected with an MTS series power system using SNMP is used. The requirements are as follows:

- Scenarios where an APM or MTS series power system is used: Before enabling SitePower Backup Policy Management, check whether the counters listed in the following table have been subscribed to during a 15-, 30-, or 60-minute measurement period.

Counter ID	Counter Name	Description
1593835640	VS.EnergyCons.BTS.Measuring.GSM	Power consumption measured by the power system of the GSM
1593835629	VS.EnergyCons.BTS.Measuring.UMTS	Power consumption measured by the power system of UMTS
1593835638	VS.EnergyCons.BTS.Measuring.LTE	Power consumption measured by the power system of the LTE
1593835805	VS.EnergyCons.BTS.Measuring.NR	Power consumption measured by the power system of NR
1593835639	VS.EnergyCons.BTS.Adding.GSM	The sum of power consumption of all boards in the GSM
1593835628	VS.EnergyCons.BTS.Adding.UMTS	The sum of power consumption of all boards in UMTS
1593835637	VS.EnergyCons.BTS.Adding.LTE	The sum of power consumption of all boards in the LTE
1593835804	VS.EnergyCons.BTS.Adding.NR	The sum of power consumption of all boards in NR

Counter ID	Counter Name	Description
1593835957	VS.EnergyCons.BTS.Adding.Service.GSM	Sum of power consumption of all boards (GSM services)
1593835958	VS.EnergyCons.BTS.Adding.Service.UMTS	Sum of power consumption of all boards (UMTS services)
1593835959	VS.EnergyCons.BTS.Adding.Service.LTE	Sum of power consumption of all boards (LTE services)
1593835961	VS.EnergyCons.BTS.Adding.Service.NR	Sum of power consumption of all boards (NR services)
73435300	VS.TCP.Usage.Mean	Average TCP Usage for Cell
1526726740	L.ChMeas.PR.B.DL.Used.Avg	Average number of used PDSCH PRBs
1526728433	L.ChMeas.PR.B.DL.Avail	Number of available downlink PRBs
1911816540	N.PR.B.DL.Used.Avg	Average number of used downlink PRBs
1911816679	N.PR.B.DL.Avail.Avg	Average number of available downlink PRBs
1593835836	VS.EnergyCons.BTSBoard (LTE eNodeB, 5G gNodeB)	Power consumption of board
1593835915	VS.PMUBoard.SolarOutputEnergy (LTE eNodeB, 5G gNodeB)	Solar Output Energy Yield
1593835976 ^a	VS.PMUBoard.DCLoadEnergy (5G gNodeB, LTE eNodeB)	DC Load Power Consumption
a: It is recommended that VS.PMUBoard.DCLoadEnergy (5G gNodeB, LTE eNodeB) be used for energy consumption measurement because the measurement using this counter is more accurate than that using VS.EnergyCons.BTS.Measuring.GSM , VS.EnergyCons.BTS.Measuring.UMTS , VS.EnergyCons.BTS.Measuring.LTE , and VS.EnergyCons.BTS.Measuring.NR .		

- Scenarios where an iSDU06D/iSDU03D interconnected with an MTS series power system using SNMP is used: Before enabling SitePower Backup Policy Management, check whether the counters listed in the following table have been subscribed to during a 15-, 30-, or 60-minute measurement period.

Counter ID	Counter Name	Description
1593835639	VS.EnergyCons.BTS.Adding.GSM	The sum of power consumption of all boards in the GSM
1593835628	VS.EnergyCons.BTS.Adding.UMTS	The sum of power consumption of all boards in UMTS
1593835637	VS.EnergyCons.BTS.Adding.LTE	The sum of power consumption of all boards in the LTE
1593835804	VS.EnergyCons.BTS.Adding.NR	The sum of power consumption of all boards in NR
1593835957	VS.EnergyCons.BTS.Adding.Service.GSM	Sum of power consumption of all boards (GSM services)
1593835958	VS.EnergyCons.BTS.Adding.Service.UMTS	Sum of power consumption of all boards (UMTS services)
1593835959	VS.EnergyCons.BTS.Adding.Service.LTE	Sum of power consumption of all boards (LTE services)
1593835961	VS.EnergyCons.BTS.Adding.Service.NR	Sum of power consumption of all boards (NR services)
73435300	VS.TCP.Usage.Mean	Average TCP Usage for Cell
1526726740	L.ChMeas.PRBDL.Used.Avg	Average number of used PDSCH PRBs
1526728433	L.ChMeas.PRBDL.Avail	Number of available downlink PRBs
1911816540	N.PRBDL.Used.Avg	Average number of used downlink PRBs
1911816679	N.PRBDL.Avail.Avg	Average number of available downlink PRBs
1593835836	VS.EnergyCons.BTSBoard (LTE eNodeB, 5G gNodeB)	Power consumption of board
1593835942	VS.EPMU.DCEnergy (LTE eNodeB, 5G gNodeB)	DC energy of the EPMU
1593835915	VS.PMUBoard.SolarOutputEnergy (LTE eNodeB, 5G gNodeB)	Solar Output Energy Yield

Counter ID	Counter Name	Description
1593835976 ^a	VS.PMUBoard.DCLoadEnergy (5G gNodeB, LTE eNodeB)	DC Load Power Consumption
a: It is recommended that VS.PMUBoard.DCLoadEnergy (5G gNodeB, LTE eNodeB) be used for energy consumption measurement because the measurement using this counter is more accurate than that using VS.EnergyCons.BTS.Measuring.GSM , VS.EnergyCons.BTS.Measuring.UMTS , VS.EnergyCons.BTS.Measuring.LTE , and VS.EnergyCons.BTS.Measuring.NR .		

- Scenarios where an MTS series power system is used with a DCDCU in an intelligent subrack: Before enabling SitePower Backup Policy Management, check whether the counters listed in the following table have been subscribed to during a 15-, 30-, or 60-minute measurement period.

Counter ID	Counter Name	Description
1593835976	VS.PMUBoard.DCLoadEnergy (5G gNodeB, LTE eNodeB)	DC Load Power Consumption
1593835639	VS.EnergyCons.BTS.Adding.GSM	The sum of power consumption of all boards in the GSM
1593835628	VS.EnergyCons.BTS.Adding.UMTS	The sum of power consumption of all boards in UMTS
1593835637	VS.EnergyCons.BTS.Adding.LTE	The sum of power consumption of all boards in the LTE
1593835804	VS.EnergyCons.BTS.Adding.NR	The sum of power consumption of all boards in NR
73435300	VS.TCP.Usage.Mean	Average TCP Usage for Cell
1526726740	L.ChMeas.PR.B.DL.Used.Avg	Average number of used PDSCH PRBs
1526728433	L.ChMeas.PR.B.DL.Avail	Number of available downlink PRBs
1911816540	N.PR.B.DL.Used.Avg	Average number of used downlink PRBs

Counter ID	Counter Name	Description
1911816679	N.PR.B.DL.Avail.Avg	Average number of available downlink PRBs
1593835836	VS.EnergyCons.BTSBoard (LTE eNodeB, 5G gNodeB)	Power consumption of board
1593835915	VS.PMUBoard.SolarOutputEnergy (LTE eNodeB, 5G gNodeB)	Solar Output Energy Yield
1593835942	VS.EPMU.DCEnergy (LTE eNodeB, 5G gNodeB)	DC energy of the EPMU

2. Obtain information about NEs at the site and identify the mapping between physical sites and logical sites.

SitePower Backup Policy Management is designed on a per-physical-site basis. That is, the entire site is powered by one power system. The MAE performs management on a per-logical-site basis. In actual scenarios, multiple NEs may be deployed at one physical site. Currently, the mapping between physical sites and logical sites can only be manually identified.

NOTE

For example, a site is configured with two BBUs, which are managed as NE 01 and NE 02 on the MAE and powered by one power system. You need to manually identify that NE 01 and NE 02 are deployed at the same physical site.

3. Configure the power supply topology.

SitePower Backup Policy Management supports two energy saving modes: module power-off and carrier shutdown. Modules are powered off through the power supply. Therefore, the power supply topology information needs to be obtained for SitePower Backup Policy Management. The methods of obtaining power supply topology information in scenarios with different power supplies are as follows:

- Scenarios where an APM power system is used: Automatic identification and reporting are not supported. You need to run the **MOD PDSATTR** command to implement manual configuration with the correct ESN used.
- Scenarios where an MTS series power system is used: Automatic identification and reporting are not supported. You need to run the **MOD PDPATTR** command to implement manual configuration with the correct ESN used.
- Scenarios where an iSDU06D/iSDU03D interconnected with an MTS series power system using SNMP is used: You need to run the **MOD PDPATTR** command with the **Topology Recognition Mode** parameter set to **AUTO** for the corresponding power supply port.

16.4.1.2 Using MML Commands

Before using MML commands, refer to [15.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency,

and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

SitePower Backup Policy Management can be configured using the MAE-Access. For details, contact Huawei engineers. When a Huawei power system or MTS series power system is used and module power-off is required, the power supply topology information must be configured.

```
//Example of configuring the power supply topology information when a Huawei power system is used
MOD PDSATTR: CN=0, SN=6,
ESNGROUP="21023000000000000001&21023000000000000002&21023000000000000003";
//Example of configuring the power supply topology information when an MTS series power system is used
MOD PDPATTR: CN=0, SRN=17, PORTNO=1, TOPORECOGMODE=MANUAL, ESN="21023000000000000001";
//Example of configuring several pieces of power supply topology information when an MTS series power
system is used
MOD PDPATTR: CN=0, SRN=17, PORTNO=1, TOPORECOGMODE=MANUAL,
ESN="21023000000000000001&21023000000000000002&21023000000000000003";
//Example of configuring the topology recognition mode of the corresponding power supply port when an
ISDU06D/iSDU03D interconnected with an MTS series power system using SNMP is used
MOD PDPATTR: CN=0, SRN=17, SN=0, PORTNO=0, TOPORECOGMODE=AUTO;
```

16.4.1.3 Using the MAE-Access

The task creation, service priority, lossless running duration, target backup power use duration, and other information related to SitePower Backup Policy Management are configured on the MAE-Access. For details, contact Huawei engineers.

16.4.2 Activation Verification

This feature takes effect when the site meets the license, software, and hardware verification requirements. If the mains supply is disconnected, the base station proactively reports that mains supply unavailability. After the MAE recognizes the mains supply disconnection, SitePower Backup Policy Management takes effect. You can view the report exported from the MAE-Access to check the software and hardware verification status and running status of the site. For details, contact Huawei engineers.

Observe whether module power-off and carrier shutdown take effect to determine whether SitePower Backup Policy Management takes effect.

Module Power-Off

The following describes how to observe whether module power-off takes effect in scenarios with different power supplies:

- Scenarios where an APM power system is used: Run the **DSP PDSATTR** command. If the value of **Power Distribution Unit Status** corresponding to the target slot is **Power-off**, the specified module has been powered off.
- Scenarios where an MTS series power system is used: Run the **DSP PDPATTR** command. If the value of **Port Status** corresponding to the target slot is **POWER_OFF**, the specified module has been powered off.

- Scenarios where an iSDU06D/iSDU03D interconnected with an MTS series power system using SNMP is used: Run the **DSP PDPATTR** command. If the value of **Port Status** corresponding to the target slot is **POWER_OFF**, the specified module has been powered off.

Carrier Shutdown

To observe whether carrier shutdown takes effect, perform the following steps:

Step 1 Check whether SitePower Backup Policy Management delivers the power saving level by running the **DSP BACKUPPWRSAVINGLV** command. If **Base Station Backup Power Saving Reason** has a value of **MAE_REPORT:YES**, SitePower Backup Policy Management has delivered the power saving level to trigger carrier shutdown.

Step 2 Check whether carrier shutdown has taken effect.

- On the NR side, carrier shutdown has taken effect if the following conditions are met:
In the **DSP NRDUCELL** command output, **Power Saving State** has a value of **Low Power Consumption Mode:Enabled**.
In the **DSP NRDUCELLPOWERSAVING** command output, the value of **Current State** corresponding to **Power Saving Type** whose value is **Low Power Consumption Mode** is **Enabled**.
- On the LTE side, carrier shutdown has taken effect if the following conditions are met:
In the **DSP CELLLOWPOWER** command output, the value of **Cell Shutdown Status** is **ENABLED**.
In the **DSP CELL** command output, the value of **Cell power save state** is **Install**.
- On the UMTS side, carrier shutdown has taken effect if the following condition is met:
In the **DSP ULOCELLPWRSAV** command output, the value of **Low Power Consumption Mode Power Saving State** is **Enable**.

----End

16.4.3 Network Monitoring

When the mains supply is disconnected, non-primary services may be shut down beyond the lossless running duration. If this occurs, the corresponding services are unavailable.

After the mains supply is disconnected, a report will be generated based on the list of sites with SitePower Backup Policy Management enabled. The report displays information such as the saved battery configuration and estimated backup power use duration. You can view the report on the MAE-Access. For details, contact Huawei engineers.

17 iSDU06D/iSDU03D Support for Connection with a Third-Party Power System

Currently, operators are gradually reducing traditional asset-heavy operations of base stations and using different vendors' auxiliary equipment except base station equipment, including site power equipment. This may lead to the phenomenon that the power supply, power distribution, and power backup equipment used for a site is provided by different vendors. To help operators perform unified monitoring and management of different power equipment, iSDU06D/iSDU03D support for connection with a third-party power system is introduced. With the function, a third-party power system is connected to the iSDU06D/iSDU03D and then connected to a wireless BBU through the iSDU06D/iSDU03D.

The base station communicates with the third-party power system through the SNMP to obtain basic information about the third-party power system for O&M.

17.1 iSDU06D/iSDU03D Support for Connection with a Third-Party Power System (Using the SNMP)

17.1.1 Principles

The base station communicates with a third-party power system through the SNMP to obtain basic information about the third-party power system for O&M.

17.1.2 Network Analysis

17.1.2.1 Benefits

The interconnection between wireless base stations and third-party power systems is implemented, streamlining dumb pipes and building a foundation for linkage channels. The basic power supply information about a third-party power system can be obtained through this interconnection.

17.1.2.2 Impacts

The impacts between this function and other functions are described in [17.1.3.2.3 Function Impacts](#).

This function has no impact on the network.

17.1.3 Requirements

17.1.3.1 Licenses

None

17.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

17.1.3.2.1 Prerequisite Functions

None

17.1.3.2.2 Mutually Exclusive Functions

The iSDU06D/iSDU03D must be connected to the third-party power system over the Ethernet port on the main control board of the base station and be configured with the IPv4 address of this port.

If the transmission configuration model (controlled by **GTRANSPARA.TRANS CFGMODE**) is an old model, run the **ADD DEVIP** command to configure the device IP address and VRF index. When the new model is used, run the **ADD IPADDR4** command to configure an IPv4 address and a VRF index, and run the **ADD INTERFACE** command to add an interface.

The physical Ethernet port on the main control board, configured IP address, configured VRF index, and configured interface (when the new model is used) can be used only for service configurations related to the **SNMPMNG** MO involved in this function.

17.1.3.2.3 Function Impacts

In a version earlier than BTS3900 V100R020C00, if the transmission configuration model is an old model (**GTRANSPARA.TRANS CFGMODE** is set to **OLD**), a device IP address with **DEVIP.PT** set to **ETH** and **DEVIP.VRFIDX** set to a value greater than **0** cannot be added by running the **ADD DEVIP** command.

Such a device IP address can be configured in BTS3900 V100R020C00 and later versions.

17.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

Boards

- Main control boards: Only the UMPTg and UMPTga series boards support this function. Physical connections require an Ethernet port on the main control board.
- Baseband processing units: All baseband processing units support this function.
- iSDU06D: Only the PMU 12B can be installed in the iSDU06D.

RF Modules

None

Third-Party Power Systems

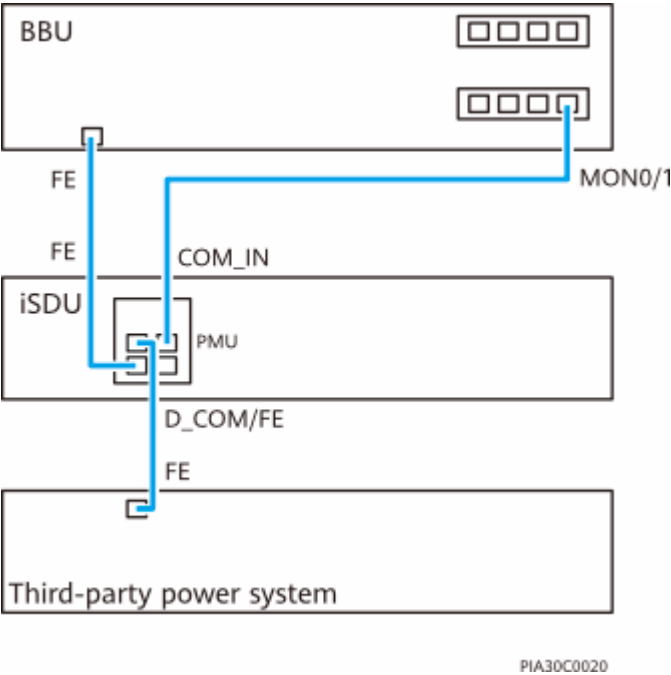
- Physical connections require a northbound monitoring network port on the third-party power system.
- This function has requirements on third-party power system capabilities. For details, contact Huawei engineers.
- The third-party power system supports only SNMPv3.

17.1.3.4 Networking

Figure 17-1 shows the networking requirement of iSDU06D/iSDU03D support for connection with a third-party power system.

- The northbound monitoring port (FE) on the third-party power system is connected to the D_COM/FE port on the iSDU06D/iSDU03D.
- The FE port on the iSDU06D/iSDU03D is connected to the FE port on the UMPT in the BBU.
- The COM_IN port on the iSDU06D/iSDU03D (used for northbound monitoring for the iSDU06D/iSDU03D) is connected to the MON0/1 port on the BBU.

Figure 17-1 Networking requirement of iSDU06D/iSDU03D support for connection with a third-party power system



17.1.3.5 Others

Whether the Function Is Supported by Base Stations in Different RATs

The following table lists whether iSDU06D/iSDU03D support for connection with a third-party power system is supported by base stations in different RATs.

Table 17-1 Whether the function is supported by base stations in different RATs

Base Station RAT Scenario	Example of Base Station RATs	Supported or Not
Multimode	GSM+UMTS+LTE+NR or a combination of some of these RATs	Supported
Single-mode (LTE-only or NR-only)	LTE or NR	Supported
Single-mode (except LTE-only or NR-only)	GSM or UMTS	Not supported

Whether the Function Is Supported by a Co-MPT Base Station and a Separate-MPT Base Station

The following table lists whether iSDU06D/iSDU03D support for connection with a third-party power system is supported by a co-MPT base station and a separate-MPT base station.

Table 17-2 Whether the function is supported by a co-MPT base station and a separate-MPT base station

Scenario	Supported or Not
Co-MPT	Supported
Separate-MPT	The third-party power system configuration and basic O&M can be performed for only one RAT.
Active and standby UMPT cold backup	Supported. The standby UMPT forwards ETH packets.

Requirements for the iSDU06D/iSDU03D to Obtain the Status of the Mains Supply to a Third-Party Power System

The iSDU06D/iSDU03D can obtain the mains status only if it is interconnected with a third-party power system using the SNMP or bound to a dry contact.

Number of Configurable EPMUs and Protocol Requirements

- An EPMU interconnected using the SNMP and an EPMU interconnected using the YDN protocol cannot be configured in the same NE.
- A maximum of one EPMU interconnected using the SNMP can be configured.

17.1.4 Operation and Maintenance

17.1.4.1 Data Configuration

17.1.4.1.1 Data Preparation

The following table describes configuration objects related to iSDU06D/iSDU03D support for connection with a third-party power system.

Table 17-3 Configuration objects related to iSDU06D/iSDU03D support for connection with a third-party power system

Object Name	MO	Description
PMU	PMU	This function requires at least one iSDU06D/iSDU03D.
EPMU	EPMU	This function requires an EPMU, which is a power monitoring module of a third-party power system.
SNMP manager	SNMPMNG	This MO is used to add and configure an SNMP manager.

Object Name	MO	Description
SNMP authentication user	SNMPAUTHUSER	This MO is used to add an SNMP authentication user and configure user authentication information in SNMP.
SNMP agent	SNMPAGT	This MO is used to add and configure an SNMP agent.

The following table describes iSDU06D/iSDU03D parameters related to iSDU06D/iSDU03D support for connection with a third-party power system.

Table 17-4 iSDU06D/iSDU03D parameters related to iSDU06D/iSDU03D support for connection with a third-party power system

Parameter Name	Parameter ID	Description
Power System Type	PMU.PTYPE	This function requires the PMU whose power system type is iSDU06D/iSDU03D.
Port Mode	PMU.PM	This parameter specifies the working mode of the D_COM/FE port. This parameter is valid only for the PMU whose power system type is iSDU06D/iSDU03D. This function requires that the iSDU06D/iSDU03D work in ETH mode.
Power Supply Group ID	PMU.PSGID	This parameter specifies the ID of the power supply group to which the PMU belongs. This parameter is valid only for the PMU whose power system type is iSDU06D/iSDU03D. The value 255 indicates an invalid value. This function requires that the iSDU06D/iSDU03D and the third-party power system (supplying power to the iSDU06D/iSDU03D) be configured with the same power supply group ID. If the PMU.PSGID parameter is set to the same value for the iSDU and third-party power system, there must be a PMU for which the PMU.PM (LTE eNodeB, 5G gNodeB) parameter is set to ETH .

The following table describes SNMP manager parameters related to iSDU06D/iSDU03D support for connection with a third-party power system.

Table 17-5 SNMP manager parameters related to iSDU06D/iSDU03D support for connection with a third-party power system

Parameter Name	Parameter ID	Description
Manager ID	SNMPMNG.MAN AGERID	This parameter specifies the ID of the SNMP manager. This function requires the planning of IP addresses of the connected ports on the base station and the third-party power system.
IP Address	SNMPMNG.IP	This parameter specifies the IP address of the SNMP manager.
VRF Index	SNMPMNG.VRFID X	This parameter specifies the index of a VRF instance.

The following table describes SNMP authentication user parameters related to iSDU06D/iSDU03D support for connection with a third-party power system.

Table 17-6 SNMP authentication user parameters related to iSDU06D/iSDU03D support for connection with a third-party power system

Parameter Name	Parameter ID	Description
User Name	SNMPAUTHUSER. USERNAME	This parameter specifies the user name used for SNMPv3 authentication.
Authenticat ion Algorithm	SNMPAUTHUSER. AUTHALG	This parameter specifies the authentication algorithm used for SNMPv3 authentication.
Encryption Algorithm	SNMPAUTHUSER. ENCRYPTALG	This parameter specifies the encryption algorithm for SNMPv3 packets.

Parameter Name	Parameter ID	Description
Authentication Password	SNMPAUTHUSER. AUTHPWD	This parameter specifies the authentication password used for SNMPv3 authentication. The password must be the same as that set on the SNMP agent (that is, the third-party device). The recommended password rules are as follows: (1) The password must contain at least two of the following: uppercase letters, lowercase letters, digits, and special characters. (2) The password must contain at least eight characters. (3) The password cannot contain duplicate character strings. (4) The password must be different from the user name. (5) The authentication password must be different from the encryption password.
Encryption Password	SNMPAUTHUSER. ENCRYPTPWD	This parameter specifies the encryption password used for SNMPv3 authentication. The password must be the same as that set on the SNMP agent (that is, the third-party device). The recommended password rules are as follows: (1) The password must contain at least two of the following: uppercase letters, lowercase letters, digits, and special characters. (2) The password must contain at least eight characters. (3) The password cannot contain duplicate character strings. (4) The password must be different from the user name. (5) The authentication password must be different from the encryption password.

The following table describes SNMP agent parameters related to iSDU06D/iSDU03D support for connection with a third-party power system.

Table 17-7 SNMP agent parameters related to iSDU06D/iSDU03D support for connection with a third-party power system

Parameter Name	Parameter ID	Description
Agent ID	SNMPAGT.AGENT ID	This parameter specifies the ID of the SNMP agent.

Parameter Name	Parameter ID	Description
Agent IP	SNMPAGT.AGENT IP	This parameter specifies the IP address of the SNMP agent. This function requires the planning of IP addresses of the connected ports on the base station and the third-party power system.
Agent Port	SNMPAGT.AGENT PORT	This parameter specifies the number of the port enabled on the SNMP agent.
User Name	SNMPAGT.USERN AME	This parameter specifies the user name used for communication between the local end and the agent.

17.1.4.1.2 Using MML Commands

Before using MML commands, refer to [17.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- Step 1** Run the **ADD CABINET** command to add a cabinet where the iSDU06D/iSDU03D to be added will be located.

```
//Adding a cabinet (For example, the cabinet is a virtual cabinet whose number is 10.)
ADD CABINET: CN=10, TYPE=VIRTUAL;
```

- Step 2** Run the **ADD PMU** command to add a PMU whose power system type is iSDU06D/iSDU03D.

```
//Adding an iSDU06D/iSDU03D directly connected to a third-party power system (For example, the port
mode of the iSDU06D/iSDU03D is ETH and the planned power supply group ID is 0.)
ADD PMU: CN=10, SRN=17, PTYPE=ISDU, MPN=0, LSDF=DISABLE, PM=ETH, PSGID=0;
```

- Step 3** Run the **ADD SNMPMNG** command to add an SNMP manager.

```
//Adding an SNMP manager (For example, the IP address is 192.168.0.20, a planned IP address of the FE
port on the base station, and the planned VRF index is 7.)
ADD SNMPMNG: MANAGERID=0, IP="192.168.0.20", VRFIDX=7;
```

- Step 4** Run the **ADD SNMPAUTHUSER** command to add an SNMP authentication user.

```
//Adding an SNMP authentication user (For example, the user name is User0, the SHA-512 authentication
algorithm and the AES-256 encryption algorithm are used, and changeme1 and changeme2 are used as the
authentication and encryption passwords, respectively.)
ADD SNMPAUTHUSER: USERNAME="User0", AUTHALG=SHA-512, ENCRYPTALG=AES-256, AUTHPWD="*****",
ENCRYPTPWD="*****";
```


Step 5 Run the **ADD SNMPAGT** command to add an SNMP agent.

```
//Adding an SNMP agent (For example, the agent ID is 65535, the agent IP address is 192.168.0.10, a  
planned IP address of the third-party power system, the port number is 161, and the user name is User0.)  
ADD SNMPAGT: AGENTID=65535, AGENTIP="192.168.0.10", AGENTPORT=161, USERNAME="User0";
```

Step 6 Run the **ADD CABINET** command to add a cabinet where the EPMU to be added will be located.

```
//Adding a cabinet (For example, the cabinet is a virtual cabinet whose number is 62.)  
ADD CABINET: CN=62, TYPE=VIRTUAL;
```

Step 7 Run the **ADD EPMU** command to add an EPMU.

```
//Adding an EPMU (For example, the cabinet number is 62, the subrack number is 41, the agent ID is  
65535, the planned power supply group ID is 0, the MIB index is 0, and the Digital Power product MTS  
power supply is used.)  
ADD EPMU: CN=62, SRN=41, DESC="MTS", AGENTID=65535 , PSGID=0, MIBINDEX=0;
```

----End

Deactivation Command Examples

Step 1 Run the **RMV EPMU** command to remove an EPMU.

```
//Removing an EPMU  
RMV EPMU: CN=62, SRN=41;
```

Step 2 Run the **RMV CABINET** command to remove the cabinet where the EPMU is located.

```
//Removing a cabinet  
RMV CABINET: CN=62;
```

Step 3 Run the **RMV SNMPAGT** command to remove an SNMP agent.

```
//Removing an SNMP agent  
RMV SNMPAGT: AGENTID=65535;
```

Step 4 Run the **RMV SNMPAUTHUSER** command to remove an SNMP authentication user.

```
//Removing an SNMP authentication user  
RMV SNMPAUTHUSER: USERNAME="User0";
```

Step 5 Run the **RMV SNMPMNG** command to remove an SNMP manager.

```
//Removing an SNMP manager  
RMV SNMPMNG: MANAGERID=0;
```

Step 6 Run the **RMV PMU** command to remove a PMU whose power system type is iSDU06D/iSDU03D.

```
//Removing a PMU whose power system type is iSDU06D/iSDU03D  
RMV PMU: CN=10, SRN=17;
```

Step 7 Run the **RMV CABINET** command to remove the cabinet where the iSDU06D/iSDU03D is located.

```
//Removing a cabinet  
RMV CABINET: CN=10;
```

----End

17.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

17.1.4.2 Activation Verification

Using MML Commands

1. To query the configuration of an EPMU, run the **LST EPMU** command. This command is used to list the configuration of an EPMU. The following is an example.

```
//Querying the configuration of an EPMU  
LST EPMU: CN=62;
```
2. To query the status of an EPMU, run the **DSP EPMU** command. This command is used to display the dynamic information about an EPMU. The following is an example. If the **DSP EPMU** command is executed successfully and no error message (for example, a message indicating that the EPMU link is unavailable) is returned, this function has taken effect.

```
//Querying the status of an EPMU  
DSP EPMU: CN=62, SRN=41;
```
3. To query the extended information of an EPMU, run the **DSP EPMUEXTINFO** command. This command is used to query the extended information of an EPMU, including its ambient temperature, ambient humidity, and fan speed. The following is an example.

```
//Querying the extended information of an EPMU  
DSP EPMUEXTINFO: CN=62, SRN=41;
```
4. To query the information about the active alarms on an EPMU, run the **DSP EPMUALM** command. This command is used to display the information about the active alarms on an EPMU. The following is an example.

```
//Querying the information about the active alarms on an EPMU  
DSP EPMUALM: CN=62, SRN=41;
```
5. To query the status of an SNMP agent, run the **DSP SNMPAGT** command. This command is used to query the status of an SNMP agent. The following is an example.

```
//Querying the information about the active alarms on an EPMU  
DSP SNMPAGT: AGENTID=65535;
```

Observing Counters

When a third-party power system works properly and is connected to the iSDU06D/iSDU03D, subscribe to the following counters and check their values on the MAE-Access. If the counter values are valid, this function has taken effect.

Table 17-8 Counters of Measurement of External Power System Power Consumption

Description	Counter Name	Meaning	Remarks
DC energy of the EPMU	VS.EPMU.DCEnergy	This counter measures the total DC output energy of the power supply module in the external power system (non-Huawei wireless power system) managed by the BBU in a measurement period.	This counter is measured when the third-party power system works properly and is connected to the iSDU06D/iSDU03D.
AC input energy of the EPMU	VS.EPMU.ACInputEnergy	This counter measures the total AC input energy of the power supply module in the external power system (non-Huawei wireless power system) managed by the BBU in a measurement period.	This counter is measured when the third-party power system works properly and is connected to the iSDU06D/iSDU03D.

17.1.4.3 Network Monitoring

- Alarm monitoring

The following table lists the device faults that can be detected by the system when a third-party power system works properly and is connected to the iSDU06D/iSDU03D.

Table 17-9 EPMU-related alarms

Alarm Name	Alarm Severity	Remarks
ALM-25600 Monitoring Device Maintenance Link Failure	Major	This alarm is reported when the maintenance link connection of a third-party power system is abnormal and the communication fails. Among the alarm parameters, if the value of Board Type is EPMU , an EPMU is used.

Alarm Name	Alarm Severity	Remarks
ALM-25622 Mains Input Out of Range	Major	This alarm is reported when a mains supply exception is detected when a third-party power system works properly and is connected to the iSDU06D/iSDU03D. Among the alarm parameters, if the value of Board Type is EPMU , an EPMU is used.
ALM-25601 Monitoring Device Hardware Fault	Major	This alarm is reported when a monitoring device fault is detected when a third-party power system works properly and is connected to the iSDU06D/iSDU03D. Among the alarm parameters, if the value of Board Type is EPMU , an EPMU is used.

- Counter monitoring

The following table lists the counters that can be subscribed to and checked on the MAE-Access when a third-party power system works properly and is connected to the iSDU06D/iSDU03D.

Table 17-10 EPMU-related counters

Description	Counter Name	Meaning
DC energy of the EPMU	VS.EPMU.DCEnergy	This counter measures the total DC output energy of the power supply module in the external power system (non-Huawei wireless power system) managed by the BBU in a measurement period.
AC input energy of the EPMU	VS.EPMU.ACInput Energy	This counter measures the total AC input energy of the power supply module in the external power system (non-Huawei wireless power system) managed by the BBU in a measurement period.

18 iSDU03D Circuit Breaker Capacity Adjustment

18.1 Principles

The iSDU03D can adjust the circuit breaker capacity of the intelligent power distribution ports LOAD0 to LOAD2. In this case, it can supply power to RF modules regardless of whether their power consumption is greater than 800 W. The circuit breaker capacity can be specified using the **PDPATTR.CBCAPC (LTE eNodeB, 5G gNodeB)** parameter based on the power consumption of RF modules connected to the intelligent power distribution ports.

- If the **PDPATTR.CBCAPC (LTE eNodeB, 5G gNodeB)** parameter is set to **65535**, the current circuit breaker capacity of the power distribution port is retained, which can be queried by running the **DSP PDPATTR** command. The current circuit breaker capacity of the power distribution port is as follows:
 - If the power distribution port is configured for the first time, its current circuit breaker capacity is 50 A.
 - If the power distribution port is not configured for the first time, its current circuit breaker capacity is the previously configured circuit breaker capacity.
- If the **PDPATTR.CBCAPC (LTE eNodeB, 5G gNodeB)** parameter is set to a value other than **65535**, this value is used as the circuit breaker capacity.

The recommended circuit breaker capacity of a power distribution port varies with the power consumption of the connected RF module. If the configured capacity is inappropriate, cables cannot be reliably protected or the RF module may be powered off.

- If the maximum power consumption of an RF module is greater than 800 W, it is recommended that the circuit breaker capacity of the corresponding port be set to 50 A.
- If the maximum power consumption of an RF module is less than or equal to 800 W, it is recommended that the circuit breaker capacity of the corresponding port be set to 25 A.

18.2 Network Analysis

18.2.1 Benefits

The iSDU03D can adjust the circuit breaker capacity. Different from the direct current distribution unit (DCDU), it can supply power to RF modules regardless of whether their power consumption is greater than 800 W.

18.2.2 Impacts

The impacts between this function and other functions are described in [18.3.2.3 Function Impacts](#).

This function has no impact on the network.

18.3 Requirements

18.3.1 Licenses

None

18.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

18.3.2.1 Prerequisite Functions

None

18.3.2.2 Mutually Exclusive Functions

None

18.3.2.3 Function Impacts

None

18.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

DBS3900 LampSite and DBS5900 LampSite

Boards

None

Power Supply

- RF modules must be powered by the iSDU03D.
- iSDU03D circuit breaker capacity adjustment applies only to intelligent power distribution ports. Ports LOAD0 to LOAD2 support this function while ports LOAD3 and LOAD4 do not support this function.

18.3.4 Others

None

18.4 Operation and Maintenance

18.4.1 Data Configuration

18.4.1.1 Data Preparation

During iSDU03D deployment, you need to obtain the power consumption, power distribution solutions, and models of the RF modules corresponding to the ports through the site survey or other methods.

The following table lists the data to be prepared for enabling timing iSDU03D port power-off.

Table 18-1 Parameters for enabling timing iSDU03D port power-off

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	PDPATTR.CN (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter based on the network plan.
Subrack No.	PDPATTR.SRN (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter based on the network plan. The circuit breaker capacity can be adjusted only when Subrack No. is set to 17 or 18 .
Slot No.	PDPATTR.SN (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Port No.	PDPATTR.PORTNO (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan.
Circuit Breaker Capacity	PDPATTR.CBCAPC (LTE eNodeB, 5G gNodeB)	Retain the default value.

18.4.1.2 Using MML Commands

NOTICE

Before running the **MOD PDPATTR** command to set the circuit breaker capacity, run the **DSP PDPATTR** command to query the actual circuit breaker capacity and compare it with that in the information of the connected RF module.

Before using MML commands, refer to [18.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Setting the circuit breaker capacity of the power distribution port whose port, slot, subrack, and cabinet numbers are 0, 0, 17, and 0, respectively, to 65535  
MOD PDPATTR: CN=0, SRN=17, SN=0, PORTNO=0, CBCAPC=65535;
```

Deactivation Command Examples

MML commands are not required for deactivating this function. You can determine whether to restore the settings of other parameters based on actual network conditions.

18.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

18.4.2 Activation Verification

Run the **DSP PDPATTR** command to query the status of power distribution ports in the power subrack and verify that the value of **Circuit Breaker Capacity(A)** is that of **PDPATTR.CBCAPC (LTE eNodeB, 5G gNodeB)**.

18.4.3 Network Monitoring

This function ensures basic network performance and does not need to be separately monitored.

19 Power Backup Perception (Trial)

19.1 Principles

Power Backup Perception predicts the backup power use duration of a site or NE based on its mains status, battery status, remaining battery power, and DC power consumption, and triggers a warning in the following scenarios for the sake of risk prediction based on the site or NE mains status and backup power use duration. Customers can arrange site visits for O&M in advance to prevent site disconnection.

1. Power outage warning: When the mains supply is unavailable, this warning is reported to notify the customers of the mains outage.
2. Potential backup power SLA risk warning: This warning is reported when the following conditions are met: (1) The mains supply is normal. (2) The battery is fully charged. (3) The site or NE backup power use duration is less than the target power backup duration set by the customers.
3. Backup power exhaustion warning: This warning is reported when the mains supply is unavailable and the remaining backup power use duration is less than one hour.

Backup Power Perception applies to GSM/UMTS base stations only when such base stations are co-sited with LTE/NR base stations.

NOTE

SLA is short for Service Level Agreement.

19.2 Network Analysis

19.2.1 Benefits

Benefits to customers	• Traffic loss prevention: Traffic loss caused by site disconnection is prevented. • O&M level improvement with pre-event prevention instead of post-event O&M: Regarding sites with backup power risks, customers are warned of out-of-service or disconnection risks in advance. Mobile diesel generators can be arranged for customers in advance to ensure that VIP sites are always online, helping customers locate the causes of base station disconnection or cell out-of-service faults. • Site visit reduction: Customers can pay accurate site visits for onsite maintenance based on backup power information or identified risky sites, which reduces site visits caused by cell or base station out-of-service for fault locating. • Precise backup power capacity expansion: Power capacity is preferentially expanded for sites with backup power risks.
Benefits to Huawei	Huawei network reliability is improved.

19.2.2 Impacts

The impacts between this function and other functions are described in [19.3.2.3 Function Impacts](#).

This function has no impact on the network.

19.3 Requirements

19.3.1 Licenses

Backup Power Perception needs to be enabled on the MAE-Access. The following table lists the licenses required on the MAE-Access.

RAT	Feature ID	Feature Name	Model	Sales Unit
LTE	MMFD-231210	Battery-caused Cell Outage Risk Prediction (LTE)	LWMSBPFPLC00	per Cell
NR	MMFD-231210	Battery-caused Cell Outage Risk Prediction (NR)	LWMSBPFPCN00	per Cell

19.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

19.3.2.1 Prerequisite Functions

None

19.3.2.2 Mutually Exclusive Functions

None

19.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
NR LTE UMTS GSM	PSU current limiting protection in case of AC circuit breaker capability insufficiency	PMU.ACCLF	<i>Power Supply Management</i>	If PSU current limiting protection in case of AC circuit breaker capability insufficiency is enabled, batteries may discharge proactively, and thus the potential backup power SLA risk warning may be mistakenly reported. If the mains supply fails, the batteries may not be fully charged. As a result, the remaining backup power use duration of the site is less than one hour when the backup power exhaustion risk warning is reported.
NR LTE UMTS GSM	Staggering electricity usage	EQUIPMENT.STAGGEREDPOWERSWITCH (LTE eNodeB, 5G gNodeB)	<i>Green Site</i>	If staggering electricity usage is enabled, batteries may discharge proactively, and thus the potential backup power SLA risk warning may be mistakenly reported. If the mains supply fails, the batteries may not be fully charged. As a result, the remaining backup power use duration of the site is less than one hour when the backup power exhaustion risk warning is reported.

RAT	Function Name	Function Switch	Reference	Description
NR LTE UMTS GSM	Automatic battery testing management	MOD BATCTPA	<i>Power Supply Management</i>	If automatic battery testing management is enabled, batteries discharge proactively, and thus the backup power SLA risk warning may be mistakenly reported. If the mains supply fails, the batteries may not be fully charged. As a result, the remaining backup power use duration of the site is less than one hour when the backup power exhaustion risk warning is reported.
NR LTE FDD LTE TDD UMTS GSM	iLiBattScan	BATCTPA.ILIB ATTSCANSW (5G gNodeB, LTE eNodeB)	<i>Power Supply Management</i>	iLiBattScan triggers battery charging and discharge. If the power supply is disconnected during discharge, the backup power use duration is reduced, and an alarm related to Power Backup Perception (indicating a backup power exhaustion warning) is reported in advance, which may affect the normal operation of the base station.

19.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

Boards

None

RF Modules

None

Power Supply

Power Supply Type	Supported Cabinet Type	Supported Battery Type	Remarks
Huawei APM power system	• APM5 930(A C) • APM5 950H • APM5 950H-L	• Lead-acid battery • Lithium battery • Hybrid use of lead-acid and lithium batteries	Only one cabinet power system is used and provides power supply for the site at a time.
iSDU06D/iSDU03D used with a third-party power system	/	• Lead-acid battery • Lithium battery • Hybrid use of lead-acid and lithium batteries	• Only one cabinet power system is used and provides power supply for the site at a time. • The third-party power system needs to be interconnected with the iSDU06D/iSDU03D through a protocol. • The third-party power system can report the following information in real time: battery type, rated battery configuration, remaining battery power, and total DC power consumption of the cabinet.

NOTICE

- For details about the requirements for a third-party power system when it is interconnected with the iSDU06D/iSDU03D through a protocol, see [17 iSDU06D/iSDU03D Support for Connection with a Third-Party Power System](#).
- If the third-party power system is a blade power module, Backup Power Perception is not supported.

19.3.4 Networking

None

19.3.5 Others

- The base station software version must be SRAN20.0 or later.
- The MAE software version must be iMaster MAE V100R024C10 or later.

19.4 Operation and Maintenance

19.4.1 Data Configuration

19.4.1.1 Data Preparation

Before enabling Backup Power Perception, subscribe to the following 15-minute counters with a subscription period of one to three weeks (three weeks is recommended). If the subscription duration is less than one week, the feature cannot be enabled.

N o.	Scenario	Counter ID	Counter Name	Description
1	Scenarios where a Huawei APM cabinet is used	1593835640	VS.EnergyCons.BTS.Measuring.GSM	Power consumption measured by the power system of the GSM
2		1593835629	VS.EnergyCons.BTS.Measuring.UMTS	Power consumption measured by the power system of UMTS
3		1593835638	VS.EnergyCons.BTS.Measuring.LTE	Power consumption measured by the power system of the LTE
4		1593835805	VS.EnergyCons.BTS.Measuring.NR	Power consumption measured by the power system of NR

N o.	Scenario	Counter ID	Counter Name	Description
5	Scenarios where the iSDU06D/iSDU03D is used with a third-party power system	1593835942	VS.EPMU.DCEnergy	DC energy of the EPMU

19.4.1.2 Using MML Commands

Before using MML commands, refer to [19.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

- Power Backup Perception needs to be activated on the MAE-Access and cannot be activated using MML commands on the base station side. For details, see [19.4.1.3 Using the MAE-Access](#).
- For details about the data configuration for the iSDU06D/iSDU03D to be interconnected with a third-party power system through a protocol, see [17 iSDU06D/iSDU03D Support for Connection with a Third-Party Power System](#).

19.4.1.3 Using the MAE-Access

Prerequisites

The required licenses have been installed.

Activation Procedure

To enable this feature on the MAE-Access, perform the following steps:

1. Log in to the MAE iFaultCare module and choose **Risk Incident Settings**.
2. In the **NE Management** area, select the NE for which the feature is to be activated.
3. In the **Switch Settings** area, turn on the **Battery-caused Cell Outage Risk Prediction Switch** to enable the feature.
4. (Optional) In the **Policy Settings** area, turn on the **Backup power SLA Incident reporting switch** to enable the setting of the backup power SLA

duration. You can set the backup power SLA duration by modifying the setting of the **Backup power SLA Incident reporting threshold (min)**.

19.4.2 Activation Verification

Observing Counters

After the feature is activated, you can choose **MAE-Access > iFaultCare > Current Risk Incidents** and click **View backup duration** to observe the backup power use duration and power supply status of NEs on which this feature has been enabled.

19.4.3 Network Monitoring

None

20 PMU Support for Associated Temperature Control of Battery Cabinets

20.1 Principles

In scenarios where class 1 and class 2 power grids are used, the power failure duration is short, batteries are in float charging state for a long time, and they basically do not heat up. (With class 1 power grid, the average AC input power failure duration for communications equipment is less than 10 hours per month; with class 2 power grid, the average AC input power failure duration for communications equipment is less than 10 hours per week.) This function controls air conditioner cooling of the air-conditioned battery cabinet based on the temperature and service status to save energy.

After this function is enabled, the default cooling start temperature of the air conditioner of the air-conditioned battery cabinet is 25°C when batteries are in charge or discharge state and 35°C when the batteries are in float charging state. Only the APM5950H and APM5950H-L power systems support this function.

This function applies to GSM, UMTS, NR, and LTE. For GSM, only the eGBTS supports this function.

20.2 Network Analysis

20.2.1 Benefits

None

20.2.2 Impacts

The impacts between this function and other functions are described in [20.3.2.3 Function Impacts](#).

This function has no impact on the network.

20.3 Requirements

20.3.1 Licenses

None

20.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

20.3.2.1 Prerequisite Functions

None

20.3.2.2 Mutually Exclusive Functions

None

20.3.2.3 Function Impacts

None

20.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

Cabinets

- APM5950H and APM5950H-L
- BBC5600A battery cabinet

Boards

Only the PMU 11C supports associated temperature control of battery cabinets.

Power Supply Systems

- For details about the requirements of this function on the power supply system type, see "Hardware" in section "Basic Power Supply Management"

Functions" of *Power Supply Management* in the feature documentation of the corresponding RAT.

- This function is supported only when the **PMU.PTYPE** parameter is set to **ETP**.

20.3.4 Others

The base station software of SRAN19.0 or later is required.

20.4 Operation and Maintenance

20.4.1 Data Configuration

20.4.1.1 Data Preparation

The following table describes the parameters used for associated temperature control of battery cabinets.

Table 20-1 Parameters used for associated temperature control of battery cabinets

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	PMU.CN (5G gNodeB, LTE eNodeB)	Set this parameter based on the actual situation.
Subrack No.	PMU.SRN (5G gNodeB, LTE eNodeB)	Generally, the default value 7 is used. For the voltage boosting power distribution subrack, the value range is 17–18 .
Slot No.	PMU.SN (5G gNodeB, LTE eNodeB)	Generally, the default value 0 is used.
Power System Type	PMU.PTYPE (5G gNodeB, LTE eNodeB)	• The power supply system type must match the physical cabinet. For details about the types that can be configured, see "Mapping Between Physical Cabinets and PTYPE Values" in "Hardware" in section "Basic Power Supply Management Functions" in <i>Power Supply Management</i>. • When BATTERY.BTYPE (LTE eNodeB, 5G gNodeB) is set to VRLA_BAT or VRLA_BAT_MIX_LI_BAT, PMU.BVR (LTE eNodeB, 5G gNodeB) must be set to 53.5V or DEFAULT for the PMU that manages the batteries.

Parameter Name	Parameter ID	Setting Notes
Temperature Control Flag	PMU.TEMPCTRLFLAG (LTE eNodeB, 5G gNodeB)	The value ENABLE is recommended.

20.4.1.2 Using MML Commands

Before using MML commands, refer to [20.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

- Step 1** (Optional) Activate the PMU software. (The base station software version must meet the requirements of this function. For details, see [20.3.4 Others](#).)

```
//Activating the PMU software
ACT SOFTWARE: OT=BOARD, CN=0, SRN=7, SN=0;
//Checking whether the PMU software has been upgraded to the target version
DSP BRDVER: CN=0,SRN=7,SN=0;
```

- Step 2** Run the **MOD PMU** command with **PMU.TEMPCTRLFLAG (LTE eNodeB, 5G gNodeB)** set to **ENABLE**.

```
MOD PMU:CN=0,SRN=7,SN=0,PTYPE=ETP,TEMPCTRLFLAG=ENABLE;
```

----End

Deactivation Command Examples

Run the **MOD PMU** command with **PMU.TEMPCTRLFLAG (LTE eNodeB, 5G gNodeB)** set to **DISABLE**.

20.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

20.4.2 Activation Verification

Run the **LST PMU** command to query the value of **Temperature Control Flag**. If the value is the same as the configured one, this function has been successfully activated.

20.5 Network Monitoring

None

21 iSDU03S/iSDU03T Intelligent Coordinated Voltage Boosting Power Distribution

21.1 Principles

iSDU03S/iSDU03T intelligent coordinated voltage boosting power distribution enables the three intelligent power distribution ports of an iSDU03S/iSDU03T to perform voltage adjustment separately, and provides different power supply policies based on the capabilities of the RRUs or AAUs connected to these ports. This function is controlled by the **PDPATTR.VOLTAGEADJUSTMENTMODE (LTE eNodeB, 5G gNodeB)** parameter.

- If the **PDPATTR.VOLTAGEADJUSTMENTMODE (LTE eNodeB, 5G gNodeB)** parameter is set to **ADAPTIVE** for an intelligent power distribution port, this port enters the adaptive mode. In this case, the iSDU03S/iSDU03T dynamically adjusts the output voltage of this port based on the power supply capability of the connected RRU or AAU. The output voltage ranges from -57 V DC to -63 V DC.

The output voltage can be dynamically adjusted only when the intelligent power distribution port of the iSDU03S/iSDU03T matches the connected RRU or AAU through topology scanning. You can run the **DSP PDPATTR** command to query the serial number of the port to check whether the topology scanning is successful.

- If the value of **Equipment Serial Number(AUTO)** is not **NULL**, the intelligent power distribution port matches the connected RRU or AAU through topology scanning and the output voltage can be dynamically adjusted.
- If the value of **Equipment Serial Number(AUTO)** is **NULL**, the intelligent power distribution port fails to match the connected RRU or AAU through topology scanning and the output voltage is fixed at -57 V DC.
- If the **PDPATTR.VOLTAGEADJUSTMENTMODE (LTE eNodeB, 5G gNodeB)** parameter is set to **FIXED**, the intelligent power distribution port enters the fixed mode and its output voltage is fixed at -57 V DC.

This function applies to GSM, UMTS, LTE, and NR.

21.2 Network Analysis

21.2.1 Benefits

The voltage adjustment mode of the iSDU03S/iSDU03T can be set based on the power distribution of RF modules. The voltage is always –57 V DC in fixed mode while the maximum output voltage is –63 V DC in adaptive mode, which is larger and thereby reduces cable loss.

21.2.2 Impacts

The impacts between this function and other functions are described in [21.3.2.3 Function Impacts](#).

This function has no impact on the network.

21.3 Requirements

21.3.1 Licenses

None

21.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

21.3.2.1 Prerequisite Functions

None

21.3.2.2 Mutually Exclusive Functions

None

21.3.2.3 Function Impacts

None

21.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

Boards

No requirements

RF Modules

- Adaptive mode: The RRU or AAU must be able to match an intelligent power distribution port of the iSDU03S/iSDU03T through topology scanning. If the RRU or AAU fails to match the intelligent power distribution port through topology scanning, the output voltage is fixed at -57 V DC. For details about RRUs and AAUs that are able to match the intelligent power distribution ports of the iSDU03S/iSDU03T through topology scanning, contact Huawei engineers.
- Fixed mode: no requirements

Power Supply Systems

- RF modules must be powered by the iSDU03S/iSDU03T.
- Only ports LOAD0 to LOAD2 on the iSDU03S/iSDU03T support this function.

21.3.4 Others

None

21.4 Operation and Maintenance

21.4.1 Data Configuration

21.4.1.1 Data Preparation

[Table 21-1](#) describes the parameters used for function activation.

Table 21-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Voltage Adjustment Mode	PDPATTR.VOLTAG EADJUSTMENTMODE (LTE eNodeB, 5G gNodeB)	Use the default value. If a PMU is configured, the value ADAPTIVE takes effect by default.

 NOTE

If no PMU is configured, the adaptive mode of iSDU03S/iSDU03T intelligent coordinated voltage boosting power distribution takes effect by default.

21.4.1.2 Using MML Commands

Before using MML commands, refer to [21.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

```
//Enabling intelligent coordinated voltage boosting power distribution
ADD PMU: CN=0, SRN=17, PTYPE=ISDU, MPN=0, LSDF=DISABLE;
MOD PDPATTR: CN=0, SRN=17, SN=0, PORTNO=1, VOLTAGEADJUSTMENTMODE=ADAPTIVE;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling intelligent coordinated voltage boosting power distribution
MOD PDPATTR: CN=0, SRN=17, SN=0, PORTNO=1, VOLTAGEADJUSTMENTMODE=FIXED;
```

21.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

21.4.2 Activation Verification

- Step 1** Run the **LST PDPATTR** command to query the configuration of intelligent power distribution ports.
- Step 2** Run the **DSP PDPATTR** command to query the output voltage of intelligent power distribution ports.

----End

21.4.3 Network Monitoring

None

22 iSDU06D/iSDU03D Topology Recognition

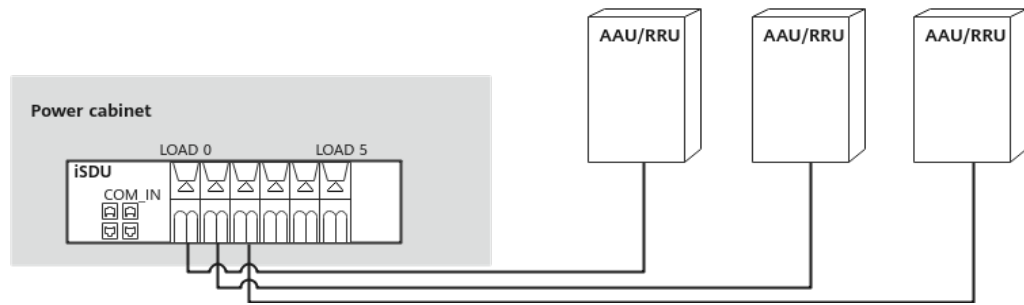
22.1 Principles

The iSDU, short for intelligent site digitalized unit, converts one –48 V DC input from a third-party power supply into multiple outputs to supply power to devices such as Huawei RRUs, AAUs, or BBUs.

The iSDU03D or iSDU06D performs topology pairing with the help of topology recognition. Topology pairing hereby refers to the pairing between power distribution ports on the iSDU and RF modules in terms of power supply. [Figure 22-1](#) shows the connection between the iSDU and RF modules. Topology recognition can be performed manually or through automatic topology scanning, which is specified by the **PDPATTR.TOPORECOGMODE (LTE eNodeB, 5G gNodeB)** parameter.

- If **PDPATTR.TOPORECOGMODE (LTE eNodeB, 5G gNodeB)** is set to **AUTO**, topology recognition is performed through automatic topology scanning. In this case, the iSDU automatically performs the pairing between power distribution ports on the iSDU and RF modules in terms of power supply.
- If **PDPATTR.TOPORECOGMODE (LTE eNodeB, 5G gNodeB)** is set to **MANUAL**, topology recognition is manually performed. In this case, **PDPATTR.ESN (5G gNodeB, LTE eNodeB)** is used to manually configure equipment serial numbers for RF modules connected with power distribution ports for pairing in terms of power supply.
- If **PDPATTR.TOPORECOGMODE (LTE eNodeB, 5G gNodeB)** is set to **CLOSE** for a port, topology recognition is not performed for this port.

Figure 22-1 Connection between the iSDU and RF modules



Automatic topology recognition supports the following functions:

- The iSDU saves the recognized topology upon successful automatic topology recognition. After a power-on reset is performed for the iSDU, the topology remains unchanged.
- If the conditions for triggering the power-off of intelligent power distribution ports on the iSDU are met, RF modules are powered off upon the port power-off. The value of **Equipment Serial Number(AUTO)** in the **DSP PDPATTR** command output still indicates the previous recognized topology.
- If the **PDPATTR.TOPORECOGMODE (LTE eNodeB, 5G gNodeB)** parameter value is changed from **MANUAL** or **CLOSE** to **AUTO** for certain intelligent power distribution ports on the iSDU, the BBU proactively queries the topology results of all ports for which this parameter is set to **AUTO** on the iSDU.

22.2 Network Analysis

22.2.1 Benefits

After topology recognition is successful, the connection between intelligent power distribution ports on the power supply device and powered RF modules can be clearly displayed.

22.2.2 Impacts

The impacts between this function and other functions are described in [22.3.2.3 Function Impacts](#).

In the case of automatic topology recognition, the following conditions must be met (In this section, a port refers to an intelligent power distribution port.):

- The RF modules and iSDU must meet installation requirements. For details, see *iSDU06D User Guide* and *iSDU03D User Guide* in *3900 & 5900 Series Base Station Product Documentation*.
- Automatic topology recognition can be performed only on the iSDU ports supplying current. The current of the ports can be queried by running the **DSP PDPATTR** command.
- If automatic topology recognition is enabled for iSDU ports but it fails after a period of time elapses, you need to manually configure the topology.

In the case of manual topology configuration, the following conditions must be met:

- If the RF module connected to a port is replaced with a new one, manual topology configuration is required again after the replacement.
- If the capacity of the RF module connected to a port is expanded, manual topology configuration is required again after the expansion.

22.3 Requirements

22.3.1 Licenses

None

22.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

22.3.2.1 Prerequisite Functions

None

22.3.2.2 Mutually Exclusive Functions

None

22.3.2.3 Function Impacts

None

22.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

Boards

None

RF Modules

Only 5000 series RF modules support topology recognition. Among them, the RRU5909 (BOM number: 02311PPP), RRU5909c (BOM number: 02314CHA), and RRU5905 (BOM number: 02312BCY) do not support automatic topology recognition performed by the iSDU.

Power Supply

Only intelligent power distribution ports support topology recognition. The intelligent power distribution ports of the iSDU03D are LOAD 0 to LOAD 2, and those of the iSDU06D are LOAD 0 to LOAD 5.

In the case of automatic topology recognition, the power cables of the RF modules connected to the iSDU must be -48 V DC shielded power cables.

22.3.4 Others

In the case of automatic topology recognition, the following conditions must be met:

- Topology recognition may fail if the iSDU is not installed according to the installation process. Specifically:
 - If the iSDU is powered on before the RF module power cables are properly connected, topology recognition may fail, and manual topology configuration is required.
 - If an RF module is connected to an intelligent power distribution port on the iSDU using the power cable inconsistent with that specified in the power supply requirements, topology recognition may fail. In this case, the power cable must be replaced with one that meets the requirements or manual topology configuration is required.
 - If the RF module connected to an intelligent power distribution port on the iSDU is replaced with a new one and the RF module is in the power-off state for less than 30 seconds, the topology cannot be updated and needs to be manually configured.
 - If an RF module is connected to an intelligent power distribution port on the iSDU using the power cable that experiences an exception (for example, the power cable is loosely connected or the sheath is pressed), topology recognition fails, and carbonization may occur. In this case, the exception needs to be rectified first.
- If specific intelligent power distribution ports on the iSDU or the entire subrack is powered off, the topology cannot be recognized. In this case, the topology cannot be updated and needs to be manually configured if any of the following conditions is met: (1) RF modules are replaced. (2) RF modules become out of service and are removed. (3) Port assignments are swapped between RF modules.

22.4 Operation and Maintenance

22.4.1 Data Configuration

22.4.1.1 Data Preparation

Before topology recognition is enabled for the iSDU03D or iSDU06D, ensure that the connected RF modules meet the corresponding requirements.

Parameter Name	Parameter ID	Setting Notes
Topology Recognition Mode	PDPATTR.TOPORECOGMODE (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan.
Equipment Serial Number	PDPATTR.ESN (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.

22.4.1.2 Using MML Commands

Before using MML commands, refer to [22.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

```
//Setting the topology recognition mode of the power distribution port whose port, slot, subrack, and cabinet numbers are 0, 0, 17, and 0, respectively, to AUTO  
MOD PDPATTR: CN=0, SRN=17, SN=0, PORTNO=0, TOPORECOGMODE=AUTO;  
//(Optional) Setting the equipment serial number of the iSDU port supplying power to an RF module that does not support automatic topology scanning, for example, manually setting the equipment serial number of port 0  
MOD PDPATTR: CN=0, SRN=17, SN=0, PORTNO=0, TOPORECOGMODE=MANUAL, ESN="XXXXXXXXXXXXXXXXXXXX";
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Setting the topology recognition mode of the power distribution port whose port, slot, subrack, and cabinet numbers are 0, 0, 17, and 0, respectively, to CLOSE  
MOD PDPATTR: CN=0, SRN=17, SN=0, PORTNO=0, TOPORECOGMODE=CLOSE;
```

22.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

22.4.2 Activation Verification

If **PDPATTR.TOPORECOGMODE (LTE eNodeB, 5G gNodeB)** is set to **AUTO**, run the **DSP PDPATTR** command to query the values of **Equipment Serial**

Number(AUTO) of power distribution ports with a specified status in the power subrack. If the RF module ESNs are displayed, automatic topology recognition is successful. If **NULL** is displayed, the recognition fails.

If **PDPATTR.TOPORECOGMODE (LTE eNodeB, 5G gNodeB)** is set to **MANUAL**, run the **DSP PDPATTR** command to query the values of **Equipment Serial Number(MANUAL)** of power distribution ports with a specified status in the power subrack. If the RF module ESNs are displayed, manual topology configuration is successful. If **NULL** is displayed, the configuration fails.

22.4.3 Network Monitoring

None

23 Coordinated Power Backup of the Entire Site in Combination of the Base Station and Microwave

23.1 Principles

If the mains supply is disconnected and the base station enters the backup power saving state, the microwave device works with the base station so that the power saving status of the base station can be transmitted to the microwave device. In this way, the microwave device also enters the power saving state, increasing the backup power use duration of the entire site.

This function is controlled by the **BKP_PSAV_EVALUATE_DATA_SW** option of the **LLDPGLOBAL.BTSMWCOSW (5G gNodeB, LTE eNodeB)** parameter on the base station side. If this option is selected and the microwave device's backup power subscription switch is turned on, the base station sends its backup power saving status to the microwave device, and the microwave device determines whether to enter the power saving state based on the backup power saving status of the base station.

The base station supports the following backup power saving functions:

- Low power consumption mode (LTE)
- Low power consumption mode (NR)
- SOC level-based low power consumption mode (LTE)

For details about principles on the microwave side, see the corresponding microwave product documentation. For details about the requirements on the microwave side, see [23.3.4 Others](#).

23.2 Network Analysis

23.2.1 Benefits

This function increases the backup power use duration.

23.2.2 Impacts

The impacts between this function and other functions are described in [23.3.2.3 Function Impacts](#).

This function has no impact on the base station.

23.3 Requirements

23.3.1 Licenses

None

23.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

23.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	SOC level-based low power consumption mode	• IPWRM.SOCLVLADJSW • SOCLVL_BASED_LOW_POWER_SWITCH option of the CellLowPower.LowPowerConsumptionModeExtSw parameter	<i>Green Site</i>	Coordinated power backup of the entire site in combination of the base station and microwave can be enabled only when low power consumption mode is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	LTE low power consumption mode	CellLowPower.LowPowerSwitch	<i>Green Site</i>	Coordinated power backup of the entire site in combination of the base station and microwave can be enabled only when low power consumption mode is in effect.
NR FDD Low-frequency NR TDD	NR low power consumption mode	LOW_PWR_MODE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Green Site</i>	Coordinated power backup of the entire site in combination of the base station and microwave can be enabled only when low power consumption mode is in effect.

23.3.2.2 Mutually Exclusive Functions

None

23.3.2.3 Function Impacts

None

23.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

Boards

No requirements

RF Modules

None

Power Supply

The base station supports only the following backup power saving functions on the condition that the power supply requirements for them are met.

- For details about the power supply requirements for low power consumption mode (LTE), see [Power Supply](#).
- For details about the power supply requirements for low power consumption mode (NR), see [Power Supply](#).
- For details about the power supply requirements for SOC level-based low power consumption mode (LTE), see [Power Supply](#).

23.3.4 Others

Coordinated power backup of the entire site in combination of the base station and microwave can be enabled only if a compatible microwave device is used and the corresponding switch on the microwave device is turned on.

 NOTE

For details about microwave-related functions, see "ECOWave" in *Feature Description* in the corresponding microwave product documentation.

23.4 Operation and Maintenance

23.4.1 Data Configuration

23.4.1.1 Data Preparation

[Table 23-1](#) describes the parameters used for function activation.

Table 23-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Base Station Microwave Collaboration Switch	LLDPGLOBAL.BTSMWCOSW (5G gNodeB, LTE eNodeB)	Select the BKP_PSAV_EVALUATE_DATA_SW option to enable coordinated power backup of the entire site in combination of the base station and microwave. If this option needs to be selected, ensure that the peer end is a directly connected Huawei microwave device and the corresponding switch on this peer device is also turned on. Otherwise, this function does not take effect.

23.4.1.2 Using MML Commands

Before using MML commands, refer to [23.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency,

and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling coordinated power backup of the entire site in combination of the base station and microwave  
SET LLDPGLOBALINFO: BTSMWCOSW=BKP_PSAV_EVALUATE_DATA_SW-1;
```

Deactivation Command Examples

```
//Disabling coordinated power backup of the entire site in combination of the base station and microwave  
SET LLDPGLOBALINFO: BTSMWCOSW=BKP_PSAV_EVALUATE_DATA_SW-0;
```

23.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

23.4.2 Activation Verification

Whether this function has taken effect is verified on the microwave side instead of the base station side.

23.4.3 Network Monitoring

None

24 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

25 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

26 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

27 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *License Control Item Lists (FDD) in eRAN Feature Documentation*
- *License Control Item Lists (CIoT) in eRAN Feature Documentation*
- *License Control Item Lists (TDD) in eRAN Feature Documentation*
- *Energy Conservation and Emission Reduction in eRAN Feature Documentation*
- *Downlink Scheduling in eRAN Feature Documentation*
- *Power Control in eRAN Feature Documentation*
- *MIMO in eRAN Feature Documentation*
- *Dynamic Power Sharing Between LTE Carriers in eRAN Feature Documentation*
- *Carrier Aggregation in eRAN Feature Documentation*
- *D-MIMO (TDD) in eRAN Feature Documentation*
- *SFN in eRAN Feature Documentation*
- *Soft Split Resource Duplex (TDD) in eRAN Feature Documentation*
- *Multi-carrier Unified Scheduling in eRAN Feature Documentation*
- *TMA (TDD) in eRAN Feature Documentation*
- *Multi-RAT Coordinated Energy Saving in eRAN Feature Documentation*
- *Synchronization in eRAN Feature Documentation*
- *Smart 8T8R (FDD) in eRAN Feature Documentation*
- *Interference Detection and Suppression in eRAN Feature Documentation*
- *Low-Band Booster in eRAN Feature Documentation*
- *Massive MIMO Enhancements (FDD) in eRAN Feature Documentation*
- *NB-IoT Basics (FDD) in eRAN Feature Documentation*
- *Seamless Intra-Band Carrier Joining (FDD) in eRAN Feature Documentation*
- *Seamless Intra-Band Carrier Joining (TDD) in eRAN Feature Documentation*
- *Intelligent Multi-Beam of 8T8R (TDD) in eRAN Feature Documentation*
- *CMAS in eRAN Feature Documentation*
- *ETWS in eRAN Feature Documentation*
- *High Speed Mobility in eRAN Feature Documentation*

- *RuralLink Solution in SRAN Feature Documentation*
- *LumiLink Solution in SRAN Feature Documentation*
- *Power Supply Management in SRAN Feature Documentation*
- *Multi-RAT Coordinated Energy Saving in SRAN Feature Documentation*
- *Base Station Equipment Reliability in SRAN Feature Documentation*
- *LTE and NR Power Allocation in SRAN Feature Documentation*
- *UMTS and LTE Dynamic Power Sharing in SRAN Feature Documentation*
- *LTE FDD and NR Inter-RAT Unified Scheduling in SRAN Feature Documentation*
- *LTE and NR Zero Bufferzone in SRAN Feature Documentation*
- *RuralCow Solution in SRAN Feature Documentation*
- *License Control Item Lists in 5G RAN Feature Documentation*
- *Energy Conservation and Emission Reduction in 5G RAN Feature Documentation*
- *MRO in 5G RAN Feature Documentation*
- *URLLC in 5G RAN Feature Documentation*
- *Remote Interference Management (Low-Frequency TDD) in 5G RAN Feature Documentation*
- *MIMO (TDD) in 5G RAN Feature Documentation*
- *NR Inter-Carrier Dynamic Power Allocation in 5G RAN Feature Documentation*
- *Beam Management in 5G RAN Feature Documentation*
- *CoMP in 5G RAN Feature Documentation*
- *Downlink Scheduling in 5G RAN Feature Documentation*
- *Smart 8T8R (FDD) in 5G RAN Feature Documentation*
- *Interference Avoidance in 5G RAN Feature Documentation*
- *Channel Management in 5G RAN Feature Documentation*
- *LTE FDD and NR Hybrid Dynamic Spectrum Sharing in 5G RAN Feature Documentation*
- *Distributed Antenna Solutions (Low-Frequency TDD) in 5G RAN Feature Documentation*
- *Emergency Communications Solution (Low-Frequency TDD) in 5G RAN Feature Documentation*
- *Green BTS in RAN Feature Documentation*
- *Narrowband Interference Suppression in RAN Feature Documentation*
- *Coverage Compensation for Site Outage in RAN Feature Documentation*
- *Intelligent Power Consumption Decrease in GBSS Feature Documentation*
- *Power Supply Management in GBSS Feature Documentation*
- *Dynamic Cell Power Off in GBSS Feature Documentation*
- *iSDU06D User Guide in 3900 & 5900 Series Base Station Product Documentation*
- *iSDU03D User Guide in 3900 & 5900 Series Base Station Product Documentation*